

BRK (00) Cycles: 7 * Size: 2		
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	** Store as OpCode. PushB = true.
1.1		Inc. PC .
1.2	Read PC	Discard.
2.1		Inc. PC .
2.2	*** Write to u8(S +0) \$0100	Write PC .H.
3.1		
3.2	*** Write to u8(S -1) \$0100	Write PC .L.
4.1		Use Interrupt flags to select Vector. If RESET, Vector = \$FFFC. Else if NMI, Vector = \$FFFA. Else if IRQ/BRK, Vector = \$FFFE. If not BRK, PushB = false.
4.2	*** Write to u8(S -2) \$0100	If PushB, Write (P (X flag)), otherwise write P .
5.1		Dec. S by 3.
5.2	Read Vector	Store as Address.L.
6.1		I flag = 1, M flag = 1, X flag = 0. Enable updates to PC (disabled by NMI/IRQ).
6.2	Read Vector + 1	Store as Address.H.
7.1		PC = Address.
7.2	Read PC	Store as OpCode.
+X.1		

* Concerning the BRK instruction, you should also note that although its second byte is basically a “don’t care” byte – that is, it can have any value - the BRK (and COP instruction as well) is a two-byte instruction, the second byte sometimes is used as a signature byte to determine the nature of the BRK being executed. When an RTI instruction is executed, control always returns to the second byte past the BRK opcode. — assembly-programming-manual-for-w65c816.pdf

** If NMI, IRQ, or RESET, OpCode is forced to 0 and writes to PC are disabled. This is done on ALL “Store as Opcode” half-cycles, but only noted here.

*** If handling a RESET, the writes turn into reads. The databus is populated but ignored.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	00	1	BRK	0006	f0	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	a0
7	0006	00	1	BRK	0006	f0	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	98
8	0007	09	1		0007	f0	4c	e8	fd	nv-BdIzc	BRK	T2	98	98	98
8	0007	09	1		0007	f0	4c	e8	fd	nv-BdIzc	BRK	T2	98	98	30
9	01fd	09	0		0008	f0	4c	e8	fd	nv-BdIzc	BRK	T3	ff	fd	30
9	01fd	00	0		0008	f0	4c	e8	fd	nv-BdIzc	BRK	T3	ff	fd	fc
10	01fc	09	0		0008	f0	4c	e8	fd	nv-BdIzc	BRK	T4	ff	fc	fc
10	01fc	08	0		0008	f0	4c	e8	fd	nv-BdIzc	BRK	T4	ff	fc	fb
11	01fb	09	0		0008	f0	4c	e8	fd	nv-BdIzc	BRK	T5	ff	fb	fb
11	01fb	34	0		0008	f0	4c	e8	fd	nv-BdIzc	BRK	T5	ff	fb	fa
12	ffff	00	1		0008	f0	4c	e8	fa	nv-BdIzc	BRK		00	ff	fa
12	ffff	00	1		0008	f0	4c	e8	fa	nv-BdIzc	BRK		00	ff	ff
13	ffff	00	1		0008	f0	4c	e8	fa	nv-BdIzc	BRK	T0	00	00	ff
13	ffff	00	1		0008	f0	4c	e8	fa	nv-BdIzc	BRK	T0	00	00	00
14	0000	a9	1	LDA #	0000	f0	4c	e8	fa	nv-BdIzc	BRK	T1	ff	00	00
14	0000	a9	1	LDA #	0000	f0	4c	e8	fa	nv-BdIzc	BRK	T1	ff	00	ff
15	0001	f0	1		0001	f0	4c	e8	fa	nv-BdIzc	LDA #	T0+T2	ff	ff	ff
15	0001	f0	1		0001	f0	4c	e8	fa	nv-BdIzc	LDA #	T0+T2	ff	ff	fe

ORA (01)		Cycles: 6	Size: 2
Indirect X (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Operand.	
2.1		Inc. PC .	
2.2	Read Operand	Discard.	
3.1		Pointer = u8(Operand + X).	
3.2	Read Pointer	Store as Address.	
4.1			
4.2	Read u8(Pointer + 1)	Store as Address.H.	
5.1			
5.2	Read Address	Store as Operand.	
6.1		A = Operand. N flag = (A & \$80), Z flag = ! A .	
6.2	Read PC	Store as OpCode.	
+X.1		* A , N flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	01	1	ORA (zp,X)	0006	f0	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	a0
7	0006	01	1	ORA (zp,X)	0006	f0	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	98
8	0007	11	1		0007	f0	4c	e8	fd	nv-BdIzc	ORA (zp,X)	T2	98	98	98
8	0007	11	1		0007	f0	4c	e8	fd	nv-BdIzc	ORA (zp,X)	T2	98	98	30
9	0011	88	1		0008	f0	4c	e8	fd	nv-BdIzc	ORA (zp,X)	T3	4c	11	30
9	0011	88	1		0008	f0	4c	e8	fd	nv-BdIzc	ORA (zp,X)	T3	4c	11	5d
10	005d	33	1		0008	f0	4c	e8	fd	nv-BdIzc	ORA (zp,X)	T4	00	5d	5d
10	005d	33	1		0008	f0	4c	e8	fd	nv-BdIzc	ORA (zp,X)	T4	00	5d	5e
11	005e	00	1		0008	f0	4c	e8	fd	nv-BdIzc	ORA (zp,X)	T5	00	33	5e
11	005e	00	1		0008	f0	4c	e8	fd	nv-BdIzc	ORA (zp,X)	T5	00	33	33
12	0033	55	1		0008	f0	4c	e8	fd	nv-BdIzc	ORA (zp,X)	T0	ff	ff	33
12	0033	55	1		0008	f0	4c	e8	fd	nv-BdIzc	ORA (zp,X)	T0	ff	ff	fe
13	0008	10	1	BPL	0008	f0	4c	e8	fd	nv-BdIzc	ORA (zp,X)	T1	f0	55	fe
13	0008	10	1	BPL	0008	f0	4c	e8	fd	nv-BdIzc	ORA (zp,X)	T1	f0	55	f5
14	0009	00	1		0009	f5	4c	e8	fd	Nv-BdIzc	BPL	T2	f5	f5	f5
14	0009	00	1		0009	f5	4c	e8	fd	Nv-BdIzc	BPL	T2	f5	f5	ea

JAM (02)	Cycles: ∞	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Discard.
2.1		Visual6502 suggests incrementing PC , but Single-Step Tests cannot be passed this way.
2.2	Read \$FFFF	Discard.
3.1		
3.2	Read \$FFFE	Discard.
4.1		
4.2	Read \$FFFE	Discard.
5.1		Repeat 5.1 and 5.2 forever.
5.2	Read \$FFFF	Discard.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	02	1	unknown	0006	f0	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	a0
7	0006	02	1	unknown	0006	f0	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	98
8	0007	11	1		0007	f0	4c	e8	fd	nv-BdIzc	unknown	T2	98	98	98
8	0007	11	1		0007	f0	4c	e8	fd	nv-BdIzc	unknown	T2	98	98	30
9	ffff	00	1		0008	f0	4c	e8	fd	nv-BdIzc	unknown	T3	ff	ff	30
9	ffff	00	1		0008	f0	4c	e8	fd	nv-BdIzc	unknown	T3	ff	ff	fe
10	fffe	00	1		0008	f0	4c	e8	fd	nv-BdIzc	unknown	T4	ff	ff	fe
10	fffe	00	1		0008	f0	4c	e8	fd	nv-BdIzc	unknown	T4	ff	ff	fe
11	fffe	00	1		0008	f0	4c	e8	fd	nv-BdIzc	unknown	T5	ff	ff	fe
11	fffe	00	1		0008	f0	4c	e8	fd	nv-BdIzc	unknown	T5	ff	ff	fe
12	ffff	00	1		0008	f0	4c	e8	fd	nv-BdIzc	unknown		ff	ff	fe
12	ffff	00	1		0008	f0	4c	e8	fd	nv-BdIzc	unknown		ff	ff	fe
13	ffff	00	1		0008	f0	4c	e8	fd	nv-BdIzc	unknown		ff	ff	fe
13	ffff	00	1		0008	f0	4c	e8	fd	nv-BdIzc	unknown		ff	ff	fe
14	ffff	00	1		0008	f0	4c	e8	fd	nv-BdIzc	unknown		ff	ff	fe
14	ffff	00	1		0008	f0	4c	e8	fd	nv-BdIzc	unknown		ff	ff	fe

SLO (03) Cycles: 8		Size: 2
Indirect X (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC .
2.2	Read Operand	Discard.
3.1		Pointer = u8(Operand + X).
3.2	Read Pointer	Store as Address.
4.1		
4.2	Read u8(Pointer + 1)	Store as Address.H.
5.1		
5.2	Read Address	Store as Operand.
6.1		
6.2	Write to Address	Write Operand.
7.1		C flag = (Operand & \$80). Operand <<= 1. A = Operand. N flag = (A & \$80), Z flag = ! A .
7.2	Write to Address	Write Operand.
8.1		
8.2	Read PC	Store as OpCode.
+X.1		* A applied.

* Setting of A is delayed by 2 cycles until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	03	1	unknown	0006	00	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	a0
7	0006	03	1	unknown	0006	00	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	98
8	0007	11	1		0007	00	4c	e8	fd	nv-BdIzc	unknown	T2	98	98	98
8	0007	11	1		0007	00	4c	e8	fd	nv-BdIzc	unknown	T2	98	98	30
9	0011	88	1		0008	00	4c	e8	fd	nv-BdIzc	unknown	T3	4c	11	30
9	0011	88	1		0008	00	4c	e8	fd	nv-BdIzc	unknown	T3	4c	11	5d
10	005d	33	1		0008	00	4c	e8	fd	nv-BdIzc	unknown	T4	00	5d	5d
10	005d	33	1		0008	00	4c	e8	fd	nv-BdIzc	unknown	T4	00	5d	5e
11	005e	00	1		0008	00	4c	e8	fd	nv-BdIzc	unknown	T5	00	33	5e
11	005e	00	1		0008	00	4c	e8	fd	nv-BdIzc	unknown	T5	00	33	33
12	0033	c0	1		0008	00	4c	e8	fd	nv-BdIzc	unknown		ff	ff	33
12	0033	c0	1		0008	00	4c	e8	fd	nv-BdIzc	unknown		ff	ff	fe
13	0033	c0	0		0008	00	4c	e8	fd	nv-BdIzc	unknown		c0	c0	fe
13	0033	c0	0		0008	00	4c	e8	fd	nv-BdIzc	unknown		c0	c0	80
14	0033	c0	0		0008	00	4c	e8	fd	Nv-BdIzc	unknown	T0	80	80	80
14	0033	80	0		0008	00	4c	e8	fd	Nv-BdIzc	unknown	T0	80	80	00
15	0008	30	1	BMI	0008	00	4c	e8	fd	Nv-BdIzc	unknown	T1	00	80	00
15	0008	30	1	BMI	0008	00	4c	e8	fd	Nv-BdIzc	unknown	T1	00	80	80
16	0009	00	1		0009	80	4c	e8	fd	Nv-BdIzc	BMI	T2	80	80	80
16	0009	00	1		0009	80	4c	e8	fd	Nv-BdIzc	BMI	T2	80	80	00

NOP (04)		Cycles: 3	Size: 2
Zero Page (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read Address	Store as Operand.	
3.1			
3.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	04	1	unknown	0006	00	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	a0
7	0006	04	1	unknown	0006	00	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	98
8	0007	11	1		0007	00	4c	e8	fd	nv-BdIzc	unknown	T2	98	98	98
8	0007	11	1		0007	00	4c	e8	fd	nv-BdIzc	unknown	T2	98	98	30
9	0011	88	1		0008	00	4c	e8	fd	nv-BdIzc	unknown	T0	ff	11	30
9	0011	88	1		0008	00	4c	e8	fd	nv-BdIzc	unknown	T0	ff	11	10
10	0008	30	1	BMI	0008	00	4c	e8	fd	nv-BdIzc	unknown	T1	ff	88	10
10	0008	30	1	BMI	0008	00	4c	e8	fd	nv-BdIzc	unknown	T1	ff	88	87
11	0009	00	1		0009	00	4c	e8	fd	nv-BdIzc	BMI	T2	87	87	87
11	0009	00	1		0009	00	4c	e8	fd	nv-BdIzc	BMI	T2	87	87	0e

ORA (05)		Cycles: 3	Size: 2
Zero Page (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read Address	Store as Operand.	
3.1		A = Operand. N flag = (A & \$80), Z flag = ! A .	
3.2	Read PC	Store as OpCode.	
+X.1		* A , N flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	05	1	ORA zp	0006	33	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	a0
7	0006	05	1	ORA zp	0006	33	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	98
8	0007	11	1		0007	33	4c	e8	fd	nv-BdIzc	ORA zp	T2	98	98	98
8	0007	11	1		0007	33	4c	e8	fd	nv-BdIzc	ORA zp	T2	98	98	30
9	0011	88	1		0008	33	4c	e8	fd	nv-BdIzc	ORA zp	T0	ff	11	30
9	0011	88	1		0008	33	4c	e8	fd	nv-BdIzc	ORA zp	T0	ff	11	10
10	0008	30	1	BMI	0008	33	4c	e8	fd	nv-BdIzc	ORA zp	T1	33	88	10
10	0008	30	1	BMI	0008	33	4c	e8	fd	nv-BdIzc	ORA zp	T1	33	88	bb
11	0009	00	1		0009	bb	4c	e8	fd	Nv-BdIzc	BMI	T2	bb	bb	bb
11	0009	00	1		0009	bb	4c	e8	fd	Nv-BdIzc	BMI	T2	bb	bb	76

ASL (06)		Cycles: 5	Size: 2
Zero Page (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read Address	Store as Operand.	
3.1			
3.2	Write to Address	Write Operand.	
4.1		C flag = (Operand & \$80). Operand <= 1. N flag = (Operand & \$80), Z flag = !Operand.	
4.2	Write to Address	Write Operand.	
5.1			
5.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	06	1	ASL zp	0006	33	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	a0
7	0006	06	1	ASL zp	0006	33	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	98
8	0007	11	1		0007	33	4c	e8	fd	nv-BdIzc	ASL zp	T2	98	98	98
8	0007	11	1		0007	33	4c	e8	fd	nv-BdIzc	ASL zp	T2	98	98	30
9	0011	7f	1		0008	33	4c	e8	fd	nv-BdIzc	ASL zp	T3	ff	11	30
9	0011	7f	1		0008	33	4c	e8	fd	nv-BdIzc	ASL zp	T3	ff	11	10
10	0011	7f	0		0008	33	4c	e8	fd	nv-BdIzc	ASL zp	T4	7f	7f	10
10	0011	7f	0		0008	33	4c	e8	fd	nv-BdIzc	ASL zp	T4	7f	7f	fe
11	0011	7f	0		0008	33	4c	e8	fd	Nv-BdIzc	ASL zp	T0	fe	fe	fe
11	0011	fe	0		0008	33	4c	e8	fd	Nv-BdIzc	ASL zp	T0	fe	fe	fc
12	0008	30	1	BMI	0008	33	4c	e8	fd	Nv-BdIzc	ASL zp	T1	ff	fe	fc
12	0008	30	1	BMI	0008	33	4c	e8	fd	Nv-BdIzc	ASL zp	T1	ff	fe	fd
13	0009	00	1		0009	33	4c	e8	fd	Nv-BdIzc	BMI	T2	fd	fd	fd
13	0009	00	1		0009	33	4c	e8	fd	Nv-BdIzc	BMI	T2	fd	fd	fa

SLO (07)	Cycles: 5	Size: 2
Zero Page (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read Address	Store as Operand.
3.1		
3.2	Write to Address	Write Operand.
4.1		C flag = (Operand & \$80). Operand <= 1. A = Operand. N flag = (A & \$80), Z flag = ! A .
4.2	Write to Address	Write Operand.
5.1		
5.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	07	1	unknown	0006	33	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	a0
7	0006	07	1	unknown	0006	33	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	98
8	0007	11	1		0007	33	4c	e8	fd	nv-BdIzc	unknown	T2	98	98	98
8	0007	11	1		0007	33	4c	e8	fd	nv-BdIzc	unknown	T2	98	98	30
9	0011	fe	1		0008	33	4c	e8	fd	nv-BdIzc	unknown	T3	ff	11	30
9	0011	fe	1		0008	33	4c	e8	fd	nv-BdIzc	unknown	T3	ff	11	10
10	0011	fe	0		0008	33	4c	e8	fd	nv-BdIzc	unknown	T4	fe	fe	10
10	0011	fe	0		0008	33	4c	e8	fd	nv-BdIzc	unknown	T4	fe	fe	fc
11	0011	fe	0		0008	33	4c	e8	fd	Nv-BdIzc	unknown	T0	fc	fc	fc
11	0011	fc	0		0008	33	4c	e8	fd	Nv-BdIzc	unknown	T0	fc	fc	f8
12	0008	8c	1	STY Abs	0008	33	4c	e8	fd	Nv-BdIzc	unknown	T1	33	fc	f8
12	0008	8c	1	STY Abs	0008	33	4c	e8	fd	Nv-BdIzc	unknown	T1	33	fc	ff
13	0009	44	1		0009	ff	4c	e8	fd	Nv-BdIzc	STY Abs	T2	ff	ff	ff
13	0009	44	1		0009	ff	4c	e8	fd	Nv-BdIzc	STY Abs	T2	ff	ff	fe

PHP (08)		Cycles: 3	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1	Read PC	Inc. PC .	
1.2		Discard.	
2.1		Operand = (P \$30).	
2.2		Write Operand.	
3.1		Dec. S by 1.	
3.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	08	1	PHP	0006	33	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	a0
7	0006	08	1	PHP	0006	33	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	98
8	0007	ea	1		0007	33	4c	e8	fd	nv-BdIzc	PHP	T2	98	98	98
8	0007	ea	1		0007	33	4c	e8	fd	nv-BdIzc	PHP	T2	98	98	30
9	01fd	ea	0		0007	33	4c	e8	fd	nv-BdIzc	PHP	T0	ff	fd	30
9	01fd	34	0		0007	33	4c	e8	fd	nv-BdIzc	PHP	T0	ff	fd	fc
10	0007	ea	1	NOP	0007	33	4c	e8	fc	nv-BdIzc	PHP	T1	00	34	fc
10	0007	ea	1	NOP	0007	33	4c	e8	fc	nv-BdIzc	PHP	T1	00	34	34
11	0008	8c	1		0008	33	4c	e8	fc	nv-BdIzc	NOP	T0+T2	34	34	34
11	0008	8c	1		0008	33	4c	e8	fc	nv-BdIzc	NOP	T0+T2	34	34	68

ORA (09)		Cycles: 2	Size: 2
Immediate			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Operand.	
2.1		Inc. PC . A = Operand. N flag = (A & \$80), Z flag = ! A .	
2.2	Read PC	Store as OpCode.	
+X.1		* A , N flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	09	1	ORA #	0006	33	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	a0
7	0006	09	1	ORA #	0006	33	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	98
8	0007	ea	1		0007	33	4c	e8	fd	nv-BdIzc	ORA #	T0+T2	98	98	98
8	0007	ea	1		0007	33	4c	e8	fd	nv-BdIzc	ORA #	T0+T2	98	98	30
9	0008	8c	1	STY Abs	0008	33	4c	e8	fd	nv-BdIzc	ORA #	T1	33	ea	30
9	0008	8c	1	STY Abs	0008	33	4c	e8	fd	nv-BdIzc	ORA #	T1	33	ea	fb
10	0009	44	1		0009	fb	4c	e8	fd	Nv-BdIzc	STY Abs	T2	fb	fb	fb
10	0009	44	1		0009	fb	4c	e8	fd	Nv-BdIzc	STY Abs	T2	fb	fb	f6

ASL (0A)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Discard.	
2.1		C flag = (A & \$80). A <= 1. N flag = (A & \$80), Z flag = ! A .	
2.2	Read PC	Store as OpCode.	
+X.1		* A , N flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	0a	1	ASL	0006	7f	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	a0
7	0006	0a	1	ASL	0006	7f	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	98
8	0007	ea	1		0007	7f	4c	e8	fd	nv-BdIzc	ASL	T0+T2	98	98	98
8	0007	ea	1		0007	7f	4c	e8	fd	nv-BdIzc	ASL	T0+T2	98	98	30
9	0007	ea	1	NOP	0007	7f	4c	e8	fd	nv-BdIzc	ASL	T1	7f	7f	30
9	0007	ea	1	NOP	0007	7f	4c	e8	fd	nv-BdIzc	ASL	T1	7f	7f	fe
10	0008	8c	1		0008	fe	4c	e8	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fe
10	0008	8c	1		0008	fe	4c	e8	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fc

ANC (0B)		Cycles: 2	Size: 2
Immediate			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1	Read PC	Inc. PC.	
1.2		Store as Operand.	
2.1		Inc. PC. A &= Operand. C flag = (A & \$80), N flag = (A & \$80), Z flag = !A.	
2.2		Store as OpCode.	
+X.1		* A, C flag, N flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.
Visual6502 omits the update of A; flags are updated based off the incorrect A value.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	0b	1	unknown	0006	81	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	a0
7	0006	0b	1	unknown	0006	81	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	98
8	0007	f0	1		0007	81	4c	e8	fd	nv-BdIzc	unknown	T0+T2	98	98	98
8	0007	f0	1		0007	81	4c	e8	fd	nv-BdIzc	unknown	T0+T2	98	98	30
9	0008	8c	1	STY Abs	0008	81	4c	e8	fd	nv-BdIzc	unknown	T1	80	80	30
9	0008	8c	1	STY Abs	0008	81	4c	e8	fd	nv-BdIzc	unknown	T1	81	81	81
10	0009	44	1		0009	81	4c	e8	fd	Nv-BdIzC	STY Abs	T2	81	81	81
10	0009	44	1		0009	81	4c	e8	fd	Nv-BdIzC	STY Abs	T2	81	81	02

ANC (ANC2, ANA, ANB)

Type: Combination of an immediate and an implied command (Sub-instructions: AND, ASL/ROL)

Op.	Mnemonic	Function	Size	Cycles	N	V	-	B	D	I	Z	C
\$0B	ANC #imm	A = A & #{imm}	2	2	o						o	o
\$2B	ANC #imm	A = A & #{imm}	2	2	o						o	o

Operation: ANDs the contents of the A register with an immediate value and then moves bit 7 of A into the Carry flag.

- This opcode works basically identically to AND #imm. except that the Carry flag is set to the same state that the Negative flag is set to. (bit 7 is put into the carry, as if the ASL/ROL would have been executed)

NOP (0C)		Cycles: 4	Size: 3
Absolute (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read PC	Store as Address.H.	
3.1		Inc. PC .	
3.2	Read Address	Store as Operand.	
4.1			
4.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	0c	1	unknown	0006	81	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	a0
7	0006	0c	1	unknown	0006	81	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	98
8	0007	f0	1		0007	81	4c	e8	fd	nv-BdIzc	unknown	T2	98	98	98
8	0007	f0	1		0007	81	4c	e8	fd	nv-BdIzc	unknown	T2	98	98	30
9	0008	b8	1		0008	81	4c	e8	fd	nv-BdIzc	unknown	T3	00	f0	30
9	0008	b8	1		0008	81	4c	e8	fd	nv-BdIzc	unknown	T3	00	f0	f0
10	b8f0	00	1		0009	81	4c	e8	fd	nv-BdIzc	unknown	T0	ff	f0	f0
10	b8f0	00	1		0009	81	4c	e8	fd	nv-BdIzc	unknown	T0	ff	f0	ef
11	0009	b8	1	CLV	0009	81	4c	e8	fd	nv-BdIzc	unknown	T1	ff	00	ef
11	0009	b8	1	CLV	0009	81	4c	e8	fd	nv-BdIzc	unknown	T1	ff	00	ff
12	000a	15	1		000a	81	4c	e8	fd	nv-BdIzc	CLV	T0+T2	ff	ff	ff
12	000a	15	1		000a	81	4c	e8	fd	nv-BdIzc	CLV	T0+T2	ff	ff	fe

ORA (0D)		Cycles: 4	Size: 3
Absolute (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read PC	Store as Address.H.	
3.1		Inc. PC .	
3.2	Read Address	Store as Operand.	
4.1		A = Operand. N flag = (A & \$80), Z flag = ! A .	
4.2	Read PC	Store as OpCode.	
+X.1		* A , N flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	0d	1	ORA Abs	0006	81	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	a0
7	0006	0d	1	ORA Abs	0006	81	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	98
8	0007	f0	1		0007	81	4c	e8	fd	nv-BdIzc	ORA Abs	T2	98	98	98
8	0007	f0	1		0007	81	4c	e8	fd	nv-BdIzc	ORA Abs	T2	98	98	30
9	0008	01	1		0008	81	4c	e8	fd	nv-BdIzc	ORA Abs	T3	00	f0	30
9	0008	01	1		0008	81	4c	e8	fd	nv-BdIzc	ORA Abs	T3	00	f0	f0
10	01f0	aa	1		0009	81	4c	e8	fd	nv-BdIzc	ORA Abs	T0	ff	ff	f0
10	01f0	aa	1		0009	81	4c	e8	fd	nv-BdIzc	ORA Abs	T0	ff	ff	fe
11	0009	b8	1	CLV	0009	81	4c	e8	fd	nv-BdIzc	ORA Abs	T1	81	aa	fe
11	0009	b8	1	CLV	0009	81	4c	e8	fd	nv-BdIzc	ORA Abs	T1	81	aa	ab
12	000a	15	1		000a	ab	4c	e8	fd	Nv-BdIzc	CLV	T0+T2	ab	ab	ab
12	000a	15	1		000a	ab	4c	e8	fd	Nv-BdIzc	CLV	T0+T2	ab	ab	56

ASL (0E)	Cycles: 6	Size: 3
Absolute (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.
5.1		C flag = (Operand & \$80). Operand <= 1. N flag = (Operand & \$80), Z flag = !Operand.
5.2	Write to Address	Write Operand.
6.1		
6.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	0e	1	ASL Abs	0006	81	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	a0
7	0006	0e	1	ASL Abs	0006	81	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	98
8	0007	f0	1		0007	81	4c	e8	fd	nv-BdIzc	ASL Abs	T2	98	98	98
8	0007	f0	1		0007	81	4c	e8	fd	nv-BdIzc	ASL Abs	T2	98	98	30
9	0008	01	1		0008	81	4c	e8	fd	nv-BdIzc	ASL Abs	T3	00	f0	30
9	0008	01	1		0008	81	4c	e8	fd	nv-BdIzc	ASL Abs	T3	00	f0	f0
10	01f0	7f	1		0009	81	4c	e8	fd	nv-BdIzc	ASL Abs	T4	ff	ff	f0
10	01f0	7f	1		0009	81	4c	e8	fd	nv-BdIzc	ASL Abs	T4	ff	ff	fe
11	01f0	7f	0		0009	81	4c	e8	fd	nv-BdIzc	ASL Abs	T5	7f	7f	fe
11	01f0	7f	0		0009	81	4c	e8	fd	nv-BdIzc	ASL Abs	T5	7f	7f	fe
12	01f0	7f	0		0009	81	4c	e8	fd	Nv-BdIzc	ASL Abs	T0	fe	fe	fe
12	01f0	fe	0		0009	81	4c	e8	fd	Nv-BdIzc	ASL Abs	T0	fe	fe	fc
13	0009	b8	1	CLV	0009	81	4c	e8	fd	Nv-BdIzc	ASL Abs	T1	ff	fe	fc
13	0009	b8	1	CLV	0009	81	4c	e8	fd	Nv-BdIzc	ASL Abs	T1	ff	fe	fd
14	000a	15	1		000a	81	4c	e8	fd	Nv-BdIzc	CLV	T0+T2	fd	fd	fd
14	000a	15	1		000a	81	4c	e8	fd	Nv-BdIzc	CLV	T0+T2	fd	fd	fa

SLO (0F)	Cycles: 6	Size: 3
Absolute (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.
5.1		C flag = (Operand & \$80). Operand <<= 1. A = Operand. N flag = (A & \$80), Z flag = ! A .
5.2	Write to Address	Write Operand.
6.1		
6.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	0e	1	ASL Abs	0006	81	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	a0
7	0006	0e	1	ASL Abs	0006	81	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	98
8	0007	f0	1		0007	81	4c	e8	fd	nv-BdIzc	ASL Abs	T2	98	98	98
8	0007	f0	1		0007	81	4c	e8	fd	nv-BdIzc	ASL Abs	T2	98	98	30
9	0008	01	1		0008	81	4c	e8	fd	nv-BdIzc	ASL Abs	T3	00	f0	30
9	0008	01	1		0008	81	4c	e8	fd	nv-BdIzc	ASL Abs	T3	00	f0	f0
10	01f0	7f	1		0009	81	4c	e8	fd	nv-BdIzc	ASL Abs	T4	ff	ff	f0
10	01f0	7f	1		0009	81	4c	e8	fd	nv-BdIzc	ASL Abs	T4	ff	ff	fe
11	01f0	7f	0		0009	81	4c	e8	fd	nv-BdIzc	ASL Abs	T5	7f	7f	fe
11	01f0	7f	0		0009	81	4c	e8	fd	nv-BdIzc	ASL Abs	T5	7f	7f	fe
12	01f0	7f	0		0009	81	4c	e8	fd	Nv-BdIzc	ASL Abs	T0	fe	fe	fe
12	01f0	fe	0		0009	81	4c	e8	fd	Nv-BdIzc	ASL Abs	T0	fe	fe	fc
13	0009	b8	1	CLV	0009	81	4c	e8	fd	Nv-BdIzc	ASL Abs	T1	ff	fe	fc
13	0009	b8	1	CLV	0009	81	4c	e8	fd	Nv-BdIzc	ASL Abs	T1	ff	fe	fd
14	000a	15	1		000a	81	4c	e8	fd	Nv-BdIzc	CLV	T0+T2	fd	fd	fd
14	000a	15	1		000a	81	4c	e8	fd	Nv-BdIzc	CLV	T0+T2	fd	fd	fa

BPL (10)	Cycles: 2-4	Size: 2
Relative (Jump)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC . Jump = ((P & \$80) == \$00).
1.2	Read PC	Store as Operand.
		If !Jump, poll interrupts.
		Otherwise Address = i16(i8(Operand)) + PC + 1.
		BoundaryCrossed = Address.H != PC .H.
		If !BoundaryCrossed, poll interrupts.
2.1		Inc. PC . If !Jump, end (next half-cycle is 4.2).
2.2	Read PC	Discard.
3.1		PC .L = Address.L. If !BoundaryCrossed, end (next half-cycle is 4.2).
3.2	Read PC	Discard. Poll interrupts.
4.1		PC .H = Address.H.
4.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	10	1	BPL	0006	81	8c	e8	fd	Nv-BdIzc	LDX #	T1	8c	8c	a0
7	0006	10	1	BPL	0006	81	8c	e8	fd	Nv-BdIzc	LDX #	T1	8c	8c	18
8	0007	10	1		0007	81	8c	e8	fd	Nv-BdIzc	BPL	T2	18	18	18
8	0007	10	1		0007	81	8c	e8	fd	Nv-BdIzc	BPL	T2	18	18	30
9	0008	01	1	ORA (zp,X)	0008	81	8c	e8	fd	Nv-BdIzc	BPL		10	08	30
9	0008	01	1	ORA (zp,X)	0008	81	8c	e8	fd	Nv-BdIzc	BPL		10	08	18
10	0009	b8	1		0009	81	8c	e8	fd	Nv-BdIzc	ORA (zp,X)	T2	18	18	18
10	0009	b8	1		0009	81	8c	e8	fd	Nv-BdIzc	ORA (zp,X)	T2	18	18	30

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	10	1	BPL	0006	81	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	a0
7	0006	10	1	BPL	0006	81	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	98
8	0007	f0	1		0007	81	4c	e8	fd	nv-BdIzc	BPL	T2	98	98	98
8	0007	f0	1		0007	81	4c	e8	fd	nv-BdIzc	BPL	T2	98	98	30
9	0008	01	1		0008	81	4c	e8	fd	nv-BdIzc	BPL	T3	10	08	30
9	0008	01	1		0008	81	4c	e8	fd	nv-BdIzc	BPL	T3	10	08	18
10	0018	48	1	PHA	0018	81	4c	e8	fd	nv-BdIzc	BPL		00	00	18
10	0018	48	1	PHA	0018	81	4c	e8	fd	nv-BdIzc	BPL		00	00	01
11	0019	ea	1		0019	81	4c	e8	fd	nv-BdIzc	PHA	T2	01	01	01
11	0019	ea	1		0019	81	4c	e8	fd	nv-BdIzc	PHA	T2	01	01	02

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	10	1	BPL	0006	81	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	a0
7	0006	10	1	BPL	0006	81	4c	e8	fd	nv-BdIzc	LDX #	T1	4c	4c	98
8	0007	f0	1		0007	81	4c	e8	fd	nv-BdIzc	BPL	T2	98	98	98
8	0007	f0	1		0007	81	4c	e8	fd	nv-BdIzc	BPL	T2	98	98	30
9	0008	01	1		0008	81	4c	e8	fd	nv-BdIzc	BPL	T3	f0	08	30
9	0008	01	1		0008	81	4c	e8	fd	nv-BdIzc	BPL	T3	f0	08	f8
10	00f8	00	1		00f8	81	4c	e8	fd	nv-BdIzc	BPL	T0	00	ff	f8
10	00f8	00	1		00f8	81	4c	e8	fd	nv-BdIzc	BPL	T0	00	ff	ff
11	fff8	00	1	BRK	fff8	81	4c	e8	fd	nv-BdIzc	BPL	T1	ff	00	ff
11	fff8	00	1	BRK	fff8	81	4c	e8	fd	nv-BdIzc	BPL	T1	ff	00	ff
12	fff9	00	1		fff9	81	4c	e8	fd	nv-BdIzc	BRK	T2	ff	ff	ff
12	fff9	00	1		fff9	81	4c	e8	fd	nv-BdIzc	BRK	T2	ff	ff	fe

ORA (11)		Cycles: 5-6	Size: 2
Indirect Y (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.L.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Address.	
3.1			
3.2	Read u8(Pointer + 1)	Store as Address.H.	
4.1		Target = Address + Y. Pointer.L = Target.L. Pointer.H = Address.H. BoundaryCrossed = Pointer.H != Target.H.	
4.2	Read Pointer	Store as Operand. If !BoundaryCrossed, skip the next cycle.	
5.1		Pointer.H = Target.H.	
5.2	Read Pointer	Store as Operand.	
6.1		A = Operand. N flag = (A & \$80), Z flag = ! A .	
6.2	Read PC	Store as OpCode.	
+X.1		* A , N flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	11	1	ORA (zp),Y	0006	59	8c	1f	fd	Nv-BdIzc	LDX #	T1	8c	8c	7c
7	0006	11	1	ORA (zp),Y	0006	59	8c	1f	fd	Nv-BdIzc	LDX #	T1	8c	8c	18
8	0007	20	1		0007	59	8c	1f	fd	Nv-BdIzc	ORA (zp),Y	T2	18	18	18
8	0007	20	1		0007	59	8c	1f	fd	Nv-BdIzc	ORA (zp),Y	T2	18	18	30
9	0020	23	1		0008	59	8c	1f	fd	Nv-BdIzc	ORA (zp),Y	T3	00	20	30
9	0020	23	1		0008	59	8c	1f	fd	Nv-BdIzc	ORA (zp),Y	T3	00	20	21
10	0021	01	1		0008	59	8c	1f	fd	Nv-BdIzc	ORA (zp),Y	T4	1f	23	21
10	0021	01	1		0008	59	8c	1f	fd	Nv-BdIzc	ORA (zp),Y	T4	1f	23	42
11	0142	33	1		0008	59	8c	1f	fd	Nv-BdIzc	ORA (zp),Y	T0	00	01	42
11	0142	33	1		0008	59	8c	1f	fd	Nv-BdIzc	ORA (zp),Y	T0	00	01	02
12	0008	ea	1	NOP	0008	59	8c	1f	fd	Nv-BdIzc	ORA (zp),Y	T1	59	33	02
12	0008	ea	1	NOP	0008	59	8c	1f	fd	Nv-BdIzc	ORA (zp),Y	T1	59	33	7b
13	0009	b8	1		0009	7b	8c	1f	fd	nv-BdIzc	NOP	T0+T2	7b	7b	7b
13	0009	b8	1		0009	7b	8c	1f	fd	nv-BdIzc	NOP	T0+T2	7b	7b	f6

JAM (12)	Cycles: ∞	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Discard.
2.1		Visual6502 suggests incrementing PC , but Single-Step Tests cannot be passed this way.
2.2	Read \$FFFF	Discard.
3.1		
3.2	Read \$FFFE	Discard.
4.1		
4.2	Read \$FFFE	Discard.
5.1		Repeat 5.1 and 5.2 forever.
5.2	Read \$FFFF	Discard.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	12	1	unknown	0006	59	8c	1f	fd	Nv-BdIzc	LDX #	T1	8c	8c	7c
7	0006	12	1	unknown	0006	59	8c	1f	fd	Nv-BdIzc	LDX #	T1	8c	8c	18
8	0007	20	1		0007	59	8c	1f	fd	Nv-BdIzc	unknown	T2	18	18	18
8	0007	20	1		0007	59	8c	1f	fd	Nv-BdIzc	unknown	T2	18	18	30
9	ffff	00	1		0008	59	8c	1f	fd	Nv-BdIzc	unknown	T3	ff	ff	30
9	ffff	00	1		0008	59	8c	1f	fd	Nv-BdIzc	unknown	T3	ff	ff	fe
10	fffe	00	1		0008	59	8c	1f	fd	Nv-BdIzc	unknown	T4	ff	ff	fe
10	fffe	00	1		0008	59	8c	1f	fd	Nv-BdIzc	unknown	T4	ff	ff	fe
11	fffe	00	1		0008	59	8c	1f	fd	Nv-BdIzc	unknown	T5	ff	ff	fe
11	fffe	00	1		0008	59	8c	1f	fd	Nv-BdIzc	unknown	T5	ff	ff	fe
12	ffff	00	1		0008	59	8c	1f	fd	Nv-BdIzc	unknown		ff	ff	fe
12	ffff	00	1		0008	59	8c	1f	fd	Nv-BdIzc	unknown		ff	ff	fe
13	ffff	00	1		0008	59	8c	1f	fd	Nv-BdIzc	unknown		ff	ff	fe
13	ffff	00	1		0008	59	8c	1f	fd	Nv-BdIzc	unknown		ff	ff	fe
14	ffff	00	1		0008	59	8c	1f	fd	Nv-BdIzc	unknown		ff	ff	fe
14	ffff	00	1		0008	59	8c	1f	fd	Nv-BdIzc	unknown		ff	ff	fe

SLO (13) Cycles: 8		Size: 2
Indirect Y (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.L.
2.1		Inc. PC .
2.2	Read Pointer	Store as Address.
3.1		
3.2	Read u8(Pointer + 1)	Store as Address.H.
4.1		Target = Address + Y. Pointer.L = Target.L. Pointer.H = Address.H.
4.2	Read Pointer	Store as Operand.
5.1		Pointer.H = Target.H.
5.2	Read Pointer	Store as Operand.
6.1		
6.2	Write to Pointer	Write Operand.
7.1		C flag = (Operand & \$80). Operand <<= 1. A = Operand. N flag = (A & \$80), Z flag = ! A .
7.2	Write to Pointer	Write Operand.
8.1		
8.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	13	1	unknown	0006	59	8c	1f	fd	Nv-BdIzc	LDX #	T1	8c	8c	7c
7	0006	13	1	unknown	0006	59	8c	1f	fd	Nv-BdIzc	LDX #	T1	8c	8c	18
8	0007	20	1		0007	59	8c	1f	fd	Nv-BdIzc	unknown	T2	18	18	18
8	0007	20	1		0007	59	8c	1f	fd	Nv-BdIzc	unknown	T2	18	18	30
9	0020	23	1		0008	59	8c	1f	fd	Nv-BdIzc	unknown	T3	00	20	30
9	0020	23	1		0008	59	8c	1f	fd	Nv-BdIzc	unknown	T3	00	20	21
10	0021	01	1		0008	59	8c	1f	fd	Nv-BdIzc	unknown	T4	1f	23	21
10	0021	01	1		0008	59	8c	1f	fd	Nv-BdIzc	unknown	T4	1f	23	42
11	0142	33	1		0008	59	8c	1f	fd	Nv-BdIzc	unknown	T5	00	01	42
11	0142	33	1		0008	59	8c	1f	fd	Nv-BdIzc	unknown	T5	00	01	02
12	0142	33	1		0008	59	8c	1f	fd	Nv-BdIzc	unknown		02	ff	02
12	0142	33	1		0008	59	8c	1f	fd	Nv-BdIzc	unknown		02	ff	01
13	0142	33	0		0008	59	8c	1f	fd	Nv-BdIzc	unknown		33	33	01
13	0142	33	0		0008	59	8c	1f	fd	Nv-BdIzc	unknown		33	33	66
14	0142	33	0		0008	59	8c	1f	fd	nv-BdIzc	unknown	T0	66	66	66
14	0142	66	0		0008	59	8c	1f	fd	nv-BdIzc	unknown	T0	66	66	cc
15	0008	ea	1	NOP	0008	59	8c	1f	fd	nv-BdIzc	unknown	T1	59	66	cc
15	0008	ea	1	NOP	0008	59	8c	1f	fd	nv-BdIzc	unknown	T1	59	66	7f
16	0009	b8	1		0009	7f	8c	1f	fd	nv-BdIzc	NOP	T0+T2	7f	7f	7f
16	0009	b8	1		0009	7f	8c	1f	fd	nv-BdIzc	NOP	T0+T2	7f	7f	fe

NOP (14)		Cycles: 4	Size: 2
Zero Page X-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.L.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Operand.	
3.1		Address = u8(Pointer + X).	
3.2	Read Address	Store as Operand.	
4.1			
4.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	14	1	unknown	0006	59	8c	1f	fd	Nv-BdIzc	LDX #	T1	8c	8c	7c
7	0006	14	1	unknown	0006	59	8c	1f	fd	Nv-BdIzc	LDX #	T1	8c	8c	18
8	0007	20	1		0007	59	8c	1f	fd	Nv-BdIzc	unknown	T2	18	18	18
8	0007	20	1		0007	59	8c	1f	fd	Nv-BdIzc	unknown	T2	18	18	30
9	0020	23	1		0008	59	8c	1f	fd	Nv-BdIzc	unknown	T3	8c	20	30
9	0020	23	1		0008	59	8c	1f	fd	Nv-BdIzc	unknown	T3	8c	20	ac
10	00ac	00	1		0008	59	8c	1f	fd	Nv-BdIzc	unknown	T0	ff	ff	ac
10	00ac	00	1		0008	59	8c	1f	fd	Nv-BdIzc	unknown	T0	ff	ff	fe
11	0008	ea	1	NOP	0008	59	8c	1f	fd	Nv-BdIzc	unknown	T1	ff	00	fe
11	0008	ea	1	NOP	0008	59	8c	1f	fd	Nv-BdIzc	unknown	T1	ff	00	ff
12	0009	b8	1		0009	59	8c	1f	fd	Nv-BdIzc	NOP	T0+T2	ff	ff	ff
12	0009	b8	1		0009	59	8c	1f	fd	Nv-BdIzc	NOP	T0+T2	ff	ff	fe

ORA (15)	Cycles: 4	Size: 2
Zero Page X-Indexed (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.L.
2.1		Inc. PC .
2.2	Read Pointer	Store as Operand.
3.1		Address = u8(Pointer + X).
3.2	Read Address	Store as Operand.
4.1		A = Operand. N flag = (A & \$80), Z flag = ! A .
4.2	Read PC	Store as OpCode.
+X.1		* A , N flag, and Z flag applied.

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	15	1	ORA zp,X	0006	70	8c	1f	fd	Nv-BdIzc	LDX #	T1	8c	8c	7c
7	0006	15	1	ORA zp,X	0006	70	8c	1f	fd	Nv-BdIzc	LDX #	T1	8c	8c	18
8	0007	0f	1		0007	70	8c	1f	fd	Nv-BdIzc	ORA zp,X	T2	18	18	18
8	0007	0f	1		0007	70	8c	1f	fd	Nv-BdIzc	ORA zp,X	T2	18	18	30
9	000f	41	1		0008	70	8c	1f	fd	Nv-BdIzc	ORA zp,X	T3	8c	0f	30
9	000f	41	1		0008	70	8c	1f	fd	Nv-BdIzc	ORA zp,X	T3	8c	0f	9b
10	009b	33	1		0008	70	8c	1f	fd	Nv-BdIzc	ORA zp,X	T0	ff	ff	9b
10	009b	33	1		0008	70	8c	1f	fd	Nv-BdIzc	ORA zp,X	T0	ff	ff	fe
11	0008	ea	1	NOP	0008	70	8c	1f	fd	Nv-BdIzc	ORA zp,X	T1	70	33	fe
11	0008	ea	1	NOP	0008	70	8c	1f	fd	Nv-BdIzc	ORA zp,X	T1	70	33	73
12	0009	b8	1		0009	73	8c	1f	fd	nv-BdIzc	NOP	T0+T2	73	73	73
12	0009	b8	1		0009	73	8c	1f	fd	nv-BdIzc	NOP	T0+T2	73	73	e6

ASL (16)		Cycles: 6	Size: 2
Zero Page X-Indexed (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.L.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Operand.	
3.1		Address = u8(Pointer + X).	
3.2	Read Address	Store as Operand.	
4.1			
4.2	Write to Address	Write Operand.	
5.1		C flag = (Operand & \$80). Operand <= 1. N flag = (Operand & \$80), Z flag = !Operand.	
5.2	Write to Address	Write Operand.	
6.1			
6.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	16	1	ASL zp,X	0006	70	8c	1f	fd	Nv-BdIzc	LDX #	T1	8c	8c	7c
7	0006	16	1	ASL zp,X	0006	70	8c	1f	fd	Nv-BdIzc	LDX #	T1	8c	8c	18
8	0007	0f	1		0007	70	8c	1f	fd	Nv-BdIzc	ASL zp,X	T2	18	18	18
8	0007	0f	1		0007	70	8c	1f	fd	Nv-BdIzc	ASL zp,X	T2	18	18	30
9	000f	41	1		0008	70	8c	1f	fd	Nv-BdIzc	ASL zp,X	T3	8c	0f	30
9	000f	41	1		0008	70	8c	1f	fd	Nv-BdIzc	ASL zp,X	T3	8c	0f	9b
10	009b	66	1		0008	70	8c	1f	fd	Nv-BdIzc	ASL zp,X	T4	ff	ff	9b
10	009b	66	1		0008	70	8c	1f	fd	Nv-BdIzc	ASL zp,X	T4	ff	ff	fe
11	009b	66	0		0008	70	8c	1f	fd	Nv-BdIzc	ASL zp,X	T5	66	66	fe
11	009b	66	0		0008	70	8c	1f	fd	Nv-BdIzc	ASL zp,X	T5	66	66	cc
12	009b	66	0		0008	70	8c	1f	fd	Nv-BdIzc	ASL zp,X	T0	cc	cc	cc
12	009b	cc	0		0008	70	8c	1f	fd	Nv-BdIzc	ASL zp,X	T0	cc	cc	98
13	0008	ea	1	NOP	0008	70	8c	1f	fd	Nv-BdIzc	ASL zp,X	T1	ff	cc	98
13	0008	ea	1	NOP	0008	70	8c	1f	fd	Nv-BdIzc	ASL zp,X	T1	ff	cc	cb
14	0009	b8	1		0009	70	8c	1f	fd	Nv-BdIzc	NOP	T0+T2	cb	cb	cb
14	0009	b8	1		0009	70	8c	1f	fd	Nv-BdIzc	NOP	T0+T2	cb	cb	96

SLO (17)	Cycles: 6	Size: 2
Zero Page X-Indexed (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.L.
2.1		Inc. PC .
2.2	Read Pointer	Store as Operand.
3.1		Address = u8(Pointer + X).
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.
5.1		C flag = (Operand & \$80). Operand <<= 1. A = Operand. N flag = (A & \$80), Z flag = ! A .
5.2	Write to Address	Write Operand.
6.1		
6.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	17	1	unknown	0006	70	8c	1f	fd	Nv-BdIzc	LDX #	T1	8c	8c	7c
7	0006	17	1	unknown	0006	70	8c	1f	fd	Nv-BdIzc	LDX #	T1	8c	8c	18
8	0007	0f	1		0007	70	8c	1f	fd	Nv-BdIzc	unknown	T2	18	18	18
8	0007	0f	1		0007	70	8c	1f	fd	Nv-BdIzc	unknown	T2	18	18	30
9	000f	41	1		0008	70	8c	1f	fd	Nv-BdIzc	unknown	T3	8c	0f	30
9	000f	41	1		0008	70	8c	1f	fd	Nv-BdIzc	unknown	T3	8c	0f	9b
10	009b	33	1		0008	70	8c	1f	fd	Nv-BdIzc	unknown	T4	ff	ff	9b
10	009b	33	1		0008	70	8c	1f	fd	Nv-BdIzc	unknown	T4	ff	ff	fe
11	009b	33	0		0008	70	8c	1f	fd	Nv-BdIzc	unknown	T5	33	33	fe
11	009b	33	0		0008	70	8c	1f	fd	Nv-BdIzc	unknown	T5	33	33	66
12	009b	33	0		0008	70	8c	1f	fd	nv-BdIzc	unknown	T0	66	66	66
12	009b	66	0		0008	70	8c	1f	fd	nv-BdIzc	unknown	T0	66	66	cc
13	0008	ea	1	NOP	0008	70	8c	1f	fd	nv-BdIzc	unknown	T1	70	66	cc
13	0008	ea	1	NOP	0008	70	8c	1f	fd	nv-BdIzc	unknown	T1	70	66	76
14	0009	b8	1		0009	76	8c	1f	fd	nv-BdIzc	NOP	T0+T2	76	76	76
14	0009	b8	1		0009	76	8c	1f	fd	nv-BdIzc	NOP	T0+T2	76	76	ec

CLC (18)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Discard.	
2.1		C flag = 0.	
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
9	0006	18	1	CLC	0006	aa	8c	1f	fd	Nv-BdIzC	LDX #	T1	8c	8c	7c
9	0006	18	1	CLC	0006	aa	8c	1f	fd	Nv-BdIzC	LDX #	T1	8c	8c	18
10	0007	ea	1		0007	aa	8c	1f	fd	Nv-BdIzC	CLC	T0+T2	18	18	18
10	0007	ea	1		0007	aa	8c	1f	fd	Nv-BdIzC	CLC	T0+T2	18	18	30
11	0007	ea	1	NOP	0007	aa	8c	1f	fd	Nv-BdIzc	CLC	T1	ff	ff	30
11	0007	ea	1	NOP	0007	aa	8c	1f	fd	Nv-BdIzc	CLC	T1	ff	ff	fe
12	0008	ea	1		0008	aa	8c	1f	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fe
12	0008	ea	1		0008	aa	8c	1f	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fc

ORA (19) Cycles: 4-5		Size: 3
Absolute Y-Indexed (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.L.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + Y. Address.L = Target.L. Address.H = Pointer.H. BoundaryCrossed = Pointer.H != Target.H.
3.2	Read Address	Store as Operand. If !BoundaryCrossed, skip the next cycle.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		A = Operand. N flag = (A & \$80), Z flag = ! A .
5.2	Read PC	Store as OpCode.
+X.1		* A , N flag, and Z flag applied.

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	19	1	ORA Abs,Y	0006	ea	11	1f	fd	nv-BdIzc	LDX #	T1	11	11	7c
7	0006	19	1	ORA Abs,Y	0006	ea	11	1f	fd	nv-BdIzc	LDX #	T1	11	11	22
8	0007	04	1		0007	ea	11	1f	fd	nv-BdIzc	ORA Abs,Y	T2	22	22	22
8	0007	04	1		0007	ea	11	1f	fd	nv-BdIzc	ORA Abs,Y	T2	22	22	44
9	0008	00	1		0008	ea	11	1f	fd	nv-BdIzc	ORA Abs,Y	T3	1f	04	44
9	0008	00	1		0008	ea	11	1f	fd	nv-BdIzc	ORA Abs,Y	T3	1f	04	23
10	0023	f0	1		0009	ea	11	1f	fd	nv-BdIzc	ORA Abs,Y	T0	00	00	23
10	0023	f0	1		0009	ea	11	1f	fd	nv-BdIzc	ORA Abs,Y	T0	00	00	01
11	0009	b8	1	CLV	0009	ea	11	1f	fd	nv-BdIzc	ORA Abs,Y	T1	ea	f0	01
11	0009	b8	1	CLV	0009	ea	11	1f	fd	nv-BdIzc	ORA Abs,Y	T1	ea	f0	fa
12	000a	15	1		000a	fa	11	1f	fd	Nv-BdIzc	CLV	T0+T2	fa	fa	fa
12	000a	15	1		000a	fa	11	1f	fd	Nv-BdIzc	CLV	T0+T2	fa	fa	f4

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	19	1	ORA Abs,Y	0006	ea	11	1f	fd	nv-BdIzc	LDX #	T1	11	11	7c
7	0006	19	1	ORA Abs,Y	0006	ea	11	1f	fd	nv-BdIzc	LDX #	T1	11	11	22
8	0007	f4	1		0007	ea	11	1f	fd	nv-BdIzc	ORA Abs,Y	T2	22	22	22
8	0007	f4	1		0007	ea	11	1f	fd	nv-BdIzc	ORA Abs,Y	T2	22	22	44
9	0008	00	1		0008	ea	11	1f	fd	nv-BdIzc	ORA Abs,Y	T3	1f	f4	44
9	0008	00	1		0008	ea	11	1f	fd	nv-BdIzc	ORA Abs,Y	T3	1f	f4	13
10	0013	0f	1		0009	ea	11	1f	fd	nv-BdIzc	ORA Abs,Y	T4	00	00	13
10	0013	0f	1		0009	ea	11	1f	fd	nv-BdIzc	ORA Abs,Y	T4	00	00	01
11	0113	01	1		0009	ea	11	1f	fd	nv-BdIzc	ORA Abs,Y	T0	01	ff	01
11	0113	01	1		0009	ea	11	1f	fd	nv-BdIzc	ORA Abs,Y	T0	01	ff	00
12	0009	b8	1	CLV	0009	ea	11	1f	fd	nv-BdIzc	ORA Abs,Y	T1	ea	01	00
12	0009	b8	1	CLV	0009	ea	11	1f	fd	nv-BdIzc	ORA Abs,Y	T1	ea	01	eb
13	000a	15	1		000a	eb	11	1f	fd	Nv-BdIzc	CLV	T0+T2	eb	eb	eb
13	000a	15	1		000a	eb	11	1f	fd	Nv-BdIzc	CLV	T0+T2	eb	eb	d6

NOP (1A)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Discard.	
2.1			
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	1a	1	unknown	0006	ea	11	1f	fd	nv-BdIzc	LDX #	T1	11	11	7c
7	0006	1a	1	unknown	0006	ea	11	1f	fd	nv-BdIzc	LDX #	T1	11	11	22
8	0007	ea	1		0007	ea	11	1f	fd	nv-BdIzc	unknown	T0+T2	22	22	22
8	0007	ea	1		0007	ea	11	1f	fd	nv-BdIzc	unknown	T0+T2	22	22	44
9	0007	ea	1	NOP	0007	ea	11	1f	fd	nv-BdIzc	unknown	T1	ff	ff	44
9	0007	ea	1	NOP	0007	ea	11	1f	fd	nv-BdIzc	unknown	T1	ff	ff	fe
10	0008	00	1		0008	ea	11	1f	fd	nv-BdIzc	NOP	T0+T2	fe	fe	fe
10	0008	00	1		0008	ea	11	1f	fd	nv-BdIzc	NOP	T0+T2	fe	fe	fc

SLO (1B)	Cycles: 7	Size: 3
Absolute Y-Indexed (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.L.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + Y . Address.L = Target.L. Address.H = Pointer.H.
3.2	Read Address	Store as Operand.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		
5.2	Write to Address	Write Operand.
6.1		C flag = (Operand & \$80). Operand <<= 1. A = Operand. N flag = (A & \$80), Z flag = ! A .
6.2	Write to Address	Write Operand.
7.1		
7.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	1b	1	unknown	0006	ea	11	1f	fd	nv-BdIzc	LDX #	T1	11	11	7c
7	0006	1b	1	unknown	0006	ea	11	1f	fd	nv-BdIzc	LDX #	T1	11	11	22
8	0007	44	1		0007	ea	11	1f	fd	nv-BdIzc	unknown	T2	22	22	22
8	0007	44	1		0007	ea	11	1f	fd	nv-BdIzc	unknown	T2	22	22	44
9	0008	01	1		0008	ea	11	1f	fd	nv-BdIzc	unknown	T3	1f	44	44
9	0008	01	1		0008	ea	11	1f	fd	nv-BdIzc	unknown	T3	1f	44	63
10	0163	55	1		0009	ea	11	1f	fd	nv-BdIzc	unknown	T4	00	01	63
10	0163	55	1		0009	ea	11	1f	fd	nv-BdIzc	unknown	T4	00	01	02
11	0163	55	1		0009	ea	11	1f	fd	nv-BdIzc	unknown	T5	02	ff	02
11	0163	55	1		0009	ea	11	1f	fd	nv-BdIzc	unknown	T5	02	ff	01
12	0163	55	0		0009	ea	11	1f	fd	nv-BdIzc	unknown		55	55	01
12	0163	55	0		0009	ea	11	1f	fd	nv-BdIzc	unknown		55	55	aa
13	0163	55	0		0009	ea	11	1f	fd	Nv-BdIzc	unknown	T0	aa	aa	aa
13	0163	aa	0		0009	ea	11	1f	fd	Nv-BdIzc	unknown	T0	aa	aa	54
14	0009	b8	1	CLV	0009	ea	11	1f	fd	Nv-BdIzc	unknown	T1	ea	aa	54
14	0009	b8	1	CLV	0009	ea	11	1f	fd	Nv-BdIzc	unknown	T1	ea	aa	ea
15	000a	15	1		000a	ea	11	1f	fd	Nv-BdIzc	CLV	T0+T2	ea	ea	ea
15	000a	15	1		000a	ea	11	1f	fd	Nv-BdIzc	CLV	T0+T2	ea	ea	d4

NOP (1C)		Cycles: 4-5	Size: 3
Absolute X-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.L.	
2.1		Inc. PC .	
2.2	Read PC	Store as Pointer.H.	
3.1		Inc. PC . Target = Pointer + X .	
		Address.L = Target.L.	
		Address.H = Pointer.H.	
		BoundaryCrossed = Pointer.H != Target.H.	
3.2	Read Address	Store as Operand. If !BoundaryCrossed, skip the next cycle.	
4.1		Address.H = Target.H.	
4.2	Read Address	Store as Operand.	
5.1			
5.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	1c	1	unknown	0006	ea	fe	1f	fd	Nv-BdIzc	LDX #	T1	fe	fe	7c
7	0006	1c	1	unknown	0006	ea	fe	1f	fd	Nv-BdIzc	LDX #	T1	fe	fe	fc
8	0007	44	1		0007	ea	fe	1f	fd	Nv-BdIzc	unknown	T2	fc	fc	fc
8	0007	44	1		0007	ea	fe	1f	fd	Nv-BdIzc	unknown	T2	fc	fc	f8
9	0008	01	1		0008	ea	fe	1f	fd	Nv-BdIzc	unknown	T3	fe	44	f8
9	0008	01	1		0008	ea	fe	1f	fd	Nv-BdIzc	unknown	T3	fe	44	42
10	0142	66	1		0009	ea	fe	1f	fd	Nv-BdIzc	unknown	T4	00	01	42
10	0142	66	1		0009	ea	fe	1f	fd	Nv-BdIzc	unknown	T4	00	01	02
11	0242	00	1		0009	ea	fe	1f	fd	Nv-BdIzc	unknown	T0	02	ff	02
11	0242	00	1		0009	ea	fe	1f	fd	Nv-BdIzc	unknown	T0	02	ff	01
12	0009	b8	1	CLV	0009	ea	fe	1f	fd	Nv-BdIzc	unknown	T1	ff	00	01
12	0009	b8	1	CLV	0009	ea	fe	1f	fd	Nv-BdIzc	unknown	T1	ff	00	ff
13	000a	15	1		000a	ea	fe	1f	fd	Nv-BdIzc	CLV	T0+T2	ff	ff	ff
13	000a	15	1		000a	ea	fe	1f	fd	Nv-BdIzc	CLV	T0+T2	ff	ff	fe

ORA (1D) Cycles: 4-5		Size: 3
Absolute X-Indexed (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.L.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + X . Address.L = Target.L. Address.H = Pointer.H. BoundaryCrossed = Pointer.H != Target.H.
3.2	Read Address	Store as Operand. If !BoundaryCrossed, skip the next cycle.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		A = Operand. N flag = (A & \$80), Z flag = ! A .
5.2	Read PC	Store as OpCode.
+X.1		* A , N flag, and Z flag applied.

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	1d	1	ORA Abs,X	0006	ea	14	1f	fd	nv-BdIzc	LDX #	T1	14	14	7c
7	0006	1d	1	ORA Abs,X	0006	ea	14	1f	fd	nv-BdIzc	LDX #	T1	14	14	28
8	0007	44	1		0007	ea	14	1f	fd	nv-BdIzc	ORA Abs,X	T2	28	28	28
8	0007	44	1		0007	ea	14	1f	fd	nv-BdIzc	ORA Abs,X	T2	28	28	50
9	0008	01	1		0008	ea	14	1f	fd	nv-BdIzc	ORA Abs,X	T3	14	44	50
9	0008	01	1		0008	ea	14	1f	fd	nv-BdIzc	ORA Abs,X	T3	14	44	58
10	0158	00	1		0009	ea	14	1f	fd	nv-BdIzc	ORA Abs,X	T0	00	01	58
10	0158	00	1		0009	ea	14	1f	fd	nv-BdIzc	ORA Abs,X	T0	00	01	02
11	0009	b8	1	CLV	0009	ea	14	1f	fd	nv-BdIzc	ORA Abs,X	T1	ea	00	02
11	0009	b8	1	CLV	0009	ea	14	1f	fd	nv-BdIzc	ORA Abs,X	T1	ea	00	ea
12	000a	15	1		000a	ea	14	1f	fd	Nv-BdIzc	CLV	T0+T2	ea	ea	ea
12	000a	15	1		000a	ea	14	1f	fd	Nv-BdIzc	CLV	T0+T2	ea	ea	d4

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	1d	1	ORA Abs,X	0006	ea	fc	1f	fd	Nv-BdIzc	LDX #	T1	fc	fc	7c
7	0006	1d	1	ORA Abs,X	0006	ea	fc	1f	fd	Nv-BdIzc	LDX #	T1	fc	fc	f8
8	0007	44	1		0007	ea	fc	1f	fd	Nv-BdIzc	ORA Abs,X	T2	f8	f8	f8
8	0007	44	1		0007	ea	fc	1f	fd	Nv-BdIzc	ORA Abs,X	T2	f8	f8	f0
9	0008	01	1		0008	ea	fc	1f	fd	Nv-BdIzc	ORA Abs,X	T3	fc	44	f0
9	0008	01	1		0008	ea	fc	1f	fd	Nv-BdIzc	ORA Abs,X	T3	fc	44	40
10	0140	00	1		0009	ea	fc	1f	fd	Nv-BdIzc	ORA Abs,X	T4	00	01	40
10	0140	00	1		0009	ea	fc	1f	fd	Nv-BdIzc	ORA Abs,X	T4	00	01	02
11	0240	00	1		0009	ea	fc	1f	fd	Nv-BdIzc	ORA Abs,X	T0	02	ff	02
11	0240	00	1		0009	ea	fc	1f	fd	Nv-BdIzc	ORA Abs,X	T0	02	ff	01
12	0009	b8	1	CLV	0009	ea	fc	1f	fd	Nv-BdIzc	ORA Abs,X	T1	ea	00	01
12	0009	b8	1	CLV	0009	ea	fc	1f	fd	Nv-BdIzc	ORA Abs,X	T1	ea	00	ea
13	000a	15	1		000a	ea	fc	1f	fd	Nv-BdIzc	CLV	T0+T2	ea	ea	ea
13	000a	15	1		000a	ea	fc	1f	fd	Nv-BdIzc	CLV	T0+T2	ea	ea	d4

ASL (1E)	Cycles: 7	Size: 3
Absolute X-Indexed (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.L.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + X . Address.L = Target.L. Address.H = Pointer.H.
3.2	Read Address	Store as Operand.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		
5.2	Write to Address	Write Operand.
6.1		C flag = (Operand & \$80). Operand <= 1. N flag = (Operand & \$80), Z flag = !Operand.
6.2	Write to Address	Write Operand.
7.1		
7.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	1e	1	ASL Abs,X	0006	ea	13	1f	fd	nv-BdIzc	LDX #	T1	13	13	7c
7	0006	1e	1	ASL Abs,X	0006	ea	13	1f	fd	nv-BdIzc	LDX #	T1	13	13	26
8	0007	44	1		0007	ea	13	1f	fd	nv-BdIzc	ASL Abs,X	T2	26	26	26
8	0007	44	1		0007	ea	13	1f	fd	nv-BdIzc	ASL Abs,X	T2	26	26	4c
9	0008	01	1		0008	ea	13	1f	fd	nv-BdIzc	ASL Abs,X	T3	13	44	4c
9	0008	01	1		0008	ea	13	1f	fd	nv-BdIzc	ASL Abs,X	T3	13	44	57
10	0157	71	1		0009	ea	13	1f	fd	nv-BdIzc	ASL Abs,X	T4	00	01	57
10	0157	71	1		0009	ea	13	1f	fd	nv-BdIzc	ASL Abs,X	T4	00	01	02
11	0157	71	1		0009	ea	13	1f	fd	nv-BdIzc	ASL Abs,X	T5	02	ff	02
11	0157	71	1		0009	ea	13	1f	fd	nv-BdIzc	ASL Abs,X	T5	02	ff	01
12	0157	71	0		0009	ea	13	1f	fd	nv-BdIzc	ASL Abs,X		71	71	01
12	0157	71	0		0009	ea	13	1f	fd	nv-BdIzc	ASL Abs,X		71	71	e2
13	0157	71	0		0009	ea	13	1f	fd	Nv-BdIzc	ASL Abs,X	T0	e2	e2	e2
13	0157	e2	0		0009	ea	13	1f	fd	Nv-BdIzc	ASL Abs,X	T0	e2	e2	c4
14	0009	b8	1	CLV	0009	ea	13	1f	fd	Nv-BdIzc	ASL Abs,X	T1	ff	e2	c4
14	0009	b8	1	CLV	0009	ea	13	1f	fd	Nv-BdIzc	ASL Abs,X	T1	ff	e2	e1
15	000a	15	1		000a	ea	13	1f	fd	Nv-BdIzc	CLV	T0+T2	e1	e1	e1
15	000a	15	1		000a	ea	13	1f	fd	Nv-BdIzc	CLV	T0+T2	e1	e1	c2

SLO (1F)	Cycles: 7	Size: 3
Absolute X-Indexed (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.L.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + X . Address.L = Target.L. Address.H = Pointer.H.
3.2	Read Address	Store as Operand.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		
5.2	Write to Address	Write Operand.
6.1		C flag = (Operand & \$80). Operand <= 1. A = Operand. N flag = (A & \$80), Z flag = ! A .
6.2	Write to Address	Write Operand.
7.1		
7.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	1f	1	unknown	0006	ea	13	1f	fd	nv-BdIzc	LDX #	T1	13	13	7c
7	0006	1f	1	unknown	0006	ea	13	1f	fd	nv-BdIzc	LDX #	T1	13	13	26
8	0007	44	1		0007	ea	13	1f	fd	nv-BdIzc	unknown	T2	26	26	26
8	0007	44	1		0007	ea	13	1f	fd	nv-BdIzc	unknown	T2	26	26	4c
9	0008	01	1		0008	ea	13	1f	fd	nv-BdIzc	unknown	T3	13	44	4c
9	0008	01	1		0008	ea	13	1f	fd	nv-BdIzc	unknown	T3	13	44	57
10	0157	e2	1		0009	ea	13	1f	fd	nv-BdIzc	unknown	T4	00	01	57
10	0157	e2	1		0009	ea	13	1f	fd	nv-BdIzc	unknown	T4	00	01	02
11	0157	e2	1		0009	ea	13	1f	fd	nv-BdIzc	unknown	T5	02	ff	02
11	0157	e2	1		0009	ea	13	1f	fd	nv-BdIzc	unknown	T5	02	ff	01
12	0157	e2	0		0009	ea	13	1f	fd	nv-BdIzc	unknown		e2	e2	01
12	0157	e2	0		0009	ea	13	1f	fd	nv-BdIzc	unknown		e2	e2	c4
13	0157	e2	0		0009	ea	13	1f	fd	Nv-BdIzc	unknown	T0	c4	c4	c4
13	0157	c4	0		0009	ea	13	1f	fd	Nv-BdIzc	unknown	T0	c4	c4	88
14	0009	b8	1	CLV	0009	ea	13	1f	fd	Nv-BdIzc	unknown	T1	ea	c4	88
14	0009	b8	1	CLV	0009	ea	13	1f	fd	Nv-BdIzc	unknown	T1	ea	c4	ee
15	000a	15	1		000a	ee	13	1f	fd	Nv-BdIzc	CLV	T0+T2	ee	ee	ee
15	000a	15	1		000a	ee	13	1f	fd	Nv-BdIzc	CLV	T0+T2	ee	ee	dc

JSR (20)	Cycles: 6	Size: 3
Absolute (Jump)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read (S \$0100)	Store as Operand.
3.1		
3.2	Write to u8(S +0) \$0100	Write PC .H.
4.1		
4.2	Write to u8(S -1) \$0100	Write PC .L.
5.1		
5.2	Read PC	Store as Address.H.
6.1		Dec. S by 2. PC = Address.
6.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	20	1	JSR Abs	0006	ea	13	1f	fd	nv-BdIzc	LDX #	T1	13	13	7c
7	0006	20	1	JSR Abs	0006	ea	13	1f	fd	nv-BdIzc	LDX #	T1	13	13	26
8	0007	44	1		0007	ea	13	1f	fd	nv-BdIzc	JSR Abs	T2	26	26	26
8	0007	44	1		0007	ea	13	1f	fd	nv-BdIzc	JSR Abs	T2	26	26	4c
9	01fd	00	1		0008	ea	13	1f	44	nv-BdIzc	JSR Abs	T3	00	fd	4c
9	01fd	00	1		0008	ea	13	1f	44	nv-BdIzc	JSR Abs	T3	00	fd	fd
10	01fd	00	0		0008	ea	13	1f	44	nv-BdIzc	JSR Abs	T4	ff	fd	fd
10	01fd	00	0		0008	ea	13	1f	44	nv-BdIzc	JSR Abs	T4	ff	fd	fc
11	01fc	00	0		0008	ea	13	1f	44	nv-BdIzc	JSR Abs	T5	ff	fc	fc
11	01fc	08	0		0008	ea	13	1f	44	nv-BdIzc	JSR Abs	T5	ff	fc	fb
12	0008	01	1		0008	ea	13	1f	44	nv-BdIzc	JSR Abs	T0	fb	00	fb
12	0008	01	1		0008	ea	13	1f	44	nv-BdIzc	JSR Abs	T0	fb	00	fb
13	0144	ea	1	NOP	0144	ea	13	1f	fb	nv-BdIzc	JSR Abs	T1	00	01	fb
13	0144	ea	1	NOP	0144	ea	13	1f	fb	nv-BdIzc	JSR Abs	T1	00	01	01
14	0145	00	1		0145	ea	13	1f	fb	nv-BdIzc	NOP	T0+T2	01	01	01
14	0145	00	1		0145	ea	13	1f	fb	nv-BdIzc	NOP	T0+T2	01	01	02

AND (21)		Cycles: 6	Size: 2
Indirect X (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Operand.	
2.1		Inc. PC .	
2.2	Read Operand	Discard.	
3.1		Pointer = u8(Operand + X).	
3.2	Read Pointer	Store as Address.	
4.1			
4.2	Read u8(Pointer + 1)	Store as Address.H.	
5.1			
5.2	Read Address	Store as Operand.	
6.1		A &= Operand. N flag = (A & \$80), Z flag = ! A .	
6.2	Read PC	Store as OpCode.	
+X.1		* A , N flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	21	1	AND (zp,X)	0006	1f	13	1f	fd	nv-BdIzc	LDX #	T1	13	13	7c
7	0006	21	1	AND (zp,X)	0006	1f	13	1f	fd	nv-BdIzc	LDX #	T1	13	13	26
8	0007	44	1		0007	1f	13	1f	fd	nv-BdIzc	AND (zp,X)	T2	26	26	26
8	0007	44	1		0007	1f	13	1f	fd	nv-BdIzc	AND (zp,X)	T2	26	26	4c
9	0044	66	1		0008	1f	13	1f	fd	nv-BdIzc	AND (zp,X)	T3	13	44	4c
9	0044	66	1		0008	1f	13	1f	fd	nv-BdIzc	AND (zp,X)	T3	13	44	57
10	0057	77	1		0008	1f	13	1f	fd	nv-BdIzc	AND (zp,X)	T4	00	57	57
10	0057	77	1		0008	1f	13	1f	fd	nv-BdIzc	AND (zp,X)	T4	00	57	58
11	0058	01	1		0008	1f	13	1f	fd	nv-BdIzc	AND (zp,X)	T5	00	77	58
11	0058	01	1		0008	1f	13	1f	fd	nv-BdIzc	AND (zp,X)	T5	00	77	77
12	0177	81	1		0008	1f	13	1f	fd	nv-BdIzc	AND (zp,X)	T0	ff	ff	77
12	0177	81	1		0008	1f	13	1f	fd	nv-BdIzc	AND (zp,X)	T0	ff	ff	fe
13	0008	ea	1	NOP	0008	1f	13	1f	fd	Nv-BdIzc	AND (zp,X)	T1	1f	81	fe
13	0008	ea	1	NOP	0008	1f	13	1f	fd	Nv-BdIzc	AND (zp,X)	T1	1f	81	01
14	0009	b8	1		0009	01	13	1f	fd	nv-BdIzc	NOP	T0+T2	01	01	01
14	0009	b8	1		0009	01	13	1f	fd	nv-BdIzc	NOP	T0+T2	01	01	02

JAM (22)	Cycles: ∞	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Discard.
2.1		Visual6502 suggests incrementing PC , but Single-Step Tests cannot be passed this way.
2.2	Read \$FFFF	Discard.
3.1		
3.2	Read \$FFFE	Discard.
4.1		
4.2	Read \$FFFE	Discard.
5.1		Repeat 5.1 and 5.2 forever.
5.2	Read \$FFFF	Discard.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	22	1	unknown	0006	1f	13	22	fd	nv-BdIzc	LDX #	T1	13	13	88
7	0006	22	1	unknown	0006	1f	13	22	fd	nv-BdIzc	LDX #	T1	13	13	26
8	0007	44	1		0007	1f	13	22	fd	nv-BdIzc	unknown	T2	26	26	26
8	0007	44	1		0007	1f	13	22	fd	nv-BdIzc	unknown	T2	26	26	4c
9	ffff	00	1		0008	1f	13	22	fd	nv-BdIzc	unknown	T3	ff	ff	4c
9	ffff	00	1		0008	1f	13	22	fd	nv-BdIzc	unknown	T3	ff	ff	fe
10	fffe	00	1		0008	1f	13	22	fd	nv-BdIzc	unknown	T4	ff	ff	fe
10	fffe	00	1		0008	1f	13	22	fd	nv-BdIzc	unknown	T4	ff	ff	fe
11	fffe	00	1		0008	1f	13	22	fd	nv-BdIzc	unknown	T5	ff	ff	fe
11	fffe	00	1		0008	1f	13	22	fd	nv-BdIzc	unknown	T5	ff	ff	fe
12	ffff	00	1		0008	1f	13	22	fd	nv-BdIzc	unknown		ff	ff	fe
12	ffff	00	1		0008	1f	13	22	fd	nv-BdIzc	unknown		ff	ff	fe
13	ffff	00	1		0008	1f	13	22	fd	nv-BdIzc	unknown		ff	ff	fe
13	ffff	00	1		0008	1f	13	22	fd	nv-BdIzc	unknown		ff	ff	fe
14	ffff	00	1		0008	1f	13	22	fd	nv-BdIzc	unknown		ff	ff	fe
14	ffff	00	1		0008	1f	13	22	fd	nv-BdIzc	unknown		ff	ff	fe

RLA (23)		Cycles: 8	Size: 2
Indirect X (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Operand.	
2.1		Inc. PC .	
2.2	Read Operand	Discard.	
3.1		Pointer = u8(Operand + X).	
3.2	Read Pointer	Store as Address.	
4.1			
4.2	Read u8(Pointer + 1)	Store as Address.H.	
5.1			
5.2	Read Address	Store as Operand.	
6.1			
6.2	Write to Address	Write Operand.	
7.1		LowBit = C flag. C flag = (Operand & \$80).	
		Operand = (Operand << 1) LowBit.	
		A &= Operand.	
		N flag = (A & \$80), Z flag = ! A .	
7.2	Write to Address	Write Operand.	
8.1			
8.2	Read PC	Store as OpCode.	
+X.1		* A applied.	

* Setting of A is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	23	1	unknown	0006	1f	13	22	fd	nv-BdIzc	LDX #	T1	13	13	88
7	0006	23	1	unknown	0006	1f	13	22	fd	nv-BdIzc	LDX #	T1	13	13	26
8	0007	44	1		0007	1f	13	22	fd	nv-BdIzc	unknown	T2	26	26	26
8	0007	44	1		0007	1f	13	22	fd	nv-BdIzc	unknown	T2	26	26	4c
9	0044	66	1		0008	1f	13	22	fd	nv-BdIzc	unknown	T3	13	44	4c
9	0044	66	1		0008	1f	13	22	fd	nv-BdIzc	unknown	T3	13	44	57
10	0057	77	1		0008	1f	13	22	fd	nv-BdIzc	unknown	T4	00	57	57
10	0057	77	1		0008	1f	13	22	fd	nv-BdIzc	unknown	T4	00	57	58
11	0058	01	1		0008	1f	13	22	fd	nv-BdIzc	unknown	T5	00	77	58
11	0058	01	1		0008	1f	13	22	fd	nv-BdIzc	unknown	T5	00	77	77
12	0177	81	1		0008	1f	13	22	fd	nv-BdIzc	unknown		ff	ff	77
12	0177	81	1		0008	1f	13	22	fd	nv-BdIzc	unknown		ff	ff	fe
13	0177	81	0		0008	1f	13	22	fd	nv-BdIzc	unknown		81	81	fe
13	0177	81	0		0008	1f	13	22	fd	nv-BdIzc	unknown		81	81	02
14	0177	81	0		0008	1f	13	22	fd	nv-BdIzc	unknown	T0	02	02	02
14	0177	02	0		0008	1f	13	22	fd	nv-BdIzc	unknown	T0	02	02	04
15	0008	ea	1	NOP	0008	1f	13	22	fd	nv-BdIzc	unknown	T1	1f	02	04
15	0008	ea	1	NOP	0008	1f	13	22	fd	nv-BdIzc	unknown	T1	1f	02	02
16	0009	b8	1		0009	02	13	22	fd	nv-BdIzc	NOP	T0+T2	02	02	02
16	0009	b8	1		0009	02	13	22	fd	nv-BdIzc	NOP	T0+T2	02	02	04

BIT (24)		Cycles: 3	Size: 2
Zero Page (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read Address	Store as Operand.	
3.1		V flag = (Operand & \$40), N flag = (Operand & \$80), Z flag = !(Operand & A).	
3.2	Read PC	Store as OpCode.	
+X.1		* Z flag applied.	

* Setting of Z flag is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	24	1	BIT zp	0006	19	13	22	fd	nv-BdIzc	LDX #	T1	13	13	88
7	0006	24	1	BIT zp	0006	19	13	22	fd	nv-BdIzc	LDX #	T1	13	13	26
8	0007	44	1		0007	19	13	22	fd	nv-BdIzc	BIT zp	T2	26	26	26
8	0007	44	1		0007	19	13	22	fd	nv-BdIzc	BIT zp	T2	26	26	4c
9	0044	e6	1		0008	19	13	22	fd	nv-BdIzc	BIT zp	T0	ff	44	4c
9	0044	e6	1		0008	19	13	22	fd	nv-BdIzc	BIT zp	T0	ff	44	43
10	0008	ea	1	NOP	0008	19	13	22	fd	NV-BdIzc	BIT zp	T1	19	e6	43
10	0008	ea	1	NOP	0008	19	13	22	fd	NV-BdIzc	BIT zp	T1	19	e6	00
11	0009	b8	1		0009	19	13	22	fd	NV-BdIZc	NOP	T0+T2	00	00	00
11	0009	b8	1		0009	19	13	22	fd	NV-BdIZc	NOP	T0+T2	00	00	00

AND (25) Cycles: 3 Size: 2		
Zero Page (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read Address	Store as Operand.
3.1		A &= Operand. N flag = (A & \$80), Z flag = ! A .
3.2	Read PC	Store as OpCode.
+X.1		* A , N flag, and Z flag applied.

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	25	1	AND zp	0006	19	13	22	fd	nv-BdIzc	LDX #	T1	13	13	88
7	0006	25	1	AND zp	0006	19	13	22	fd	nv-BdIzc	LDX #	T1	13	13	26
8	0007	44	1		0007	19	13	22	fd	nv-BdIzc	AND zp	T2	26	26	26
8	0007	44	1		0007	19	13	22	fd	nv-BdIzc	AND zp	T2	26	26	4c
9	0044	e6	1		0008	19	13	22	fd	nv-BdIzc	AND zp	T0	ff	44	4c
9	0044	e6	1		0008	19	13	22	fd	nv-BdIzc	AND zp	T0	ff	44	43
10	0008	ea	1	NOP	0008	19	13	22	fd	Nv-BdIzc	AND zp	T1	19	e6	43
10	0008	ea	1	NOP	0008	19	13	22	fd	Nv-BdIzc	AND zp	T1	19	e6	00
11	0009	b8	1		0009	00	13	22	fd	nv-BdIZc	NOP	T0+T2	00	00	00
11	0009	b8	1		0009	00	13	22	fd	nv-BdIZc	NOP	T0+T2	00	00	00

ROL (26)	Cycles: 5	Size: 2
Zero Page (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read Address	Store as Operand.
3.1		
3.2	Write to Address	Write Operand.
4.1		LowBit = C flag. C flag = (Operand & \$80). Operand = (Operand << 1) LowBit. N flag = (Operand & \$80), Z flag = !Operand.
4.2	Write to Address	Write Operand.
5.1		
5.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	26	1	ROL zp	0006	19	13	22	fd	nv-BdIzc	LDX #	T1	13	13	88
7	0006	26	1	ROL zp	0006	19	13	22	fd	nv-BdIzc	LDX #	T1	13	13	26
8	0007	44	1		0007	19	13	22	fd	nv-BdIzc	ROL zp	T2	26	26	26
8	0007	44	1		0007	19	13	22	fd	nv-BdIzc	ROL zp	T2	26	26	4c
9	0044	e6	1		0008	19	13	22	fd	nv-BdIzc	ROL zp	T3	ff	44	4c
9	0044	e6	1		0008	19	13	22	fd	nv-BdIzc	ROL zp	T3	ff	44	43
10	0044	e6	0		0008	19	13	22	fd	nv-BdIzc	ROL zp	T4	e6	e6	43
10	0044	e6	0		0008	19	13	22	fd	nv-BdIzc	ROL zp	T4	e6	e6	cc
11	0044	e6	0		0008	19	13	22	fd	Nv-BdIzc	ROL zp	T0	cc	cc	cc
11	0044	cc	0		0008	19	13	22	fd	Nv-BdIzc	ROL zp	T0	cc	cc	98
12	0008	ea	1	NOP	0008	19	13	22	fd	Nv-BdIzc	ROL zp	T1	ff	cc	98
12	0008	ea	1	NOP	0008	19	13	22	fd	Nv-BdIzc	ROL zp	T1	ff	cc	cc
13	0009	b8	1		0009	19	13	22	fd	Nv-BdIzc	NOP	T0+T2	cc	cc	cc
13	0009	b8	1		0009	19	13	22	fd	Nv-BdIzc	NOP	T0+T2	cc	cc	98

RLA (27)		Cycles: 5	Size: 2
Zero Page (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read Address	Store as Operand.	
3.1			
3.2	Write to Address	Write Operand.	
4.1		LowBit = C flag. C flag = (Operand & \$80). Operand = (Operand << 1) LowBit. A &= Operand. N flag = (A & \$80), Z flag = ! A .	
4.2	Write to Address	Write Operand.	
5.1			
5.2	Read PC	Store as OpCode.	
+X.1		* N flag applied.	

* N flag initially set based off Operand in 4.1, then set again based off A with a delay of 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	27	1	unknown	0006	19	13	22	fd	nv-BdIzc	LDX #	T1	13	13	88
7	0006	27	1	unknown	0006	19	13	22	fd	nv-BdIzc	LDX #	T1	13	13	26
8	0007	44	1		0007	19	13	22	fd	nv-BdIzc	unknown	T2	26	26	26
8	0007	44	1		0007	19	13	22	fd	nv-BdIzc	unknown	T2	26	26	4c
9	0044	cc	1		0008	19	13	22	fd	nv-BdIzc	unknown	T3	ff	44	4c
9	0044	cc	1		0008	19	13	22	fd	nv-BdIzc	unknown	T3	ff	44	43
10	0044	cc	0		0008	19	13	22	fd	nv-BdIzc	unknown	T4	cc	cc	43
10	0044	cc	0		0008	19	13	22	fd	nv-BdIzc	unknown	T4	cc	cc	98
11	0044	cc	0		0008	19	13	22	fd	Nv-BdIzc	unknown	T0	98	98	98
11	0044	98	0		0008	19	13	22	fd	Nv-BdIzc	unknown	T0	98	98	30
12	0008	ea	1	NOP	0008	19	13	22	fd	Nv-BdIzc	unknown	T1	19	98	30
12	0008	ea	1	NOP	0008	19	13	22	fd	Nv-BdIzc	unknown	T1	19	98	18
13	0009	b8	1		0009	18	13	22	fd	nv-BdIzc	NOP	T0+T2	18	18	18
13	0009	b8	1		0009	18	13	22	fd	nv-BdIzc	NOP	T0+T2	18	18	30

PLP (28)	Cycles: 4	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Discard.
2.1		
2.2	Read (S \$0100)	Store as Operand.
3.1		Inc. S by 1.
3.2	Read (S \$0100)	Store as Operand.
4.1		P = ((Operand & ~(X flag)) (M flag)).
4.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	28	1	PLP	0006	19	13	22	fd	nv-BdIzc	LDX #	T1	13	13	88
7	0006	28	1	PLP	0006	19	13	22	fd	nv-BdIzc	LDX #	T1	13	13	26
8	0007	ea	1		0007	19	13	22	fd	nv-BdIzc	PLP	T2	26	26	26
8	0007	ea	1		0007	19	13	22	fd	nv-BdIzc	PLP	T2	26	26	4c
9	01fd	00	1		0007	19	13	22	fd	nv-BdIzc	PLP	T3	00	fd	4c
9	01fd	00	1		0007	19	13	22	fd	nv-BdIzc	PLP	T3	00	fd	fe
10	01fe	00	1		0007	19	13	22	fe	nv-BdIzc	PLP	T0	00	fe	fe
10	01fe	00	1		0007	19	13	22	fe	nv-BdIzc	PLP	T0	00	fe	ff
11	0007	ea	1	NOP	0007	19	13	22	fe	nv-Bdizc	PLP	T1	00	00	ff
11	0007	ea	1	NOP	0007	19	13	22	fe	nv-Bdizc	PLP	T1	00	00	01
12	0008	ea	1		0008	19	13	22	fe	nv-Bdizc	NOP	T0+T2	00	00	00
12	0008	ea	1		0008	19	13	22	fe	nv-Bdizc	NOP	T0+T2	00	01	02

AND (29) Cycles: 2 Size: 2		
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . A &= Operand. N flag = (A & \$80), Z flag = ! A .
2.2	Read PC	Store as OpCode.
+X.1		* A , N flag, and Z flag applied.

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	29	1	AND #	0006	fe	13	22	fd	nv-BdIzc	LDX #	T1	13	13	88
7	0006	29	1	AND #	0006	fe	13	22	fd	nv-BdIzc	LDX #	T1	13	13	26
8	0007	83	1		0007	fe	13	22	fd	nv-BdIzc	AND #	T0+T2	26	26	26
8	0007	83	1		0007	fe	13	22	fd	nv-BdIzc	AND #	T0+T2	26	26	4c
9	0008	ea	1	NOP	0008	fe	13	22	fd	Nv-BdIzc	AND #	T1	fe	83	4c
9	0008	ea	1	NOP	0008	fe	13	22	fd	Nv-BdIzc	AND #	T1	fe	83	82
10	0009	b8	1		0009	82	13	22	fd	Nv-BdIzc	NOP	T0+T2	82	82	82
10	0009	b8	1		0009	82	13	22	fd	Nv-BdIzc	NOP	T0+T2	82	82	04

ROL (2A)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1	Read PC	Inc. PC .	
1.2		Discard.	
2.1		LowBit = C flag. C flag = (A & \$80). A = (A << 1) LowBit.	
2.2	Read PC	N flag = (A & \$80), Z flag = ! A . Store as OpCode.	
+X.1		* A , N flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	2a	1	ROL	0006	fe	13	22	fd	nv-BdIzc	LDX #	T1	13	13	88
7	0006	2a	1	ROL	0006	fe	13	22	fd	nv-BdIzc	LDX #	T1	13	13	26
8	0007	ea	1		0007	fe	13	22	fd	nv-BdIzc	ROL	T0+T2	26	26	26
8	0007	ea	1		0007	fe	13	22	fd	nv-BdIzc	ROL	T0+T2	26	26	4c
9	0007	ea	1	NOP	0007	fe	13	22	fd	nv-BdIzc	ROL	T1	fe	fe	4c
9	0007	ea	1	NOP	0007	fe	13	22	fd	nv-BdIzc	ROL	T1	fe	fe	fc
10	0008	a4	1		0008	fc	13	22	fd	Nv-BdIzC	NOP	T0+T2	fc	fc	fc
10	0008	a4	1		0008	fc	13	22	fd	Nv-BdIzC	NOP	T0+T2	fc	fc	f8

ANC (2B)		Cycles: 2	Size: 2
Immediate			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1	Read PC	Inc. PC .	
1.2		Store as Operand.	
2.1		Inc. PC . A &= Operand. C flag = (A & \$80), N flag = (A & \$80), Z flag = ! A .	
2.2		Store as OpCode.	
+X.1		* A , C flag, N flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

Visual6502 omits the update of A; flags are updated based off the incorrect A value.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	2b	1	unknown	0006	31	13	22	fd	nv-BdIzc	LDX #	T1	13	13	88
7	0006	2b	1	unknown	0006	31	13	22	fd	nv-BdIzc	LDX #	T1	13	13	26
8	0007	11	1	NOP	0007	31	13	22	fd	nv-BdIzc	unknown	T0+T2	26	26	26
8	0007	11	1		0007	31	13	22	fd	nv-BdIzc	unknown	T0+T2	26	26	4c
9	0008	ea	1		0008	31	13	22	fd	nv-BdIzc	unknown	T1	11	11	4c
9	0008	ea	1		0008	31	13	22	fd	nv-BdIzc	unknown	T1	31	31	31
10	0009	b8	1		0009	31	13	22	fd	nv-BdIzc	NOP	T0+T2	31	31	31
10	0009	b8	1		0009	31	13	22	fd	nv-BdIzc	NOP	T0+T2	31	31	62

ANC (ANC2, ANA, ANB)

Type: Combination of an immediate and an implied command (Sub-instructions: AND, ASL/ROL)

OpC.	Mnemonic	Function	Size	Cycles	N	V	-	B	D	I	Z	C
\$0B	ANC #imm	A = A & #{imm}	2	2	o						o	o
\$2B	ANC #imm	A = A & #{imm}	2	2	o						o	o

Operation: ANDs the contents of the A register with an immediate value and then moves bit 7 of A into the Carry flag.

- This opcode works basically identically to AND #imm. except that the Carry flag is set to the same state that the Negative flag is set to. (bit 7 is put into the carry, as if the ASL/ROL would have been executed)

— NoMoreSecrets-NMOS6510UnintendedOpCodes-20242412.pdf

BIT (2C)	Cycles: 4	Size: 3
Absolute (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address	Store as Operand.
4.1		V flag = (Operand & \$40), N flag = (Operand & \$80), Z flag = !(Operand & A).
4.2	Read PC	Store as OpCode.
+X.1		* Z flag applied.

* Setting of Z flag is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	2c	1	BIT Abs	0006	99	f0	22	fd	Nv-BdIzc	LDX #	T1	f0	f0	88
7	0006	2c	1	BIT Abs	0006	99	f0	22	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	11	1		0007	99	f0	22	fd	Nv-BdIzc	BIT Abs	T2	e0	e0	e0
8	0007	11	1		0007	99	f0	22	fd	Nv-BdIzc	BIT Abs	T2	e0	e0	c0
9	0008	01	1		0008	99	f0	22	fd	Nv-BdIzc	BIT Abs	T3	00	11	c0
9	0008	01	1		0008	99	f0	22	fd	Nv-BdIzc	BIT Abs	T3	00	11	11
10	0111	66	1		0009	99	f0	22	fd	Nv-BdIzc	BIT Abs	T0	ff	11	11
10	0111	66	1		0009	99	f0	22	fd	Nv-BdIzc	BIT Abs	T0	ff	11	10
11	0009	ea	1	NOP	0009	99	f0	22	fd	nV-BdIzc	BIT Abs	T1	99	66	10
11	0009	ea	1	NOP	0009	99	f0	22	fd	nV-BdIzc	BIT Abs	T1	99	66	00
12	000a	15	1		000a	99	f0	22	fd	nV-BdIzc	NOP	T0+T2	00	00	00
12	000a	15	1		000a	99	f0	22	fd	nV-BdIzc	NOP	T0+T2	00	00	00

AND (2D)		Cycles: 4	Size: 3
Absolute (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read PC	Store as Address.H.	
3.1		Inc. PC .	
3.2	Read Address	Store as Operand.	
4.1		A &= Operand. N flag = (A & \$80), Z flag = ! A .	
4.2	Read PC	Store as OpCode.	
+X.1		* A , N flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	2d	1	AND Abs	0006	99	f0	22	fd	Nv-BdIzc	LDX #	T1	f0	f0	88
7	0006	2d	1	AND Abs	0006	99	f0	22	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	11	1		0007	99	f0	22	fd	Nv-BdIzc	AND Abs	T2	e0	e0	e0
8	0007	11	1		0007	99	f0	22	fd	Nv-BdIzc	AND Abs	T2	e0	e0	c0
9	0008	01	1		0008	99	f0	22	fd	Nv-BdIzc	AND Abs	T3	00	11	c0
9	0008	01	1		0008	99	f0	22	fd	Nv-BdIzc	AND Abs	T3	00	11	11
10	0111	66	1		0009	99	f0	22	fd	Nv-BdIzc	AND Abs	T0	ff	ff	11
10	0111	66	1		0009	99	f0	22	fd	Nv-BdIzc	AND Abs	T0	ff	ff	fe
11	0009	ea	1	NOP	0009	99	f0	22	fd	nv-BdIzc	AND Abs	T1	99	66	fe
11	0009	ea	1	NOP	0009	99	f0	22	fd	nv-BdIzc	AND Abs	T1	99	66	00
12	000a	15	1		000a	00	f0	22	fd	nv-BdIzc	NOP	T0+T2	00	00	00
12	000a	15	1		000a	00	f0	22	fd	nv-BdIzc	NOP	T0+T2	00	00	00

ROL (2E)		Cycles: 6	Size: 3
Absolute (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read PC	Store as Address.H.	
3.1		Inc. PC .	
3.2	Read Address	Store as Operand.	
4.1			
4.2	Write to Address	Write Operand.	
5.1		LowBit = C flag. C flag = (Operand & \$80).	
		Operand = (Operand << 1) LowBit.	
		N flag = (Operand & \$80), Z flag = !Operand.	
5.2	Write to Address	Write Operand.	
6.1			
6.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	2e	1	ROL Abs	0006	99	7f	22	fd	nv-BdIzc	LDX #	T1	7f	7f	88
7	0006	2e	1	ROL Abs	0006	99	7f	22	fd	nv-BdIzc	LDX #	T1	7f	7f	fe
8	0007	11	1		0007	99	7f	22	fd	nv-BdIzc	ROL Abs	T2	fe	fe	fe
8	0007	11	1		0007	99	7f	22	fd	nv-BdIzc	ROL Abs	T2	fe	fe	fc
9	0008	01	1		0008	99	7f	22	fd	nv-BdIzc	ROL Abs	T3	00	11	fc
9	0008	01	1		0008	99	7f	22	fd	nv-BdIzc	ROL Abs	T3	00	11	11
10	0111	cc	1		0009	99	7f	22	fd	nv-BdIzc	ROL Abs	T4	ff	ff	11
10	0111	cc	1		0009	99	7f	22	fd	nv-BdIzc	ROL Abs	T4	ff	ff	fe
11	0111	cc	0		0009	99	7f	22	fd	nv-BdIzc	ROL Abs	T5	cc	cc	fe
11	0111	cc	0		0009	99	7f	22	fd	nv-BdIzc	ROL Abs	T5	cc	cc	98
12	0111	cc	0		0009	99	7f	22	fd	Nv-BdIzc	ROL Abs	T0	98	98	98
12	0111	98	0		0009	99	7f	22	fd	Nv-BdIzc	ROL Abs	T0	98	98	30
13	0009	ea	1	NOP	0009	99	7f	22	fd	Nv-BdIzc	ROL Abs	T1	ff	98	30
13	0009	ea	1	NOP	0009	99	7f	22	fd	Nv-BdIzc	ROL Abs	T1	ff	98	98
14	000a	15	1		000a	99	7f	22	fd	Nv-BdIzc	NOP	T0+T2	98	98	98
14	000a	15	1		000a	99	7f	22	fd	Nv-BdIzc	NOP	T0+T2	98	98	30

RLA (2F)	Cycles: 6	Size: 3
Absolute (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.
5.1		LowBit = C flag. C flag = (Operand & \$80). Operand = (Operand << 1) LowBit. A &= Operand. N flag = (A & \$80), Z flag = ! A .
5.2	Write to Address	Write Operand.
6.1		
6.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	2f	1	unknown	0006	99	cc	22	fd	Nv-BdIzc	LDX #	T1	cc	cc	88
7	0006	2f	1	unknown	0006	99	cc	22	fd	Nv-BdIzc	LDX #	T1	cc	cc	98
8	0007	11	1		0007	99	cc	22	fd	Nv-BdIzc	unknown	T2	98	98	98
8	0007	11	1		0007	99	cc	22	fd	Nv-BdIzc	unknown	T2	98	98	30
9	0008	01	1		0008	99	cc	22	fd	Nv-BdIzc	unknown	T3	00	11	30
9	0008	01	1		0008	99	cc	22	fd	Nv-BdIzc	unknown	T3	00	11	11
10	0111	88	1		0009	99	cc	22	fd	Nv-BdIzc	unknown	T4	ff	ff	11
10	0111	88	1		0009	99	cc	22	fd	Nv-BdIzc	unknown	T4	ff	ff	fe
11	0111	88	0		0009	99	cc	22	fd	Nv-BdIzc	unknown	T5	88	88	fe
11	0111	88	0		0009	99	cc	22	fd	Nv-BdIzc	unknown	T5	88	88	10
12	0111	88	0		0009	99	cc	22	fd	nv-BdIzC	unknown	T0	10	10	10
12	0111	10	0		0009	99	cc	22	fd	nv-BdIzC	unknown	T0	10	10	20
13	0009	ea	1	NOP	0009	99	cc	22	fd	nv-BdIzC	unknown	T1	99	10	20
13	0009	ea	1	NOP	0009	99	cc	22	fd	nv-BdIzC	unknown	T1	99	10	10
14	000a	15	1		000a	10	cc	22	fd	nv-BdIzC	NOP	T0+T2	10	10	10
14	000a	15	1		000a	10	cc	22	fd	nv-BdIzC	NOP	T0+T2	10	10	20

BMI (30)	Cycles: 2-4	Size: 2
Relative (Jump)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC . Jump = ((P & \$80) == \$80).
1.2	Read PC	Store as Operand.
		If !Jump, poll interrupts.
		Otherwise Address = i16(i8(Operand)) + PC + 1.
		BoundaryCrossed = Address.H != (PC + 1).H.
		If !BoundaryCrossed, poll interrupts.
2.1		Inc. PC . If !Jump, end (next half-cycle is 4.2).
2.2	Read PC	Discard.
3.1		PC .L = Address.L. If !BoundaryCrossed, end (next half-cycle is 4.2).
3.2	Read PC	Discard. Poll interrupts.
4.1		PC .H = Address.H.
4.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	30	1	BMI	0006	99	7f	22	fd	nv-BdIzc	LDX #	T1	7f	7f	88
7	0006	30	1	BMI	0006	99	7f	22	fd	nv-BdIzc	LDX #	T1	7f	7f	fe
8	0007	11	1		0007	99	7f	22	fd	nv-BdIzc	BMI	T2	fe	fe	fe
8	0007	11	1		0007	99	7f	22	fd	nv-BdIzc	BMI	T2	fe	fe	fc
9	0008	01	1	ORA (zp,X)	0008	99	7f	22	fd	nv-BdIzc	BMI		11	08	fc
9	0008	01	1	ORA (zp,X)	0008	99	7f	22	fd	nv-BdIzc	BMI		11	08	19
10	0009	ea	1		0009	99	7f	22	fd	nv-BdIzc	ORA (zp,X)	T2	19	19	19
10	0009	ea	1		0009	99	7f	22	fd	nv-BdIzc	ORA (zp,X)	T2	19	19	32

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	30	1	BMI	0006	99	cc	22	fd	Nv-BdIzc	LDX #	T1	cc	cc	88
7	0006	30	1	BMI	0006	99	cc	22	fd	Nv-BdIzc	LDX #	T1	cc	cc	98
8	0007	11	1		0007	99	cc	22	fd	Nv-BdIzc	BMI	T2	98	98	98
8	0007	11	1		0007	99	cc	22	fd	Nv-BdIzc	BMI	T2	98	98	30
9	0008	01	1		0008	99	cc	22	fd	Nv-BdIzc	BMI	T3	11	08	30
9	0008	01	1		0008	99	cc	22	fd	Nv-BdIzc	BMI	T3	11	08	19
10	0019	ea	1	NOP	0019	99	cc	22	fd	Nv-BdIzc	BMI		00	00	19
10	0019	ea	1	NOP	0019	99	cc	22	fd	Nv-BdIzc	BMI		00	00	01
11	001a	00	1		001a	99	cc	22	fd	Nv-BdIzc	NOP	T0+T2	01	01	01
11	001a	00	1		001a	99	cc	22	fd	Nv-BdIzc	NOP	T0+T2	01	01	02

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	30	1	BMI	0006	99	cc	22	fd	Nv-BdIzc	LDX #	T1	cc	cc	88
7	0006	30	1	BMI	0006	99	cc	22	fd	Nv-BdIzc	LDX #	T1	cc	cc	98
8	0007	f4	1		0007	99	cc	22	fd	Nv-BdIzc	BMI	T2	98	98	98
8	0007	f4	1		0007	99	cc	22	fd	Nv-BdIzc	BMI	T2	98	98	30
9	0008	01	1		0008	99	cc	22	fd	Nv-BdIzc	BMI	T3	f4	08	30
9	0008	01	1		0008	99	cc	22	fd	Nv-BdIzc	BMI	T3	f4	08	fc
10	00fc	00	1		00fc	99	cc	22	fd	Nv-BdIzc	BMI	T0	00	ff	fc
10	00fc	00	1		00fc	99	cc	22	fd	Nv-BdIzc	BMI	T0	00	ff	ff
11	ffff	00	1	BRK	ffff	99	cc	22	fd	Nv-BdIzc	BMI	T1	ff	00	ff
11	ffff	00	1	BRK	ffff	99	cc	22	fd	Nv-BdIzc	BMI	T1	ff	00	ff
12	ffff	00	1		ffff	99	cc	22	fd	Nv-BdIzc	BRK	T2	ff	ff	ff
12	ffff	00	1		ffff	99	cc	22	fd	Nv-BdIzc	BRK	T2	ff	ff	fe

AND (31)		Cycles: 5-6	Size: 2
Indirect Y (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.L.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Address.	
3.1			
3.2	Read u8(Pointer + 1)	Store as Address.H.	
4.1		Target = Address + Y. Pointer.L = Target.L. Pointer.H = Address.H. BoundaryCrossed = Pointer.H != Target.H.	
4.2	Read Pointer	Store as Operand. If !BoundaryCrossed, skip the next cycle.	
5.1		Pointer.H = Target.H.	
5.2	Read Pointer	Store as Operand.	
6.1		A &= Operand. N flag = (A & \$80), Z flag = ! A .	
6.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	31	1	AND (zp),Y	0006	99	cc	22	fd	Nv-BdIZc	LDX #	T1	cc	cc	88
7	0006	31	1	AND (zp),Y	0006	99	cc	22	fd	Nv-BdIZc	LDX #	T1	cc	cc	98
8	0007	f4	1		0007	99	cc	22	fd	Nv-BdIZc	AND (zp),Y	T2	98	98	98
8	0007	f4	1		0007	99	cc	22	fd	Nv-BdIZc	AND (zp),Y	T2	98	98	30
9	00f4	0f	1		0008	99	cc	22	fd	Nv-BdIZc	AND (zp),Y	T3	00	f4	30
9	00f4	0f	1		0008	99	cc	22	fd	Nv-BdIZc	AND (zp),Y	T3	00	f4	f5
10	00f5	01	1		0008	99	cc	22	fd	Nv-BdIZc	AND (zp),Y	T4	22	0f	f5
10	00f5	01	1		0008	99	cc	22	fd	Nv-BdIZc	AND (zp),Y	T4	22	0f	31
11	0131	00	1		0008	99	cc	22	fd	Nv-BdIZc	AND (zp),Y	T0	00	01	31
11	0131	00	1		0008	99	cc	22	fd	Nv-BdIZc	AND (zp),Y	T0	00	01	02
12	0008	01	1	ORA (zp,X)	0008	99	cc	22	fd	nv-BdIZc	AND (zp),Y	T1	99	00	02
12	0008	01	1	ORA (zp,X)	0008	99	cc	22	fd	nv-BdIZc	AND (zp),Y	T1	99	00	00
13	0009	ea	1		0009	00	cc	22	fd	nv-BdIZc	ORA (zp,X)	T2	00	00	00
13	0009	ea	1		0009	00	cc	22	fd	nv-BdIZc	ORA (zp,X)	T2	00	00	00

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	31	1	AND (zp),Y	0006	99	cc	ee	fd	Nv-BdIZc	LDX #	T1	cc	cc	b8
7	0006	31	1	AND (zp),Y	0006	99	cc	ee	fd	Nv-BdIZc	LDX #	T1	cc	cc	98
8	0007	f4	1		0007	99	cc	ee	fd	Nv-BdIZc	AND (zp),Y	T2	98	98	98
8	0007	f4	1		0007	99	cc	ee	fd	Nv-BdIZc	AND (zp),Y	T2	98	98	30
9	00f4	f0	1		0008	99	cc	ee	fd	Nv-BdIZc	AND (zp),Y	T3	00	f4	30
9	00f4	f0	1		0008	99	cc	ee	fd	Nv-BdIZc	AND (zp),Y	T3	00	f4	f5
10	00f5	00	1		0008	99	cc	ee	fd	Nv-BdIZc	AND (zp),Y	T4	ee	f0	f5
10	00f5	00	1		0008	99	cc	ee	fd	Nv-BdIZc	AND (zp),Y	T4	ee	f0	de
11	00de	00	1		0008	99	cc	ee	fd	Nv-BdIZc	AND (zp),Y	T5	00	00	de
11	00de	00	1		0008	99	cc	ee	fd	Nv-BdIZc	AND (zp),Y	T5	00	00	01
12	01de	00	1		0008	99	cc	ee	fd	Nv-BdIZc	AND (zp),Y	T0	01	ff	01
12	01de	00	1		0008	99	cc	ee	fd	Nv-BdIZc	AND (zp),Y	T0	01	ff	00
13	0008	01	1	ORA (zp,X)	0008	99	cc	ee	fd	nv-BdIZc	AND (zp),Y	T1	99	00	00
13	0008	01	1	ORA (zp,X)	0008	99	cc	ee	fd	nv-BdIZc	AND (zp),Y	T1	99	00	00
14	0009	ea	1		0009	00	cc	ee	fd	nv-BdIZc	ORA (zp,X)	T2	00	00	00
14	0009	ea	1		0009	00	cc	ee	fd	nv-BdIZc	ORA (zp,X)	T2	00	00	00

JAM (32)	Cycles: ∞	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Discard.
2.1		Visual6502 suggests incrementing PC , but Single-Step Tests cannot be passed this way.
2.2	Read \$FFFF	Discard.
3.1		
3.2	Read \$FFFE	Discard.
4.1		
4.2	Read \$FFFE	Discard.
5.1		Repeat 5.1 and 5.2 forever.
5.2	Read \$FFFF	Discard.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	32	1	unknown	0006	99	cc	ee	fd	Nv-BdIzc	LDX #	T1	cc	cc	b8
7	0006	32	1	unknown	0006	99	cc	ee	fd	Nv-BdIzc	LDX #	T1	cc	cc	98
8	0007	f4	1		0007	99	cc	ee	fd	Nv-BdIzc	unknown	T2	98	98	98
8	0007	f4	1		0007	99	cc	ee	fd	Nv-BdIzc	unknown	T2	98	98	30
9	ffff	00	1		0008	99	cc	ee	fd	Nv-BdIzc	unknown	T3	ff	ff	30
9	ffff	00	1		0008	99	cc	ee	fd	Nv-BdIzc	unknown	T3	ff	ff	fe
10	fffe	00	1		0008	99	cc	ee	fd	Nv-BdIzc	unknown	T4	ff	ff	fe
10	fffe	00	1		0008	99	cc	ee	fd	Nv-BdIzc	unknown	T4	ff	ff	fe
11	fffe	00	1		0008	99	cc	ee	fd	Nv-BdIzc	unknown	T5	ff	ff	fe
11	fffe	00	1		0008	99	cc	ee	fd	Nv-BdIzc	unknown	T5	ff	ff	fe
12	ffff	00	1		0008	99	cc	ee	fd	Nv-BdIzc	unknown		ff	ff	fe
12	ffff	00	1		0008	99	cc	ee	fd	Nv-BdIzc	unknown		ff	ff	fe
13	ffff	00	1		0008	99	cc	ee	fd	Nv-BdIzc	unknown		ff	ff	fe
13	ffff	00	1		0008	99	cc	ee	fd	Nv-BdIzc	unknown		ff	ff	fe
14	ffff	00	1		0008	99	cc	ee	fd	Nv-BdIzc	unknown		ff	ff	fe
14	ffff	00	1		0008	99	cc	ee	fd	Nv-BdIzc	unknown		ff	ff	fe

RLA (33)		Cycles: 8	Size: 2
Indirect Y (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.L.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Address.	
3.1			
3.2	Read u8(Pointer + 1)	Store as Address.H.	
4.1		Target = Address + Y. Pointer.L = Target.L. Pointer.H = Address.H.	
4.2	Read Pointer	Store as Operand.	
5.1		Pointer.H = Target.H.	
5.2	Read Pointer	Store as Operand.	
6.1			
6.2	Write to Pointer	Write Operand.	
7.1		LowBit = C flag. C flag = (Operand & \$80). Operand = (Operand << 1) LowBit. A &= Operand. N flag = (A & \$80), Z flag = ! A .	
7.2	Write to Pointer	Write Operand.	
8.1			
8.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	33	1	unknown	0006	99	cc	ee	fd	Nv-BdIzc	LDX #	T1	cc	cc	b8
7	0006	33	1	unknown	0006	99	cc	ee	fd	Nv-BdIzc	LDX #	T1	cc	cc	98
8	0007	f4	1		0007	99	cc	ee	fd	Nv-BdIzc	unknown	T2	98	98	98
8	0007	f4	1		0007	99	cc	ee	fd	Nv-BdIzc	unknown	T2	98	98	30
9	00f4	f0	1		0008	99	cc	ee	fd	Nv-BdIzc	unknown	T3	00	f4	30
9	00f4	f0	1		0008	99	cc	ee	fd	Nv-BdIzc	unknown	T3	00	f4	f5
10	00f5	00	1		0008	99	cc	ee	fd	Nv-BdIzc	unknown	T4	ee	f0	f5
10	00f5	00	1		0008	99	cc	ee	fd	Nv-BdIzc	unknown	T4	ee	f0	de
11	00de	00	1		0008	99	cc	ee	fd	Nv-BdIzc	unknown	T5	00	00	de
11	00de	00	1		0008	99	cc	ee	fd	Nv-BdIzc	unknown	T5	00	00	01
12	01de	00	1		0008	99	cc	ee	fd	Nv-BdIzc	unknown		01	ff	01
12	01de	00	1		0008	99	cc	ee	fd	Nv-BdIzc	unknown		01	ff	00
13	01de	00	0		0008	99	cc	ee	fd	Nv-BdIzc	unknown		00	00	00
13	01de	00	0		0008	99	cc	ee	fd	Nv-BdIzc	unknown		00	00	00
14	01de	00	0		0008	99	cc	ee	fd	nv-BdIzc	unknown	T0	00	00	00
14	01de	00	0		0008	99	cc	ee	fd	nv-BdIzc	unknown	T0	00	00	00
15	0008	01	1	ORA (zp,X)	0008	99	cc	ee	fd	nv-BdIzc	unknown	T1	99	00	00
15	0008	01	1	ORA (zp,X)	0008	99	cc	ee	fd	nv-BdIzc	unknown	T1	99	00	00
16	0009	ea	1		0009	00	cc	ee	fd	nv-BdIzc	ORA (zp,X)	T2	00	00	00
16	0009	ea	1		0009	00	cc	ee	fd	nv-BdIzc	ORA (zp,X)	T2	00	00	00

NOP (34)		Cycles: 4	Size: 2
Zero Page X-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.L.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Operand.	
3.1		Address = u8(Pointer + X).	
3.2	Read Address	Store as Operand.	
4.1			
4.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	34	1	unknown	0006	99	cc	ee	fd	Nv-BdIzc	LDX #	T1	cc	cc	b8
7	0006	34	1	unknown	0006	99	cc	ee	fd	Nv-BdIzc	LDX #	T1	cc	cc	98
8	0007	f4	1		0007	99	cc	ee	fd	Nv-BdIzc	unknown	T2	98	98	98
8	0007	f4	1		0007	99	cc	ee	fd	Nv-BdIzc	unknown	T2	98	98	30
9	00f4	f0	1		0008	99	cc	ee	fd	Nv-BdIzc	unknown	T3	cc	f4	30
9	00f4	f0	1		0008	99	cc	ee	fd	Nv-BdIzc	unknown	T3	cc	f4	c0
10	00c0	00	1		0008	99	cc	ee	fd	Nv-BdIzc	unknown	T0	ff	ff	c0
10	00c0	00	1		0008	99	cc	ee	fd	Nv-BdIzc	unknown	T0	ff	ff	fe
11	0008	01	1	ORA (zp,X)	0008	99	cc	ee	fd	Nv-BdIzc	unknown	T1	ff	00	fe
11	0008	01	1	ORA (zp,X)	0008	99	cc	ee	fd	Nv-BdIzc	unknown	T1	ff	00	ff
12	0009	ea	1		0009	99	cc	ee	fd	Nv-BdIzc	ORA (zp,X)	T2	ff	ff	ff
12	0009	ea	1		0009	99	cc	ee	fd	Nv-BdIzc	ORA (zp,X)	T2	ff	ff	fe

AND (35)		Cycles: 4	Size: 2
Zero Page X-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.L.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Operand.	
3.1		Address = u8(Pointer + X).	
3.2	Read Address	Store as Operand.	
4.1		A &= Operand. N flag = (A & \$80), Z flag = ! A .	
4.2	Read PC	Store as OpCode.	
+X.1		* A , N flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	35	1	AND zp,X	0006	99	cc	ee	fd	Nv-BdIZc	LDX #	T1	cc	cc	b8
7	0006	35	1	AND zp,X	0006	99	cc	ee	fd	Nv-BdIZc	LDX #	T1	cc	cc	98
8	0007	f4	1		0007	99	cc	ee	fd	Nv-BdIZc	AND zp,X	T2	98	98	98
8	0007	f4	1		0007	99	cc	ee	fd	Nv-BdIZc	AND zp,X	T2	98	98	30
9	00f4	f0	1		0008	99	cc	ee	fd	Nv-BdIZc	AND zp,X	T3	cc	f4	30
9	00f4	f0	1		0008	99	cc	ee	fd	Nv-BdIZc	AND zp,X	T3	cc	f4	c0
10	00c0	00	1		0008	99	cc	ee	fd	Nv-BdIZc	AND zp,X	T0	ff	ff	c0
10	00c0	00	1		0008	99	cc	ee	fd	Nv-BdIZc	AND zp,X	T0	ff	ff	fe
11	0008	01	1	ORA (zp,X)	0008	99	cc	ee	fd	nv-BdIZc	AND zp,X	T1	99	00	fe
11	0008	01	1	ORA (zp,X)	0008	99	cc	ee	fd	nv-BdIZc	AND zp,X	T1	99	00	00
12	0009	ea	1		0009	00	cc	ee	fd	nv-BdIZc	ORA (zp,X)	T2	00	00	00
12	0009	ea	1		0009	00	cc	ee	fd	nv-BdIZc	ORA (zp,X)	T2	00	00	00

ROL (36)		Cycles: 6	Size: 2
Zero Page X-Indexed (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.L.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Operand.	
3.1		Address = u8(Pointer + X).	
3.2	Read Address	Store as Operand.	
4.1			
4.2	Write to Address	Write Operand.	
5.1		LowBit = C flag. C flag = (Operand & \$80).	
		Operand = (Operand << 1) LowBit.	
		N flag = (Operand & \$80), Z flag = !Operand.	
5.2	Write to Address	Write Operand.	
6.1			
6.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	36	1	ROL zp,X	0006	66	cc	ee	fd	Nv-BdIzc	LDX #	T1	cc	cc	b8
7	0006	36	1	ROL zp,X	0006	66	cc	ee	fd	Nv-BdIzc	LDX #	T1	cc	cc	98
8	0007	f4	1		0007	66	cc	ee	fd	Nv-BdIzc	ROL zp,X	T2	98	98	98
8	0007	f4	1		0007	66	cc	ee	fd	Nv-BdIzc	ROL zp,X	T2	98	98	30
9	00f4	f0	1		0008	66	cc	ee	fd	Nv-BdIzc	ROL zp,X	T3	cc	f4	30
9	00f4	f0	1		0008	66	cc	ee	fd	Nv-BdIzc	ROL zp,X	T3	cc	f4	c0
10	00c0	99	1		0008	66	cc	ee	fd	Nv-BdIzc	ROL zp,X	T4	ff	ff	c0
10	00c0	99	1		0008	66	cc	ee	fd	Nv-BdIzc	ROL zp,X	T4	ff	ff	fe
11	00c0	99	0		0008	66	cc	ee	fd	Nv-BdIzc	ROL zp,X	T5	99	99	fe
11	00c0	99	0		0008	66	cc	ee	fd	Nv-BdIzc	ROL zp,X	T5	99	99	32
12	00c0	99	0		0008	66	cc	ee	fd	nv-BdIzc	ROL zp,X	T0	32	32	32
12	00c0	32	0		0008	66	cc	ee	fd	nv-BdIzc	ROL zp,X	T0	32	32	64
13	0008	01	1	ORA (zp,X)	0008	66	cc	ee	fd	nv-BdIzc	ROL zp,X	T1	ff	32	64
13	0008	01	1	ORA (zp,X)	0008	66	cc	ee	fd	nv-BdIzc	ROL zp,X	T1	ff	32	32
14	0009	ea	1		0009	66	cc	ee	fd	nv-BdIzc	ORA (zp,X)	T2	32	32	32
14	0009	ea	1		0009	66	cc	ee	fd	nv-BdIzc	ORA (zp,X)	T2	32	32	64

RLA (37)	Cycles: 6	Size: 2
Zero Page X-Indexed (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.L.
2.1		Inc. PC .
2.2	Read Pointer	Store as Operand.
3.1		Address = u8(Pointer + X).
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.
5.1		LowBit = C flag. C flag = (Operand & \$80). Operand = (Operand << 1) LowBit. A &= Operand. N flag = (A & \$80), Z flag = ! A .
5.2	Write to Address	Write Operand.
6.1		
6.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	37	1	unknown	0006	66	cc	ee	fd	Nv-BdIzc	LDX #	T1	cc	cc	b8
7	0006	37	1	unknown	0006	66	cc	ee	fd	Nv-BdIzc	LDX #	T1	cc	cc	98
8	0007	f4	1		0007	66	cc	ee	fd	Nv-BdIzc	unknown	T2	98	98	98
8	0007	f4	1		0007	66	cc	ee	fd	Nv-BdIzc	unknown	T2	98	98	30
9	00f4	f0	1		0008	66	cc	ee	fd	Nv-BdIzc	unknown	T3	cc	f4	30
9	00f4	f0	1		0008	66	cc	ee	fd	Nv-BdIzc	unknown	T3	cc	f4	c0
10	00c0	32	1		0008	66	cc	ee	fd	Nv-BdIzc	unknown	T4	ff	ff	c0
10	00c0	32	1		0008	66	cc	ee	fd	Nv-BdIzc	unknown	T4	ff	ff	fe
11	00c0	32	0		0008	66	cc	ee	fd	Nv-BdIzc	unknown	T5	32	32	fe
11	00c0	32	0		0008	66	cc	ee	fd	Nv-BdIzc	unknown	T5	32	32	64
12	00c0	32	0		0008	66	cc	ee	fd	nv-BdIzc	unknown	T0	64	64	64
12	00c0	64	0		0008	66	cc	ee	fd	nv-BdIzc	unknown	T0	64	64	c8
13	0008	01	1	ORA (zp,X)	0008	66	cc	ee	fd	nv-BdIzc	unknown	T1	66	64	c8
13	0008	01	1	ORA (zp,X)	0008	66	cc	ee	fd	nv-BdIzc	unknown	T1	66	64	64
14	0009	ea	1		0009	64	cc	ee	fd	nv-BdIzc	ORA (zp,X)	T2	64	64	64
14	0009	ea	1		0009	64	cc	ee	fd	nv-BdIzc	ORA (zp,X)	T2	64	64	c8

SEC (38)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Discard.	
2.1		C flag = 1.	
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	38	1	SEC	0006	66	cc	ee	fd	Nv-BdIzc	LDX #	T1	cc	cc	b8
7	0006	38	1	SEC	0006	66	cc	ee	fd	Nv-BdIzc	LDX #	T1	cc	cc	98
8	0007	ea	1		0007	66	cc	ee	fd	Nv-BdIzc	SEC	T0+T2	98	98	98
8	0007	ea	1		0007	66	cc	ee	fd	Nv-BdIzc	SEC	T0+T2	98	98	30
9	0007	ea	1	NOP	0007	66	cc	ee	fd	Nv-BdIzc	SEC	T1	ff	ff	30
9	0007	ea	1	NOP	0007	66	cc	ee	fd	Nv-BdIzc	SEC	T1	ff	ff	fe
10	0008	01	1		0008	66	cc	ee	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fe
10	0008	01	1		0008	66	cc	ee	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fc

AND (39)		Cycles: 4-5	Size: 3
Absolute Y-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.L.	
2.1		Inc. PC .	
2.2	Read PC	Store as Pointer.H.	
3.1		Inc. PC . Target = Pointer + Y. Address.L = Target.L. Address.H = Pointer.H. BoundaryCrossed = Pointer.H != Target.H.	
3.2	Read Address	Store as Operand. If !BoundaryCrossed, skip the next cycle.	
4.1		Address.H = Target.H.	
4.2	Read Address	Store as Operand.	
5.1		A &= Operand. N flag = (A & \$80), Z flag = ! A .	
5.2	Read PC	Store as OpCode.	
+X.1		* A , N flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	39	1	AND Abs,Y	0006	66	cc	44	fd	Nv-BdIzc	LDX #	T1	cc	cc	10
7	0006	39	1	AND Abs,Y	0006	66	cc	44	fd	Nv-BdIzc	LDX #	T1	cc	cc	98
8	0007	33	1		0007	66	cc	44	fd	Nv-BdIzc	AND Abs,Y	T2	98	98	98
8	0007	33	1		0007	66	cc	44	fd	Nv-BdIzc	AND Abs,Y	T2	98	98	30
9	0008	01	1		0008	66	cc	44	fd	Nv-BdIzc	AND Abs,Y	T3	44	33	30
9	0008	01	1		0008	66	cc	44	fd	Nv-BdIzc	AND Abs,Y	T3	44	33	77
10	0177	02	1		0009	66	cc	44	fd	Nv-BdIzc	AND Abs,Y	T0	00	01	77
10	0177	02	1		0009	66	cc	44	fd	Nv-BdIzc	AND Abs,Y	T0	00	01	02
11	0009	ea	1	NOP	0009	66	cc	44	fd	nv-BdIzc	AND Abs,Y	T1	66	02	02
11	0009	ea	1	NOP	0009	66	cc	44	fd	nv-BdIzc	AND Abs,Y	T1	66	02	02
12	000a	15	1		000a	02	cc	44	fd	nv-BdIzc	NOP	T0+T2	02	02	02
12	000a	15	1		000a	02	cc	44	fd	nv-BdIzc	NOP	T0+T2	02	02	04

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	39	1	AND Abs,Y	0006	66	cc	44	fd	Nv-BdIzc	LDX #	T1	cc	cc	10
7	0006	39	1	AND Abs,Y	0006	66	cc	44	fd	Nv-BdIzc	LDX #	T1	cc	cc	98
8	0007	ee	1		0007	66	cc	44	fd	Nv-BdIzc	AND Abs,Y	T2	98	98	98
8	0007	ee	1		0007	66	cc	44	fd	Nv-BdIzc	AND Abs,Y	T2	98	98	30
9	0008	00	1		0008	66	cc	44	fd	Nv-BdIzc	AND Abs,Y	T3	44	ee	30
9	0008	00	1		0008	66	cc	44	fd	Nv-BdIzc	AND Abs,Y	T3	44	ee	32
10	0032	00	1		0009	66	cc	44	fd	Nv-BdIzc	AND Abs,Y	T4	00	00	32
10	0032	00	1		0009	66	cc	44	fd	Nv-BdIzc	AND Abs,Y	T4	00	00	01
11	0132	99	1		0009	66	cc	44	fd	Nv-BdIzc	AND Abs,Y	T0	01	ff	01
11	0132	99	1		0009	66	cc	44	fd	Nv-BdIzc	AND Abs,Y	T0	01	ff	00
12	0009	ea	1	NOP	0009	66	cc	44	fd	Nv-BdIzc	AND Abs,Y	T1	66	99	00
12	0009	ea	1	NOP	0009	66	cc	44	fd	Nv-BdIzc	AND Abs,Y	T1	66	99	00
13	000a	15	1		000a	00	cc	44	fd	nv-BdIzc	NOP	T0+T2	00	00	00
13	000a	15	1		000a	00	cc	44	fd	nv-BdIzc	NOP	T0+T2	00	00	00

NOP (3A)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Discard.	
2.1			
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	3a	1	unknown	0006	66	cc	44	fd	Nv-BdIzc	LDX #	T1	cc	cc	10
7	0006	3a	1	unknown	0006	66	cc	44	fd	Nv-BdIzc	LDX #	T1	cc	cc	98
8	0007	ea	1		0007	66	cc	44	fd	Nv-BdIzc	unknown	T0+T2	98	98	98
8	0007	ea	1		0007	66	cc	44	fd	Nv-BdIzc	unknown	T0+T2	98	98	30
9	0007	ea	1	NOP	0007	66	cc	44	fd	Nv-BdIzc	unknown	T1	ff	ff	30
9	0007	ea	1	NOP	0007	66	cc	44	fd	Nv-BdIzc	unknown	T1	ff	ff	fe
10	0008	00	1		0008	66	cc	44	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fe
10	0008	00	1		0008	66	cc	44	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fc

RLA (3B)	Cycles: 7	Size: 3
Absolute Y-Indexed		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.L.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + Y . Address.L = Target.L. Address.H = Pointer.H.
3.2	Read Address	Store as Operand.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		
5.2	Write to Address	Write Operand.
6.1		LowBit = C flag. C flag = (Operand & \$80). Operand = (Operand << 1) LowBit. A &= Operand. N flag = (A & \$80), Z flag = ! A .
6.2	Write to Address	Write Operand.
7.1		
7.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	3b	1	unknown	0006	66	cc	44	fd	Nv-BdIzc	LDX #	T1	cc	cc	10
7	0006	3b	1	unknown	0006	66	cc	44	fd	Nv-BdIzc	LDX #	T1	cc	cc	98
8	0007	ea	1		0007	66	cc	44	fd	Nv-BdIzc	unknown	T2	98	98	98
8	0007	ea	1		0007	66	cc	44	fd	Nv-BdIzc	unknown	T2	98	98	30
9	0008	00	1		0008	66	cc	44	fd	Nv-BdIzc	unknown	T3	44	ea	30
9	0008	00	1		0008	66	cc	44	fd	Nv-BdIzc	unknown	T3	44	ea	2e
10	002e	00	1		0009	66	cc	44	fd	Nv-BdIzc	unknown	T4	00	00	2e
10	002e	00	1		0009	66	cc	44	fd	Nv-BdIzc	unknown	T4	00	00	01
11	012e	1f	1		0009	66	cc	44	fd	Nv-BdIzc	unknown	T5	01	ff	01
11	012e	1f	1		0009	66	cc	44	fd	Nv-BdIzc	unknown	T5	01	ff	00
12	012e	1f	0		0009	66	cc	44	fd	Nv-BdIzc	unknown		1f	1f	00
12	012e	1f	0		0009	66	cc	44	fd	Nv-BdIzc	unknown		1f	1f	3e
13	012e	1f	0		0009	66	cc	44	fd	nv-BdIzc	unknown	T0	3e	3e	3e
13	012e	3e	0		0009	66	cc	44	fd	nv-BdIzc	unknown	T0	3e	3e	7c
14	0009	ea	1	NOP	0009	66	cc	44	fd	nv-BdIzc	unknown	T1	66	3e	7c
14	0009	ea	1	NOP	0009	66	cc	44	fd	nv-BdIzc	unknown	T1	66	3e	26
15	000a	15	1		000a	26	cc	44	fd	nv-BdIzc	NOP	T0+T2	26	26	26
15	000a	15	1		000a	26	cc	44	fd	nv-BdIzc	NOP	T0+T2	26	26	4c

NOP (3C)		Cycles: 4-5	Size: 3
Absolute X-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.L.	
2.1		Inc. PC .	
2.2	Read PC	Store as Pointer.H.	
3.1		Inc. PC . Target = Pointer + X .	
		Address.L = Target.L.	
		Address.H = Pointer.H.	
		BoundaryCrossed = Pointer.H != Target.H.	
3.2	Read Address	Store as Operand. If !BoundaryCrossed, skip the next cycle.	
4.1		Address.H = Target.H.	
4.2	Read Address	Store as Operand.	
5.1			
5.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	3c	1	unknown	0006	66	cc	44	fd	Nv-BdIzc	LDX #	T1	cc	cc	10
7	0006	3c	1	unknown	0006	66	cc	44	fd	Nv-BdIzc	LDX #	T1	cc	cc	98
8	0007	ea	1		0007	66	cc	44	fd	Nv-BdIzc	unknown	T2	98	98	98
8	0007	ea	1		0007	66	cc	44	fd	Nv-BdIzc	unknown	T2	98	98	30
9	0008	00	1		0008	66	cc	44	fd	Nv-BdIzc	unknown	T3	cc	ea	30
9	0008	00	1		0008	66	cc	44	fd	Nv-BdIzc	unknown	T3	cc	ea	b6
10	00b6	00	1		0009	66	cc	44	fd	Nv-BdIzc	unknown	T4	00	00	b6
10	00b6	00	1		0009	66	cc	44	fd	Nv-BdIzc	unknown	T4	00	00	01
11	01b6	00	1		0009	66	cc	44	fd	Nv-BdIzc	unknown	T0	01	ff	01
11	01b6	00	1		0009	66	cc	44	fd	Nv-BdIzc	unknown	T0	01	ff	00
12	0009	ea	1	NOP	0009	66	cc	44	fd	Nv-BdIzc	unknown	T1	ff	00	00
12	0009	ea	1	NOP	0009	66	cc	44	fd	Nv-BdIzc	unknown	T1	ff	00	ff
13	000a	15	1		000a	66	cc	44	fd	Nv-BdIzc	NOP	T0+T2	ff	ff	ff
13	000a	15	1		000a	66	cc	44	fd	Nv-BdIzc	NOP	T0+T2	ff	ff	fe

AND (3D) Cycles: 4-5		Size: 3
Absolute X-Indexed (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.L.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + X . Address.L = Target.L. Address.H = Pointer.H. BoundaryCrossed = Pointer.H != Target.H.
3.2	Read Address	Store as Operand. If !BoundaryCrossed, skip the next cycle.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		A &= Operand. N flag = (A & \$80), Z flag = ! A .
5.2	Read PC	Store as OpCode.
+X.1		* A , N flag, and Z flag applied.

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	3d	1	AND Abs,X	0006	99	11	44	fd	nv-BdIzc	LDX #	T1	11	11	10
7	0006	3d	1	AND Abs,X	0006	99	11	44	fd	nv-BdIzc	LDX #	T1	11	11	22
8	0007	ea	1		0007	99	11	44	fd	nv-BdIzc	AND Abs,X	T2	22	22	22
8	0007	ea	1		0007	99	11	44	fd	nv-BdIzc	AND Abs,X	T2	22	22	44
9	0008	00	1		0008	99	11	44	fd	nv-BdIzc	AND Abs,X	T3	11	ea	44
9	0008	00	1		0008	99	11	44	fd	nv-BdIzc	AND Abs,X	T3	11	ea	fb
10	00fb	66	1		0009	99	11	44	fd	nv-BdIzc	AND Abs,X	T0	00	00	fb
10	00fb	66	1		0009	99	11	44	fd	nv-BdIzc	AND Abs,X	T0	00	00	01
11	0009	ea	1	NOP	0009	99	11	44	fd	nv-BdIzc	AND Abs,X	T1	99	66	01
11	0009	ea	1	NOP	0009	99	11	44	fd	nv-BdIzc	AND Abs,X	T1	99	66	00
12	000a	15	1		000a	00	11	44	fd	nv-BdIzc	NOP	T0+T2	00	00	00
12	000a	15	1		000a	00	11	44	fd	nv-BdIzc	NOP	T0+T2	00	00	00

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	3d	1	AND Abs,X	0006	66	cc	44	fd	Nv-BdIzc	LDX #	T1	cc	cc	10
7	0006	3d	1	AND Abs,X	0006	66	cc	44	fd	Nv-BdIzc	LDX #	T1	cc	cc	98
8	0007	ea	1		0007	66	cc	44	fd	Nv-BdIzc	AND Abs,X	T2	98	98	98
8	0007	ea	1		0007	66	cc	44	fd	Nv-BdIzc	AND Abs,X	T2	98	98	30
9	0008	00	1		0008	66	cc	44	fd	Nv-BdIzc	AND Abs,X	T3	cc	ea	30
9	0008	00	1		0008	66	cc	44	fd	Nv-BdIzc	AND Abs,X	T3	cc	ea	b6
10	00b6	00	1		0009	66	cc	44	fd	Nv-BdIzc	AND Abs,X	T4	00	00	b6
10	00b6	00	1		0009	66	cc	44	fd	Nv-BdIzc	AND Abs,X	T4	00	00	01
11	01b6	99	1		0009	66	cc	44	fd	Nv-BdIzc	AND Abs,X	T0	01	ff	01
11	01b6	99	1		0009	66	cc	44	fd	Nv-BdIzc	AND Abs,X	T0	01	ff	00
12	0009	ea	1	NOP	0009	66	cc	44	fd	Nv-BdIzc	AND Abs,X	T1	66	99	00
12	0009	ea	1	NOP	0009	66	cc	44	fd	Nv-BdIzc	AND Abs,X	T1	66	99	00
13	000a	15	1		000a	00	cc	44	fd	nv-BdIzc	NOP	T0+T2	00	00	00
13	000a	15	1		000a	00	cc	44	fd	nv-BdIzc	NOP	T0+T2	00	00	00

ROL (3E) Cycles: 7		Size: 3
Absolute X-Indexed (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.L.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + X . Address.L = Target.L. Address.H = Pointer.H.
3.2	Read Address	Store as Operand.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		
5.2	Write to Address	Write Operand.
6.1		LowBit = C flag. C flag = (Operand & \$80). Operand = (Operand << 1) LowBit. N flag = (Operand & \$80), Z flag = !Operand.
6.2	Write to Address	Write Operand.
7.1		
7.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	3e	1	ROL Abs,X	0006	99	11	44	fd	nv-BdIzc	LDX #	T1	11	11	10
7	0006	3e	1	ROL Abs,X	0006	99	11	44	fd	nv-BdIzc	LDX #	T1	11	11	22
8	0007	ea	1		0007	99	11	44	fd	nv-BdIzc	ROL Abs,X	T2	22	22	22
8	0007	ea	1		0007	99	11	44	fd	nv-BdIzc	ROL Abs,X	T2	22	22	44
9	0008	00	1		0008	99	11	44	fd	nv-BdIzc	ROL Abs,X	T3	11	ea	44
9	0008	00	1		0008	99	11	44	fd	nv-BdIzc	ROL Abs,X	T3	11	ea	fb
10	00fb	66	1		0009	99	11	44	fd	nv-BdIzc	ROL Abs,X	T4	00	00	fb
10	00fb	66	1		0009	99	11	44	fd	nv-BdIzc	ROL Abs,X	T4	00	00	01
11	00fb	66	1		0009	99	11	44	fd	nv-BdIzc	ROL Abs,X	T5	01	ff	01
11	00fb	66	1		0009	99	11	44	fd	nv-BdIzc	ROL Abs,X	T5	01	ff	00
12	00fb	66	0		0009	99	11	44	fd	nv-BdIzc	ROL Abs,X		66	66	00
12	00fb	66	0		0009	99	11	44	fd	nv-BdIzc	ROL Abs,X		66	66	cc
13	00fb	66	0		0009	99	11	44	fd	Nv-BdIzc	ROL Abs,X	T0	cc	cc	cc
13	00fb	cc	0		0009	99	11	44	fd	Nv-BdIzc	ROL Abs,X	T0	cc	cc	98
14	0009	ea	1	NOP	0009	99	11	44	fd	Nv-BdIzc	ROL Abs,X	T1	ff	cc	98
14	0009	ea	1	NOP	0009	99	11	44	fd	Nv-BdIzc	ROL Abs,X	T1	ff	cc	cb
15	000a	15	1		000a	99	11	44	fd	Nv-BdIzc	NOP	T0+T2	cb	cb	cb
15	000a	15	1		000a	99	11	44	fd	Nv-BdIzc	NOP	T0+T2	cb	cb	96

RLA (3F)		Cycles: 7	Size: 3
Absolute X-Indexed (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.L.	
2.1		Inc. PC .	
2.2	Read PC	Store as Pointer.H.	
3.1		Inc. PC . Target = Pointer + X .	
		Address.L = Target.L.	
		Address.H = Pointer.H.	
3.2	Read Address	Store as Operand.	
4.1		Address.H = Target.H.	
4.2	Read Address	Store as Operand.	
5.1			
5.2	Write to Address	Write Operand.	
6.1		LowBit = C flag. C flag = (Operand & \$80).	
		Operand = (Operand << 1) LowBit.	
		A &= Operand.	
		N flag = (A & \$80), Z flag = ! A .	
6.2	Write to Address	Write Operand.	
7.1			
7.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	3f	1	unknown	0006	99	11	44	fd	nv-BdIzc	LDX #	T1	11	11	10
7	0006	3f	1	unknown	0006	99	11	44	fd	nv-BdIzc	LDX #	T1	11	11	22
8	0007	ea	1		0007	99	11	44	fd	nv-BdIzc	unknown	T2	22	22	22
8	0007	ea	1		0007	99	11	44	fd	nv-BdIzc	unknown	T2	22	22	44
9	0008	00	1		0008	99	11	44	fd	nv-BdIzc	unknown	T3	11	ea	44
9	0008	00	1		0008	99	11	44	fd	nv-BdIzc	unknown	T3	11	ea	fb
10	00fb	cc	1		0009	99	11	44	fd	nv-BdIzc	unknown	T4	00	00	fb
10	00fb	cc	1		0009	99	11	44	fd	nv-BdIzc	unknown	T4	00	00	01
11	00fb	cc	1		0009	99	11	44	fd	nv-BdIzc	unknown	T5	01	ff	01
11	00fb	cc	1		0009	99	11	44	fd	nv-BdIzc	unknown	T5	01	ff	00
12	00fb	cc	0		0009	99	11	44	fd	nv-BdIzc	unknown		cc	cc	00
12	00fb	cc	0		0009	99	11	44	fd	nv-BdIzc	unknown		cc	cc	98
13	00fb	cc	0		0009	99	11	44	fd	Nv-BdIzc	unknown	T0	98	98	98
13	00fb	98	0		0009	99	11	44	fd	Nv-BdIzc	unknown	T0	98	98	30
14	0009	ea	1	NOP	0009	99	11	44	fd	Nv-BdIzc	unknown	T1	99	98	30
14	0009	ea	1	NOP	0009	99	11	44	fd	Nv-BdIzc	unknown	T1	99	98	98
15	000a	15	1		000a	98	11	44	fd	Nv-BdIzc	NOP	T0+T2	98	98	98
15	000a	15	1		000a	98	11	44	fd	Nv-BdIzc	NOP	T0+T2	98	98	30

RTI (40)	Cycles: 6	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC .
2.2	Read u8(S +0) \$0100	Store as Operand.
3.1		
3.2	Read u8(S +1) \$0100	Store as Operand.
4.1		Mask = (M flag X flag). P = (Operand & ~Mask) (P & Mask).
4.2	Read u8(S +2) \$0100	Store as Target.L.
5.1		Inc. S by 3.
5.2	Read u8(S +0) \$0100	Store as Target.H.
6.1		PC = Target.
6.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	40	1	RTI	0006	99	11	44	fd	nv-BdIzc	LDX #	T1	11	11	10
7	0006	40	1	RTI	0006	99	11	44	fd	nv-BdIzc	LDX #	T1	11	11	22
8	0007	ea	1		0007	99	11	44	fd	nv-BdIzc	RTI	T2	22	22	22
8	0007	ea	1		0007	99	11	44	fd	nv-BdIzc	RTI	T2	22	22	44
9	01fd	00	1		0008	99	11	44	fd	nv-BdIzc	RTI	T3	00	fd	44
9	01fd	00	1		0008	99	11	44	fd	nv-BdIzc	RTI	T3	00	fd	fe
10	01fe	00	1		0008	99	11	44	fd	nv-BdIzc	RTI	T4	00	fe	fe
10	01fe	00	1		0008	99	11	44	fd	nv-BdIzc	RTI	T4	00	fe	ff
11	01ff	00	1		0008	99	11	44	fd	nv-Bdizc	RTI	T5	00	ff	ff
11	01ff	00	1		0008	99	11	44	fd	nv-Bdizc	RTI	T5	00	ff	00
12	0100	00	1		0008	99	11	44	00	nv-Bdizc	RTI	T0	00	00	00
12	0100	00	1		0008	99	11	44	00	nv-Bdizc	RTI	T0	00	00	00
13	0000	a9	1	LDA #	0000	99	11	44	00	nv-Bdizc	RTI	T1	00	00	00
13	0000	a9	1	LDA #	0000	99	11	44	00	nv-Bdizc	RTI	T1	00	00	00
14	0001	99	1		0001	99	11	44	00	nv-Bdizc	LDA #	T0+T2	00	00	00
14	0001	99	1		0001	99	11	44	00	nv-Bdizc	LDA #	T0+T2	00	00	00

EOR (41)		Cycles: 6	Size: 2
Indirect X (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Operand.	
2.1		Inc. PC .	
2.2	Read Operand	Discard.	
3.1		Pointer = u8(Operand + X).	
3.2	Read Pointer	Store as Address.	
4.1			
4.2	Read u8(Pointer + 1)	Store as Address.H.	
5.1			
5.2	Read Address	Store as Operand.	
6.1		A ^= Operand. N flag = (A & \$80), Z flag = ! A .	
6.2	Read PC	Store as OpCode.	
+X.1		* A , N flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	41	1	EOR (zp,X)	0006	99	11	44	fd	nv-BdIzc	LDX #	T1	11	11	10
7	0006	41	1	EOR (zp,X)	0006	99	11	44	fd	nv-BdIzc	LDX #	T1	11	11	22
8	0007	dd	1		0007	99	11	44	fd	nv-BdIzc	EOR (zp,X)	T2	22	22	22
8	0007	dd	1		0007	99	11	44	fd	nv-BdIzc	EOR (zp,X)	T2	22	22	44
9	00dd	00	1		0008	99	11	44	fd	nv-BdIzc	EOR (zp,X)	T3	11	dd	44
9	00dd	00	1		0008	99	11	44	fd	nv-BdIzc	EOR (zp,X)	T3	11	dd	ee
10	00ee	77	1		0008	99	11	44	fd	nv-BdIzc	EOR (zp,X)	T4	00	ee	ee
10	00ee	77	1		0008	99	11	44	fd	nv-BdIzc	EOR (zp,X)	T4	00	ee	ef
11	00ef	01	1		0008	99	11	44	fd	nv-BdIzc	EOR (zp,X)	T5	00	77	ef
11	00ef	01	1		0008	99	11	44	fd	nv-BdIzc	EOR (zp,X)	T5	00	77	77
12	0177	66	1		0008	99	11	44	fd	nv-BdIzc	EOR (zp,X)	T0	ff	ff	77
12	0177	66	1		0008	99	11	44	fd	nv-BdIzc	EOR (zp,X)	T0	ff	ff	fe
13	0008	ea	1	NOP	0008	99	11	44	fd	nv-BdIzc	EOR (zp,X)	T1	99	66	fe
13	0008	ea	1	NOP	0008	99	11	44	fd	nv-BdIzc	EOR (zp,X)	T1	99	66	ff
14	0009	ea	1		0009	ff	11	44	fd	Nv-BdIzc	NOP	T0+T2	ff	ff	ff
14	0009	ea	1		0009	ff	11	44	fd	Nv-BdIzc	NOP	T0+T2	ff	ff	fe

JAM (42)	Cycles: ∞	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Discard.
2.1		Visual6502 suggests incrementing PC , but Single-Step Tests cannot be passed this way.
2.2	Read \$FFFF	Discard.
3.1		
3.2	Read \$FFFE	Discard.
4.1		
4.2	Read \$FFFE	Discard.
5.1		Repeat 5.1 and 5.2 forever.
5.2	Read \$FFFF	Discard.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	42	1	unknown	0006	99	11	44	fd	nv-BdIzc	LDX #	T1	11	11	10
7	0006	42	1	unknown	0006	99	11	44	fd	nv-BdIzc	LDX #	T1	11	11	22
8	0007	dd	1		0007	99	11	44	fd	nv-BdIzc	unknown	T2	22	22	22
8	0007	dd	1		0007	99	11	44	fd	nv-BdIzc	unknown	T2	22	22	44
9	ffff	00	1		0008	99	11	44	fd	nv-BdIzc	unknown	T3	ff	ff	44
9	ffff	00	1		0008	99	11	44	fd	nv-BdIzc	unknown	T3	ff	ff	fe
10	fffe	00	1		0008	99	11	44	fd	nv-BdIzc	unknown	T4	ff	ff	fe
10	fffe	00	1		0008	99	11	44	fd	nv-BdIzc	unknown	T4	ff	ff	fe
11	fffe	00	1		0008	99	11	44	fd	nv-BdIzc	unknown	T5	ff	ff	fe
11	fffe	00	1		0008	99	11	44	fd	nv-BdIzc	unknown	T5	ff	ff	fe
12	ffff	00	1		0008	99	11	44	fd	nv-BdIzc	unknown		ff	ff	fe
12	ffff	00	1		0008	99	11	44	fd	nv-BdIzc	unknown		ff	ff	fe
13	ffff	00	1		0008	99	11	44	fd	nv-BdIzc	unknown		ff	ff	fe
13	ffff	00	1		0008	99	11	44	fd	nv-BdIzc	unknown		ff	ff	fe
14	ffff	00	1		0008	99	11	44	fd	nv-BdIzc	unknown		ff	ff	fe
14	ffff	00	1		0008	99	11	44	fd	nv-BdIzc	unknown		ff	ff	fe

SRE (43)	Cycles: 8	Size: 2
Indirect X (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC .
2.2	Read Operand	Discard.
3.1		Pointer = u8(Operand + X).
3.2	Read Pointer	Store as Address.
4.1		
4.2	Read u8(Pointer + 1)	Store as Address.H.
5.1		
5.2	Read Address	Store as Operand.
6.1		
6.2	Write to Address	Write Operand.
7.1		C flag = (Operand & \$01). Operand >>= 1. A ^= Operand. N flag = (A & \$80), Z flag = ! A .
7.2	Write to Address	Write Operand.
8.1		
8.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	43	1	unknown	0006	99	11	44	fd	nv-BdIzc	LDX #	T1	11	11	10
7	0006	43	1	unknown	0006	99	11	44	fd	nv-BdIzc	LDX #	T1	11	11	22
8	0007	dd	1		0007	99	11	44	fd	nv-BdIzc	unknown	T2	22	22	22
8	0007	dd	1		0007	99	11	44	fd	nv-BdIzc	unknown	T2	22	22	44
9	00dd	00	1		0008	99	11	44	fd	nv-BdIzc	unknown	T3	11	dd	44
9	00dd	00	1		0008	99	11	44	fd	nv-BdIzc	unknown	T3	11	dd	ee
10	00ee	77	1		0008	99	11	44	fd	nv-BdIzc	unknown	T4	00	ee	ee
10	00ee	77	1		0008	99	11	44	fd	nv-BdIzc	unknown	T4	00	ee	ef
11	00ef	01	1		0008	99	11	44	fd	nv-BdIzc	unknown	T5	00	77	ef
11	00ef	01	1		0008	99	11	44	fd	nv-BdIzc	unknown	T5	00	77	77
12	0177	66	1		0008	99	11	44	fd	nv-BdIzc	unknown		ff	ff	77
12	0177	66	1		0008	99	11	44	fd	nv-BdIzc	unknown		ff	ff	fe
13	0177	66	0		0008	99	11	44	fd	nv-BdIzc	unknown		ff	66	fe
13	0177	66	0		0008	99	11	44	fd	nv-BdIzc	unknown		ff	66	33
14	0177	66	0		0008	99	11	44	fd	nv-BdIzc	unknown	T0	33	33	33
14	0177	33	0		0008	99	11	44	fd	nv-BdIzc	unknown	T0	33	33	66
15	0008	ea	1	NOP	0008	99	11	44	fd	nv-BdIzc	unknown	T1	99	33	66
15	0008	ea	1	NOP	0008	99	11	44	fd	nv-BdIzc	unknown	T1	99	33	aa
16	0009	ea	1		0009	aa	11	44	fd	Nv-BdIzc	NOP	T0+T2	aa	aa	aa
16	0009	ea	1		0009	aa	11	44	fd	Nv-BdIzc	NOP	T0+T2	aa	aa	54

NOP (44)		Cycles: 3	Size: 2
Zero Page (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read Address	Store as Operand.	
3.1			
3.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	44	1	unknown	0006	99	11	44	fd	nv-BdIzc	LDX #	T1	11	11	10
7	0006	44	1	unknown	0006	99	11	44	fd	nv-BdIzc	LDX #	T1	11	11	22
8	0007	dd	1		0007	99	11	44	fd	nv-BdIzc	unknown	T2	22	22	22
8	0007	dd	1		0007	99	11	44	fd	nv-BdIzc	unknown	T2	22	22	44
9	00dd	00	1		0008	99	11	44	fd	nv-BdIzc	unknown	T0	ff	dd	44
9	00dd	00	1		0008	99	11	44	fd	nv-BdIzc	unknown	T0	ff	dd	dc
10	0008	ea	1	NOP	0008	99	11	44	fd	nv-BdIzc	unknown	T1	ff	00	dc
10	0008	ea	1	NOP	0008	99	11	44	fd	nv-BdIzc	unknown	T1	ff	00	ff
11	0009	ea	1		0009	99	11	44	fd	nv-BdIzc	NOP	T0+T2	ff	ff	ff
11	0009	ea	1		0009	99	11	44	fd	nv-BdIzc	NOP	T0+T2	ff	ff	fe

EOR (45)		Cycles: 3	Size: 2
Zero Page (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read Address	Store as Operand.	
3.1		A ^= Operand. N flag = (A & \$80), Z flag = ! A .	
3.2	Read PC	Store as OpCode.	
+X.1		* A , N flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	45	1	EOR zp	0006	99	11	44	fd	nv-BdIzc	LDX #	T1	11	11	10
7	0006	45	1	EOR zp	0006	99	11	44	fd	nv-BdIzc	LDX #	T1	11	11	22
8	0007	dd	1		0007	99	11	44	fd	nv-BdIzc	EOR zp	T2	22	22	22
8	0007	dd	1		0007	99	11	44	fd	nv-BdIzc	EOR zp	T2	22	22	44
9	00dd	66	1		0008	99	11	44	fd	nv-BdIzc	EOR zp	T0	ff	dd	44
9	00dd	66	1		0008	99	11	44	fd	nv-BdIzc	EOR zp	T0	ff	dd	dc
10	0008	ea	1	NOP	0008	99	11	44	fd	nv-BdIzc	EOR zp	T1	99	66	dc
10	0008	ea	1	NOP	0008	99	11	44	fd	nv-BdIzc	EOR zp	T1	99	66	ff
11	0009	ea	1		0009	ff	11	44	fd	Nv-BdIzc	NOP	T0+T2	ff	ff	ff
11	0009	ea	1		0009	ff	11	44	fd	Nv-BdIzc	NOP	T0+T2	ff	ff	fe

LSR (46)		Cycles: 5	Size: 2
Zero Page (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read Address	Store as Operand.	
3.1			
3.2	Write to Address	Write Operand.	
4.1		C flag = (Operand & \$01). Operand >>= 1. N flag = 0, Z flag = !Operand.	
4.2	Write to Address	Write Operand.	
5.1			
5.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	46	1	LSR zp	0006	99	11	44	fd	nv-BdIzc	LDX #	T1	11	11	10
7	0006	46	1	LSR zp	0006	99	11	44	fd	nv-BdIzc	LDX #	T1	11	11	22
8	0007	dd	1		0007	99	11	44	fd	nv-BdIzc	LSR zp	T2	22	22	22
8	0007	dd	1		0007	99	11	44	fd	nv-BdIzc	LSR zp	T2	22	22	44
9	00dd	66	1		0008	99	11	44	fd	nv-BdIzc	LSR zp	T3	ff	dd	44
9	00dd	66	1		0008	99	11	44	fd	nv-BdIzc	LSR zp	T3	ff	dd	dc
10	00dd	66	0		0008	99	11	44	fd	nv-BdIzc	LSR zp	T4	ff	66	dc
10	00dd	66	0		0008	99	11	44	fd	nv-BdIzc	LSR zp	T4	ff	66	33
11	00dd	66	0		0008	99	11	44	fd	nv-BdIzc	LSR zp	T0	33	33	33
11	00dd	33	0		0008	99	11	44	fd	nv-BdIzc	LSR zp	T0	33	33	66
12	0008	ea	1	NOP	0008	99	11	44	fd	nv-BdIzc	LSR zp	T1	ff	33	66
12	0008	ea	1	NOP	0008	99	11	44	fd	nv-BdIzc	LSR zp	T1	ff	33	32
13	0009	ea	1		0009	99	11	44	fd	nv-BdIzc	NOP	T0+T2	32	32	32
13	0009	ea	1		0009	99	11	44	fd	nv-BdIzc	NOP	T0+T2	32	32	64

SRE (47)		Cycles: 5	Size: 2
Zero Page (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read Address	Store as Operand.	
3.1			
3.2	Write to Address	Write Operand.	
4.1		C flag = (Operand & \$01). Operand >>= 1. A ^= Operand. N flag = (A & \$80), Z flag = ! A .	
4.2	Write to Address	Write Operand.	
5.1			
5.2	Read PC	Store as OpCode.	
+X.1		* A , N flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	47	1	unknown	0006	99	11	44	fd	nv-BdIzc	LDX #	T1	11	11	10
7	0006	47	1	unknown	0006	99	11	44	fd	nv-BdIzc	LDX #	T1	11	11	22
8	0007	dd	1		0007	99	11	44	fd	nv-BdIzc	unknown	T2	22	22	22
8	0007	dd	1		0007	99	11	44	fd	nv-BdIzc	unknown	T2	22	22	44
9	00dd	33	1		0008	99	11	44	fd	nv-BdIzc	unknown	T3	ff	dd	44
9	00dd	33	1		0008	99	11	44	fd	nv-BdIzc	unknown	T3	ff	dd	dc
10	00dd	33	0		0008	99	11	44	fd	nv-BdIzc	unknown	T4	ff	33	dc
10	00dd	33	0		0008	99	11	44	fd	nv-BdIzc	unknown	T4	ff	33	19
11	00dd	33	0		0008	99	11	44	fd	nv-BdIzc	unknown	T0	19	19	19
11	00dd	19	0		0008	99	11	44	fd	nv-BdIzc	unknown	T0	19	19	32
12	0008	ea	1	NOP	0008	99	11	44	fd	nv-BdIzc	unknown	T1	99	19	32
12	0008	ea	1	NOP	0008	99	11	44	fd	nv-BdIzc	unknown	T1	99	19	80
13	0009	ea	1		0009	80	11	44	fd	Nv-BdIzc	NOP	T0+T2	80	80	80
13	0009	ea	1		0009	80	11	44	fd	Nv-BdIzc	NOP	T0+T2	80	80	00

PHA (48)		Cycles: 3	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1	Read PC	Inc. PC .	
1.2		Discard.	
2.1			
2.2		Write A .	
3.1		Dec. S by 1.	
3.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	48	1	PHA	0006	99	11	44	fd	nv-BdIzc	LDX #	T1	11	11	10
7	0006	48	1	PHA	0006	99	11	44	fd	nv-BdIzc	LDX #	T1	11	11	22
8	0007	dd	1		0007	99	11	44	fd	nv-BdIzc	PHA	T2	22	22	22
8	0007	dd	1		0007	99	11	44	fd	nv-BdIzc	PHA	T2	22	22	44
9	01fd	dd	0		0007	99	11	44	fd	nv-BdIzc	PHA	T0	ff	fd	44
9	01fd	99	0		0007	99	11	44	fd	nv-BdIzc	PHA	T0	ff	fd	fc
10	0007	dd	1	CMP Abs,X	0007	99	11	44	fc	nv-BdIzc	PHA	T1	00	99	fc
10	0007	dd	1	CMP Abs,X	0007	99	11	44	fc	nv-BdIzc	PHA	T1	00	99	99
11	0008	ea	1		0008	99	11	44	fc	nv-BdIzc	CMP Abs,X	T2	99	99	99
11	0008	ea	1		0008	99	11	44	fc	nv-BdIzc	CMP Abs,X	T2	99	99	32

EOR (49)		Cycles: 2	Size: 2
Immediate			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Operand.	
2.1		Inc. PC . A ^= Operand. N flag = (A & \$80), Z flag = ! A .	
2.2	Read PC	Store as OpCode.	
+X.1		* A , N flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	49	1	EOR #	0006	99	00	44	fd	nv-BdIZc	LDX #	T1	00	00	10
7	0006	49	1	EOR #	0006	99	00	44	fd	nv-BdIZc	LDX #	T1	00	00	00
8	0007	66	1		0007	99	00	44	fd	nv-BdIZc	EOR #	T0+T2	00	00	00
8	0007	66	1		0007	99	00	44	fd	nv-BdIZc	EOR #	T0+T2	00	00	00
9	0008	ea	1	NOP	0008	99	00	44	fd	nv-BdIZc	EOR #	T1	99	66	00
9	0008	ea	1	NOP	0008	99	00	44	fd	nv-BdIZc	EOR #	T1	99	66	ff
10	0009	ea	1		0009	ff	00	44	fd	Nv-BdIZc	NOP	T0+T2	ff	ff	ff
10	0009	ea	1		0009	ff	00	44	fd	Nv-BdIZc	NOP	T0+T2	ff	ff	fe

LSR (4A)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Discard.	
2.1		C flag = (A & \$01). A >>= 1. N flag = 0, Z flag = ! A .	
2.2	Read PC	Store as OpCode.	
+X.1		* A , C flag, N flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	4a	1	LSR	0006	99	00	44	fd	nv-BdIZc	LDX #	T1	00	00	10
7	0006	4a	1	LSR	0006	99	00	44	fd	nv-BdIZc	LDX #	T1	00	00	00
8	0007	ea	1		0007	99	00	44	fd	nv-BdIZc	LSR	T0+T2	00	00	00
8	0007	ea	1		0007	99	00	44	fd	nv-BdIZc	LSR	T0+T2	00	00	00
9	0007	ea	1	NOP	0007	99	00	44	fd	nv-BdIZC	LSR	T1	99	99	00
9	0007	ea	1	NOP	0007	99	00	44	fd	nv-BdIZC	LSR	T1	99	99	4c
10	0008	ea	1		0008	4c	00	44	fd	nv-BdIzC	NOP	T0+T2	4c	4c	4c
10	0008	ea	1		0008	4c	00	44	fd	nv-BdIzC	NOP	T0+T2	4c	4c	98

ALR (4B)	Cycles: 2	Size: 2
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . A &= Operand. C flag = (A & \$01). A = (A >> 1). N flag = 0, Z flag = ! A .
2.2	Read PC	Store as OpCode.
+X.1		* A , C flag, N flag, and Z flag applied.

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.
Visual6502 omits (A &= Operand); flags are updated based off the incorrect A value.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	4b	1	unknown	0006	99	ff	44	fd	Nv-BdIzc	LDX #	T1	ff	ff	10
7	0006	4b	1	unknown	0006	99	ff	44	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	0f	1		0007	99	ff	44	fd	Nv-BdIzc	unknown	T0+T2	fe	fe	fe
8	0007	0f	1		0007	99	ff	44	fd	Nv-BdIzc	unknown	T0+T2	fe	fe	fc
9	0008	ea	1	NOP	0008	99	ff	44	fd	Nv-BdIzc	unknown	T1	09	09	fc
9	0008	ea	1	NOP	0008	99	ff	44	fd	Nv-BdIzc	unknown	T1	99	99	4c
10	0009	ea	1		0009	4c	ff	44	fd	nv-BdIzc	NOP	T0+T2	4c	4c	4c
10	0009	ea	1		0009	4c	ff	44	fd	nv-BdIzc	NOP	T0+T2	4c	4c	98

ALR (ASR)

Type: Combination of an immediate and an implied command (Sub-instructions: AND, LSR)

Op.	Mnemonic	Function	Size	Cycles	N	V	B	D	I	Z	C
\$4B	ALR #imm	A = (A & #{imm}) / 2	2	2	o					o	o

Operation: AND the contents of the A register with an immediate value and then LSRs the result.

- Bit 1 (after the AND) is shifted into the carry flag
- N and Z are set after the shift

JMP (4C)		Cycles: 3	Size: 3
Absolute (Jump)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read PC	Store as Address.H.	
3.1		PC = Address.	
3.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	4c	1	JMP Abs	0006	99	ff	44	fd	Nv-BdIzc	LDX #	T1	ff	ff	10
7	0006	4c	1	JMP Abs	0006	99	ff	44	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	33	1		0007	99	ff	44	fd	Nv-BdIzc	JMP Abs	T2	fe	fe	fe
8	0007	33	1		0007	99	ff	44	fd	Nv-BdIzc	JMP Abs	T2	fe	fe	fc
9	0008	01	1		0008	99	ff	44	fd	Nv-BdIzc	JMP Abs	T0	00	33	fc
9	0008	01	1		0008	99	ff	44	fd	Nv-BdIzc	JMP Abs	T0	00	33	33
10	0133	ea	1	NOP	0133	99	ff	44	fd	Nv-BdIzc	JMP Abs	T1	00	01	33
10	0133	ea	1	NOP	0133	99	ff	44	fd	Nv-BdIzc	JMP Abs	T1	00	01	01
11	0134	00	1		0134	99	ff	44	fd	Nv-BdIzc	NOP	T0+T2	00	01	01
11	0134	00	1		0134	99	ff	44	fd	Nv-BdIzc	NOP	T0+T2	00	01	01

EOR (4D)		Cycles: 4	Size: 3
Absolute (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read PC	Store as Address.H.	
3.1		Inc. PC .	
3.2	Read Address	Store as Operand.	
4.1		A ^= Operand. N flag = (A & \$80), Z flag = ! A .	
4.2	Read PC	Store as OpCode.	
+X.1		* A , N flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	4d	1	EOR Abs	0006	99	ff	44	fd	Nv-BdIzc	LDX #	T1	ff	ff	10
7	0006	4d	1	EOR Abs	0006	99	ff	44	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	33	1		0007	99	ff	44	fd	Nv-BdIzc	EOR Abs	T2	fe	fe	fe
8	0007	33	1		0007	99	ff	44	fd	Nv-BdIzc	EOR Abs	T2	fe	fe	fc
9	0008	01	1		0008	99	ff	44	fd	Nv-BdIzc	EOR Abs	T3	00	33	fc
9	0008	01	1		0008	99	ff	44	fd	Nv-BdIzc	EOR Abs	T3	00	33	33
10	0133	99	1		0009	99	ff	44	fd	Nv-BdIzc	EOR Abs	T0	ff	ff	33
10	0133	99	1		0009	99	ff	44	fd	Nv-BdIzc	EOR Abs	T0	ff	ff	fe
11	0009	ea	1	NOP	0009	99	ff	44	fd	Nv-BdIzc	EOR Abs	T1	99	99	fe
11	0009	ea	1	NOP	0009	99	ff	44	fd	Nv-BdIzc	EOR Abs	T1	99	99	00
12	000a	15	1		000a	00	ff	44	fd	nv-BdIZc	NOP	T0+T2	00	00	00
12	000a	15	1		000a	00	ff	44	fd	nv-BdIZc	NOP	T0+T2	00	00	00

LSR (4E)	Cycles: 6	Size: 3
Absolute (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.
5.1		C flag = (Operand & \$01). Operand >>= 1. N flag = 0, Z flag = !Operand.
5.2	Write to Address	Write Operand.
6.1		
6.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	4e	1	LSR Abs	0006	99	ff	44	fd	Nv-BdIzc	LDX #	T1	ff	ff	10
7	0006	4e	1	LSR Abs	0006	99	ff	44	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	33	1		0007	99	ff	44	fd	Nv-BdIzc	LSR Abs	T2	fe	fe	fe
8	0007	33	1		0007	99	ff	44	fd	Nv-BdIzc	LSR Abs	T2	fe	fe	fc
9	0008	01	1		0008	99	ff	44	fd	Nv-BdIzc	LSR Abs	T3	00	33	fc
9	0008	01	1		0008	99	ff	44	fd	Nv-BdIzc	LSR Abs	T3	00	33	33
10	0133	99	1		0009	99	ff	44	fd	Nv-BdIzc	LSR Abs	T4	ff	ff	33
10	0133	99	1		0009	99	ff	44	fd	Nv-BdIzc	LSR Abs	T4	ff	ff	fe
11	0133	99	0		0009	99	ff	44	fd	Nv-BdIzc	LSR Abs	T5	ff	99	fe
11	0133	99	0		0009	99	ff	44	fd	Nv-BdIzc	LSR Abs	T5	ff	99	4c
12	0133	99	0		0009	99	ff	44	fd	nv-BdIzc	LSR Abs	T0	4c	4c	4c
12	0133	4c	0		0009	99	ff	44	fd	nv-BdIzc	LSR Abs	T0	4c	4c	98
13	0009	ea	1	NOP	0009	99	ff	44	fd	nv-BdIzc	LSR Abs	T1	ff	4c	98
13	0009	ea	1	NOP	0009	99	ff	44	fd	nv-BdIzc	LSR Abs	T1	ff	4c	4b
14	000a	15	1		000a	99	ff	44	fd	nv-BdIzc	NOP	T0+T2	4b	4b	4b
14	000a	15	1		000a	99	ff	44	fd	nv-BdIzc	NOP	T0+T2	4b	4b	96

SRE (4F)	Cycles: 6	Size: 3
Absolute (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.
5.1		C flag = (Operand & \$01). Operand >>= 1. A ^= Operand. N flag = (A & \$80), Z flag = ! A .
5.2	Write to Address	Write Operand.
6.1		
6.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	4f	1	unknown	0006	99	ff	44	fd	Nv-BdIzc	LDX #	T1	ff	ff	10
7	0006	4f	1	unknown	0006	99	ff	44	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	33	1		0007	99	ff	44	fd	Nv-BdIzc	unknown	T2	fe	fe	fe
8	0007	33	1		0007	99	ff	44	fd	Nv-BdIzc	unknown	T2	fe	fe	fc
9	0008	01	1		0008	99	ff	44	fd	Nv-BdIzc	unknown	T3	00	33	fc
9	0008	01	1		0008	99	ff	44	fd	Nv-BdIzc	unknown	T3	00	33	33
10	0133	4c	1		0009	99	ff	44	fd	Nv-BdIzc	unknown	T4	ff	ff	33
10	0133	4c	1		0009	99	ff	44	fd	Nv-BdIzc	unknown	T4	ff	ff	fe
11	0133	4c	0		0009	99	ff	44	fd	Nv-BdIzc	unknown	T5	ff	4c	fe
11	0133	4c	0		0009	99	ff	44	fd	Nv-BdIzc	unknown	T5	ff	4c	26
12	0133	4c	0		0009	99	ff	44	fd	nv-BdIzc	unknown	T0	26	26	26
12	0133	26	0		0009	99	ff	44	fd	nv-BdIzc	unknown	T0	26	26	4c
13	0009	ea	1	NOP	0009	99	ff	44	fd	nv-BdIzc	unknown	T1	99	26	4c
13	0009	ea	1	NOP	0009	99	ff	44	fd	nv-BdIzc	unknown	T1	99	26	bf
14	000a	15	1		000a	bf	ff	44	fd	Nv-BdIzc	NOP	T0+T2	bf	bf	bf
14	000a	15	1		000a	bf	ff	44	fd	Nv-BdIzc	NOP	T0+T2	bf	bf	7e

BVC (50)		Cycles: 2-4	Size: 2
Relative (Jump)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC . Jump = ((P & \$40) == \$00).	
1.2	Read PC	Store as Operand.	
		If !Jump, poll interrupts.	
		Otherwise Address = i16(i8(Operand)) + PC + 1.	
		BoundaryCrossed = Address.H != (PC + 1).H.	
		If !BoundaryCrossed, poll interrupts.	
2.1		Inc. PC . If !Jump, end (next half-cycle is 4.2).	
2.2	Read PC	Discard.	
3.1		PC .L = Address.L. If !BoundaryCrossed, end (next half-cycle is 4.2).	
3.2	Read PC	Discard. Poll interrupts.	
4.1		PC .H = Address.H.	
4.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	50	1	BVC	0006	7f	00	44	fd	nv-BdIzc	ADC #	T1	7f	01	10
7	0006	50	1	BVC	0006	7f	00	44	fd	nv-BdIzc	ADC #	T1	7f	01	80
8	0007	cc	1		0007	80	00	44	fd	nv-BdIzc	BVC	T2	80	80	80
8	0007	cc	1		0007	80	00	44	fd	nv-BdIzc	BVC	T2	80	80	00
9	0008	01	1	ORA (zp,X)	0008	80	00	44	fd	nv-BdIzc	BVC		cc	08	00
9	0008	01	1	ORA (zp,X)	0008	80	00	44	fd	nv-BdIzc	BVC		cc	08	d4
10	0009	ea	1		0009	80	00	44	fd	nv-BdIzc	ORA (zp,X)	T2	d4	d4	d4
10	0009	ea	1		0009	80	00	44	fd	nv-BdIzc	ORA (zp,X)	T2	d4	d4	a8

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	50	1	BVC	0006	99	ff	44	fd	Nv-BdIzc	LDX #	T1	ff	ff	10
7	0006	50	1	BVC	0006	99	ff	44	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	33	1		0007	99	ff	44	fd	Nv-BdIzc	BVC	T2	fe	fe	fe
8	0007	33	1		0007	99	ff	44	fd	Nv-BdIzc	BVC	T2	fe	fe	fc
9	0008	01	1		0008	99	ff	44	fd	Nv-BdIzc	BVC	T3	33	08	fc
9	0008	01	1		0008	99	ff	44	fd	Nv-BdIzc	BVC	T3	33	08	3b
10	003b	ea	1	NOP	003b	99	ff	44	fd	Nv-BdIzc	BVC		00	00	3b
10	003b	ea	1	NOP	003b	99	ff	44	fd	Nv-BdIzc	BVC		00	00	01
11	003c	00	1		003c	99	ff	44	fd	Nv-BdIzc	NOP	T0+T2	01	01	01
11	003c	00	1		003c	99	ff	44	fd	Nv-BdIzc	NOP	T0+T2	01	01	02

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	50	1	BVC	0006	99	ff	44	fd	Nv-BdIzc	LDX #	T1	ff	ff	10
7	0006	50	1	BVC	0006	99	ff	44	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	cc	1		0007	99	ff	44	fd	Nv-BdIzc	BVC	T2	fe	fe	fe
8	0007	cc	1		0007	99	ff	44	fd	Nv-BdIzc	BVC	T2	fe	fe	fc
9	0008	01	1		0008	99	ff	44	fd	Nv-BdIzc	BVC	T3	cc	08	fc
9	0008	01	1		0008	99	ff	44	fd	Nv-BdIzc	BVC	T3	cc	08	d4
10	00d4	00	1		00d4	99	ff	44	fd	Nv-BdIzc	BVC	T0	00	ff	d4
10	00d4	00	1		00d4	99	ff	44	fd	Nv-BdIzc	BVC	T0	00	ff	ff
11	ffd4	00	1	BRK	ffd4	99	ff	44	fd	Nv-BdIzc	BVC	T1	ff	00	ff
11	ffd4	00	1	BRK	ffd4	99	ff	44	fd	Nv-BdIzc	BVC	T1	ff	00	ff
12	ffd5	00	1		ffd5	99	ff	44	fd	Nv-BdIzc	BRK	T2	ff	ff	ff
12	ffd5	00	1		ffd5	99	ff	44	fd	Nv-BdIzc	BRK	T2	ff	ff	fe

EOR (51)	Cycles: 5-6	Size: 2
Indirect Y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read Pointer	Store as Address.
3.1		
3.2	Read u8(Pointer + 1)	Store as Address.H.
4.1		Target = Address + Y. Pointer.L = Target.L. Pointer.H = Address.H. BoundaryCrossed = Pointer.H != Target.H.
4.2	Read Pointer	Store as Operand. If !BoundaryCrossed, skip the next cycle.
5.1		Pointer.H = Target.H.
5.2	Read Pointer	Store as Operand.
6.1		A ^= Operand. N flag = (A & \$80), Z flag = ! A .
6.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	51	1	EOR (zp),Y	0006	7f	ff	44	fd	Nv-BdIzc	LDX #	T1	ff	ff	10
7	0006	51	1	EOR (zp),Y	0006	7f	ff	44	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	cc	1		0007	7f	ff	44	fd	Nv-BdIzc	EOR (zp),Y	T2	fe	fe	fe
8	0007	cc	1		0007	7f	ff	44	fd	Nv-BdIzc	EOR (zp),Y	T2	fe	fe	fc
9	00cc	11	1		0008	7f	ff	44	fd	Nv-BdIzc	EOR (zp),Y	T3	00	cc	fc
9	00cc	11	1		0008	7f	ff	44	fd	Nv-BdIzc	EOR (zp),Y	T3	00	cc	cd
10	00cd	00	1		0008	7f	ff	44	fd	Nv-BdIzc	EOR (zp),Y	T4	44	11	cd
10	00cd	00	1		0008	7f	ff	44	fd	Nv-BdIzc	EOR (zp),Y	T4	44	11	55
11	0055	0f	1		0008	7f	ff	44	fd	Nv-BdIzc	EOR (zp),Y	T0	00	00	55
11	0055	0f	1		0008	7f	ff	44	fd	Nv-BdIzc	EOR (zp),Y	T0	00	00	01
12	0008	ea	1	NOP	0008	7f	ff	44	fd	Nv-BdIzc	EOR (zp),Y	T1	7f	0f	01
12	0008	ea	1	NOP	0008	7f	ff	44	fd	Nv-BdIzc	EOR (zp),Y	T1	7f	0f	70
13	0009	ea	1		0009	70	ff	44	fd	nv-BdIzc	NOP	T0+T2	70	70	70
13	0009	ea	1		0009	70	ff	44	fd	nv-BdIzc	NOP	T0+T2	70	70	e0

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	51	1	EOR (zp),Y	0006	7f	ff	ef	fd	Nv-BdIzc	LDX #	T1	ff	ff	bc
7	0006	51	1	EOR (zp),Y	0006	7f	ff	ef	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	cc	1		0007	7f	ff	ef	fd	Nv-BdIzc	EOR (zp),Y	T2	fe	fe	fe
8	0007	cc	1		0007	7f	ff	ef	fd	Nv-BdIzc	EOR (zp),Y	T2	fe	fe	fc
9	00cc	11	1		0008	7f	ff	ef	fd	Nv-BdIzc	EOR (zp),Y	T3	00	cc	fc
9	00cc	11	1		0008	7f	ff	ef	fd	Nv-BdIzc	EOR (zp),Y	T3	00	cc	cd
10	00cd	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	EOR (zp),Y	T4	ef	11	cd
10	00cd	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	EOR (zp),Y	T4	ef	11	00
11	0000	a9	1		0008	7f	ff	ef	fd	Nv-BdIzc	EOR (zp),Y	T5	00	00	00
11	0000	a9	1		0008	7f	ff	ef	fd	Nv-BdIzc	EOR (zp),Y	T5	00	00	01
12	0100	80	1		0008	7f	ff	ef	fd	Nv-BdIzc	EOR (zp),Y	T0	01	ff	01
12	0100	80	1		0008	7f	ff	ef	fd	Nv-BdIzc	EOR (zp),Y	T0	01	ff	00
13	0008	ea	1	NOP	0008	7f	ff	ef	fd	Nv-BdIzc	EOR (zp),Y	T1	7f	80	00
13	0008	ea	1	NOP	0008	7f	ff	ef	fd	Nv-BdIzc	EOR (zp),Y	T1	7f	80	ff
14	0009	ea	1		0009	ff	ff	ef	fd	Nv-BdIzc	NOP	T0+T2	ff	ff	ff
14	0009	ea	1		0009	ff	ff	ef	fd	Nv-BdIzc	NOP	T0+T2	ff	ff	fe

JAM (52)	Cycles: ∞	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Discard.
2.1		Visual6502 suggests incrementing PC , but Single-Step Tests cannot be passed this way.
2.2	Read \$FFFF	Discard.
3.1		
3.2	Read \$FFFE	Discard.
4.1		
4.2	Read \$FFFE	Discard.
5.1		Repeat 5.1 and 5.2 forever.
5.2	Read \$FFFF	Discard.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	52	1	unknown	0006	7f	ff	ef	fd	Nv-BdIzc	LDX #	T1	ff	ff	bc
7	0006	52	1	unknown	0006	7f	ff	ef	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	cc	1		0007	7f	ff	ef	fd	Nv-BdIzc	unknown	T2	fe	fe	fe
8	0007	cc	1		0007	7f	ff	ef	fd	Nv-BdIzc	unknown	T2	fe	fe	fc
9	ffff	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown	T3	ff	ff	fc
9	ffff	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown	T3	ff	ff	fe
10	fffe	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown	T4	ff	ff	fe
10	fffe	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown	T4	ff	ff	fe
11	fffe	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown	T5	ff	ff	fe
11	fffe	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown	T5	ff	ff	fe
12	ffff	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown		ff	ff	fe
12	ffff	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown		ff	ff	fe
13	ffff	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown		ff	ff	fe
13	ffff	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown		ff	ff	fe
14	ffff	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown		ff	ff	fe
14	ffff	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown		ff	ff	fe

SRE (53)		Cycles: 8	Size: 2
Indirect Y (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Address.	
3.1			
3.2	Read u8(Pointer + 1)	Store as Address.H.	
4.1		Target = Address + Y. Pointer.L = Target.L. Pointer.H = Address.H.	
4.2	Read Pointer	Store as Operand.	
5.1		Pointer.H = Target.H.	
5.2	Read Pointer	Store as Operand.	
6.1			
6.2	Write to Pointer	Write Operand.	
7.1		C flag = (Operand & \$01). Operand >>= 1. A ^= Operand. N flag = (A & \$80), Z flag = ! A .	
7.2	Write to Pointer	Write Operand.	
8.1			
8.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	53	1	unknown	0006	7f	ff	ef	fd	Nv-BdIzc	LDX #	T1	ff	ff	bc
7	0006	53	1	unknown	0006	7f	ff	ef	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	cd	1		0007	7f	ff	ef	fd	Nv-BdIzc	unknown	T2	fe	fe	fe
8	0007	cd	1		0007	7f	ff	ef	fd	Nv-BdIzc	unknown	T2	fe	fe	fc
9	00cd	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown	T3	00	cd	fc
9	00cd	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown	T3	00	cd	ce
10	00ce	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown	T4	ef	00	ce
10	00ce	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown	T4	ef	00	ef
11	00ef	01	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown	T5	00	00	ef
11	00ef	01	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown	T5	00	00	01
12	00ef	01	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown		01	ff	01
12	00ef	01	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown		01	ff	00
13	00ef	01	0		0008	7f	ff	ef	fd	Nv-BdIzc	unknown		ff	01	00
13	00ef	01	0		0008	7f	ff	ef	fd	Nv-BdIzc	unknown		ff	01	00
14	00ef	01	0		0008	7f	ff	ef	fd	nv-BdIZC	unknown	T0	00	00	00
14	00ef	00	0		0008	7f	ff	ef	fd	nv-BdIZC	unknown	T0	00	00	00
15	0008	ea	1	NOP	0008	7f	ff	ef	fd	nv-BdIZC	unknown	T1	7f	00	00
15	0008	ea	1	NOP	0008	7f	ff	ef	fd	nv-BdIZC	unknown	T1	7f	00	7f
16	0009	ea	1		0009	7f	ff	ef	fd	nv-BdIZC	NOP	T0+T2	7f	7f	7f
16	0009	ea	1		0009	7f	ff	ef	fd	nv-BdIZC	NOP	T0+T2	7f	7f	fe

NOP (54)		Cycles: 4	Size: 2
Zero Page X-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Operand.	
3.1		Address = u8(Pointer + X).	
3.2	Read Address	Store as Operand.	
4.1			
4.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	54	1	unknown	0006	7f	ff	ef	fd	Nv-BdIzc	LDX #	T1	ff	ff	bc
7	0006	54	1	unknown	0006	7f	ff	ef	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	cd	1		0007	7f	ff	ef	fd	Nv-BdIzc	unknown	T2	fe	fe	fe
8	0007	cd	1		0007	7f	ff	ef	fd	Nv-BdIzc	unknown	T2	fe	fe	fc
9	00cd	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown	T3	ff	cd	fc
9	00cd	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown	T3	ff	cd	cc
10	00cc	11	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown	T0	ff	ff	cc
10	00cc	11	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown	T0	ff	ff	fe
11	0008	ea	1	NOP	0008	7f	ff	ef	fd	Nv-BdIzc	unknown	T1	ff	11	fe
11	0008	ea	1	NOP	0008	7f	ff	ef	fd	Nv-BdIzc	unknown	T1	ff	11	10
12	0009	ea	1		0009	7f	ff	ef	fd	Nv-BdIzc	NOP	T0+T2	10	10	10
12	0009	ea	1		0009	7f	ff	ef	fd	Nv-BdIzc	NOP	T0+T2	10	10	20

EOR (55)		Cycles: 4	Size: 2
Zero Page X-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Operand.	
3.1		Address = u8(Pointer + X).	
3.2	Read Address	Store as Operand.	
4.1		A ^= Operand. N flag = (A & \$80), Z flag = ! A .	
4.2	Read PC	Store as OpCode.	
+X.1		* A , N flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	55	1	EOR zp,X	0006	7f	ff	ef	fd	Nv-BdIzc	LDX #	T1	ff	ff	bc
7	0006	55	1	EOR zp,X	0006	7f	ff	ef	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	cd	1		0007	7f	ff	ef	fd	Nv-BdIzc	EOR zp,X	T2	fe	fe	fe
8	0007	cd	1		0007	7f	ff	ef	fd	Nv-BdIzc	EOR zp,X	T2	fe	fe	fc
9	00cd	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	EOR zp,X	T3	ff	cd	fc
9	00cd	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	EOR zp,X	T3	ff	cd	cc
10	00cc	7f	1		0008	7f	ff	ef	fd	Nv-BdIzc	EOR zp,X	T0	ff	ff	cc
10	00cc	7f	1		0008	7f	ff	ef	fd	Nv-BdIzc	EOR zp,X	T0	ff	ff	fe
11	0008	ea	1	NOP	0008	7f	ff	ef	fd	Nv-BdIzc	EOR zp,X	T1	7f	7f	fe
11	0008	ea	1	NOP	0008	7f	ff	ef	fd	Nv-BdIzc	EOR zp,X	T1	7f	7f	00
12	0009	ea	1		0009	00	ff	ef	fd	nv-BdIZc	NOP	T0+T2	00	00	00
12	0009	ea	1		0009	00	ff	ef	fd	nv-BdIZc	NOP	T0+T2	00	00	00

LSR (56)	Cycles: 6	Size: 2
Zero Page X-Indexed (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read Pointer	Store as Operand.
3.1		Address = u8(Pointer + X).
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.
5.1		C flag = (Operand & \$01). Operand >>= 1. N flag = 0, Z flag = !Operand.
5.2	Write to Address	Write Operand.
6.1		
6.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	56	1	LSR zp,X	0006	7f	ff	ef	fd	Nv-BdIzc	LDX #	T1	ff	ff	bc
7	0006	56	1	LSR zp,X	0006	7f	ff	ef	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	cd	1		0007	7f	ff	ef	fd	Nv-BdIzc	LSR zp,X	T2	fe	fe	fe
8	0007	cd	1		0007	7f	ff	ef	fd	Nv-BdIzc	LSR zp,X	T2	fe	fe	fc
9	00cd	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	LSR zp,X	T3	ff	cd	fc
9	00cd	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	LSR zp,X	T3	ff	cd	cc
10	00cc	01	1		0008	7f	ff	ef	fd	Nv-BdIzc	LSR zp,X	T4	ff	ff	cc
10	00cc	01	1		0008	7f	ff	ef	fd	Nv-BdIzc	LSR zp,X	T4	ff	ff	fe
11	00cc	01	0		0008	7f	ff	ef	fd	Nv-BdIzc	LSR zp,X	T5	ff	01	fe
11	00cc	01	0		0008	7f	ff	ef	fd	Nv-BdIzc	LSR zp,X	T5	ff	01	00
12	00cc	01	0		0008	7f	ff	ef	fd	nv-BdIzc	LSR zp,X	T0	00	00	00
12	00cc	00	0		0008	7f	ff	ef	fd	nv-BdIzc	LSR zp,X	T0	00	00	00
13	0008	ea	1	NOP	0008	7f	ff	ef	fd	nv-BdIzc	LSR zp,X	T1	ff	00	00
13	0008	ea	1	NOP	0008	7f	ff	ef	fd	nv-BdIzc	LSR zp,X	T1	ff	00	ff
14	0009	ea	1		0009	7f	ff	ef	fd	nv-BdIzc	NOP	T0+T2	ff	ff	ff
14	0009	ea	1		0009	7f	ff	ef	fd	nv-BdIzc	NOP	T0+T2	ff	ff	fe

SRE (57)	Cycles: 6	Size: 2
Zero Page X-Indexed (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read Pointer	Store as Operand.
3.1		Address = u8(Pointer + X).
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.
5.1		C flag = (Operand & \$01). Operand >>= 1. A ^= Operand. N flag = (A & \$80), Z flag = ! A .
5.2	Write to Address	Write Operand.
6.1		
6.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	57	1	unknown	0006	7f	ff	ef	fd	Nv-BdIzc	LDX #	T1	ff	ff	bc
7	0006	57	1	unknown	0006	7f	ff	ef	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	cd	1		0007	7f	ff	ef	fd	Nv-BdIzc	unknown	T2	fe	fe	fe
8	0007	cd	1		0007	7f	ff	ef	fd	Nv-BdIzc	unknown	T2	fe	fe	fc
9	00cd	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown	T3	ff	cd	fc
9	00cd	00	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown	T3	ff	cd	cc
10	00cc	01	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown	T4	ff	ff	cc
10	00cc	01	1		0008	7f	ff	ef	fd	Nv-BdIzc	unknown	T4	ff	ff	fe
11	00cc	01	0		0008	7f	ff	ef	fd	Nv-BdIzc	unknown	T5	ff	01	fe
11	00cc	01	0		0008	7f	ff	ef	fd	Nv-BdIzc	unknown	T5	ff	01	00
12	00cc	01	0		0008	7f	ff	ef	fd	nv-BdIZC	unknown	T0	00	00	00
12	00cc	00	0		0008	7f	ff	ef	fd	nv-BdIZC	unknown	T0	00	00	00
13	0008	ea	1	NOP	0008	7f	ff	ef	fd	nv-BdIZC	unknown	T1	7f	00	00
13	0008	ea	1	NOP	0008	7f	ff	ef	fd	nv-BdIZC	unknown	T1	7f	00	7f
14	0009	ea	1		0009	7f	ff	ef	fd	nv-BdIZC	NOP	T0+T2	7f	7f	7f
14	0009	ea	1		0009	7f	ff	ef	fd	nv-BdIZC	NOP	T0+T2	7f	7f	fe

CLI (58)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Discard.
2.1		I flag = 0.
2.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	58	1	CLI	0006	7f	ff	ef	fd	Nv-BdIzc	LDX #	T1	ff	ff	bc
7	0006	58	1	CLI	0006	7f	ff	ef	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	ea	1		0007	7f	ff	ef	fd	Nv-BdIzc	CLI	T0+T2	fe	fe	fe
8	0007	ea	1		0007	7f	ff	ef	fd	Nv-BdIzc	CLI	T0+T2	fe	fe	fc
9	0007	ea	1	NOP	0007	7f	ff	ef	fd	Nv-Bdizc	CLI	T1	ff	ff	fc
9	0007	ea	1	NOP	0007	7f	ff	ef	fd	Nv-Bdizc	CLI	T1	ff	ff	fe
10	0008	ea	1		0008	7f	ff	ef	fd	Nv-Bdizc	NOP	T0+T2	fe	fe	fe
10	0008	ea	1		0008	7f	ff	ef	fd	Nv-Bdizc	NOP	T0+T2	fe	fe	fc

EOR (59)	Cycles: 4-5	Size: 3
Absolute Y-Indexed (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + Y. Address.L = Target.L. Address.H = Pointer.H. BoundaryCrossed = Pointer.H != Target.H.
3.2	Read Address	Store as Operand. If !BoundaryCrossed, skip the next cycle.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		A ^= Operand. N flag = (A & \$80), Z flag = ! A .
5.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	59	1	EOR Abs,Y	0006	7f	ff	11	fd	Nv-BdIzc	LDX #	T1	ff	ff	44
7	0006	59	1	EOR Abs,Y	0006	7f	ff	11	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	cd	1		0007	7f	ff	11	fd	Nv-BdIzc	EOR Abs,Y	T2	fe	fe	fe
8	0007	cd	1		0007	7f	ff	11	fd	Nv-BdIzc	EOR Abs,Y	T2	fe	fe	fc
9	0008	00	1		0008	7f	ff	11	fd	Nv-BdIzc	EOR Abs,Y	T3	11	cd	fc
9	0008	00	1		0008	7f	ff	11	fd	Nv-BdIzc	EOR Abs,Y	T3	11	cd	de
10	00de	7f	1		0009	7f	ff	11	fd	Nv-BdIzc	EOR Abs,Y	T0	00	00	de
10	00de	7f	1		0009	7f	ff	11	fd	Nv-BdIzc	EOR Abs,Y	T0	00	00	01
11	0009	ea	1	NOP	0009	7f	ff	11	fd	Nv-BdIzc	EOR Abs,Y	T1	7f	7f	01
11	0009	ea	1	NOP	0009	7f	ff	11	fd	Nv-BdIzc	EOR Abs,Y	T1	7f	7f	00
12	000a	15	1		000a	00	ff	11	fd	nv-BdIzc	NOP	T0+T2	00	00	00
12	000a	15	1		000a	00	ff	11	fd	nv-BdIzc	NOP	T0+T2	00	00	00

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	59	1	EOR Abs,Y	0006	7f	ff	ef	fd	Nv-BdIzc	LDX #	T1	ff	ff	bc
7	0006	59	1	EOR Abs,Y	0006	7f	ff	ef	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	cc	1		0007	7f	ff	ef	fd	Nv-BdIzc	EOR Abs,Y	T2	fe	fe	fe
8	0007	cc	1		0007	7f	ff	ef	fd	Nv-BdIzc	EOR Abs,Y	T2	fe	fe	fc
9	0008	01	1		0008	7f	ff	ef	fd	Nv-BdIzc	EOR Abs,Y	T3	ef	cc	fc
9	0008	01	1		0008	7f	ff	ef	fd	Nv-BdIzc	EOR Abs,Y	T3	ef	cc	bb
10	01bb	00	1		0009	7f	ff	ef	fd	Nv-BdIzc	EOR Abs,Y	T4	00	01	bb
10	01bb	00	1		0009	7f	ff	ef	fd	Nv-BdIzc	EOR Abs,Y	T4	00	01	02
11	02bb	00	1		0009	7f	ff	ef	fd	Nv-BdIzc	EOR Abs,Y	T0	02	ff	02
11	02bb	00	1		0009	7f	ff	ef	fd	Nv-BdIzc	EOR Abs,Y	T0	02	ff	01
12	0009	ea	1	NOP	0009	7f	ff	ef	fd	Nv-BdIzc	EOR Abs,Y	T1	7f	00	01
12	0009	ea	1	NOP	0009	7f	ff	ef	fd	Nv-BdIzc	EOR Abs,Y	T1	7f	00	7f
13	000a	15	1		000a	7f	ff	ef	fd	nv-BdIzc	NOP	T0+T2	7f	7f	7f
13	000a	15	1		000a	7f	ff	ef	fd	nv-BdIzc	NOP	T0+T2	7f	7f	fe

NOP (5A)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Discard.	
2.1			
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	5a	1	unknown	0006	7f	ff	11	fd	Nv-BdIzc	LDX #	T1	ff	ff	44
7	0006	5a	1	unknown	0006	7f	ff	11	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	ea	1		0007	7f	ff	11	fd	Nv-BdIzc	unknown	T0+T2	fe	fe	fe
8	0007	ea	1		0007	7f	ff	11	fd	Nv-BdIzc	unknown	T0+T2	fe	fe	fc
9	0007	ea	1	NOP	0007	7f	ff	11	fd	Nv-BdIzc	unknown	T1	ff	ff	fc
9	0007	ea	1	NOP	0007	7f	ff	11	fd	Nv-BdIzc	unknown	T1	ff	ff	fe
10	0008	00	1		0008	7f	ff	11	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fe
10	0008	00	1		0008	7f	ff	11	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fc

SRE (5B)	Cycles: 7	Size: 3
Absolute Y-Indexed (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + Y. Address.L = Target.L. Address.H = Pointer.H.
3.2	Read Address	Store as Operand.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		
5.2	Write to Address	Write Operand.
6.1		C flag = (Operand & \$01). Operand >>= 1. A ^= Operand. N flag = (A & \$80), Z flag = ! A .
6.2	Write to Address	Write Operand.
7.1		
7.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	5b	1	unknown	0006	7f	ff	11	fd	Nv-BdIzc	LDX #	T1	ff	ff	44
7	0006	5b	1	unknown	0006	7f	ff	11	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	ea	1		0007	7f	ff	11	fd	Nv-BdIzc	unknown	T2	fe	fe	fe
8	0007	ea	1		0007	7f	ff	11	fd	Nv-BdIzc	unknown	T2	fe	fe	fc
9	0008	00	1		0008	7f	ff	11	fd	Nv-BdIzc	unknown	T3	11	ea	fc
9	0008	00	1		0008	7f	ff	11	fd	Nv-BdIzc	unknown	T3	11	ea	fb
10	00fb	98	1		0009	7f	ff	11	fd	Nv-BdIzc	unknown	T4	00	00	fb
10	00fb	98	1		0009	7f	ff	11	fd	Nv-BdIzc	unknown	T4	00	00	01
11	00fb	98	1		0009	7f	ff	11	fd	Nv-BdIzc	unknown	T5	01	ff	01
11	00fb	98	1		0009	7f	ff	11	fd	Nv-BdIzc	unknown	T5	01	ff	00
12	00fb	98	0		0009	7f	ff	11	fd	Nv-BdIzc	unknown		ff	98	00
12	00fb	98	0		0009	7f	ff	11	fd	Nv-BdIzc	unknown		ff	98	4c
13	00fb	98	0		0009	7f	ff	11	fd	nv-BdIzc	unknown	T0	4c	4c	4c
13	00fb	4c	0		0009	7f	ff	11	fd	nv-BdIzc	unknown	T0	4c	4c	98
14	0009	ea	1	NOP	0009	7f	ff	11	fd	nv-BdIzc	unknown	T1	7f	4c	98
14	0009	ea	1	NOP	0009	7f	ff	11	fd	nv-BdIzc	unknown	T1	7f	4c	33
15	000a	15	1		000a	33	ff	11	fd	nv-BdIzc	NOP	T0+T2	33	33	33
15	000a	15	1		000a	33	ff	11	fd	nv-BdIzc	NOP	T0+T2	33	33	66

NOP (5C)		Cycles: 4-5	Size: 3
Absolute X-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read PC	Store as Pointer.H.	
3.1		Inc. PC . Target = Pointer + X .	
		Address.L = Target.L.	
		Address.H = Pointer.H.	
		BoundaryCrossed = Pointer.H != Target.H.	
3.2	Read Address	Store as Operand. If !BoundaryCrossed, skip the next cycle.	
4.1		Address.H = Target.H.	
4.2	Read Address	Store as Operand.	
5.1			
5.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	5c	1	unknown	0006	7f	ff	11	fd	Nv-BdIzc	LDX #	T1	ff	ff	44
7	0006	5c	1	unknown	0006	7f	ff	11	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	ea	1		0007	7f	ff	11	fd	Nv-BdIzc	unknown	T2	fe	fe	fe
8	0007	ea	1		0007	7f	ff	11	fd	Nv-BdIzc	unknown	T2	fe	fe	fc
9	0008	00	1		0008	7f	ff	11	fd	Nv-BdIzc	unknown	T3	ff	ea	fc
9	0008	00	1		0008	7f	ff	11	fd	Nv-BdIzc	unknown	T3	ff	ea	e9
10	00e9	00	1		0009	7f	ff	11	fd	Nv-BdIzc	unknown	T4	00	00	e9
10	00e9	00	1		0009	7f	ff	11	fd	Nv-BdIzc	unknown	T4	00	00	01
11	01e9	00	1		0009	7f	ff	11	fd	Nv-BdIzc	unknown	T0	01	ff	01
11	01e9	00	1		0009	7f	ff	11	fd	Nv-BdIzc	unknown	T0	01	ff	00
12	0009	ea	1	NOP	0009	7f	ff	11	fd	Nv-BdIzc	unknown	T1	ff	00	00
12	0009	ea	1	NOP	0009	7f	ff	11	fd	Nv-BdIzc	unknown	T1	ff	00	ff
13	000a	15	1		000a	7f	ff	11	fd	Nv-BdIzc	NOP	T0+T2	ff	ff	ff
13	000a	15	1		000a	7f	ff	11	fd	Nv-BdIzc	NOP	T0+T2	ff	ff	fe

EOR (5D)		Cycles: 4-5	Size: 3
Absolute X-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read PC	Store as Pointer.H.	
3.1		Inc. PC . Target = Pointer + X .	
		Address.L = Target.L.	
		Address.H = Pointer.H.	
		BoundaryCrossed = Pointer.H != Target.H.	
3.2	Read Address	Store as Operand. If !BoundaryCrossed, skip the next cycle.	
4.1		Address.H = Target.H.	
4.2	Read Address	Store as Operand.	
5.1		A ^= Operand. N flag = (A & \$80), Z flag = ! A .	
5.2	Read PC	Store as OpCode.	
+X.1		* A , N flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	5d	1	EOR Abs,X	0006	7f	05	11	fd	nv-BdIzc	LDX #	T1	05	05	44
7	0006	5d	1	EOR Abs,X	0006	7f	05	11	fd	nv-BdIzc	LDX #	T1	05	05	0a
8	0007	ea	1		0007	7f	05	11	fd	nv-BdIzc	EOR Abs,X	T2	0a	0a	0a
8	0007	ea	1		0007	7f	05	11	fd	nv-BdIzc	EOR Abs,X	T2	0a	0a	14
9	0008	00	1		0008	7f	05	11	fd	nv-BdIzc	EOR Abs,X	T3	05	ea	14
9	0008	00	1		0008	7f	05	11	fd	nv-BdIzc	EOR Abs,X	T3	05	ea	ef
10	00ef	ff	1		0009	7f	05	11	fd	nv-BdIzc	EOR Abs,X	T0	00	00	ef
10	00ef	ff	1		0009	7f	05	11	fd	nv-BdIzc	EOR Abs,X	T0	00	00	01
11	0009	ea	1	NOP	0009	7f	05	11	fd	nv-BdIzc	EOR Abs,X	T1	7f	ff	01
11	0009	ea	1	NOP	0009	7f	05	11	fd	nv-BdIzc	EOR Abs,X	T1	7f	ff	80
12	000a	15	1		000a	80	05	11	fd	Nv-BdIzc	NOP	T0+T2	80	80	80
12	000a	15	1		000a	80	05	11	fd	Nv-BdIzc	NOP	T0+T2	80	80	00

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	5d	1	EOR Abs,X	0006	7f	ff	11	fd	Nv-BdIzc	LDX #	T1	ff	ff	44
7	0006	5d	1	EOR Abs,X	0006	7f	ff	11	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	ea	1		0007	7f	ff	11	fd	Nv-BdIzc	EOR Abs,X	T2	fe	fe	fe
8	0007	ea	1		0007	7f	ff	11	fd	Nv-BdIzc	EOR Abs,X	T2	fe	fe	fc
9	0008	00	1		0008	7f	ff	11	fd	Nv-BdIzc	EOR Abs,X	T3	ff	ea	fc
9	0008	00	1		0008	7f	ff	11	fd	Nv-BdIzc	EOR Abs,X	T3	ff	ea	e9
10	00e9	00	1		0009	7f	ff	11	fd	Nv-BdIzc	EOR Abs,X	T4	00	00	e9
10	00e9	00	1		0009	7f	ff	11	fd	Nv-BdIzc	EOR Abs,X	T4	00	00	01
11	01e9	7f	1		0009	7f	ff	11	fd	Nv-BdIzc	EOR Abs,X	T0	01	ff	01
11	01e9	7f	1		0009	7f	ff	11	fd	Nv-BdIzc	EOR Abs,X	T0	01	ff	00
12	0009	ea	1	NOP	0009	7f	ff	11	fd	Nv-BdIzc	EOR Abs,X	T1	7f	7f	00
12	0009	ea	1	NOP	0009	7f	ff	11	fd	Nv-BdIzc	EOR Abs,X	T1	7f	7f	00
13	000a	15	1		000a	00	ff	11	fd	nv-BdIzc	NOP	T0+T2	00	00	00
13	000a	15	1		000a	00	ff	11	fd	nv-BdIzc	NOP	T0+T2	00	00	00

LSR (5E)	Cycles: 7	Size: 3
Absolute X-Indexed (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + X . Address.L = Target.L. Address.H = Pointer.H.
3.2	Read Address	Store as Operand.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		
5.2	Write to Address	Write Operand.
6.1		C flag = (Operand & \$01). Operand >>= 1. N flag = 0, Z flag = !Operand.
6.2	Write to Address	Write Operand.
7.1		
7.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	5e	1	LSR Abs,X	0006	7f	ff	11	fd	Nv-BdIzc	LDX #	T1	ff	ff	44
7	0006	5e	1	LSR Abs,X	0006	7f	ff	11	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	ea	1		0007	7f	ff	11	fd	Nv-BdIzc	LSR Abs,X	T2	fe	fe	fe
8	0007	ea	1		0007	7f	ff	11	fd	Nv-BdIzc	LSR Abs,X	T2	fe	fe	fc
9	0008	00	1		0008	7f	ff	11	fd	Nv-BdIzc	LSR Abs,X	T3	ff	ea	fc
9	0008	00	1		0008	7f	ff	11	fd	Nv-BdIzc	LSR Abs,X	T3	ff	ea	e9
10	00e9	00	1		0009	7f	ff	11	fd	Nv-BdIzc	LSR Abs,X	T4	00	00	e9
10	00e9	00	1		0009	7f	ff	11	fd	Nv-BdIzc	LSR Abs,X	T4	00	00	01
11	01e9	7f	1		0009	7f	ff	11	fd	Nv-BdIzc	LSR Abs,X	T5	01	ff	01
11	01e9	7f	1		0009	7f	ff	11	fd	Nv-BdIzc	LSR Abs,X	T5	01	ff	00
12	01e9	7f	0		0009	7f	ff	11	fd	Nv-BdIzc	LSR Abs,X		ff	7f	00
12	01e9	7f	0		0009	7f	ff	11	fd	Nv-BdIzc	LSR Abs,X		ff	7f	3f
13	01e9	7f	0		0009	7f	ff	11	fd	nv-BdIzc	LSR Abs,X	T0	3f	3f	3f
13	01e9	3f	0		0009	7f	ff	11	fd	nv-BdIzc	LSR Abs,X	T0	3f	3f	7e
14	0009	ea	1	NOP	0009	7f	ff	11	fd	nv-BdIzc	LSR Abs,X	T1	ff	3f	7e
14	0009	ea	1	NOP	0009	7f	ff	11	fd	nv-BdIzc	LSR Abs,X	T1	ff	3f	3e
15	000a	15	1		000a	7f	ff	11	fd	nv-BdIzc	NOP	T0+T2	3e	3e	3e
15	000a	15	1		000a	7f	ff	11	fd	nv-BdIzc	NOP	T0+T2	3e	3e	7c

SRE (5F)	Cycles: 7	Size: 3
Absolute X-Indexed (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + X . Address.L = Target.L. Address.H = Pointer.H.
3.2	Read Address	Store as Operand.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		
5.2	Write to Address	Write Operand.
6.1		C flag = (Operand & \$01). Operand >>= 1. A ^= Operand. N flag = (A & \$80), Z flag = ! A .
6.2	Write to Address	Write Operand.
7.1		
7.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	5f	1	unknown	0006	7f	ff	11	fd	Nv-BdIzc	LDX #	T1	ff	ff	44
7	0006	5f	1	unknown	0006	7f	ff	11	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	ea	1		0007	7f	ff	11	fd	Nv-BdIzc	unknown	T2	fe	fe	fe
8	0007	ea	1		0007	7f	ff	11	fd	Nv-BdIzc	unknown	T2	fe	fe	fc
9	0008	00	1		0008	7f	ff	11	fd	Nv-BdIzc	unknown	T3	ff	ea	fc
9	0008	00	1		0008	7f	ff	11	fd	Nv-BdIzc	unknown	T3	ff	ea	e9
10	00e9	00	1		0009	7f	ff	11	fd	Nv-BdIzc	unknown	T4	00	00	e9
10	00e9	00	1		0009	7f	ff	11	fd	Nv-BdIzc	unknown	T4	00	00	01
11	01e9	3f	1		0009	7f	ff	11	fd	Nv-BdIzc	unknown	T5	01	ff	01
11	01e9	3f	1		0009	7f	ff	11	fd	Nv-BdIzc	unknown	T5	01	ff	00
12	01e9	3f	0		0009	7f	ff	11	fd	Nv-BdIzc	unknown		ff	3f	00
12	01e9	3f	0		0009	7f	ff	11	fd	Nv-BdIzc	unknown		ff	3f	1f
13	01e9	3f	0		0009	7f	ff	11	fd	nv-BdIzc	unknown	T0	1f	1f	1f
13	01e9	1f	0		0009	7f	ff	11	fd	nv-BdIzc	unknown	T0	1f	1f	3e
14	0009	ea	1	NOP	0009	7f	ff	11	fd	nv-BdIzc	unknown	T1	7f	1f	3e
14	0009	ea	1	NOP	0009	7f	ff	11	fd	nv-BdIzc	unknown	T1	7f	1f	60
15	000a	15	1		000a	60	ff	11	fd	nv-BdIzc	NOP	T0+T2	60	60	60
15	000a	15	1		000a	60	ff	11	fd	nv-BdIzc	NOP	T0+T2	60	60	c0

RTS (60)		Cycles: 6	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Operand.	
2.1		Inc. PC .	
2.2	Read u8(S +0) \$0100	Store as Operand.	
3.1			
3.2	Read u8(S +1) \$0100	Store as Target.L.	
4.1		Inc. S by 2.	
4.2	Read u8(S +0) \$0100	Store as Target.H.	
5.1		PC = Target.	
5.2	Read PC	Store as Operand.	
6.1		Inc. PC .	
6.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	60	1	RTS	0006	7f	ff	11	fd	Nv-BdIzc	LDX #	T1	ff	ff	44
7	0006	60	1	RTS	0006	7f	ff	11	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	ea	1		0007	7f	ff	11	fd	Nv-BdIzc	RTS	T2	fe	fe	fe
8	0007	ea	1		0007	7f	ff	11	fd	Nv-BdIzc	RTS	T2	fe	fe	fc
9	01fd	99	1		0008	7f	ff	11	fd	Nv-BdIzc	RTS	T3	00	fd	fc
9	01fd	99	1		0008	7f	ff	11	fd	Nv-BdIzc	RTS	T3	00	fd	fe
10	01fe	06	1		0008	7f	ff	11	fd	Nv-BdIzc	RTS	T4	00	fe	fe
10	01fe	06	1		0008	7f	ff	11	fd	Nv-BdIzc	RTS	T4	00	fe	ff
11	01ff	00	1		0008	7f	ff	11	ff	Nv-BdIzc	RTS	T5	00	06	ff
11	01ff	00	1		0008	7f	ff	11	ff	Nv-BdIzc	RTS	T5	00	06	06
12	0006	60	1		0006	7f	ff	11	ff	Nv-BdIzc	RTS	T0	00	00	06
12	0006	60	1		0006	7f	ff	11	ff	Nv-BdIzc	RTS	T0	00	00	00
13	0007	ea	1	NOP	0007	7f	ff	11	ff	Nv-BdIzc	RTS	T1	00	60	00
13	0007	ea	1	NOP	0007	7f	ff	11	ff	Nv-BdIzc	RTS	T1	00	60	60
14	0008	00	1		0008	7f	ff	11	ff	Nv-BdIzc	NOP	T0+T2	00	60	60
14	0008	00	1		0008	7f	ff	11	ff	Nv-BdIzc	NOP	T0+T2	00	60	60

ADC (61)		Cycles: 6	Size: 2
Indirect X (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Operand.	
2.1		Inc. PC .	
2.2	Read Operand	Discard.	
3.1		Pointer = u8(Operand + X).	
3.2	Read Pointer	Store as Address.	
4.1			
4.2	Read u8(Pointer + 1)	Store as Address.H.	
5.1			
5.2	Read Address	Store as Operand.	
6.1		Result16 = u16(A) + u16(Operand) + C flag. V flag = (~(u16(A) ^ u16(Operand)) & (u16(A) ^ Result16) & 0x0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (Result16 & \$80), Z flag = (A == \$00).	
6.2	Read PC	Store as OpCode.	
+X.1		* A, C flag, N flag, V flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	61	1	ADC (zp,X)	0006	7f	33	11	fd	nv-BdIzc	LDX #	T1	33	33	44
7	0006	61	1	ADC (zp,X)	0006	7f	33	11	fd	nv-BdIzc	LDX #	T1	33	33	66
8	0007	55	1		0007	7f	33	11	fd	nv-BdIzc	ADC (zp,X)	T2	66	66	66
8	0007	55	1		0007	7f	33	11	fd	nv-BdIzc	ADC (zp,X)	T2	66	66	cc
9	0055	0f	1		0008	7f	33	11	fd	nv-BdIzc	ADC (zp,X)	T3	33	55	cc
9	0055	0f	1		0008	7f	33	11	fd	nv-BdIzc	ADC (zp,X)	T3	33	55	88
10	0088	22	1		0008	7f	33	11	fd	nv-BdIzc	ADC (zp,X)	T4	00	88	88
10	0088	22	1		0008	7f	33	11	fd	nv-BdIzc	ADC (zp,X)	T4	00	88	89
11	0089	01	1		0008	7f	33	11	fd	nv-BdIzc	ADC (zp,X)	T5	00	22	89
11	0089	01	1		0008	7f	33	11	fd	nv-BdIzc	ADC (zp,X)	T5	00	22	22
12	0122	01	1		0008	7f	33	11	fd	nv-BdIzc	ADC (zp,X)	T0	ff	ff	22
12	0122	01	1		0008	7f	33	11	fd	nv-BdIzc	ADC (zp,X)	T0	ff	ff	fe
13	0008	ea	1	NOP	0008	7f	33	11	fd	nv-BdIzc	ADC (zp,X)	T1	7f	01	fe
13	0008	ea	1	NOP	0008	7f	33	11	fd	nv-BdIzc	ADC (zp,X)	T1	7f	01	80
14	0009	ea	1		0009	80	33	11	fd	NV-BdIzc	NOP	T0+T2	80	80	80
14	0009	ea	1		0009	80	33	11	fd	NV-BdIzc	NOP	T0+T2	80	80	00

JAM (62)	Cycles: ∞	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Discard.
2.1		Visual6502 suggests incrementing PC , but Single-Step Tests cannot be passed this way.
2.2	Read \$FFFF	Discard.
3.1		
3.2	Read \$FFFE	Discard.
4.1		
4.2	Read \$FFFE	Discard.
5.1		Repeat 5.1 and 5.2 forever.
5.2	Read \$FFFF	Discard.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	62	1	unknown	0006	7f	33	11	fd	nv-BdIzc	LDX #	T1	33	33	44
7	0006	62	1	unknown	0006	7f	33	11	fd	nv-BdIzc	LDX #	T1	33	33	66
8	0007	55	1		0007	7f	33	11	fd	nv-BdIzc	unknown	T2	66	66	66
8	0007	55	1		0007	7f	33	11	fd	nv-BdIzc	unknown	T2	66	66	cc
9	ffff	00	1		0008	7f	33	11	fd	nv-BdIzc	unknown	T3	ff	ff	cc
9	ffff	00	1		0008	7f	33	11	fd	nv-BdIzc	unknown	T3	ff	ff	fe
10	fffe	00	1		0008	7f	33	11	fd	nv-BdIzc	unknown	T4	ff	ff	fe
10	fffe	00	1		0008	7f	33	11	fd	nv-BdIzc	unknown	T4	ff	ff	fe
11	fffe	00	1		0008	7f	33	11	fd	nv-BdIzc	unknown	T5	ff	ff	fe
11	fffe	00	1		0008	7f	33	11	fd	nv-BdIzc	unknown	T5	ff	ff	fe
12	ffff	00	1		0008	7f	33	11	fd	nv-BdIzc	unknown		ff	ff	fe
12	ffff	00	1		0008	7f	33	11	fd	nv-BdIzc	unknown		ff	ff	fe
13	ffff	00	1		0008	7f	33	11	fd	nv-BdIzc	unknown		ff	ff	fe
13	ffff	00	1		0008	7f	33	11	fd	nv-BdIzc	unknown		ff	ff	fe
14	ffff	00	1		0008	7f	33	11	fd	nv-BdIzc	unknown		ff	ff	fe
14	ffff	00	1		0008	7f	33	11	fd	nv-BdIzc	unknown		ff	ff	fe

RRA (63)		Cycles: 8	Size: 2
Indirect X (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Operand.	
2.1		Inc. PC .	
2.2	Read Operand	Discard.	
3.1		Pointer = u8(Operand + X).	
3.2	Read Pointer	Store as Address.	
4.1			
4.2	Read u8(Pointer + 1)	Store as Address.H.	
5.1			
5.2	Read Address	Store as Operand.	
6.1			
6.2	Write to Address	Write Operand.	
7.1		HiBit = (C flag) << 7. C flag = (Operand & \$01). Operand = (Operand >> 1) HiBit. Result16 = u16(A) + u16(Operand) + C flag. V flag = (~(u16(A) ^ u16(Operand)) & (u16(A) ^ Result16) & \$0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (A & \$80), Z flag = ! A .	
7.2	Write to Address	Write Operand.	
8.1			
8.2	Read PC	Store as OpCode.	
+X.1		* A , C flag, N flag, V flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	63	1	unknown	0006	7f	33	11	fd	nv-BdIzc	LDX #	T1	33	33	44
7	0006	63	1	unknown	0006	7f	33	11	fd	nv-BdIzc	LDX #	T1	33	33	66
8	0007	55	1		0007	7f	33	11	fd	nv-BdIzc	unknown	T2	66	66	66
8	0007	55	1		0007	7f	33	11	fd	nv-BdIzc	unknown	T2	66	66	cc
9	0055	0f	1		0008	7f	33	11	fd	nv-BdIzc	unknown	T3	33	55	cc
9	0055	0f	1		0008	7f	33	11	fd	nv-BdIzc	unknown	T3	33	55	88
10	0088	22	1		0008	7f	33	11	fd	nv-BdIzc	unknown	T4	00	88	88
10	0088	22	1		0008	7f	33	11	fd	nv-BdIzc	unknown	T4	00	88	89
11	0089	01	1		0008	7f	33	11	fd	nv-BdIzc	unknown	T5	00	22	89
11	0089	01	1		0008	7f	33	11	fd	nv-BdIzc	unknown	T5	00	22	22
12	0122	01	1		0008	7f	33	11	fd	nv-BdIzc	unknown		ff	ff	22
12	0122	01	1		0008	7f	33	11	fd	nv-BdIzc	unknown		ff	ff	fe
13	0122	01	0		0008	7f	33	11	fd	nv-BdIzc	unknown		ff	01	fe
13	0122	01	0		0008	7f	33	11	fd	nv-BdIzc	unknown		ff	01	00
14	0122	01	0		0008	7f	33	11	fd	nv-BdIzc	unknown	T0	00	00	00
14	0122	00	0		0008	7f	33	11	fd	nv-BdIzc	unknown	T0	00	00	00
15	0008	ea	1	NOP	0008	7f	33	11	fd	nv-BdIzc	unknown	T1	7f	00	00
15	0008	ea	1	NOP	0008	7f	33	11	fd	nv-BdIzc	unknown	T1	7f	00	80
16	0009	ea	1		0009	80	33	11	fd	NV-BdIzc	NOP	T0+T2	80	80	80
16	0009	ea	1		0009	80	33	11	fd	NV-BdIzc	NOP	T0+T2	80	80	00

NOP (64)		Cycles: 3	Size: 2
Zero Page (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read Address	Store as Operand.	
3.1			
3.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	64	1	unknown	0006	7f	33	11	fd	nv-BdIzc	LDX #	T1	33	33	44
7	0006	64	1	unknown	0006	7f	33	11	fd	nv-BdIzc	LDX #	T1	33	33	66
8	0007	55	1		0007	7f	33	11	fd	nv-BdIzc	unknown	T2	66	66	66
8	0007	55	1		0007	7f	33	11	fd	nv-BdIzc	unknown	T2	66	66	cc
9	0055	0f	1		0008	7f	33	11	fd	nv-BdIzc	unknown	T0	ff	55	cc
9	0055	0f	1		0008	7f	33	11	fd	nv-BdIzc	unknown	T0	ff	55	54
10	0008	ea	1	NOP	0008	7f	33	11	fd	nv-BdIzc	unknown	T1	ff	0f	54
10	0008	ea	1	NOP	0008	7f	33	11	fd	nv-BdIzc	unknown	T1	ff	0f	0e
11	0009	ea	1		0009	7f	33	11	fd	nv-BdIzc	NOP	T0+T2	0e	0e	0e
11	0009	ea	1		0009	7f	33	11	fd	nv-BdIzc	NOP	T0+T2	0e	0e	1c

ADC (65)		Cycles: 3	Size: 2
Zero Page (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1	Read PC	Inc. PC .	
1.2		Store as Address.	
2.1		Inc. PC .	
2.2		Store as Operand.	
3.1	Read Address	Result16 = u16(A) + u16(Operand) + C flag.	
		V flag = $(\sim(u16(\mathbf{A}) \wedge u16(\text{Operand})) \& (u16(\mathbf{A}) \wedge \text{Result16}) \& 0x0080) \neq 0$.	
		A = u8(Result16).	
		C flag = (Result16 > \$FF), N flag = (Result16 & \$80), Z flag = (A == \$00).	
3.2	Read PC	Store as OpCode.	
+X.1		* A , C flag, N flag, V flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	65	1	ADC zp	0006	7f	33	11	fd	nv-BdIzc	LDX #	T1	33	33	44
7	0006	65	1	ADC zp	0006	7f	33	11	fd	nv-BdIzc	LDX #	T1	33	33	66
8	0007	55	1		0007	7f	33	11	fd	nv-BdIzc	ADC zp	T2	66	66	66
8	0007	55	1		0007	7f	33	11	fd	nv-BdIzc	ADC zp	T2	66	66	cc
9	0055	0f	1		0008	7f	33	11	fd	nv-BdIzc	ADC zp	T0	ff	55	cc
9	0055	0f	1		0008	7f	33	11	fd	nv-BdIzc	ADC zp	T0	ff	55	54
10	0008	ea	1	NOP	0008	7f	33	11	fd	nv-BdIzc	ADC zp	T1	7f	0f	54
10	0008	ea	1	NOP	0008	7f	33	11	fd	nv-BdIzc	ADC zp	T1	7f	0f	8e
11	0009	ea	1		0009	8e	33	11	fd	NV-BdIzc	NOP	T0+T2	8e	8e	8e
11	0009	ea	1		0009	8e	33	11	fd	NV-BdIzc	NOP	T0+T2	8e	8e	1c

ROR (66)		Cycles: 5	Size: 2
Zero Page (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read Address	Store as Operand.	
3.1			
3.2	Write to Address	Write Operand.	
4.1		HiBit = (C flag) << 7. C flag = (Operand & \$01).	
		Operand = (Operand >> 1) HiBit.	
		N flag = (Operand & \$80), Z flag = !Operand.	
4.2	Write to Address	Write Operand.	
5.1			
5.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	66	1	ROR zp	0006	7f	ff	cc	fd	Nv-BdIzc	LDX #	T1	ff	ff	30
7	0006	66	1	ROR zp	0006	7f	ff	cc	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	55	1		0007	7f	ff	cc	fd	Nv-BdIzc	ROR zp	T2	fe	fe	fe
8	0007	55	1		0007	7f	ff	cc	fd	Nv-BdIzc	ROR zp	T2	fe	fe	fc
9	0055	01	1		0008	7f	ff	cc	fd	Nv-BdIzc	ROR zp	T3	ff	55	fc
9	0055	01	1		0008	7f	ff	cc	fd	Nv-BdIzc	ROR zp	T3	ff	55	54
10	0055	01	0		0008	7f	ff	cc	fd	Nv-BdIzc	ROR zp	T4	ff	01	54
10	0055	01	0		0008	7f	ff	cc	fd	Nv-BdIzc	ROR zp	T4	ff	01	00
11	0055	01	0		0008	7f	ff	cc	fd	nv-BdIZC	ROR zp	T0	00	00	00
11	0055	00	0		0008	7f	ff	cc	fd	nv-BdIZC	ROR zp	T0	00	00	00
12	0008	ea	1	NOP	0008	7f	ff	cc	fd	nv-BdIZC	ROR zp	T1	ff	00	00
12	0008	ea	1	NOP	0008	7f	ff	cc	fd	nv-BdIZC	ROR zp	T1	ff	00	ff
13	0009	ea	1		0009	7f	ff	cc	fd	nv-BdIZC	NOP	T0+T2	ff	ff	ff
13	0009	ea	1		0009	7f	ff	cc	fd	nv-BdIZC	NOP	T0+T2	ff	ff	fe

RRA (67)	Cycles: 5	Size: 2
Zero Page (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read Address	Store as Operand.
3.1		
3.2	Write to Address	Write Operand.
4.1		$HiBit = (C\text{ flag}) \ll 7$. $C\text{ flag} = (Operand \& \$01)$. $Operand = (Operand \gg 1) HiBit$. $Result16 = u16(A) + u16(Operand) + C\text{ flag}$. $V\text{ flag} = (\sim(u16(A) \wedge u16(Operand)) \& (u16(A) \wedge Result16) \& \$0080) \neq 0$. $A = u8(Result16)$. $C\text{ flag} = (Result16 > \$FF)$, $N\text{ flag} = (A \& \$80)$, $Z\text{ flag} = !A$.
4.2	Write to Address	Write Operand.
5.1		
5.2	Read PC	Store as OpCode.
+X.1		* A , C flag, N flag, V flag, and Z flag applied.

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	67	1	unknown	0006	7f	ff	cc	fd	Nv-BdIzc	LDX #	T1	ff	ff	30
7	0006	67	1	unknown	0006	7f	ff	cc	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	55	1		0007	7f	ff	cc	fd	Nv-BdIzc	unknown	T2	fe	fe	fe
8	0007	55	1		0007	7f	ff	cc	fd	Nv-BdIzc	unknown	T2	fe	fe	fc
9	0055	80	1		0008	7f	ff	cc	fd	Nv-BdIzc	unknown	T3	ff	55	fc
9	0055	80	1		0008	7f	ff	cc	fd	Nv-BdIzc	unknown	T3	ff	55	54
10	0055	80	0		0008	7f	ff	cc	fd	Nv-BdIzc	unknown	T4	ff	80	54
10	0055	80	0		0008	7f	ff	cc	fd	Nv-BdIzc	unknown	T4	ff	80	40
11	0055	80	0		0008	7f	ff	cc	fd	nv-BdIzc	unknown	T0	40	40	40
11	0055	40	0		0008	7f	ff	cc	fd	nv-BdIzc	unknown	T0	40	40	80
12	0008	ea	1	NOP	0008	7f	ff	cc	fd	nv-BdIzc	unknown	T1	7f	40	80
12	0008	ea	1	NOP	0008	7f	ff	cc	fd	nv-BdIzc	unknown	T1	7f	40	bf
13	0009	ea	1		0009	bf	ff	cc	fd	NV-BdIzc	NOP	T0+T2	bf	bf	bf
13	0009	ea	1		0009	bf	ff	cc	fd	NV-BdIzc	NOP	T0+T2	bf	bf	7e

PLA (68)	Cycles: 4	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Discard.
2.1		
2.2	Read (S \$0100)	Store as Operand.
3.1		Inc. S by 1.
3.2	Read (S \$0100)	Store as Operand.
4.1		A = Operand. N flag = (A & \$80), Z flag = ! A .
4.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	68	1	PLA	0006	7f	ff	cc	fd	Nv-BdIzc	LDX #	T1	ff	ff	30
7	0006	68	1	PLA	0006	7f	ff	cc	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	ea	1		0007	7f	ff	cc	fd	Nv-BdIzc	PLA	T2	fe	fe	fe
8	0007	ea	1		0007	7f	ff	cc	fd	Nv-BdIzc	PLA	T2	fe	fe	fc
9	01fd	99	1		0007	7f	ff	cc	fd	Nv-BdIzc	PLA	T3	00	fd	fc
9	01fd	99	1		0007	7f	ff	cc	fd	Nv-BdIzc	PLA	T3	00	fd	fe
10	01fe	06	1		0007	7f	ff	cc	fe	Nv-BdIzc	PLA	T0	00	fe	fe
10	01fe	06	1		0007	7f	ff	cc	fe	Nv-BdIzc	PLA	T0	00	fe	ff
11	0007	ea	1	NOP	0007	06	ff	cc	fe	nv-BdIzc	PLA	T1	00	06	ff
11	0007	ea	1	NOP	0007	06	ff	cc	fe	nv-BdIzc	PLA	T1	00	06	07
12	0008	ea	1		0008	06	ff	cc	fe	nv-BdIzc	NOP	T0+T2	00	02	02
12	0008	ea	1		0008	06	ff	cc	fe	nv-BdIzc	NOP	T0+T2	00	07	08

ADC (69)		Cycles: 2	Size: 2
Immediate			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1	Read PC	Inc. PC .	
1.2		Store as Operand.	
2.1		Inc. PC . Result16 = u16(A) + u16(Operand) + C flag. V flag = (~(u16(A) ^ u16(Operand)) & (u16(A) ^ Result16) & 0x0080) != 0. A = u8(Result16).	
2.2		C flag = (Result16 > \$FF), N flag = (Result16 & \$80), Z flag = (A == \$00). Store as OpCode.	
+X.1		* A , C flag, N flag, V flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	69	1	ADC #	0006	7f	ff	cc	fd	Nv-BdIzc	LDX #	T1	ff	ff	30
7	0006	69	1	ADC #	0006	7f	ff	cc	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	cc	1		0007	7f	ff	cc	fd	Nv-BdIzc	ADC #	T0+T2	fe	fe	fe
8	0007	cc	1		0007	7f	ff	cc	fd	Nv-BdIzc	ADC #	T0+T2	fe	fe	fc
9	0008	ea	1	NOP	0008	7f	ff	cc	fd	Nv-BdIzc	ADC #	T1	7f	cc	fc
9	0008	ea	1	NOP	0008	7f	ff	cc	fd	Nv-BdIzc	ADC #	T1	7f	cc	4b
10	0009	ea	1		0009	4b	ff	cc	fd	nv-BdIzC	NOP	T0+T2	4b	4b	4b
10	0009	ea	1		0009	4b	ff	cc	fd	nv-BdIzC	NOP	T0+T2	4b	4b	96

ROR (6A)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1	Read PC	Inc. PC .	
1.2		Discard.	
2.1		HiBit = (C flag) << 7. C flag = (A & \$01).	
		A = (A >> 1) HiBit.	
		N flag = (A & \$80), Z flag = ! A .	
2.2	Read PC	Store as OpCode.	
+X.1		* A , N flag and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	6a	1	ROR	0006	01	ff	cc	fd	Nv-BdIzc	LDX #	T1	ff	ff	30
7	0006	6a	1	ROR	0006	01	ff	cc	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	ea	1		0007	01	ff	cc	fd	Nv-BdIzc	ROR	T0+T2	fe	fe	fe
8	0007	ea	1		0007	01	ff	cc	fd	Nv-BdIzc	ROR	T0+T2	fe	fe	fc
9	0007	ea	1	NOP	0007	01	ff	cc	fd	Nv-BdIzc	ROR	T1	01	01	fc
9	0007	ea	1	NOP	0007	01	ff	cc	fd	Nv-BdIzc	ROR	T1	01	01	00
10	0008	ea	1		0008	00	ff	cc	fd	nv-BdIZC	NOP	T0+T2	00	00	00
10	0008	ea	1		0008	00	ff	cc	fd	nv-BdIZC	NOP	T0+T2	00	00	00

ARR (6B)	Cycles: 2	Size: 2
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . A &= Operand. HighBit = (C flag) << 7. C flag = (A & \$80). A = (A >> 1) HighBit. V flag = (C flag ^ ((A >> 5) & \$01)). N flag = (A & \$80), Z flag = ! A .
2.2	Read PC	Store as OpCode.
+X.1		* A , C flag, N flag, V flag, and Z flag applied.

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.
Visual6502 omits (A &= Operand); flags are updated based off the incorrect A value.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	6b	1	unknown	0006	01	ff	cc	fd	Nv-BdIzc	LDX #	T1	ff	ff	30
7	0006	6b	1	unknown	0006	01	ff	cc	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	ea	1		0007	01	ff	cc	fd	Nv-BdIzc	unknown	T0+T2	fe	fe	fe
8	0007	ea	1		0007	01	ff	cc	fd	Nv-BdIzc	unknown	T0+T2	fe	fe	fc
9	0008	ea	1	NOP	0008	01	ff	cc	fd	Nv-BdIzc	unknown	T1	00	00	fc
9	0008	ea	1	NOP	0008	01	ff	cc	fd	Nv-BdIzc	unknown	T1	01	01	00
10	0009	ea	1		0009	00	ff	cc	fd	nv-BdIZc	NOP	T0+T2	00	00	00
10	0009	ea	1		0009	00	ff	cc	fd	nv-BdIZc	NOP	T0+T2	00	00	00

JMP (6C) Cycles: 5 Size: 3		
Absolute Indirect (Jump)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC .
3.2	Read Pointer	Store as Address.
4.1		
4.2	* Read (Pointer.H << 8) u8(Pointer.L + 1)	Store as Address.H.
5.1		PC = Address.
5.2	Read PC	Store as OpCode.
+X.1		

* Read operation does not cross into a new page if (Pointer+1) causes overflow, hence the high byte of Pointer is retained while "+ 1" is applied only to the low byte.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	6c	1	JMP (Abs)	0006	01	ff	cc	fd	Nv-BdIzc	LDX #	T1	ff	ff	30
7	0006	6c	1	JMP (Abs)	0006	01	ff	cc	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	ff	1		0007	01	ff	cc	fd	Nv-BdIzc	JMP (Abs)	T2	fe	fe	fe
8	0007	ff	1		0007	01	ff	cc	fd	Nv-BdIzc	JMP (Abs)	T2	fe	fe	fc
9	0008	01	1		0008	01	ff	cc	fd	Nv-BdIzc	JMP (Abs)	T3	00	ff	fc
9	0008	01	1		0008	01	ff	cc	fd	Nv-BdIzc	JMP (Abs)	T3	00	ff	ff
10	01ff	55	1		0009	01	ff	cc	fd	Nv-BdIzc	JMP (Abs)	T4	00	ff	ff
10	01ff	55	1		0009	01	ff	cc	fd	Nv-BdIzc	JMP (Abs)	T4	00	ff	00
11	0100	01	1		0009	01	ff	cc	fd	Nv-BdIzc	JMP (Abs)	T0	00	55	00
11	0100	01	1		0009	01	ff	cc	fd	Nv-BdIzc	JMP (Abs)	T0	00	55	55
12	0155	ea	1	NOP	0155	01	ff	cc	fd	Nv-BdIzc	JMP (Abs)	T1	00	01	55
12	0155	ea	1	NOP	0155	01	ff	cc	fd	Nv-BdIzc	JMP (Abs)	T1	00	01	01
13	0156	00	1		0156	01	ff	cc	fd	Nv-BdIzc	NOP	T0+T2	00	01	01
13	0156	00	1		0156	01	ff	cc	fd	Nv-BdIzc	NOP	T0+T2	00	01	01

ADC (6D)		Cycles: 4	Size: 3
Absolute (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read PC	Store as Address.H.	
3.1		Inc. PC .	
3.2	Read Address	Store as Operand.	
4.1		Result16 = u16(A) + u16(Operand) + C flag. V flag = (~(u16(A) ^ u16(Operand)) & (u16(A) ^ Result16) & 0x0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (Result16 & \$80), Z flag = (A == \$00).	
4.2	Read PC	Store as OpCode.	
+X.1		* A , C flag, N flag, V flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	6d	1	ADC Abs	0006	7f	44	cc	fd	nv-BdIzc	LDX #	T1	44	44	30
7	0006	6d	1	ADC Abs	0006	7f	44	cc	fd	nv-BdIzc	LDX #	T1	44	44	88
8	0007	ff	1		0007	7f	44	cc	fd	nv-BdIzc	ADC Abs	T2	88	88	88
8	0007	ff	1		0007	7f	44	cc	fd	nv-BdIzc	ADC Abs	T2	88	88	10
9	0008	01	1		0008	7f	44	cc	fd	nv-BdIzc	ADC Abs	T3	00	ff	10
9	0008	01	1		0008	7f	44	cc	fd	nv-BdIzc	ADC Abs	T3	00	ff	ff
10	01ff	55	1		0009	7f	44	cc	fd	nv-BdIzc	ADC Abs	T0	ff	ff	ff
10	01ff	55	1		0009	7f	44	cc	fd	nv-BdIzc	ADC Abs	T0	ff	ff	fe
11	0009	ea	1	NOP	0009	7f	44	cc	fd	nv-BdIzc	ADC Abs	T1	7f	55	fe
11	0009	ea	1	NOP	0009	7f	44	cc	fd	nv-BdIzc	ADC Abs	T1	7f	55	d4
12	000a	15	1		000a	d4	44	cc	fd	NV-BdIzc	NOP	T0+T2	d4	d4	d4
12	000a	15	1		000a	d4	44	cc	fd	NV-BdIzc	NOP	T0+T2	d4	d4	a8

ROR (6E)		Cycles: 6	Size: 3
Absolute (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read PC	Store as Address.H.	
3.1		Inc. PC .	
3.2	Read Address	Store as Operand.	
4.1			
4.2	Write to Address	Write Operand.	
5.1		HiBit = (C flag) << 7. C flag = (Operand & \$01).	
		Operand = (Operand >> 1) HiBit.	
		N flag = (Operand & \$80), Z flag = !Operand.	
5.2	Write to Address	Write Operand.	
6.1			
6.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	6e	1	ROR Abs	0006	7f	fe	cc	fd	Nv-BdIzc	LDX #	T1	fe	fe	30
7	0006	6e	1	ROR Abs	0006	7f	fe	cc	fd	Nv-BdIzc	LDX #	T1	fe	fe	fc
8	0007	ff	1		0007	7f	fe	cc	fd	Nv-BdIzc	ROR Abs	T2	fc	fc	fc
8	0007	ff	1		0007	7f	fe	cc	fd	Nv-BdIzc	ROR Abs	T2	fc	fc	f8
9	0008	01	1		0008	7f	fe	cc	fd	Nv-BdIzc	ROR Abs	T3	00	ff	f8
9	0008	01	1		0008	7f	fe	cc	fd	Nv-BdIzc	ROR Abs	T3	00	ff	ff
10	01ff	01	1		0009	7f	fe	cc	fd	Nv-BdIzc	ROR Abs	T4	ff	ff	ff
10	01ff	01	1		0009	7f	fe	cc	fd	Nv-BdIzc	ROR Abs	T4	ff	ff	fe
11	01ff	01	0		0009	7f	fe	cc	fd	Nv-BdIzc	ROR Abs	T5	ff	01	fe
11	01ff	01	0		0009	7f	fe	cc	fd	Nv-BdIzc	ROR Abs	T5	ff	01	00
12	01ff	01	0		0009	7f	fe	cc	fd	nv-BdIZC	ROR Abs	T0	00	00	00
12	01ff	00	0		0009	7f	fe	cc	fd	nv-BdIZC	ROR Abs	T0	00	00	00
13	0009	ea	1	NOP	0009	7f	fe	cc	fd	nv-BdIZC	ROR Abs	T1	ff	00	00
13	0009	ea	1	NOP	0009	7f	fe	cc	fd	nv-BdIZC	ROR Abs	T1	ff	00	ff
14	000a	15	1		000a	7f	fe	cc	fd	nv-BdIZC	NOP	T0+T2	ff	ff	ff
14	000a	15	1		000a	7f	fe	cc	fd	nv-BdIZC	NOP	T0+T2	ff	ff	fe

RRA (6F)		Cycles: 6	Size: 3
Absolute (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read PC	Store as Address.H.	
3.1		Inc. PC .	
3.2	Read Address	Store as Operand.	
4.1			
4.2	Write to Address	Write Operand.	
5.1		HiBit = (C flag) << 7. C flag = (Operand & \$01). Operand = (Operand >> 1) HiBit. Result16 = u16(A) + u16(Operand) + C flag. V flag = (~(u16(A) ^ u16(Operand)) & (u16(A) ^ Result16) & \$0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (A & \$80), Z flag = ! A .	
5.2	Write to Address	Write Operand.	
6.1			
6.2	Read PC	Store as OpCode.	
+X.1		* A , C flag, N flag, V flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	6f	1	unknown	0006	7f	fe	cc	fd	Nv-BdIzc	LDX #	T1	fe	fe	30
7	0006	6f	1	unknown	0006	7f	fe	cc	fd	Nv-BdIzc	LDX #	T1	fe	fe	fc
8	0007	ff	1		0007	7f	fe	cc	fd	Nv-BdIzc	unknown	T2	fc	fc	fc
8	0007	ff	1		0007	7f	fe	cc	fd	Nv-BdIzc	unknown	T2	fc	fc	f8
9	0008	01	1		0008	7f	fe	cc	fd	Nv-BdIzc	unknown	T3	00	ff	f8
9	0008	01	1		0008	7f	fe	cc	fd	Nv-BdIzc	unknown	T3	00	ff	ff
10	01ff	01	1		0009	7f	fe	cc	fd	Nv-BdIzc	unknown	T4	ff	ff	ff
10	01ff	01	1		0009	7f	fe	cc	fd	Nv-BdIzc	unknown	T4	ff	ff	fe
11	01ff	01	0		0009	7f	fe	cc	fd	Nv-BdIzc	unknown	T5	ff	01	fe
11	01ff	01	0		0009	7f	fe	cc	fd	Nv-BdIzc	unknown	T5	ff	01	00
12	01ff	01	0		0009	7f	fe	cc	fd	nv-BdIZC	unknown	T0	00	00	00
12	01ff	00	0		0009	7f	fe	cc	fd	nv-BdIZC	unknown	T0	00	00	00
13	0009	ea	1	NOP	0009	7f	fe	cc	fd	nv-BdIZC	unknown	T1	7f	00	00
13	0009	ea	1	NOP	0009	7f	fe	cc	fd	nv-BdIZC	unknown	T1	7f	00	80
14	000a	15	1		000a	80	fe	cc	fd	NV-BdIzc	NOP	T0+T2	80	80	80
14	000a	15	1		000a	80	fe	cc	fd	NV-BdIzc	NOP	T0+T2	80	80	00

BVS (70)		Cycles: 2-4	Size: 2
Relative (Jump)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC . Jump = ((P & \$40) == \$40).	
1.2	Read PC	Store as Operand.	
		If !Jump, poll interrupts.	
		Otherwise Address = i16(i8(Operand)) + PC + 1.	
		BoundaryCrossed = Address.H != (PC + 1).H.	
		If !BoundaryCrossed, poll interrupts.	
2.1		Inc. PC . If !Jump, end (next half-cycle is 4.2).	
2.2	Read PC	Discard.	
3.1		PC .L = Address.L. If !BoundaryCrossed, end (next half-cycle is 4.2).	
3.2	Read PC	Discard. Poll interrupts.	
4.1		PC .H = Address.H.	
4.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	70	1	BVS	0006	7f	fe	cc	fd	Nv-BdIzc	LDX #	T1	fe	fe	30
7	0006	70	1	BVS	0006	7f	fe	cc	fd	Nv-BdIzc	LDX #	T1	fe	fe	fc
8	0007	ff	1		0007	7f	fe	cc	fd	Nv-BdIzc	BVS	T2	fc	fc	fc
8	0007	ff	1		0007	7f	fe	cc	fd	Nv-BdIzc	BVS	T2	fc	fc	f8
9	0008	01	1	ORA (zp,X)	0008	7f	fe	cc	fd	Nv-BdIzc	BVS		ff	08	f8
9	0008	01	1	ORA (zp,X)	0008	7f	fe	cc	fd	Nv-BdIzc	BVS		ff	08	07
10	0009	ea	1		0009	7f	fe	cc	fd	Nv-BdIzc	ORA (zp,X)	T2	07	07	07
10	0009	ea	1		0009	7f	fe	cc	fd	Nv-BdIzc	ORA (zp,X)	T2	07	07	0e

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	70	1	BVS	0006	7f	00	cc	fd	Nv-BdIzc	ADC #	T1	7f	01	30
7	0006	70	1	BVS	0006	7f	00	cc	fd	Nv-BdIzc	ADC #	T1	7f	01	80
8	0007	44	1		0007	80	00	cc	fd	Nv-BdIzc	BVS	T2	80	80	80
8	0007	44	1		0007	80	00	cc	fd	Nv-BdIzc	BVS	T2	80	80	00
9	0008	ea	1		0008	80	00	cc	fd	Nv-BdIzc	BVS	T3	44	08	00
9	0008	ea	1		0008	80	00	cc	fd	Nv-BdIzc	BVS	T3	44	08	4c
10	004c	ea	1	NOP	004c	80	00	cc	fd	Nv-BdIzc	BVS		00	00	4c
10	004c	ea	1	NOP	004c	80	00	cc	fd	Nv-BdIzc	BVS		00	00	01
11	004d	ea	1		004d	80	00	cc	fd	Nv-BdIzc	NOP	T0+T2	01	01	01
11	004d	ea	1		004d	80	00	cc	fd	Nv-BdIzc	NOP	T0+T2	01	01	02

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	70	1	BVS	0006	7f	00	cc	fd	Nv-BdIzc	ADC #	T1	7f	01	30
7	0006	70	1	BVS	0006	7f	00	cc	fd	Nv-BdIzc	ADC #	T1	7f	01	80
8	0007	e0	1		0007	80	00	cc	fd	Nv-BdIzc	BVS	T2	80	80	80
8	0007	e0	1		0007	80	00	cc	fd	Nv-BdIzc	BVS	T2	80	80	00
9	0008	ea	1		0008	80	00	cc	fd	Nv-BdIzc	BVS	T3	e0	08	00
9	0008	ea	1		0008	80	00	cc	fd	Nv-BdIzc	BVS	T3	e0	08	e8
10	00e8	00	1		00e8	80	00	cc	fd	Nv-BdIzc	BVS	T0	00	ff	e8
10	00e8	00	1		00e8	80	00	cc	fd	Nv-BdIzc	BVS	T0	00	ff	ff
11	ffe8	00	1	BRK	ffe8	80	00	cc	fd	Nv-BdIzc	BVS	T1	ff	00	ff
11	ffe8	00	1	BRK	ffe8	80	00	cc	fd	Nv-BdIzc	BVS	T1	ff	00	ff
12	ffe9	00	1		ffe9	80	00	cc	fd	Nv-BdIzc	BRK	T2	ff	ff	ff
12	ffe9	00	1		ffe9	80	00	cc	fd	Nv-BdIzc	BRK	T2	ff	ff	fe

ADC (71)		Cycles: 5-6	Size: 2
Indirect Y (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Address.	
3.1			
3.2	Read u8(Pointer + 1)	Store as Address.H.	
4.1		Target = Address + Y. Pointer.L = Target.L. Pointer.H = Address.H. BoundaryCrossed = Pointer.H != Target.H.	
4.2	Read Pointer	Store as Operand. If !BoundaryCrossed, skip the next cycle.	
5.1		Pointer.H = Target.H.	
5.2	Read Pointer	Store as Operand.	
6.1		Result16 = u16(A) + u16(Operand) + C flag. V flag = (~(u16(A) ^ u16(Operand)) & (u16(A) ^ Result16) & 0x0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (Result16 & \$80), Z flag = (A == \$00).	
6.2	Read PC	Store as OpCode.	
+X.1		* A, C flag, N flag, V flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	71	1	ADC (zp),Y	0006	7f	01	cc	fd	nv-BdIzc	LDX #	T1	01	01	30
7	0006	71	1	ADC (zp),Y	0006	7f	01	cc	fd	nv-BdIzc	LDX #	T1	01	01	02
8	0007	22	1		0007	7f	01	cc	fd	nv-BdIzc	ADC (zp),Y	T2	02	02	02
8	0007	22	1		0007	7f	01	cc	fd	nv-BdIzc	ADC (zp),Y	T2	02	02	04
9	0022	22	1		0008	7f	01	cc	fd	nv-BdIzc	ADC (zp),Y	T3	00	22	04
9	0022	22	1		0008	7f	01	cc	fd	nv-BdIzc	ADC (zp),Y	T3	00	22	23
10	0023	01	1		0008	7f	01	cc	fd	nv-BdIzc	ADC (zp),Y	T4	cc	22	23
10	0023	01	1		0008	7f	01	cc	fd	nv-BdIzc	ADC (zp),Y	T4	cc	22	ee
11	01ee	01	1		0008	7f	01	cc	fd	nv-BdIzc	ADC (zp),Y	T0	00	01	ee
11	01ee	01	1		0008	7f	01	cc	fd	nv-BdIzc	ADC (zp),Y	T0	00	01	02
12	0008	ea	1	NOP	0008	7f	01	cc	fd	nv-BdIzc	ADC (zp),Y	T1	7f	01	02
12	0008	ea	1	NOP	0008	7f	01	cc	fd	nv-BdIzc	ADC (zp),Y	T1	7f	01	80
13	0009	ea	1		0009	80	01	cc	fd	NV-BdIzc	NOP	T0+T2	80	80	80
13	0009	ea	1		0009	80	01	cc	fd	NV-BdIzc	NOP	T0+T2	80	80	00

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	71	1	ADC (zp),Y	0006	7f	fe	ff	fd	Nv-BdIzc	LDX #	T1	fe	fe	fc
7	0006	71	1	ADC (zp),Y	0006	7f	fe	ff	fd	Nv-BdIzc	LDX #	T1	fe	fe	fc
8	0007	22	1		0007	7f	fe	ff	fd	Nv-BdIzc	ADC (zp),Y	T2	fc	fc	fc
8	0007	22	1		0007	7f	fe	ff	fd	Nv-BdIzc	ADC (zp),Y	T2	fc	fc	f8
9	0022	22	1		0008	7f	fe	ff	fd	Nv-BdIzc	ADC (zp),Y	T3	00	22	f8
9	0022	22	1		0008	7f	fe	ff	fd	Nv-BdIzc	ADC (zp),Y	T3	00	22	23
10	0023	dd	1		0008	7f	fe	ff	fd	Nv-BdIzc	ADC (zp),Y	T4	ff	22	23
10	0023	dd	1		0008	7f	fe	ff	fd	Nv-BdIzc	ADC (zp),Y	T4	ff	22	21
11	dd21	00	1		0008	7f	fe	ff	fd	Nv-BdIzc	ADC (zp),Y	T5	00	dd	21
11	dd21	00	1		0008	7f	fe	ff	fd	Nv-BdIzc	ADC (zp),Y	T5	00	dd	de
12	de21	00	1		0008	7f	fe	ff	fd	Nv-BdIzc	ADC (zp),Y	T0	de	ff	de
12	de21	00	1		0008	7f	fe	ff	fd	Nv-BdIzc	ADC (zp),Y	T0	de	ff	dd
13	0008	ea	1	NOP	0008	7f	fe	ff	fd	Nv-BdIzc	ADC (zp),Y	T1	7f	00	dd
13	0008	ea	1	NOP	0008	7f	fe	ff	fd	Nv-BdIzc	ADC (zp),Y	T1	7f	00	7f
14	0009	ea	1		0009	7f	fe	ff	fd	nv-BdIzc	NOP	T0+T2	7f	7f	7f
14	0009	ea	1		0009	7f	fe	ff	fd	nv-BdIzc	NOP	T0+T2	7f	7f	fe

JAM (72)	Cycles: ∞	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Discard.
2.1		Visual6502 suggests incrementing PC , but Single-Step Tests cannot be passed this way.
2.2	Read \$FFFF	Discard.
3.1		
3.2	Read \$FFFE	Discard.
4.1		
4.2	Read \$FFFE	Discard.
5.1		Repeat 5.1 and 5.2 forever.
5.2	Read \$FFFF	Discard.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	72	1	unknown	0006	7f	fe	ff	fd	Nv-BdIzc	LDX #	T1	fe	fe	fc
7	0006	72	1	unknown	0006	7f	fe	ff	fd	Nv-BdIzc	LDX #	T1	fe	fe	fc
8	0007	22	1		0007	7f	fe	ff	fd	Nv-BdIzc	unknown	T2	fc	fc	fc
8	0007	22	1		0007	7f	fe	ff	fd	Nv-BdIzc	unknown	T2	fc	fc	f8
9	ffff	00	1		0008	7f	fe	ff	fd	Nv-BdIzc	unknown	T3	ff	ff	f8
9	ffff	00	1		0008	7f	fe	ff	fd	Nv-BdIzc	unknown	T3	ff	ff	fe
10	fffe	00	1		0008	7f	fe	ff	fd	Nv-BdIzc	unknown	T4	ff	ff	fe
10	fffe	00	1		0008	7f	fe	ff	fd	Nv-BdIzc	unknown	T4	ff	ff	fe
11	fffe	00	1		0008	7f	fe	ff	fd	Nv-BdIzc	unknown	T5	ff	ff	fe
11	fffe	00	1		0008	7f	fe	ff	fd	Nv-BdIzc	unknown	T5	ff	ff	fe
12	ffff	00	1		0008	7f	fe	ff	fd	Nv-BdIzc	unknown		ff	ff	fe
12	ffff	00	1		0008	7f	fe	ff	fd	Nv-BdIzc	unknown		ff	ff	fe
13	ffff	00	1		0008	7f	fe	ff	fd	Nv-BdIzc	unknown		ff	ff	fe
13	ffff	00	1		0008	7f	fe	ff	fd	Nv-BdIzc	unknown		ff	ff	fe
14	ffff	00	1		0008	7f	fe	ff	fd	Nv-BdIzc	unknown		ff	ff	fe
14	ffff	00	1		0008	7f	fe	ff	fd	Nv-BdIzc	unknown		ff	ff	fe

RRA (73)		Cycles: 8	Size: 2
Indirect Y (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Address.	
3.1			
3.2	Read u8(Pointer + 1)	Store as Address.H.	
4.1		Target = Address + Y. Pointer.L = Target.L. Pointer.H = Address.H.	
4.2	Read Pointer	Store as Operand.	
5.1		Pointer.H = Target.H.	
5.2	Read Pointer	Store as Operand.	
6.1			
6.2	Write to Pointer	Write Operand.	
7.1		HiBit = (C flag) << 7. C flag = (Operand & \$01). Operand = (Operand >> 1) HiBit. Result16 = u16(A) + u16(Operand) + C flag. V flag = ~(u16(A) ^ u16(Operand)) & (u16(A) ^ Result16) & \$0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (A & \$80), Z flag = ! A .	
7.2	Write to Pointer	Write Operand.	
8.1			
8.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	73	1	unknown	0006	7f	fe	f0	fd	Nv-BdIzc	LDX #	T1	fe	fe	c0
7	0006	73	1	unknown	0006	7f	fe	f0	fd	Nv-BdIzc	LDX #	T1	fe	fe	fc
8	0007	22	1		0007	7f	fe	f0	fd	Nv-BdIzc	unknown	T2	fc	fc	fc
8	0007	22	1		0007	7f	fe	f0	fd	Nv-BdIzc	unknown	T2	fc	fc	f8
9	0022	22	1		0008	7f	fe	f0	fd	Nv-BdIzc	unknown	T3	00	22	f8
9	0022	22	1		0008	7f	fe	f0	fd	Nv-BdIzc	unknown	T3	00	22	23
10	0023	01	1		0008	7f	fe	f0	fd	Nv-BdIzc	unknown	T4	f0	22	23
10	0023	01	1		0008	7f	fe	f0	fd	Nv-BdIzc	unknown	T4	f0	22	12
11	0112	00	1		0008	7f	fe	f0	fd	Nv-BdIzc	unknown	T5	00	01	12
11	0112	00	1		0008	7f	fe	f0	fd	Nv-BdIzc	unknown	T5	00	01	02
12	0212	00	1		0008	7f	fe	f0	fd	Nv-BdIzc	unknown		02	ff	02
12	0212	00	1		0008	7f	fe	f0	fd	Nv-BdIzc	unknown		02	ff	01
13	0212	00	0		0008	7f	fe	f0	fd	Nv-BdIzc	unknown		ff	00	01
13	0212	00	0		0008	7f	fe	f0	fd	Nv-BdIzc	unknown		ff	00	00
14	0212	00	0		0008	7f	fe	f0	fd	nv-BdIzc	unknown	T0	00	00	00
14	0212	00	0		0008	7f	fe	f0	fd	nv-BdIzc	unknown	T0	00	00	00
15	0008	ea	1	NOP	0008	7f	fe	f0	fd	nv-BdIzc	unknown	T1	7f	00	00
15	0008	ea	1	NOP	0008	7f	fe	f0	fd	nv-BdIzc	unknown	T1	7f	00	7f
16	0009	ea	1		0009	7f	fe	f0	fd	nv-BdIzc	NOP	T0+T2	7f	7f	7f
16	0009	ea	1		0009	7f	fe	f0	fd	nv-BdIzc	NOP	T0+T2	7f	7f	fe

NOP (74)		Cycles: 4	Size: 2
Zero Page X-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Operand.	
3.1		Address = u8(Pointer + X).	
3.2	Read Address	Store as Operand.	
4.1			
4.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	74	1	unknown	0006	7f	fe	f0	fd	Nv-BdIzc	LDX #	T1	fe	fe	c0
7	0006	74	1	unknown	0006	7f	fe	f0	fd	Nv-BdIzc	LDX #	T1	fe	fe	fc
8	0007	22	1		0007	7f	fe	f0	fd	Nv-BdIzc	unknown	T2	fc	fc	fc
8	0007	22	1		0007	7f	fe	f0	fd	Nv-BdIzc	unknown	T2	fc	fc	f8
9	0022	22	1		0008	7f	fe	f0	fd	Nv-BdIzc	unknown	T3	fe	22	f8
9	0022	22	1		0008	7f	fe	f0	fd	Nv-BdIzc	unknown	T3	fe	22	20
10	0020	23	1		0008	7f	fe	f0	fd	Nv-BdIzc	unknown	T0	ff	ff	20
10	0020	23	1		0008	7f	fe	f0	fd	Nv-BdIzc	unknown	T0	ff	ff	fe
11	0008	ea	1	NOP	0008	7f	fe	f0	fd	Nv-BdIzc	unknown	T1	ff	23	fe
11	0008	ea	1	NOP	0008	7f	fe	f0	fd	Nv-BdIzc	unknown	T1	ff	23	22
12	0009	ea	1		0009	7f	fe	f0	fd	Nv-BdIzc	NOP	T0+T2	22	22	22
12	0009	ea	1		0009	7f	fe	f0	fd	Nv-BdIzc	NOP	T0+T2	22	22	44

ADC (75)		Cycles: 4	Size: 2
Zero Page X-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Operand.	
3.1		Address = u8(Pointer + X).	
3.2	Read Address	Store as Operand.	
4.1		Result16 = u16(A) + u16(Operand) + C flag. V flag = (~(u16(A) ^ u16(Operand)) & (u16(A) ^ Result16) & 0x0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (Result16 & \$80), Z flag = (A == \$00).	
4.2	Read PC	Store as OpCode.	
+X.1		* A , C flag, N flag, V flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	75	1	ADC zp,X	0006	7f	3f	f0	fd	nv-BdIzc	LDX #	T1	3f	3f	c0
7	0006	75	1	ADC zp,X	0006	7f	3f	f0	fd	nv-BdIzc	LDX #	T1	3f	3f	7e
8	0007	22	1		0007	7f	3f	f0	fd	nv-BdIzc	ADC zp,X	T2	7e	7e	7e
8	0007	22	1		0007	7f	3f	f0	fd	nv-BdIzc	ADC zp,X	T2	7e	7e	fc
9	0022	24	1		0008	7f	3f	f0	fd	nv-BdIzc	ADC zp,X	T3	3f	22	fc
9	0022	24	1		0008	7f	3f	f0	fd	nv-BdIzc	ADC zp,X	T3	3f	22	61
10	0061	01	1		0008	7f	3f	f0	fd	nv-BdIzc	ADC zp,X	T0	ff	ff	61
10	0061	01	1		0008	7f	3f	f0	fd	nv-BdIzc	ADC zp,X	T0	ff	ff	fe
11	0008	ea	1	NOP	0008	7f	3f	f0	fd	nv-BdIzc	ADC zp,X	T1	7f	01	fe
11	0008	ea	1	NOP	0008	7f	3f	f0	fd	nv-BdIzc	ADC zp,X	T1	7f	01	80
12	0009	ea	1		0009	80	3f	f0	fd	NV-BdIzc	NOP	T0+T2	80	80	80
12	0009	ea	1		0009	80	3f	f0	fd	NV-BdIzc	NOP	T0+T2	80	80	00

ROR (76) Cycles: 6 Size: 2		
Zero Page X-Indexed (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read Pointer	Store as Operand.
3.1		Address = u8(Pointer + X).
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.
5.1		HiBit = (C flag) << 7. C flag = (Operand & \$01). Operand = (Operand >> 1) HiBit. N flag = (Operand & \$80), Z flag = !Operand.
5.2	Write to Address	Write Operand.
6.1		
6.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	76	1	ROR zp,X	0006	7f	3f	f0	fd	nv-BdIzc	LDX #	T1	3f	3f	c0
7	0006	76	1	ROR zp,X	0006	7f	3f	f0	fd	nv-BdIzc	LDX #	T1	3f	3f	7e
8	0007	22	1		0007	7f	3f	f0	fd	nv-BdIzc	ROR zp,X	T2	7e	7e	7e
8	0007	22	1		0007	7f	3f	f0	fd	nv-BdIzc	ROR zp,X	T2	7e	7e	fc
9	0022	24	1		0008	7f	3f	f0	fd	nv-BdIzc	ROR zp,X	T3	3f	22	fc
9	0022	24	1		0008	7f	3f	f0	fd	nv-BdIzc	ROR zp,X	T3	3f	22	61
10	0061	01	1		0008	7f	3f	f0	fd	nv-BdIzc	ROR zp,X	T4	ff	ff	61
10	0061	01	1		0008	7f	3f	f0	fd	nv-BdIzc	ROR zp,X	T4	ff	ff	fe
11	0061	01	0		0008	7f	3f	f0	fd	nv-BdIzc	ROR zp,X	T5	ff	01	fe
11	0061	01	0		0008	7f	3f	f0	fd	nv-BdIzc	ROR zp,X	T5	ff	01	00
12	0061	01	0		0008	7f	3f	f0	fd	nv-BdIzc	ROR zp,X	T0	00	00	00
12	0061	00	0		0008	7f	3f	f0	fd	nv-BdIzc	ROR zp,X	T0	00	00	00
13	0008	ea	1	NOP	0008	7f	3f	f0	fd	nv-BdIzc	ROR zp,X	T1	ff	00	00
13	0008	ea	1	NOP	0008	7f	3f	f0	fd	nv-BdIzc	ROR zp,X	T1	ff	00	ff
14	0009	ea	1		0009	7f	3f	f0	fd	nv-BdIzc	NOP	T0+T2	ff	ff	ff
14	0009	ea	1		0009	7f	3f	f0	fd	nv-BdIzc	NOP	T0+T2	ff	ff	fe

RRA (77)		Cycles: 6	Size: 2
Zero Page X-Indexed (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Operand.	
3.1		Address = u8(Pointer + X).	
3.2	Read Address	Store as Operand.	
4.1			
4.2	Write to Address	Write Operand.	
5.1		HiBit = (C flag) << 7. C flag = (Operand & \$01). Operand = (Operand >> 1) HiBit. Result16 = u16(A) + u16(Operand) + C flag. V flag = (~(u16(A) ^ u16(Operand)) & (u16(A) ^ Result16) & \$0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (A & \$80), Z flag = ! A .	
5.2	Write to Address	Write Operand.	
6.1			
6.2	Read PC	Store as OpCode.	
+X.1		* A , C flag, N flag, V flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	77	1	unknown	0006	7f	3f	f0	fd	nv-BdIzc	LDX #	T1	3f	3f	c0
7	0006	77	1	unknown	0006	7f	3f	f0	fd	nv-BdIzc	LDX #	T1	3f	3f	7e
8	0007	22	1		0007	7f	3f	f0	fd	nv-BdIzc	unknown	T2	7e	7e	7e
8	0007	22	1		0007	7f	3f	f0	fd	nv-BdIzc	unknown	T2	7e	7e	fc
9	0022	24	1		0008	7f	3f	f0	fd	nv-BdIzc	unknown	T3	3f	22	fc
9	0022	24	1		0008	7f	3f	f0	fd	nv-BdIzc	unknown	T3	3f	22	61
10	0061	01	1		0008	7f	3f	f0	fd	nv-BdIzc	unknown	T4	ff	ff	61
10	0061	01	1		0008	7f	3f	f0	fd	nv-BdIzc	unknown	T4	ff	ff	fe
11	0061	01	0		0008	7f	3f	f0	fd	nv-BdIzc	unknown	T5	ff	01	fe
11	0061	01	0		0008	7f	3f	f0	fd	nv-BdIzc	unknown	T5	ff	01	00
12	0061	01	0		0008	7f	3f	f0	fd	nv-BdIzc	unknown	T0	00	00	00
12	0061	00	0		0008	7f	3f	f0	fd	nv-BdIzc	unknown	T0	00	00	00
13	0008	ea	1	NOP	0008	7f	3f	f0	fd	nv-BdIzc	unknown	T1	7f	00	00
13	0008	ea	1	NOP	0008	7f	3f	f0	fd	nv-BdIzc	unknown	T1	7f	00	80
14	0009	ea	1		0009	80	3f	f0	fd	NV-BdIzc	NOP	T0+T2	80	80	80
14	0009	ea	1		0009	80	3f	f0	fd	NV-BdIzc	NOP	T0+T2	80	80	00

SEI (78)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Discard.
2.1		I flag = 1.
2.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	78	1	SEI	0006	7f	3f	f0	fd	nv-BdIzc	LDX #	T1	3f	3f	c0
7	0006	78	1	SEI	0006	7f	3f	f0	fd	nv-BdIzc	LDX #	T1	3f	3f	7e
8	0007	ea	1		0007	7f	3f	f0	fd	nv-BdIzc	SEI	T0+T2	7e	7e	7e
8	0007	ea	1		0007	7f	3f	f0	fd	nv-BdIzc	SEI	T0+T2	7e	7e	fc
9	0007	ea	1	NOP	0007	7f	3f	f0	fd	nv-BdIzc	SEI	T1	ff	ff	fc
9	0007	ea	1	NOP	0007	7f	3f	f0	fd	nv-BdIzc	SEI	T1	ff	ff	fe
10	0008	ea	1		0008	7f	3f	f0	fd	nv-BdIzc	NOP	T0+T2	fe	fe	fe
10	0008	ea	1		0008	7f	3f	f0	fd	nv-BdIzc	NOP	T0+T2	fe	fe	fc

ADC (79)		Cycles: 4-5	Size: 3
Absolute Y-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read PC	Store as Pointer.H.	
3.1		Inc. PC . Target = Pointer + Y.	
		Address.L = Target.L.	
		Address.H = Pointer.H.	
		BoundaryCrossed = Pointer.H != Target.H.	
3.2	Read Address	Store as Operand. If !BoundaryCrossed, skip the next cycle.	
4.1		Address.H = Target.H.	
4.2	Read Address	Store as Operand.	
5.1		Result16 = u16(A) + u16(Operand) + C flag.	
		V flag = $(\sim(u16(A) \wedge u16(Operand)) \& (u16(A) \wedge Result16) \& 0x0080) \neq 0$.	
		A = u8(Result16).	
		C flag = (Result16 > \$FF), N flag = (Result16 & \$80), Z flag = (A == \$00).	
5.2	Read PC	Store as OpCode.	
+X.1		* A, C flag, N flag, V flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	79	1	ADC Abs,Y	0006	7f	3f	05	fd	nv-BdIzc	LDX #	T1	3f	3f	14
7	0006	79	1	ADC Abs,Y	0006	7f	3f	05	fd	nv-BdIzc	LDX #	T1	3f	3f	7e
8	0007	ea	1		0007	7f	3f	05	fd	nv-BdIzc	ADC Abs,Y	T2	7e	7e	7e
8	0007	ea	1		0007	7f	3f	05	fd	nv-BdIzc	ADC Abs,Y	T2	7e	7e	fc
9	0008	01	1		0008	7f	3f	05	fd	nv-BdIzc	ADC Abs,Y	T3	05	ea	fc
9	0008	01	1		0008	7f	3f	05	fd	nv-BdIzc	ADC Abs,Y	T3	05	ea	ef
10	01ef	01	1		0009	7f	3f	05	fd	nv-BdIzc	ADC Abs,Y	T0	00	01	ef
10	01ef	01	1		0009	7f	3f	05	fd	nv-BdIzc	ADC Abs,Y	T0	00	01	02
11	0009	ea	1	NOP	0009	7f	3f	05	fd	nv-BdIzc	ADC Abs,Y	T1	7f	01	02
11	0009	ea	1	NOP	0009	7f	3f	05	fd	nv-BdIzc	ADC Abs,Y	T1	7f	01	80
12	000a	15	1		000a	80	3f	05	fd	NV-BdIzc	NOP	T0+T2	80	80	80
12	000a	15	1		000a	80	3f	05	fd	NV-BdIzc	NOP	T0+T2	80	80	00

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	79	1	ADC Abs,Y	0006	7f	80	25	fd	Nv-BdIzc	LDX #	T1	80	80	94
7	0006	79	1	ADC Abs,Y	0006	7f	80	25	fd	Nv-BdIzc	LDX #	T1	80	80	00
8	0007	ea	1		0007	7f	80	25	fd	Nv-BdIzc	ADC Abs,Y	T2	00	00	00
8	0007	ea	1		0007	7f	80	25	fd	Nv-BdIzc	ADC Abs,Y	T2	00	00	00
9	0008	01	1		0008	7f	80	25	fd	Nv-BdIzc	ADC Abs,Y	T3	25	ea	00
9	0008	01	1		0008	7f	80	25	fd	Nv-BdIzc	ADC Abs,Y	T3	25	ea	0f
10	010f	66	1		0009	7f	80	25	fd	Nv-BdIzc	ADC Abs,Y	T4	00	01	0f
10	010f	66	1		0009	7f	80	25	fd	Nv-BdIzc	ADC Abs,Y	T4	00	01	02
11	020f	00	1		0009	7f	80	25	fd	Nv-BdIzc	ADC Abs,Y	T0	02	ff	02
11	020f	00	1		0009	7f	80	25	fd	Nv-BdIzc	ADC Abs,Y	T0	02	ff	01
12	0009	ea	1	NOP	0009	7f	80	25	fd	Nv-BdIzc	ADC Abs,Y	T1	7f	00	01
12	0009	ea	1	NOP	0009	7f	80	25	fd	Nv-BdIzc	ADC Abs,Y	T1	7f	00	7f
13	000a	15	1		000a	7f	80	25	fd	nv-BdIzc	NOP	T0+T2	7f	7f	7f
13	000a	15	1		000a	7f	80	25	fd	nv-BdIzc	NOP	T0+T2	7f	7f	fe

NOP (7A)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Discard.	
2.1			
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	7a	1	unknown	0006	7f	80	25	fd	Nv-BdIzc	LDX #	T1	80	80	94
7	0006	7a	1	unknown	0006	7f	80	25	fd	Nv-BdIzc	LDX #	T1	80	80	00
8	0007	ea	1		0007	7f	80	25	fd	Nv-BdIzc	unknown	T0+T2	00	00	00
8	0007	ea	1		0007	7f	80	25	fd	Nv-BdIzc	unknown	T0+T2	00	00	00
9	0007	ea	1	NOP	0007	7f	80	25	fd	Nv-BdIzc	unknown	T1	ff	ff	00
9	0007	ea	1	NOP	0007	7f	80	25	fd	Nv-BdIzc	unknown	T1	ff	ff	fe
10	0008	01	1		0008	7f	80	25	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fe
10	0008	01	1		0008	7f	80	25	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fc

RRA (7B) Cycles: 7		Size: 3
Absolute Y-Indexed (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + Y. Address.L = Target.L. Address.H = Pointer.H.
3.2	Read Address	Store as Operand.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		
5.2	Write to Address	Write Operand.
6.1		HiBit = (C flag) << 7. C flag = (Operand & \$01). Operand = (Operand >> 1) HiBit. Result16 = u16(A) + u16(Operand) + C flag. V flag = (~(u16(A) ^ u16(Operand)) & (u16(A) ^ Result16) & \$0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (A & \$80), Z flag = ! A .
6.2	Write to Address	Write Operand.
7.1		
7.2	Read PC	Store as OpCode.
+X.1		* A , C flag, N flag, V flag, and Z flag applied.

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	7b	1	unknown	0006	7f	80	05	fd	Nv-BdIzc	LDX #	T1	80	80	14
7	0006	7b	1	unknown	0006	7f	80	05	fd	Nv-BdIzc	LDX #	T1	80	80	00
8	0007	ea	1		0007	7f	80	05	fd	Nv-BdIzc	unknown	T2	00	00	00
8	0007	ea	1		0007	7f	80	05	fd	Nv-BdIzc	unknown	T2	00	00	00
9	0008	01	1		0008	7f	80	05	fd	Nv-BdIzc	unknown	T3	05	ea	00
9	0008	01	1		0008	7f	80	05	fd	Nv-BdIzc	unknown	T3	05	ea	ef
10	01ef	01	1		0009	7f	80	05	fd	Nv-BdIzc	unknown	T4	00	01	ef
10	01ef	01	1		0009	7f	80	05	fd	Nv-BdIzc	unknown	T4	00	01	02
11	01ef	01	1		0009	7f	80	05	fd	Nv-BdIzc	unknown	T5	02	ff	02
11	01ef	01	1		0009	7f	80	05	fd	Nv-BdIzc	unknown	T5	02	ff	01
12	01ef	01	0		0009	7f	80	05	fd	Nv-BdIzc	unknown		ff	01	01
12	01ef	01	0		0009	7f	80	05	fd	Nv-BdIzc	unknown		ff	01	00
13	01ef	01	0		0009	7f	80	05	fd	nv-BdIZC	unknown	T0	00	00	00
13	01ef	00	0		0009	7f	80	05	fd	nv-BdIZC	unknown	T0	00	00	00
14	0009	ea	1	NOP	0009	7f	80	05	fd	nv-BdIZC	unknown	T1	7f	00	00
14	0009	ea	1	NOP	0009	7f	80	05	fd	nv-BdIZC	unknown	T1	7f	00	80
15	000a	15	1		000a	80	80	05	fd	NV-BdIzc	NOP	T0+T2	80	80	80
15	000a	15	1		000a	80	80	05	fd	NV-BdIzc	NOP	T0+T2	80	80	00

NOP (7C)		Cycles: 4-5	Size: 3
Absolute X-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read PC	Store as Pointer.H.	
3.1		Inc. PC . Target = Pointer + X .	
		Address.L = Target.L.	
		Address.H = Pointer.H.	
		BoundaryCrossed = Pointer.H != Target.H.	
3.2	Read Address	Store as Operand. If !BoundaryCrossed, skip the next cycle.	
4.1		Address.H = Target.H.	
4.2	Read Address	Store as Operand.	
5.1			
5.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	7c	1	unknown	0006	7f	80	05	fd	Nv-BdIzc	LDX #	T1	80	80	14
7	0006	7c	1	unknown	0006	7f	80	05	fd	Nv-BdIzc	LDX #	T1	80	80	00
8	0007	ea	1		0007	7f	80	05	fd	Nv-BdIzc	unknown	T2	00	00	00
8	0007	ea	1		0007	7f	80	05	fd	Nv-BdIzc	unknown	T2	00	00	00
9	0008	01	1		0008	7f	80	05	fd	Nv-BdIzc	unknown	T3	80	ea	00
9	0008	01	1		0008	7f	80	05	fd	Nv-BdIzc	unknown	T3	80	ea	6a
10	016a	00	1		0009	7f	80	05	fd	Nv-BdIzc	unknown	T4	00	01	6a
10	016a	00	1		0009	7f	80	05	fd	Nv-BdIzc	unknown	T4	00	01	02
11	026a	00	1		0009	7f	80	05	fd	Nv-BdIzc	unknown	T0	02	ff	02
11	026a	00	1		0009	7f	80	05	fd	Nv-BdIzc	unknown	T0	02	ff	01
12	0009	ea	1	NOP	0009	7f	80	05	fd	Nv-BdIzc	unknown	T1	ff	00	01
12	0009	ea	1	NOP	0009	7f	80	05	fd	Nv-BdIzc	unknown	T1	ff	00	ff
13	000a	15	1		000a	7f	80	05	fd	Nv-BdIzc	NOP	T0+T2	ff	ff	ff
13	000a	15	1		000a	7f	80	05	fd	Nv-BdIzc	NOP	T0+T2	ff	ff	fe

ADC (7D) Cycles: 4-5		Size: 3
Absolute X-Indexed (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + X . Address.L = Target.L. Address.H = Pointer.H. BoundaryCrossed = Pointer.H != Target.H.
3.2	Read Address	Store as Operand. If !BoundaryCrossed, skip the next cycle.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		Result16 = u16(A) + u16(Operand) + C flag. V flag = $\sim(u16(\mathbf{A}) \wedge u16(\text{Operand})) \wedge (u16(\mathbf{A}) \wedge \text{Result16}) \& 0x0080) \neq 0$. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (Result16 & \$80), Z flag = (A == \$00).
5.2	Read PC	Store as OpCode.
+X.1		* A , C flag, N flag, V flag, and Z flag applied.

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	7d	1	ADC Abs,X	0006	7f	04	05	fd	nv-BdIzc	LDX #	T1	04	04	14
7	0006	7d	1	ADC Abs,X	0006	7f	04	05	fd	nv-BdIzc	LDX #	T1	04	04	08
8	0007	ea	1		0007	7f	04	05	fd	nv-BdIzc	ADC Abs,X	T2	08	08	08
8	0007	ea	1		0007	7f	04	05	fd	nv-BdIzc	ADC Abs,X	T2	08	08	10
9	0008	01	1		0008	7f	04	05	fd	nv-BdIzc	ADC Abs,X	T3	04	ea	10
9	0008	01	1		0008	7f	04	05	fd	nv-BdIzc	ADC Abs,X	T3	04	ea	ee
10	01ee	01	1		0009	7f	04	05	fd	nv-BdIzc	ADC Abs,X	T0	00	01	ee
10	01ee	01	1		0009	7f	04	05	fd	nv-BdIzc	ADC Abs,X	T0	00	01	02
11	0009	ea	1	NOP	0009	7f	04	05	fd	nv-BdIzc	ADC Abs,X	T1	7f	01	02
11	0009	ea	1	NOP	0009	7f	04	05	fd	nv-BdIzc	ADC Abs,X	T1	7f	01	80
12	000a	15	1		000a	80	04	05	fd	NV-BdIzc	NOP	T0+T2	80	80	80
12	000a	15	1		000a	80	04	05	fd	NV-BdIzc	NOP	T0+T2	80	80	00

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	7d	1	ADC Abs,X	0006	7f	86	05	fd	Nv-BdIzc	LDX #	T1	86	86	14
7	0006	7d	1	ADC Abs,X	0006	7f	86	05	fd	Nv-BdIzc	LDX #	T1	86	86	0c
8	0007	ea	1		0007	7f	86	05	fd	Nv-BdIzc	ADC Abs,X	T2	0c	0c	0c
8	0007	ea	1		0007	7f	86	05	fd	Nv-BdIzc	ADC Abs,X	T2	0c	0c	18
9	0008	01	1		0008	7f	86	05	fd	Nv-BdIzc	ADC Abs,X	T3	86	ea	18
9	0008	01	1		0008	7f	86	05	fd	Nv-BdIzc	ADC Abs,X	T3	86	ea	70
10	0170	00	1		0009	7f	86	05	fd	Nv-BdIzc	ADC Abs,X	T4	00	01	70
10	0170	00	1		0009	7f	86	05	fd	Nv-BdIzc	ADC Abs,X	T4	00	01	02
11	0270	00	1		0009	7f	86	05	fd	Nv-BdIzc	ADC Abs,X	T0	02	ff	02
11	0270	00	1		0009	7f	86	05	fd	Nv-BdIzc	ADC Abs,X	T0	02	ff	01
12	0009	ea	1	NOP	0009	7f	86	05	fd	Nv-BdIzc	ADC Abs,X	T1	7f	00	01
12	0009	ea	1	NOP	0009	7f	86	05	fd	Nv-BdIzc	ADC Abs,X	T1	7f	00	7f
13	000a	15	1		000a	7f	86	05	fd	nv-BdIzc	NOP	T0+T2	7f	7f	7f
13	000a	15	1		000a	7f	86	05	fd	nv-BdIzc	NOP	T0+T2	7f	7f	fe

ROR (7E) Cycles: 7		Size: 3
Absolute X-Indexed (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + X . Address.L = Target.L. Address.H = Pointer.H.
3.2	Read Address	Store as Operand.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		
5.2	Write to Address	Write Operand.
6.1		HiBit = (C flag) << 7. C flag = (Operand & \$01). Operand = (Operand >> 1) HiBit. N flag = (Operand & \$80), Z flag = !Operand.
6.2	Write to Address	Write Operand.
7.1		
7.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	7e	1	ROR Abs,X	0006	7f	06	05	fd	nv-BdIzc	LDX #	T1	06	06	14
7	0006	7e	1	ROR Abs,X	0006	7f	06	05	fd	nv-BdIzc	LDX #	T1	06	06	0c
8	0007	ea	1		0007	7f	06	05	fd	nv-BdIzc	ROR Abs,X	T2	0c	0c	0c
8	0007	ea	1		0007	7f	06	05	fd	nv-BdIzc	ROR Abs,X	T2	0c	0c	18
9	0008	01	1		0008	7f	06	05	fd	nv-BdIzc	ROR Abs,X	T3	06	ea	18
9	0008	01	1		0008	7f	06	05	fd	nv-BdIzc	ROR Abs,X	T3	06	ea	f0
10	01f0	ff	1		0009	7f	06	05	fd	nv-BdIzc	ROR Abs,X	T4	00	01	f0
10	01f0	ff	1		0009	7f	06	05	fd	nv-BdIzc	ROR Abs,X	T4	00	01	02
11	01f0	ff	1		0009	7f	06	05	fd	nv-BdIzc	ROR Abs,X	T5	02	ff	02
11	01f0	ff	1		0009	7f	06	05	fd	nv-BdIzc	ROR Abs,X	T5	02	ff	01
12	01f0	ff	0		0009	7f	06	05	fd	nv-BdIzc	ROR Abs,X		ff	ff	01
12	01f0	ff	0		0009	7f	06	05	fd	nv-BdIzc	ROR Abs,X		ff	ff	7f
13	01f0	ff	0		0009	7f	06	05	fd	nv-BdIzc	ROR Abs,X	T0	7f	7f	7f
13	01f0	7f	0		0009	7f	06	05	fd	nv-BdIzc	ROR Abs,X	T0	7f	7f	fe
14	0009	ea	1	NOP	0009	7f	06	05	fd	nv-BdIzc	ROR Abs,X	T1	ff	7f	fe
14	0009	ea	1	NOP	0009	7f	06	05	fd	nv-BdIzc	ROR Abs,X	T1	ff	7f	7e
15	000a	15	1		000a	7f	06	05	fd	nv-BdIzc	NOP	T0+T2	7e	7e	7e
15	000a	15	1		000a	7f	06	05	fd	nv-BdIzc	NOP	T0+T2	7e	7e	fc

RRA (7F)		Cycles: 7	Size: 3
Absolute X-Indexed (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read PC	Store as Pointer.H.	
3.1		Inc. PC . Target = Pointer + X .	
		Address.L = Target.L.	
		Address.H = Pointer.H.	
3.2	Read Address	Store as Operand.	
4.1		Address.H = Target.H.	
4.2	Read Address	Store as Operand.	
5.1			
5.2	Write to Address	Write Operand.	
6.1		HiBit = (C flag) << 7. C flag = (Operand & \$01).	
		Operand = (Operand >> 1) HiBit.	
		Result16 = u16(A) + u16(Operand) + C flag.	
		V flag = (~(u16(A) ^ u16(Operand)) & (u16(A) ^ Result16) & \$0080) != 0.	
		A = u8(Result16).	
		C flag = (Result16 > \$FF), N flag = (A & \$80), Z flag = ! A .	
6.2	Write to Address	Write Operand.	
7.1			
7.2	Read PC	Store as OpCode.	
+X.1		* A , C flag, N flag, V flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	7f	1	unknown	0006	7f	06	05	fd	nv-BdIzc	LDX #	T1	06	06	14
7	0006	7f	1	unknown	0006	7f	06	05	fd	nv-BdIzc	LDX #	T1	06	06	0c
8	0007	ea	1		0007	7f	06	05	fd	nv-BdIzc	unknown	T2	0c	0c	0c
8	0007	ea	1		0007	7f	06	05	fd	nv-BdIzc	unknown	T2	0c	0c	18
9	0008	01	1		0008	7f	06	05	fd	nv-BdIzc	unknown	T3	06	ea	18
9	0008	01	1		0008	7f	06	05	fd	nv-BdIzc	unknown	T3	06	ea	f0
10	01f0	7f	1		0009	7f	06	05	fd	nv-BdIzc	unknown	T4	00	01	f0
10	01f0	7f	1		0009	7f	06	05	fd	nv-BdIzc	unknown	T4	00	01	02
11	01f0	7f	1		0009	7f	06	05	fd	nv-BdIzc	unknown	T5	02	ff	02
11	01f0	7f	1		0009	7f	06	05	fd	nv-BdIzc	unknown	T5	02	ff	01
12	01f0	7f	0		0009	7f	06	05	fd	nv-BdIzc	unknown		ff	7f	01
12	01f0	7f	0		0009	7f	06	05	fd	nv-BdIzc	unknown		ff	7f	3f
13	01f0	7f	0		0009	7f	06	05	fd	nv-BdIzc	unknown	T0	3f	3f	3f
13	01f0	3f	0		0009	7f	06	05	fd	nv-BdIzc	unknown	T0	3f	3f	7e
14	0009	ea	1	NOP	0009	7f	06	05	fd	nv-BdIzc	unknown	T1	7f	3f	7e
14	0009	ea	1	NOP	0009	7f	06	05	fd	nv-BdIzc	unknown	T1	7f	3f	bf
15	000a	15	1		000a	bf	06	05	fd	NV-BdIzc	NOP	T0+T2	bf	bf	bf
15	000a	15	1		000a	bf	06	05	fd	NV-BdIzc	NOP	T0+T2	bf	bf	7e

NOP (80)		Cycles: 2	Size: 2
Immediate			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Operand.	
2.1		Inc. PC .	
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	80	1	unknown	0006	7f	06	05	fd	nv-BdIzc	LDX #	T1	06	06	14
7	0006	80	1	unknown	0006	7f	06	05	fd	nv-BdIzc	LDX #	T1	06	06	0c
8	0007	01	1		0007	7f	06	05	fd	nv-BdIzc	unknown	T0+T2	0c	0c	0c
8	0007	01	1		0007	7f	06	05	fd	nv-BdIzc	unknown	T0+T2	0c	0c	18
9	0008	ea	1	NOP	0008	7f	06	05	fd	nv-BdIzc	unknown	T1	ff	01	18
9	0008	ea	1	NOP	0008	7f	06	05	fd	nv-BdIzc	unknown	T1	ff	01	00
10	0009	ea	1		0009	7f	06	05	fd	nv-BdIzc	NOP	T0+T2	00	00	00
10	0009	ea	1		0009	7f	06	05	fd	nv-BdIzc	NOP	T0+T2	00	00	00

STA (81)	Cycles: 6	Size: 2
Indirect X (Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC .
2.2	Read Operand	Discard.
3.1		Pointer = u8(Operand + X).
3.2	Read Pointer	Store as Address.
4.1		
4.2	Read u8(Pointer + 1)	Store as Address.H.
5.1		
5.2	Write to Address	Write A .
6.1		
6.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	81	1	STA (zp,X)	0006	7f	20	05	fd	nv-BdIzc	LDX #	T1	20	20	14
7	0006	81	1	STA (zp,X)	0006	7f	20	05	fd	nv-BdIzc	LDX #	T1	20	20	40
8	0007	ff	1		0007	7f	20	05	fd	nv-BdIzc	STA (zp,X)	T2	40	40	40
8	0007	ff	1		0007	7f	20	05	fd	nv-BdIzc	STA (zp,X)	T2	40	40	80
9	00ff	22	1		0008	7f	20	05	fd	nv-BdIzc	STA (zp,X)	T3	20	ff	80
9	00ff	22	1		0008	7f	20	05	fd	nv-BdIzc	STA (zp,X)	T3	20	ff	1f
10	001f	02	1		0008	7f	20	05	fd	nv-BdIzc	STA (zp,X)	T4	00	1f	1f
10	001f	02	1		0008	7f	20	05	fd	nv-BdIzc	STA (zp,X)	T4	00	1f	20
11	0020	01	1		0008	7f	20	05	fd	nv-BdIzc	STA (zp,X)	T5	00	02	20
11	0020	01	1		0008	7f	20	05	fd	nv-BdIzc	STA (zp,X)	T5	00	02	02
12	0102	01	0		0008	7f	20	05	fd	nv-BdIzc	STA (zp,X)	T0	ff	7f	02
12	0102	7f	0		0008	7f	20	05	fd	nv-BdIzc	STA (zp,X)	T0	ff	7f	7e
13	0008	ea	1	NOP	0008	7f	20	05	fd	nv-BdIzc	STA (zp,X)	T1	7f	7f	7e
13	0008	ea	1	NOP	0008	7f	20	05	fd	nv-BdIzc	STA (zp,X)	T1	7f	7f	fe
14	0009	ea	1		0009	7f	20	05	fd	nv-BdIzc	NOP	T0+T2	fe	fe	fe
14	0009	ea	1		0009	7f	20	05	fd	nv-BdIzc	NOP	T0+T2	fe	fe	fc

NOP (82)		Cycles: 2	Size: 2
Immediate			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Operand.	
2.1		Inc. PC .	
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	82	1	unknown	0006	7f	20	05	fd	nv-BdIzc	LDX #	T1	20	20	14
7	0006	82	1	unknown	0006	7f	20	05	fd	nv-BdIzc	LDX #	T1	20	20	40
8	0007	ff	1		0007	7f	20	05	fd	nv-BdIzc	unknown	T0+T2	40	40	40
8	0007	ff	1		0007	7f	20	05	fd	nv-BdIzc	unknown	T0+T2	40	40	80
9	0008	ea	1	NOP	0008	7f	20	05	fd	nv-BdIzc	unknown	T1	ff	ff	80
9	0008	ea	1	NOP	0008	7f	20	05	fd	nv-BdIzc	unknown	T1	ff	ff	fe
10	0009	ea	1		0009	7f	20	05	fd	nv-BdIzc	NOP	T0+T2	fe	fe	fe
10	0009	ea	1		0009	7f	20	05	fd	nv-BdIzc	NOP	T0+T2	fe	fe	fc

SAX (83)	Cycles: 6	Size: 2
Indirect X (Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC .
2.2	Read Operand	Discard.
3.1		Pointer = u8(Operand + X).
3.2	Read Pointer	Store as Address.
4.1		
4.2	Read u8(Pointer + 1)	Store as Address.H.
5.1		
5.2	Write to Address	Write (A & X).
6.1		
6.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	83	1	unknown	0006	7f	f0	05	fd	Nv-BdIzc	LDX #	T1	f0	f0	14
7	0006	83	1	unknown	0006	7f	f0	05	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ff	1		0007	7f	f0	05	fd	Nv-BdIzc	unknown	T2	e0	e0	e0
8	0007	ff	1		0007	7f	f0	05	fd	Nv-BdIzc	unknown	T2	e0	e0	c0
9	00ff	22	1		0008	7f	f0	05	fd	Nv-BdIzc	unknown	T3	f0	ff	c0
9	00ff	22	1		0008	7f	f0	05	fd	Nv-BdIzc	unknown	T3	f0	ff	ef
10	00ef	ff	1		0008	7f	f0	05	fd	Nv-BdIzc	unknown	T4	00	ef	ef
10	00ef	ff	1		0008	7f	f0	05	fd	Nv-BdIzc	unknown	T4	00	ef	f0
11	00f0	00	1		0008	7f	f0	05	fd	Nv-BdIzc	unknown	T5	00	ff	f0
11	00f0	00	1		0008	7f	f0	05	fd	Nv-BdIzc	unknown	T5	00	ff	ff
12	00ff	00	0		0008	7f	f0	05	fd	Nv-BdIzc	unknown	T0	70	70	ff
12	00ff	70	0		0008	7f	f0	05	fd	Nv-BdIzc	unknown	T0	70	70	e0
13	0008	ea	1	NOP	0008	7f	f0	05	fd	Nv-BdIzc	unknown	T1	7f	70	e0
13	0008	ea	1	NOP	0008	7f	f0	05	fd	Nv-BdIzc	unknown	T1	7f	70	ef
14	0009	ea	1		0009	7f	f0	05	fd	Nv-BdIzc	NOP	T0+T2	ef	ef	ef
14	0009	ea	1		0009	7f	f0	05	fd	Nv-BdIzc	NOP	T0+T2	ef	ef	de

STY (84)	Cycles: 3	Size: 2
Zero Page (Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Write to Address	Write Y .
3.1		
3.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	84	1	STY zp	0006	7f	f0	05	fd	Nv-BdIzc	LDX #	T1	f0	f0	14
7	0006	84	1	STY zp	0006	7f	f0	05	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ff	1		0007	7f	f0	05	fd	Nv-BdIzc	STY zp	T2	e0	e0	e0
8	0007	ff	1		0007	7f	f0	05	fd	Nv-BdIzc	STY zp	T2	e0	e0	c0
9	00ff	ff	0		0008	7f	f0	05	fd	Nv-BdIzc	STY zp	T0	05	ff	c0
9	00ff	05	0		0008	7f	f0	05	fd	Nv-BdIzc	STY zp	T0	05	ff	04
10	0008	ea	1	NOP	0008	7f	f0	05	fd	Nv-BdIzc	STY zp	T1	ff	05	04
10	0008	ea	1	NOP	0008	7f	f0	05	fd	Nv-BdIzc	STY zp	T1	ff	05	04
11	0009	ea	1		0009	7f	f0	05	fd	Nv-BdIzc	NOP	T0+T2	04	04	04
11	0009	ea	1		0009	7f	f0	05	fd	Nv-BdIzc	NOP	T0+T2	04	04	08

STA (85)		Cycles: 3	Size: 2
Zero Page (Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1	Read PC	Inc. PC .	
1.2		Store as Address.	
2.1		Inc. PC .	
2.2		Write A .	
3.1			
3.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	85	1	STA zp	0006	7f	f0	05	fd	Nv-BdIzc	LDX #	T1	f0	f0	14
7	0006	85	1	STA zp	0006	7f	f0	05	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ff	1		0007	7f	f0	05	fd	Nv-BdIzc	STA zp	T2	e0	e0	e0
8	0007	ff	1		0007	7f	f0	05	fd	Nv-BdIzc	STA zp	T2	e0	e0	c0
9	00ff	ff	0		0008	7f	f0	05	fd	Nv-BdIzc	STA zp	T0	ff	ff	c0
9	00ff	7f	0		0008	7f	f0	05	fd	Nv-BdIzc	STA zp	T0	ff	ff	fe
10	0008	ea	1	NOP	0008	7f	f0	05	fd	Nv-BdIzc	STA zp	T1	7f	7f	fe
10	0008	ea	1	NOP	0008	7f	f0	05	fd	Nv-BdIzc	STA zp	T1	7f	7f	fe
11	0009	ea	1		0009	7f	f0	05	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fe
11	0009	ea	1		0009	7f	f0	05	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fc

STX (86)	Cycles: 3	Size: 2
Zero Page (Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Write to Address	Write X .
3.1		
3.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	86	1	STX zp	0006	7f	f0	05	fd	Nv-BdIzc	LDX #	T1	f0	f0	14
7	0006	86	1	STX zp	0006	7f	f0	05	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ff	1		0007	7f	f0	05	fd	Nv-BdIzc	STX zp	T2	e0	e0	e0
8	0007	ff	1		0007	7f	f0	05	fd	Nv-BdIzc	STX zp	T2	e0	e0	c0
9	00ff	ff	0		0008	7f	f0	05	fd	Nv-BdIzc	STX zp	T0	f0	ff	c0
9	00ff	f0	0		0008	7f	f0	05	fd	Nv-BdIzc	STX zp	T0	f0	ff	ef
10	0008	ea	1	NOP	0008	7f	f0	05	fd	Nv-BdIzc	STX zp	T1	ff	f0	ef
10	0008	ea	1	NOP	0008	7f	f0	05	fd	Nv-BdIzc	STX zp	T1	ff	f0	ef
11	0009	ea	1		0009	7f	f0	05	fd	Nv-BdIzc	NOP	T0+T2	ef	ef	ef
11	0009	ea	1		0009	7f	f0	05	fd	Nv-BdIzc	NOP	T0+T2	ef	ef	de

SAX (87)	Cycles: 3	Size: 2
Zero Page (Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Write to Address	Write (A & X).
3.1		
3.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	87	1	unknown	0006	7f	f0	05	fd	Nv-BdIzc	LDX #	T1	f0	f0	14
7	0006	87	1	unknown	0006	7f	f0	05	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ff	1		0007	7f	f0	05	fd	Nv-BdIzc	unknown	T2	e0	e0	e0
8	0007	ff	1		0007	7f	f0	05	fd	Nv-BdIzc	unknown	T2	e0	e0	c0
9	00ff	ff	0		0008	7f	f0	05	fd	Nv-BdIzc	unknown	T0	70	ff	c0
9	00ff	70	0		0008	7f	f0	05	fd	Nv-BdIzc	unknown	T0	70	ff	6f
10	0008	ea	1	NOP	0008	7f	f0	05	fd	Nv-BdIzc	unknown	T1	7f	70	6f
10	0008	ea	1	NOP	0008	7f	f0	05	fd	Nv-BdIzc	unknown	T1	7f	70	ef
11	0009	ea	1		0009	7f	f0	05	fd	Nv-BdIzc	NOP	T0+T2	ef	ef	ef
11	0009	ea	1		0009	7f	f0	05	fd	Nv-BdIzc	NOP	T0+T2	ef	ef	de

DEY (88)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Discard.	
2.1		Y -= 1. N flag = (Y & \$80), Z flag = ! Y .	
2.2	Read PC	Store as OpCode.	
+X.1		* Y , N flag, and Z flag applied.	

* Setting of Y/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	88	1	DEY	0006	7f	f0	01	fd	Nv-BdIzc	LDX #	T1	f0	f0	04
7	0006	88	1	DEY	0006	7f	f0	01	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ea	1		0007	7f	f0	01	fd	Nv-BdIzc	DEY	T0+T2	e0	e0	e0
8	0007	ea	1		0007	7f	f0	01	fd	Nv-BdIzc	DEY	T0+T2	e0	e0	c0
9	0007	ea	1	NOP	0007	7f	f0	01	fd	Nv-BdIzc	DEY	T1	01	ff	c0
9	0007	ea	1	NOP	0007	7f	f0	01	fd	Nv-BdIzc	DEY	T1	01	ff	00
10	0008	ea	1		0008	7f	f0	00	fd	nv-BdIZc	NOP	T0+T2	00	00	00
10	0008	ea	1		0008	7f	f0	00	fd	nv-BdIZc	NOP	T0+T2	00	00	00

NOP (89)		Cycles: 2	Size: 2
Immediate			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Operand.	
2.1		Inc. PC .	
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	89	1	unknown	0006	7f	f0	01	fd	Nv-BdIzc	LDX #	T1	f0	f0	04
7	0006	89	1	unknown	0006	7f	f0	01	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ea	1		0007	7f	f0	01	fd	Nv-BdIzc	unknown	T0+T2	e0	e0	e0
8	0007	ea	1		0007	7f	f0	01	fd	Nv-BdIzc	unknown	T0+T2	e0	e0	c0
9	0008	ea	1	NOP	0008	7f	f0	01	fd	Nv-BdIzc	unknown	T1	7f	ea	c0
9	0008	ea	1	NOP	0008	7f	f0	01	fd	Nv-BdIzc	unknown	T1	7f	ea	69
10	0009	ea	1		0009	7f	f0	01	fd	Nv-BdIzc	NOP	T0+T2	69	69	69
10	0009	ea	1		0009	7f	f0	01	fd	Nv-BdIzc	NOP	T0+T2	69	69	d2

TXA (8A)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Discard.	
2.1		A = X . N flag = (A & \$80), Z flag = ! A .	
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	8a	1	TXA	0006	7f	f0	01	fd	nv-BdIzc	LDA #	T1	7f	7f	04
7	0006	8a	1	TXA	0006	7f	f0	01	fd	nv-BdIzc	LDA #	T1	7f	7f	fe
8	0007	ea	1		0007	7f	f0	01	fd	nv-BdIzc	TXA	T0+T2	fe	fe	fe
8	0007	ea	1		0007	7f	f0	01	fd	nv-BdIzc	TXA	T0+T2	fe	fe	fc
9	0007	ea	1	NOP	0007	f0	f0	01	fd	Nv-BdIzc	TXA	T1	f0	f0	fc
9	0007	ea	1	NOP	0007	f0	f0	01	fd	Nv-BdIzc	TXA	T1	f0	f0	e0
10	0008	ea	1		0008	f0	f0	01	fd	Nv-BdIzc	NOP	T0+T2	e0	e0	e0
10	0008	ea	1		0008	f0	f0	01	fd	Nv-BdIzc	NOP	T0+T2	e0	e0	c0

ANE (8B)		Cycles: 2	Size: 2
Immediate			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC.	
1.2	Read PC	Store as Operand.	
2.1		Inc. * PC. A = ((A CONST) & X & Operand). N flag = (A & \$80), Z flag = !A.	
2.2	Read PC	Store as OpCode.	
+X.1			

* FF, EE, and 00 are common values for CONST. Actual value may depend on RDY and chip/temperature, but EE passes all known tests. Visual6502 uses 00 for CON

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	8b	1	unknown	0006	f3	fe	01	fd	Nv-BdIzc	LDA #	T1	f3	f3	04
7	0006	8b	1	unknown	0006	f3	fe	01	fd	Nv-BdIzc	LDA #	T1	f3	f3	e6
8	0007	7f	1		0007	f3	fe	01	fd	Nv-BdIzc	unknown	T0+T2	e6	e6	e6
8	0007	7f	1		0007	f3	fe	01	fd	Nv-BdIzc	unknown	T0+T2	e6	e6	cc
9	0008	ea	1	NOP	0008	72	fe	01	fd	nv-BdIzc	unknown	T1	72	72	cc
9	0008	ea	1	NOP	0008	72	fe	01	fd	nv-BdIzc	unknown	T1	72	72	e4
10	0009	ea	1		0009	72	fe	01	fd	nv-BdIzc	NOP	T0+T2	e4	e4	e4
10	0009	ea	1		0009	72	fe	01	fd	nv-BdIzc	NOP	T0+T2	e4	e4	c8

ANE (XAA, AXM)

Type: Combination of an immediate and an implied command

Opc.	Mnemonic	Function	Size	Cycles	N	V	-	B	D	I	Z	C
\$8B	ANE #imm	A = (A {CONST}) & X & #{imm}	2	2	o						o	

Operation: This opcode ORs the A register with CONST, ANDs the result with X. ANDs the result with an immediate value, and then stores the result in A.

Instability: CONST is chip- and/or temperature dependent (common values may be \$ee, \$00, \$ff ...). Some dependency on the RDY line. Bit 0 and Bit 4 are “weaker” than the other bits, and may drop to 0 in the first cycle of DMA when RDY goes low.

Do not use ANE with any immediate value other than 0, or when the accumulator value is \$ff (both take the magic constant out of the equation)! (Or, more accurately, these are safe if all bits that could be 0 in A are 0 in either the immediate value or X or both.)

STY (8C)		Cycles: 4	Size: 3
Absolute (Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read PC	Store as Address.H.	
3.1		Inc. PC .	
3.2	Write to Address	Write Y .	
4.1			
4.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	8c	1	STY Abs	0006	f3	fe	01	fd	Nv-BdIzc	LDA #	T1	f3	f3	04
7	0006	8c	1	STY Abs	0006	f3	fe	01	fd	Nv-BdIzc	LDA #	T1	f3	f3	e6
8	0007	33	1		0007	f3	fe	01	fd	Nv-BdIzc	STY Abs	T2	e6	e6	e6
8	0007	33	1		0007	f3	fe	01	fd	Nv-BdIzc	STY Abs	T2	e6	e6	cc
9	0008	03	1		0008	f3	fe	01	fd	Nv-BdIzc	STY Abs	T3	00	33	cc
9	0008	03	1		0008	f3	fe	01	fd	Nv-BdIzc	STY Abs	T3	00	33	33
10	0333	03	0		0009	f3	fe	01	fd	Nv-BdIzc	STY Abs	T0	01	01	33
10	0333	01	0		0009	f3	fe	01	fd	Nv-BdIzc	STY Abs	T0	01	01	02
11	0009	ea	1	NOP	0009	f3	fe	01	fd	Nv-BdIzc	STY Abs	T1	ff	01	02
11	0009	ea	1	NOP	0009	f3	fe	01	fd	Nv-BdIzc	STY Abs	T1	ff	01	00
12	000a	15	1		000a	f3	fe	01	fd	Nv-BdIzc	NOP	T0+T2	00	00	00
12	000a	15	1		000a	f3	fe	01	fd	Nv-BdIzc	NOP	T0+T2	00	00	00

STA (8D)		Cycles: 4	Size: 3
Absolute (Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read PC	Store as Address.H.	
3.1		Inc. PC .	
3.2	Write to Address	Write A .	
4.1			
4.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	8d	1	STA Abs	0006	f3	fe	01	fd	Nv-BdIzc	LDA #	T1	f3	f3	04
7	0006	8d	1	STA Abs	0006	f3	fe	01	fd	Nv-BdIzc	LDA #	T1	f3	f3	e6
8	0007	33	1		0007	f3	fe	01	fd	Nv-BdIzc	STA Abs	T2	e6	e6	e6
8	0007	33	1		0007	f3	fe	01	fd	Nv-BdIzc	STA Abs	T2	e6	e6	cc
9	0008	03	1		0008	f3	fe	01	fd	Nv-BdIzc	STA Abs	T3	00	33	cc
9	0008	03	1		0008	f3	fe	01	fd	Nv-BdIzc	STA Abs	T3	00	33	33
10	0333	03	0		0009	f3	fe	01	fd	Nv-BdIzc	STA Abs	T0	ff	f3	33
10	0333	f3	0		0009	f3	fe	01	fd	Nv-BdIzc	STA Abs	T0	ff	f3	f2
11	0009	ea	1	NOP	0009	f3	fe	01	fd	Nv-BdIzc	STA Abs	T1	f3	f3	f2
11	0009	ea	1	NOP	0009	f3	fe	01	fd	Nv-BdIzc	STA Abs	T1	f3	f3	e6
12	000a	15	1		000a	f3	fe	01	fd	Nv-BdIzc	NOP	T0+T2	e6	e6	e6
12	000a	15	1		000a	f3	fe	01	fd	Nv-BdIzc	NOP	T0+T2	e6	e6	cc

STX (8E)	Cycles: 4	Size: 3
Absolute (Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Write to Address	Write X .
4.1		
4.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	8e	1	STX Abs	0006	f3	fe	01	fd	Nv-BdIzc	LDA #	T1	f3	f3	04
7	0006	8e	1	STX Abs	0006	f3	fe	01	fd	Nv-BdIzc	LDA #	T1	f3	f3	e6
8	0007	33	1		0007	f3	fe	01	fd	Nv-BdIzc	STX Abs	T2	e6	e6	e6
8	0007	33	1		0007	f3	fe	01	fd	Nv-BdIzc	STX Abs	T2	e6	e6	cc
9	0008	03	1		0008	f3	fe	01	fd	Nv-BdIzc	STX Abs	T3	00	33	cc
9	0008	03	1		0008	f3	fe	01	fd	Nv-BdIzc	STX Abs	T3	00	33	33
10	0333	03	0		0009	f3	fe	01	fd	Nv-BdIzc	STX Abs	T0	fe	fe	33
10	0333	fe	0		0009	f3	fe	01	fd	Nv-BdIzc	STX Abs	T0	fe	fe	fc
11	0009	ea	1	NOP	0009	f3	fe	01	fd	Nv-BdIzc	STX Abs	T1	ff	fe	fc
11	0009	ea	1	NOP	0009	f3	fe	01	fd	Nv-BdIzc	STX Abs	T1	ff	fe	fd
12	000a	15	1		000a	f3	fe	01	fd	Nv-BdIzc	NOP	T0+T2	fd	fd	fd
12	000a	15	1		000a	f3	fe	01	fd	Nv-BdIzc	NOP	T0+T2	fd	fd	fa

SAX (8F)	Cycles: 4	Size: 3
Absolute (Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Write to Address	Write (A & X).
4.1		
4.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	8f	1	unknown	0006	f3	fe	01	fd	Nv-BdIzc	LDA #	T1	f3	f3	04
7	0006	8f	1	unknown	0006	f3	fe	01	fd	Nv-BdIzc	LDA #	T1	f3	f3	e6
8	0007	33	1		0007	f3	fe	01	fd	Nv-BdIzc	unknown	T2	e6	e6	e6
8	0007	33	1		0007	f3	fe	01	fd	Nv-BdIzc	unknown	T2	e6	e6	cc
9	0008	03	1		0008	f3	fe	01	fd	Nv-BdIzc	unknown	T3	00	33	cc
9	0008	03	1		0008	f3	fe	01	fd	Nv-BdIzc	unknown	T3	00	33	33
10	0333	03	0		0009	f3	fe	01	fd	Nv-BdIzc	unknown	T0	f2	f2	33
10	0333	f2	0		0009	f3	fe	01	fd	Nv-BdIzc	unknown	T0	f2	f2	e4
11	0009	ea	1	NOP	0009	f3	fe	01	fd	Nv-BdIzc	unknown	T1	f3	f2	e4
11	0009	ea	1	NOP	0009	f3	fe	01	fd	Nv-BdIzc	unknown	T1	f3	f2	e5
12	000a	15	1		000a	f3	fe	01	fd	Nv-BdIzc	NOP	T0+T2	e5	e5	e5
12	000a	15	1		000a	f3	fe	01	fd	Nv-BdIzc	NOP	T0+T2	e5	e5	ca

BCC (90) Cycles: 2-4 Size: 2		
Relative (Jump)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC . Jump = ((P & \$01) == \$00).
1.2	Read PC	Store as Operand.
		If !Jump, poll interrupts.
		Otherwise Address = i16(i8(Operand)) + PC + 1.
		BoundaryCrossed = Address.H != (PC + 1).H.
		If !BoundaryCrossed, poll interrupts.
2.1		Inc. PC . If !Jump, end (next half-cycle is 4.2).
2.2	Read PC	Discard.
3.1		PC .L = Address.L. If !BoundaryCrossed, end (next half-cycle is 4.2).
3.2	Read PC	Discard. Poll interrupts.
4.1		PC .H = Address.H.
4.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
8	0006	90	1	BCC	0006	fe	ea	00	fd	Nv-BdIzC	LDX #	T1	ea	ea	fc
8	0006	90	1	BCC	0006	fe	ea	00	fd	Nv-BdIzC	LDX #	T1	ea	ea	d4
9	0007	ee	1		0007	fe	ea	00	fd	Nv-BdIzC	BCC	T2	d4	d4	d4
9	0007	ee	1		0007	fe	ea	00	fd	Nv-BdIzC	BCC	T2	d4	d4	a8
10	0008	ea	1	NOP	0008	fe	ea	00	fd	Nv-BdIzC	BCC		ee	08	a8
10	0008	ea	1	NOP	0008	fe	ea	00	fd	Nv-BdIzC	BCC		ee	08	f6
11	0009	ea	1		0009	fe	ea	00	fd	Nv-BdIzC	NOP	T0+T2	f6	f6	f6
11	0009	ea	1		0009	fe	ea	00	fd	Nv-BdIzC	NOP	T0+T2	f6	f6	ec

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	90	1	BCC	0006	f3	fe	01	fd	Nv-BdIzC	LDA #	T1	f3	f3	04
7	0006	90	1	BCC	0006	f3	fe	01	fd	Nv-BdIzC	LDA #	T1	f3	f3	e6
8	0007	33	1		0007	f3	fe	01	fd	Nv-BdIzC	BCC	T2	e6	e6	e6
8	0007	33	1		0007	f3	fe	01	fd	Nv-BdIzC	BCC	T2	e6	e6	cc
9	0008	03	1		0008	f3	fe	01	fd	Nv-BdIzC	BCC	T3	33	08	cc
9	0008	03	1		0008	f3	fe	01	fd	Nv-BdIzC	BCC	T3	33	08	3b
10	003b	ea	1	NOP	003b	f3	fe	01	fd	Nv-BdIzC	BCC		00	00	3b
10	003b	ea	1	NOP	003b	f3	fe	01	fd	Nv-BdIzC	BCC		00	00	01
11	003c	00	1		003c	f3	fe	01	fd	Nv-BdIzC	NOP	T0+T2	01	01	01
11	003c	00	1		003c	f3	fe	01	fd	Nv-BdIzC	NOP	T0+T2	01	01	02

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	90	1	BCC	0006	dd	fe	01	fd	Nv-BdIzC	LDA #	T1	dd	dd	04
7	0006	90	1	BCC	0006	dd	fe	01	fd	Nv-BdIzC	LDA #	T1	dd	dd	ba
8	0007	ee	1		0007	dd	fe	01	fd	Nv-BdIzC	BCC	T2	ba	ba	ba
8	0007	ee	1		0007	dd	fe	01	fd	Nv-BdIzC	BCC	T2	ba	ba	74
9	0008	03	1		0008	dd	fe	01	fd	Nv-BdIzC	BCC	T3	ee	08	74
9	0008	03	1		0008	dd	fe	01	fd	Nv-BdIzC	BCC	T3	ee	08	f6
10	00f6	00	1		00f6	dd	fe	01	fd	Nv-BdIzC	BCC	T0	00	ff	f6
10	00f6	00	1		00f6	dd	fe	01	fd	Nv-BdIzC	BCC	T0	00	ff	ff
11	fff6	00	1	BRK	fff6	dd	fe	01	fd	Nv-BdIzC	BCC	T1	ff	00	ff
11	fff6	00	1	BRK	fff6	dd	fe	01	fd	Nv-BdIzC	BCC	T1	ff	00	ff
12	fff7	00	1		fff7	dd	fe	01	fd	Nv-BdIzC	BRK	T2	ff	ff	ff
12	fff7	00	1		fff7	dd	fe	01	fd	Nv-BdIzC	BRK	T2	ff	ff	fe

STA (91)	Cycles: 6	Size: 2
Indirect Y (Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read Pointer	Store as Address.
3.1		
3.2	Read u8(Pointer + 1)	Store as Address.H.
4.1		Target = Address + Y. Pointer.L = Target.L. Pointer.H = Address.H.
4.2	Read Pointer	Store as Operand.
5.1		Pointer.H = Target.H.
5.2	Write to Pointer	Write A .
6.1		
6.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	91	1	STA (zp),Y	0006	ff	ea	80	fd	Nv-BdIzc	LDX #	T1	ea	ea	00
7	0006	91	1	STA (zp),Y	0006	ff	ea	80	fd	Nv-BdIzc	LDX #	T1	ea	ea	d4
8	0007	ee	1		0007	ff	ea	80	fd	Nv-BdIzc	STA (zp),Y	T2	d4	d4	d4
8	0007	ee	1		0007	ff	ea	80	fd	Nv-BdIzc	STA (zp),Y	T2	d4	d4	a8
9	00ee	77	1		0008	ff	ea	80	fd	Nv-BdIzc	STA (zp),Y	T3	00	ee	a8
9	00ee	77	1		0008	ff	ea	80	fd	Nv-BdIzc	STA (zp),Y	T3	00	ee	ef
10	00ef	ff	1		0008	ff	ea	80	fd	Nv-BdIzc	STA (zp),Y	T4	80	77	ef
10	00ef	ff	1		0008	ff	ea	80	fd	Nv-BdIzc	STA (zp),Y	T4	80	77	f7
11	fff7	00	1		0008	ff	ea	80	fd	Nv-BdIzc	STA (zp),Y	T5	00	ff	f7
11	fff7	00	1		0008	ff	ea	80	fd	Nv-BdIzc	STA (zp),Y	T5	00	ff	00
12	fff7	00	0		0008	ff	ea	80	fd	Nv-BdIzc	STA (zp),Y	T0	00	ff	00
12	fff7	ff	0		0008	ff	ea	80	fd	Nv-BdIzc	STA (zp),Y	T0	00	ff	ff
13	0008	ea	1	NOP	0008	ff	ea	80	fd	Nv-BdIzc	STA (zp),Y	T1	ff	ff	ff
13	0008	ea	1	NOP	0008	ff	ea	80	fd	Nv-BdIzc	STA (zp),Y	T1	ff	ff	fe
14	0009	ea	1		0009	ff	ea	80	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fe
14	0009	ea	1		0009	ff	ea	80	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fc

JAM (92)	Cycles: ∞	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Discard.
2.1		Visual6502 suggests incrementing PC , but Single-Step Tests cannot be passed this way.
2.2	Read \$FFFF	Discard.
3.1		
3.2	Read \$FFFE	Discard.
4.1		
4.2	Read \$FFFE	Discard.
5.1		Repeat 5.1 and 5.2 forever.
5.2	Read \$FFFF	Discard.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	92	1	unknown	0006	ff	ea	80	fd	Nv-BdIzc	LDX #	T1	ea	ea	00
7	0006	92	1	unknown	0006	ff	ea	80	fd	Nv-BdIzc	LDX #	T1	ea	ea	d4
8	0007	ee	1		0007	ff	ea	80	fd	Nv-BdIzc	unknown	T2	d4	d4	d4
8	0007	ee	1		0007	ff	ea	80	fd	Nv-BdIzc	unknown	T2	d4	d4	a8
9	ffff	00	1		0008	ff	ea	80	fd	Nv-BdIzc	unknown	T3	ff	ff	a8
9	ffff	00	1		0008	ff	ea	80	fd	Nv-BdIzc	unknown	T3	ff	ff	fe
10	fffe	00	1		0008	ff	ea	80	fd	Nv-BdIzc	unknown	T4	ff	ff	fe
10	fffe	00	1		0008	ff	ea	80	fd	Nv-BdIzc	unknown	T4	ff	ff	fe
11	fffe	00	1		0008	ff	ea	80	fd	Nv-BdIzc	unknown	T5	ff	ff	fe
11	fffe	00	1		0008	ff	ea	80	fd	Nv-BdIzc	unknown	T5	ff	ff	fe
12	ffff	00	1		0008	ff	ea	80	fd	Nv-BdIzc	unknown		ff	ff	fe
12	ffff	00	1		0008	ff	ea	80	fd	Nv-BdIzc	unknown		ff	ff	fe
13	ffff	00	1		0008	ff	ea	80	fd	Nv-BdIzc	unknown		ff	ff	fe
13	ffff	00	1		0008	ff	ea	80	fd	Nv-BdIzc	unknown		ff	ff	fe
14	ffff	00	1		0008	ff	ea	80	fd	Nv-BdIzc	unknown		ff	ff	fe
14	ffff	00	1		0008	ff	ea	80	fd	Nv-BdIzc	unknown		ff	ff	fe

SHA (93)		Cycles: 6	Size: 2
Indirect Y (Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Address.	
3.1			
3.2	Read u8(Pointer + 1)	Store as Address.H.	
4.1		Target = Address + Y. Pointer.L = Target.L. Pointer.H = Address.H. BoundaryCrossed = Pointer.H != Target.H.	
4.2	Read Pointer	Store as Operand.	
5.1		Pointer.H = Target.H.	
5.2		High = Pointer.H. If !BoundaryCrossed, High = (High + 1). If RDY just went low on the previous cycle, High = \$FFFF. * Val = u8(High & A & X).	
		* If BoundaryCrossed, write to (Pointer.L (Val << 8)) Otherwise write to Pointer	
6.1		Write Val.	
6.2	Read PC	Store as OpCode.	
+X.1			

* The Single-Step Test tests are unable to encapsulate the full behavior of this instruction, so it is expected that many (but not all) tests will fail with this implementation.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	93	1	unknown	0006	1f	ea	80	fd	Nv-BdIzc	LDX #	T1	ea	ea	00
7	0006	93	1	unknown	0006	1f	ea	80	fd	Nv-BdIzc	LDX #	T1	ea	ea	d4
8	0007	ff	1		0007	1f	ea	80	fd	Nv-BdIzc	unknown	T2	d4	d4	d4
8	0007	ff	1		0007	1f	ea	80	fd	Nv-BdIzc	unknown	T2	d4	d4	a8
9	00ff	70	1		0008	1f	ea	80	fd	Nv-BdIzc	unknown	T3	00	ff	a8
9	00ff	70	1		0008	1f	ea	80	fd	Nv-BdIzc	unknown	T3	00	ff	00
10	0000	a9	1		0008	1f	ea	80	fd	Nv-BdIzc	unknown	T4	80	70	00
10	0000	a9	1		0008	1f	ea	80	fd	Nv-BdIzc	unknown	T4	80	70	f0
11	a9f0	aa	1		0008	1f	ea	80	fd	Nv-BdIzc	unknown	T5	00	a9	f0
11	a9f0	aa	1		0008	1f	ea	80	fd	Nv-BdIzc	unknown	T5	00	a9	aa
12	a9f0	aa	0		0008	1f	ea	80	fd	Nv-BdIzc	unknown	T0	0a	0a	0a
12	a9f0	0a	0		0008	1f	ea	80	fd	Nv-BdIzc	unknown	T0	0a	0a	14
13	0008	ea	1	NOP	0008	1f	ea	80	fd	Nv-BdIzc	unknown	T1	1f	0a	14
13	0008	ea	1	NOP	0008	1f	ea	80	fd	Nv-BdIzc	unknown	T1	1f	0a	29
14	0009	ea	1		0009	1f	ea	80	fd	Nv-BdIzc	NOP	T0+T2	29	29	29
14	0009	ea	1		0009	1f	ea	80	fd	Nv-BdIzc	NOP	T0+T2	29	29	52

SHA (AXA, AHX, TEA)

Type: Combinations of STA/STX/STY

OpC.	Mnemonic	Function	Size	Cycles	N	V	-	B	D	I	Z	C
\$93	SHA (zp), y	{addr} = A & X & {H+1}	2	6								
\$9F	SHA abs, y	{addr} = A & X & {H+1}	3	5								

Operation: This opcode stores the result of A AND X AND the high byte of the target address of the operand +1 in memory.

Instabilities:

- The value written is ANDed with &{H+1}, except when the RDY line goes low in the 4th (opcode \$9f) or 5th (opcode \$93) cycle.
- When adding Y to the target address causes a page boundary crossing, the highbyte of the target address is incremented by one (as expected), and then ANDed with (A & X).

STY (94)	Cycles: 4	Size: 2
Zero Page X-Indexed (Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read Pointer	Store as Operand.
3.1		Address = u8(Pointer + X).
3.2	Write to Address	Write Y .
4.1		
4.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	94	1	STY zp,X	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	94	1	STY zp,X	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ff	1		0007	1f	f0	80	fd	Nv-BdIzc	STY zp,X	T2	e0	e0	e0
8	0007	ff	1		0007	1f	f0	80	fd	Nv-BdIzc	STY zp,X	T2	e0	e0	c0
9	00ff	70	1		0008	1f	f0	80	fd	Nv-BdIzc	STY zp,X	T3	f0	ff	c0
9	00ff	70	1		0008	1f	f0	80	fd	Nv-BdIzc	STY zp,X	T3	f0	ff	ef
10	00ef	70	0		0008	1f	f0	80	fd	Nv-BdIzc	STY zp,X	T0	80	80	ef
10	00ef	80	0		0008	1f	f0	80	fd	Nv-BdIzc	STY zp,X	T0	80	80	00
11	0008	ea	1	NOP	0008	1f	f0	80	fd	Nv-BdIzc	STY zp,X	T1	ff	80	00
11	0008	ea	1	NOP	0008	1f	f0	80	fd	Nv-BdIzc	STY zp,X	T1	ff	80	7f
12	0009	ea	1		0009	1f	f0	80	fd	Nv-BdIzc	NOP	T0+T2	7f	7f	7f
12	0009	ea	1		0009	1f	f0	80	fd	Nv-BdIzc	NOP	T0+T2	7f	7f	fe

STA (95)	Cycles: 4	Size: 2
Zero Page X-Indexed (Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read Pointer	Store as Operand.
3.1		Address = u8(Pointer + X).
3.2	Write to Address	Write A .
4.1		
4.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	95	1	STA zp,X	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	95	1	STA zp,X	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ff	1		0007	1f	f0	80	fd	Nv-BdIzc	STA zp,X	T2	e0	e0	e0
8	0007	ff	1		0007	1f	f0	80	fd	Nv-BdIzc	STA zp,X	T2	e0	e0	c0
9	00ff	70	1		0008	1f	f0	80	fd	Nv-BdIzc	STA zp,X	T3	f0	ff	c0
9	00ff	70	1		0008	1f	f0	80	fd	Nv-BdIzc	STA zp,X	T3	f0	ff	ef
10	00ef	70	0		0008	1f	f0	80	fd	Nv-BdIzc	STA zp,X	T0	ff	1f	ef
10	00ef	1f	0		0008	1f	f0	80	fd	Nv-BdIzc	STA zp,X	T0	ff	1f	1e
11	0008	ea	1	NOP	0008	1f	f0	80	fd	Nv-BdIzc	STA zp,X	T1	1f	1f	1e
11	0008	ea	1	NOP	0008	1f	f0	80	fd	Nv-BdIzc	STA zp,X	T1	1f	1f	3e
12	0009	ea	1		0009	1f	f0	80	fd	Nv-BdIzc	NOP	T0+T2	3e	3e	3e
12	0009	ea	1		0009	1f	f0	80	fd	Nv-BdIzc	NOP	T0+T2	3e	3e	7c

STX (96)	Cycles: 4	Size: 2
Zero Page Y-Indexed (Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read Pointer	Store as Operand.
3.1		Target = Pointer + Y . Address.L = Target.L. Address.H = Pointer.H.
3.2	Write to Address	Write X .
4.1		
4.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	96	1	STX zp,Y	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	96	1	STX zp,Y	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ff	1		0007	1f	f0	80	fd	Nv-BdIzc	STX zp,Y	T2	e0	e0	e0
8	0007	ff	1		0007	1f	f0	80	fd	Nv-BdIzc	STX zp,Y	T2	e0	e0	c0
9	00ff	70	1		0008	1f	f0	80	fd	Nv-BdIzc	STX zp,Y	T3	80	ff	c0
9	00ff	70	1		0008	1f	f0	80	fd	Nv-BdIzc	STX zp,Y	T3	80	ff	7f
10	007f	70	0		0008	1f	f0	80	fd	Nv-BdIzc	STX zp,Y	T0	f0	f0	7f
10	007f	f0	0		0008	1f	f0	80	fd	Nv-BdIzc	STX zp,Y	T0	f0	f0	e0
11	0008	ea	1	NOP	0008	1f	f0	80	fd	Nv-BdIzc	STX zp,Y	T1	ff	f0	e0
11	0008	ea	1	NOP	0008	1f	f0	80	fd	Nv-BdIzc	STX zp,Y	T1	ff	f0	ef
12	0009	ea	1		0009	1f	f0	80	fd	Nv-BdIzc	NOP	T0+T2	ef	ef	ef
12	0009	ea	1		0009	1f	f0	80	fd	Nv-BdIzc	NOP	T0+T2	ef	ef	de

SAX (97)	Cycles: 4	Size: 2
Zero Page Y-Indexed (Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read Pointer	Store as Operand.
3.1		Target = Pointer + Y . Address.L = Target.L. Address.H = Pointer.H.
3.2	Write to Address	Write (A & X).
4.1		
4.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	97	1	unknown	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	97	1	unknown	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ff	1		0007	1f	f0	80	fd	Nv-BdIzc	unknown	T2	e0	e0	e0
8	0007	ff	1		0007	1f	f0	80	fd	Nv-BdIzc	unknown	T2	e0	e0	c0
9	00ff	70	1		0008	1f	f0	80	fd	Nv-BdIzc	unknown	T3	80	ff	c0
9	00ff	70	1		0008	1f	f0	80	fd	Nv-BdIzc	unknown	T3	80	ff	7f
10	007f	70	0		0008	1f	f0	80	fd	Nv-BdIzc	unknown	T0	10	10	7f
10	007f	10	0		0008	1f	f0	80	fd	Nv-BdIzc	unknown	T0	10	10	20
11	0008	ea	1	NOP	0008	1f	f0	80	fd	Nv-BdIzc	unknown	T1	1f	10	20
11	0008	ea	1	NOP	0008	1f	f0	80	fd	Nv-BdIzc	unknown	T1	1f	10	2f
12	0009	ea	1		0009	1f	f0	80	fd	Nv-BdIzc	NOP	T0+T2	2f	2f	2f
12	0009	ea	1		0009	1f	f0	80	fd	Nv-BdIzc	NOP	T0+T2	2f	2f	5e

TYA (98)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Discard.	
2.1		A = Y . N flag = (A & \$80), Z flag = ! A .	
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	98	1	TYA	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	98	1	TYA	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ea	1		0007	1f	f0	80	fd	Nv-BdIzc	TYA	T0+T2	e0	e0	e0
8	0007	ea	1		0007	1f	f0	80	fd	Nv-BdIzc	TYA	T0+T2	e0	e0	c0
9	0007	ea	1	NOP	0007	80	f0	80	fd	Nv-BdIzc	TYA	T1	80	80	c0
9	0007	ea	1	NOP	0007	80	f0	80	fd	Nv-BdIzc	TYA	T1	80	80	00
10	0008	ea	1		0008	80	f0	80	fd	Nv-BdIzc	NOP	T0+T2	00	00	00
10	0008	ea	1		0008	80	f0	80	fd	Nv-BdIzc	NOP	T0+T2	00	00	00

STA (99)	Cycles: 5	Size: 3
Absolute Y-Indexed (Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + Y. Address.L = Target.L. Address.H = Pointer.H.
3.2	Read Address	Store as Operand.
4.1		Address.H = Target.H.
4.2	Write to Address	Write A .
5.1		
5.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	99	1	STA Abs,Y	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	99	1	STA Abs,Y	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	44	1		0007	1f	f0	80	fd	Nv-BdIzc	STA Abs,Y	T2	e0	e0	e0
8	0007	44	1		0007	1f	f0	80	fd	Nv-BdIzc	STA Abs,Y	T2	e0	e0	c0
9	0008	22	1		0008	1f	f0	80	fd	Nv-BdIzc	STA Abs,Y	T3	80	44	c0
9	0008	22	1		0008	1f	f0	80	fd	Nv-BdIzc	STA Abs,Y	T3	80	44	c4
10	22c4	00	1		0009	1f	f0	80	fd	Nv-BdIzc	STA Abs,Y	T4	00	22	c4
10	22c4	00	1		0009	1f	f0	80	fd	Nv-BdIzc	STA Abs,Y	T4	00	22	23
11	22c4	00	0		0009	1f	f0	80	fd	Nv-BdIzc	STA Abs,Y	T0	23	1f	23
11	22c4	1f	0		0009	1f	f0	80	fd	Nv-BdIzc	STA Abs,Y	T0	23	1f	42
12	0009	ea	1	NOP	0009	1f	f0	80	fd	Nv-BdIzc	STA Abs,Y	T1	1f	1f	42
12	0009	ea	1	NOP	0009	1f	f0	80	fd	Nv-BdIzc	STA Abs,Y	T1	1f	1f	3e
13	000a	15	1		000a	1f	f0	80	fd	Nv-BdIzc	NOP	T0+T2	3e	3e	3e
13	000a	15	1		000a	1f	f0	80	fd	Nv-BdIzc	NOP	T0+T2	3e	3e	7c

TXS (9A)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Discard.	
2.1		S = X.	
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	9a	1	TXS	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	9a	1	TXS	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ea	1		0007	1f	f0	80	fd	Nv-BdIzc	TXS	T0+T2	e0	e0	e0
8	0007	ea	1		0007	1f	f0	80	fd	Nv-BdIzc	TXS	T0+T2	e0	e0	c0
9	0007	ea	1	NOP	0007	1f	f0	80	f0	Nv-BdIzc	TXS	T1	f0	ff	c0
9	0007	ea	1	NOP	0007	1f	f0	80	f0	Nv-BdIzc	TXS	T1	f0	ff	ef
10	0008	22	1		0008	1f	f0	80	f0	Nv-BdIzc	NOP	T0+T2	ef	ef	ef
10	0008	22	1		0008	1f	f0	80	f0	Nv-BdIzc	NOP	T0+T2	ef	ef	de

TAS (9B)		Cycles: 5	Size: 3
Absolute Y-Indexed (Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC.	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC.	
2.2	Read PC	Store as Pointer.H.	
3.1		Inc. PC. Target = Pointer + Y.	
		Address.L = Target.L.	
		Address.H = Pointer.H.	
		BoundaryCrossed = Pointer.H != Target.H.	
3.2	Read Address	Store as Operand.	
4.1		Address.H = Target.H.	
4.2		If BoundaryCrossed, Val = u8(Address.H & A & X).	
		Otherwise Val = u8((Address.H + 1) & A & X).	
	If BoundaryCrossed, write to (Address.L (Val << 8))		
	Otherwise write to Address	Write Val.	
5.1			
5.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	9b	1	unknown	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	9b	1	unknown	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ea	1		0007	1f	f0	80	fd	Nv-BdIzc	unknown	T2	e0	e0	e0
8	0007	ea	1		0007	1f	f0	80	fd	Nv-BdIzc	unknown	T2	e0	e0	c0
9	0008	22	1		0008	1f	f0	80	fd	Nv-BdIzc	unknown	T3	80	ea	c0
9	0008	22	1		0008	1f	f0	80	fd	Nv-BdIzc	unknown	T3	80	ea	6a
10	226a	00	1		0009	1f	f0	80	fd	Nv-BdIzc	unknown	T4	00	22	6a
10	226a	00	1		0009	1f	f0	80	fd	Nv-BdIzc	unknown	T4	00	22	23
11	006a	00	0		0009	1f	f0	80	fd	Nv-BdIzc	unknown	T0	00	00	00
11	006a	00	0		0009	1f	f0	80	fd	Nv-BdIzc	unknown	T0	00	00	00
12	0009	ea	1	NOP	0009	1f	f0	80	10	Nv-BdIzc	unknown	T1	10	00	00
12	0009	ea	1	NOP	0009	1f	f0	80	10	Nv-BdIzc	unknown	T1	10	00	10
13	000a	15	1		000a	1f	f0	80	10	Nv-BdIzc	NOP	T0+T2	10	10	10
13	000a	15	1		000a	1f	f0	80	10	Nv-BdIzc	NOP	T0+T2	10	10	20

TAS (XAS, SHS)

Type: Combinations of STA/TXS and LDA/TSX

OpC.	Mnemonic	Function	Size	Cycles	N	V	-	B	D	I	Z	C
\$9B	TAS abs., y	SP = A & X {addr} = A & X & {H+1}	3	5								

Operation: This opcode ANDs the contents of the A and X registers (without changing the contents of either register) and transfers the result to the stack pointer. It then ANDs that result with the contents of the high byte of the target address of the operand +1 and stores that final result in memory.

Instabilities:

- The value written is ANDed with &{H+1}, except when the RDY line goes low in the 4th cycle.
- When adding Y to the target address causes a page boundary crossing, the highbyte of the target address is incremented by one (as expected), and then ANDed with (A & X).

SHY (9C)		Cycles: 5	Size: 3
Absolute X-Indexed (Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1	Read PC	Inc. PC .	
1.2		Store as Pointer.	
2.1		Inc. PC .	
2.2		Store as Pointer.H.	
3.1		Inc. PC . Target = Pointer + X .	
	Read PC	Address.L = Target.L.	
		Address.H = Pointer.H.	
		BoundaryCrossed = Pointer.H != Target.H.	
3.2		Store as Operand.	
4.1		Address.H = Target.H.	
4.2	Read Address	If BoundaryCrossed, Val = (Address.H & Y).	
		Otherwise Val = u8((Address.H + 1) & Y).	
		If BoundaryCrossed, write to (Address.L (Val << 8))	
		Otherwise write to Address	
		Write Val.	
5.1	Read PC		
5.2		Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	9c	1	unknown	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	9c	1	unknown	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ea	1		0007	1f	f0	80	fd	Nv-BdIzc	unknown	T2	e0	e0	e0
8	0007	ea	1		0007	1f	f0	80	fd	Nv-BdIzc	unknown	T2	e0	e0	c0
9	0008	22	1		0008	1f	f0	80	fd	Nv-BdIzc	unknown	T3	f0	ea	c0
9	0008	22	1		0008	1f	f0	80	fd	Nv-BdIzc	unknown	T3	f0	ea	da
10	22da	00	1		0009	1f	f0	80	fd	Nv-BdIzc	unknown	T4	00	22	da
10	22da	00	1		0009	1f	f0	80	fd	Nv-BdIzc	unknown	T4	00	22	23
11	00da	00	0		0009	1f	f0	80	fd	Nv-BdIzc	unknown	T0	00	00	00
11	00da	00	0		0009	1f	f0	80	fd	Nv-BdIzc	unknown	T0	00	00	00
12	0009	ea	1	NOP	0009	1f	f0	80	fd	Nv-BdIzc	unknown	T1	ff	00	00
12	0009	ea	1	NOP	0009	1f	f0	80	fd	Nv-BdIzc	unknown	T1	ff	00	ff
13	000a	15	1		000a	1f	f0	80	fd	Nv-BdIzc	NOP	T0+T2	ff	ff	ff
13	000a	15	1		000a	1f	f0	80	fd	Nv-BdIzc	NOP	T0+T2	ff	ff	fe

SHY (A11, SYA, SAY, TEY)

Type: Combinations of STA/STX/STY

OpC.	Mnemonic	Function	Size	Cycles	N	V	-	B	D	I	Z	C
\$9C	SHY abs, x	{addr} = Y & {H+1}	3	5								

Operation: AND Y register with the high byte of the target address of the argument + 1. Store the result in memory.

Instabilities:

- The value written is ANDed with &{H+1}, except when the RDY line goes low in the 4th cycle.
- When adding X to the target address causes a page boundary crossing, the highbyte of the target address is incremented by one (as expected), and then ANDed with Y.

STA (9D)		Cycles: 5	Size: 3
Absolute X-Indexed (Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read PC	Store as Pointer.H.	
3.1		Inc. PC . Target = Pointer + X .	
		Address.L = Target.L.	
		Address.H = Pointer.H.	
3.2	Read Address	Store as Operand.	
4.1		Address.H = Target.H.	
4.2	Write to Address	Write A .	
5.1			
5.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	9d	1	STA Abs,X	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	9d	1	STA Abs,X	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ea	1		0007	1f	f0	80	fd	Nv-BdIzc	STA Abs,X	T2	e0	e0	e0
8	0007	ea	1		0007	1f	f0	80	fd	Nv-BdIzc	STA Abs,X	T2	e0	e0	c0
9	0008	22	1		0008	1f	f0	80	fd	Nv-BdIzc	STA Abs,X	T3	f0	ea	c0
9	0008	22	1		0008	1f	f0	80	fd	Nv-BdIzc	STA Abs,X	T3	f0	ea	da
10	22da	00	1		0009	1f	f0	80	fd	Nv-BdIzc	STA Abs,X	T4	00	22	da
10	22da	00	1		0009	1f	f0	80	fd	Nv-BdIzc	STA Abs,X	T4	00	22	23
11	23da	00	0		0009	1f	f0	80	fd	Nv-BdIzc	STA Abs,X	T0	23	1f	23
11	23da	1f	0		0009	1f	f0	80	fd	Nv-BdIzc	STA Abs,X	T0	23	1f	42
12	0009	ea	1	NOP	0009	1f	f0	80	fd	Nv-BdIzc	STA Abs,X	T1	1f	1f	42
12	0009	ea	1	NOP	0009	1f	f0	80	fd	Nv-BdIzc	STA Abs,X	T1	1f	1f	3e
13	000a	15	1		000a	1f	f0	80	fd	Nv-BdIzc	NOP	T0+T2	3e	3e	3e
13	000a	15	1		000a	1f	f0	80	fd	Nv-BdIzc	NOP	T0+T2	3e	3e	7c

SHX (9E)		Cycles: 5	Size: 3
Absolute Y-Indexed (Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1	Read PC	Inc. PC .	
1.2		Store as Pointer.	
2.1		Inc. PC .	
2.2		Store as Pointer.H.	
3.1	Read PC	Inc. PC . Target = Pointer + Y .	
		Address.L = Target.L.	
		Address.H = Pointer.H.	
		BoundaryCrossed = Pointer.H != Target.H.	
3.2	Read Address	Store as Operand.	
4.1		Address.H = Target.H.	
4.2		If BoundaryCrossed, Val = (Address.H & X).	
		Otherwise Val = u8((Address.H + 1) & X).	
		If BoundaryCrossed, write to (Address.L (Val << 8))	
	Otherwise write to Address	Write Val.	
5.1	Read PC		
5.2		Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	9e	1	unknown	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	9e	1	unknown	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ea	1		0007	1f	f0	80	fd	Nv-BdIzc	unknown	T2	e0	e0	e0
8	0007	ea	1		0007	1f	f0	80	fd	Nv-BdIzc	unknown	T2	e0	e0	c0
9	0008	22	1		0008	1f	f0	80	fd	Nv-BdIzc	unknown	T3	80	ea	c0
9	0008	22	1		0008	1f	f0	80	fd	Nv-BdIzc	unknown	T3	80	ea	6a
10	226a	00	1		0009	1f	f0	80	fd	Nv-BdIzc	unknown	T4	00	22	6a
10	226a	00	1		0009	1f	f0	80	fd	Nv-BdIzc	unknown	T4	00	22	23
11	206a	00	0		0009	1f	f0	80	fd	Nv-BdIzc	unknown	T0	20	20	20
11	206a	20	0		0009	1f	f0	80	fd	Nv-BdIzc	unknown	T0	20	20	40
12	0009	ea	1	NOP	0009	1f	f0	80	fd	Nv-BdIzc	unknown	T1	ff	20	40
12	0009	ea	1	NOP	0009	1f	f0	80	fd	Nv-BdIzc	unknown	T1	ff	20	1f
13	000a	15	1		000a	1f	f0	80	fd	Nv-BdIzc	NOP	T0+T2	1f	1f	1f
13	000a	15	1		000a	1f	f0	80	fd	Nv-BdIzc	NOP	T0+T2	1f	1f	3e

SHX (A11, SXA, XAS, TEX)

Type: Combinations of STA/STX/STY

OpC.	Mnemonic	Function	Size	Cycles	N	V	-	B	D	I	Z	C
\$9E	SHX abs. y	{addr} = X & {H+1}	3	5								

Operation: AND X register with the high byte of the target address of the argument + 1. Store the result in memory.

Instabilities:

- The value written is ANDed with &{H+1}, except when the RDY line goes low in the 4th cycle.
- When adding Y to the target address causes a page boundary crossing, the highbyte of the target address is incremented by one (as expected), and then ANDed with X.

SHA (9F)		Cycles: 5	Size: 3
Absolute Y-Indexed (Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1	Read PC	Inc. PC .	
1.2		Store as Pointer.	
2.1	Read PC	Inc. PC .	
2.2		Store as Pointer.H.	
3.1	Read Address	Inc. PC . Target = Pointer + Y.	
		Address.L = Target.L.	
		Address.H = Pointer.H.	
		BoundaryCrossed = Pointer.H != Target.H.	
3.2		Store as Operand.	
4.1	Read PC	Address.H = Target.H.	
4.2		High = Address.H.	
	If BoundaryCrossed, write to (Address.L (Val << 8))	If !BoundaryCrossed, High = (High + 1).	
		If RDY just went low on the previous cycle, High = \$FFFF.	
	Otherwise write to Address	Val = u8(High & A & X).	
		Write Val.	
5.1	Read PC		
5.2		Store as OpCode.	
+X.1			

* The Single-Step Test tests are unable to encapsulate the full behavior of this instruction, so it is expected that many (but not all) tests will fail with this implementation.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	9f	1	unknown	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	9f	1	unknown	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ea	1		0007	1f	f0	80	fd	Nv-BdIzc	unknown	T2	e0	e0	e0
8	0007	ea	1		0007	1f	f0	80	fd	Nv-BdIzc	unknown	T2	e0	e0	c0
9	0008	22	1		0008	1f	f0	80	fd	Nv-BdIzc	unknown	T3	80	ea	c0
9	0008	22	1		0008	1f	f0	80	fd	Nv-BdIzc	unknown	T3	80	ea	6a
10	226a	00	1		0009	1f	f0	80	fd	Nv-BdIzc	unknown	T4	00	22	6a
10	226a	00	1		0009	1f	f0	80	fd	Nv-BdIzc	unknown	T4	00	22	23
11	006a	00	0		0009	1f	f0	80	fd	Nv-BdIzc	unknown	T0	00	00	00
11	006a	00	0		0009	1f	f0	80	fd	Nv-BdIzc	unknown	T0	00	00	00
12	0009	ea	1	NOP	0009	1f	f0	80	fd	Nv-BdIzc	unknown	T1	1f	00	00
12	0009	ea	1	NOP	0009	1f	f0	80	fd	Nv-BdIzc	unknown	T1	1f	00	1f
13	000a	15	1		000a	1f	f0	80	fd	Nv-BdIzc	NOP	T0+T2	1f	1f	1f
13	000a	15	1		000a	1f	f0	80	fd	Nv-BdIzc	NOP	T0+T2	1f	1f	3e

SHA (AXA, AHX, TEA)

Type: Combinations of STA/STX/STY

OpC.	Mnemonic	Function	Size	Cycles	N	V	-	B	D	I	Z	C
\$93	SHA (zp), y	{addr} = A & X & {H+1}	2	6								
\$9F	SHA abs, y	{addr} = A & X & {H+1}	3	5								

Operation: This opcode stores the result of A AND X AND the high byte of the target address of the operand +1 in memory.

Instabilities:

- The value written is ANDed with &{H+1}, except when the RDY line goes low in the 4th (opcode \$9f) or 5th (opcode \$93) cycle.
- When adding Y to the target address causes a page boundary crossing, the highbyte of the target address is incremented by one (as expected), and then ANDed with (A & X).

LDY (A0)		Cycles: 2	Size: 2
Immediate			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Operand.	
2.1		Inc. PC . Y = Operand. N flag = (Y & \$80), Z flag = !Y .	
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	a0	1	LDY #	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	a0	1	LDY #	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	0f	1		0007	1f	f0	80	fd	Nv-BdIzc	LDY #	T0+T2	e0	e0	e0
8	0007	0f	1		0007	1f	f0	80	fd	Nv-BdIzc	LDY #	T0+T2	e0	e0	c0
9	0008	ea	1	NOP	0008	1f	f0	0f	fd	nv-BdIzc	LDY #	T1	0f	0f	c0
9	0008	ea	1	NOP	0008	1f	f0	0f	fd	nv-BdIzc	LDY #	T1	0f	0f	1e
10	0009	ea	1		0009	1f	f0	0f	fd	nv-BdIzc	NOP	T0+T2	1e	1e	1e
10	0009	ea	1		0009	1f	f0	0f	fd	nv-BdIzc	NOP	T0+T2	1e	1e	3c

LDA (A1)	Cycles: 6	Size: 2
Indirect X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC .
2.2	Read Operand	Discard.
3.1		Pointer = u8(Operand + X).
3.2	Read Pointer	Store as Address.
4.1		
4.2	Read u8(Pointer + 1)	Store as Address.H.
5.1		
5.2	Read Address	Store as Operand.
6.1		A = Operand. N flag = (A & \$80), Z flag = ! A .
6.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	a1	1	LDA (zp,X)	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	a1	1	LDA (zp,X)	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	0e	1		0007	1f	f0	80	fd	Nv-BdIzc	LDA (zp,X)	T2	e0	e0	e0
8	0007	0e	1		0007	1f	f0	80	fd	Nv-BdIzc	LDA (zp,X)	T2	e0	e0	c0
9	000e	00	1		0008	1f	f0	80	fd	Nv-BdIzc	LDA (zp,X)	T3	f0	0e	c0
9	000e	00	1		0008	1f	f0	80	fd	Nv-BdIzc	LDA (zp,X)	T3	f0	0e	fe
10	00fe	70	1		0008	1f	f0	80	fd	Nv-BdIzc	LDA (zp,X)	T4	00	fe	fe
10	00fe	70	1		0008	1f	f0	80	fd	Nv-BdIzc	LDA (zp,X)	T4	00	fe	ff
11	00ff	01	1		0008	1f	f0	80	fd	Nv-BdIzc	LDA (zp,X)	T5	00	70	ff
11	00ff	01	1		0008	1f	f0	80	fd	Nv-BdIzc	LDA (zp,X)	T5	00	70	70
12	0170	66	1		0008	1f	f0	80	fd	Nv-BdIzc	LDA (zp,X)	T0	ff	ff	70
12	0170	66	1		0008	1f	f0	80	fd	Nv-BdIzc	LDA (zp,X)	T0	ff	ff	fe
13	0008	ea	1	NOP	0008	66	f0	80	fd	nv-BdIzc	LDA (zp,X)	T1	66	66	fe
13	0008	ea	1	NOP	0008	66	f0	80	fd	nv-BdIzc	LDA (zp,X)	T1	66	66	cc
14	0009	ea	1		0009	66	f0	80	fd	nv-BdIzc	NOP	T0+T2	cc	cc	cc
14	0009	ea	1		0009	66	f0	80	fd	nv-BdIzc	NOP	T0+T2	cc	cc	98

LDX (A2)		Cycles: 2	Size: 2
Immediate			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Operand.	
2.1		Inc. PC . X = Operand. N flag = (X & \$80), Z flag = !X .	
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	a2	1	LDX #	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	a2	1	LDX #	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	0e	1		0007	1f	f0	80	fd	Nv-BdIzc	LDX #	T0+T2	e0	e0	e0
8	0007	0e	1		0007	1f	f0	80	fd	Nv-BdIzc	LDX #	T0+T2	e0	e0	c0
9	0008	ea	1	NOP	0008	1f	0e	80	fd	nv-BdIzc	LDX #	T1	0e	0e	c0
9	0008	ea	1	NOP	0008	1f	0e	80	fd	nv-BdIzc	LDX #	T1	0e	0e	1c
10	0009	ea	1		0009	1f	0e	80	fd	nv-BdIzc	NOP	T0+T2	1c	1c	1c
10	0009	ea	1		0009	1f	0e	80	fd	nv-BdIzc	NOP	T0+T2	1c	1c	38

LAX (A3)	Cycles: 6	Size: 2
Indirect X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC .
2.2	Read Operand	Discard.
3.1		Pointer = u8(Operand + X).
3.2	Read Pointer	Store as Address.
4.1		
4.2	Read u8(Pointer + 1)	Store as Address.H.
5.1		
5.2	Read Address	Store as Operand.
6.1		A = X = Operand. N flag = (A & \$80), Z flag = ! A .
6.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	a3	1	unknown	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	a3	1	unknown	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	0e	1		0007	1f	f0	80	fd	Nv-BdIzc	unknown	T2	e0	e0	e0
8	0007	0e	1		0007	1f	f0	80	fd	Nv-BdIzc	unknown	T2	e0	e0	c0
9	000e	00	1		0008	1f	f0	80	fd	Nv-BdIzc	unknown	T3	f0	0e	c0
9	000e	00	1		0008	1f	f0	80	fd	Nv-BdIzc	unknown	T3	f0	0e	fe
10	00fe	70	1		0008	1f	f0	80	fd	Nv-BdIzc	unknown	T4	00	fe	fe
10	00fe	70	1		0008	1f	f0	80	fd	Nv-BdIzc	unknown	T4	00	fe	ff
11	00ff	01	1		0008	1f	f0	80	fd	Nv-BdIzc	unknown	T5	00	70	ff
11	00ff	01	1		0008	1f	f0	80	fd	Nv-BdIzc	unknown	T5	00	70	70
12	0170	66	1		0008	1f	f0	80	fd	Nv-BdIzc	unknown	T0	ff	ff	70
12	0170	66	1		0008	1f	f0	80	fd	Nv-BdIzc	unknown	T0	ff	ff	fe
13	0008	ea	1	NOP	0008	66	66	80	fd	nv-BdIzc	unknown	T1	66	66	fe
13	0008	ea	1	NOP	0008	66	66	80	fd	nv-BdIzc	unknown	T1	66	66	cc
14	0009	ea	1		0009	66	66	80	fd	nv-BdIzc	NOP	T0+T2	cc	cc	cc
14	0009	ea	1		0009	66	66	80	fd	nv-BdIzc	NOP	T0+T2	cc	cc	98

LDY (A4)		Cycles: 3	Size: 2
Zero Page (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read Address	Store as Operand.	
3.1		Y = Operand. N flag = (Y & \$80), Z flag = ! Y .	
3.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	a4	1	LDY zp	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	a4	1	LDY zp	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	0e	1		0007	1f	f0	80	fd	Nv-BdIzc	LDY zp	T2	e0	e0	e0
8	0007	0e	1		0007	1f	f0	80	fd	Nv-BdIzc	LDY zp	T2	e0	e0	c0
9	000e	55	1		0008	1f	f0	80	fd	Nv-BdIzc	LDY zp	T0	ff	0e	c0
9	000e	55	1		0008	1f	f0	80	fd	Nv-BdIzc	LDY zp	T0	ff	0e	0d
10	0008	ea	1	NOP	0008	1f	f0	55	fd	nv-BdIzc	LDY zp	T1	55	55	0d
10	0008	ea	1	NOP	0008	1f	f0	55	fd	nv-BdIzc	LDY zp	T1	55	55	aa
11	0009	ea	1		0009	1f	f0	55	fd	nv-BdIzc	NOP	T0+T2	aa	aa	aa
11	0009	ea	1		0009	1f	f0	55	fd	nv-BdIzc	NOP	T0+T2	aa	aa	54

LDA (A5)	Cycles: 3	Size: 2
Zero Page (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read Address	Store as Operand.
3.1		A = Operand. N flag = (A & \$80), Z flag = ! A .
3.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	a5	1	LDA zp	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	a5	1	LDA zp	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	0e	1		0007	1f	f0	80	fd	Nv-BdIzc	LDA zp	T2	e0	e0	e0
8	0007	0e	1		0007	1f	f0	80	fd	Nv-BdIzc	LDA zp	T2	e0	e0	c0
9	000e	55	1		0008	1f	f0	80	fd	Nv-BdIzc	LDA zp	T0	ff	0e	c0
9	000e	55	1		0008	1f	f0	80	fd	Nv-BdIzc	LDA zp	T0	ff	0e	0d
10	0008	ea	1	NOP	0008	55	f0	80	fd	nv-BdIzc	LDA zp	T1	55	55	0d
10	0008	ea	1	NOP	0008	55	f0	80	fd	nv-BdIzc	LDA zp	T1	55	55	aa
11	0009	ea	1		0009	55	f0	80	fd	nv-BdIzc	NOP	T0+T2	aa	aa	aa
11	0009	ea	1		0009	55	f0	80	fd	nv-BdIzc	NOP	T0+T2	aa	aa	54

LDX (A6)		Cycles: 3	Size: 2
Zero Page (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read Address	Store as Operand.	
3.1		X = Operand. N flag = (X & \$80), Z flag = ! X .	
3.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	a6	1	LDX zp	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	a6	1	LDX zp	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	0e	1		0007	1f	f0	80	fd	Nv-BdIzc	LDX zp	T2	e0	e0	e0
8	0007	0e	1		0007	1f	f0	80	fd	Nv-BdIzc	LDX zp	T2	e0	e0	c0
9	000e	55	1		0008	1f	f0	80	fd	Nv-BdIzc	LDX zp	T0	ff	0e	c0
9	000e	55	1		0008	1f	f0	80	fd	Nv-BdIzc	LDX zp	T0	ff	0e	0d
10	0008	ea	1	NOP	0008	1f	55	80	fd	nv-BdIzc	LDX zp	T1	55	55	0d
10	0008	ea	1	NOP	0008	1f	55	80	fd	nv-BdIzc	LDX zp	T1	55	55	aa
11	0009	ea	1		0009	1f	55	80	fd	nv-BdIzc	NOP	T0+T2	aa	aa	aa
11	0009	ea	1		0009	1f	55	80	fd	nv-BdIzc	NOP	T0+T2	aa	aa	54

LAX (A7)	Cycles: 3	Size: 2
Zero Page (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read Address	Store as Operand.
3.1		A = X = Operand. N flag = (A & \$80), Z flag = ! A .
3.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	a7	1	unknown	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	a7	1	unknown	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	0f	1		0007	1f	f0	80	fd	Nv-BdIzc	unknown	T2	e0	e0	e0
8	0007	0f	1		0007	1f	f0	80	fd	Nv-BdIzc	unknown	T2	e0	e0	c0
9	000f	41	1		0008	1f	f0	80	fd	Nv-BdIzc	unknown	T0	ff	0f	c0
9	000f	41	1		0008	1f	f0	80	fd	Nv-BdIzc	unknown	T0	ff	0f	0e
10	0008	ea	1	NOP	0008	41	41	80	fd	nv-BdIzc	unknown	T1	41	41	0e
10	0008	ea	1	NOP	0008	41	41	80	fd	nv-BdIzc	unknown	T1	41	41	82
11	0009	ea	1		0009	41	41	80	fd	nv-BdIzc	NOP	T0+T2	82	82	82
11	0009	ea	1		0009	41	41	80	fd	nv-BdIzc	NOP	T0+T2	82	82	04

TAY (A8)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Discard.	
2.1		Y = A . N flag = (Y & \$80), Y flag = !X .	
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	a8	1	TAY	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	a8	1	TAY	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ea	1		0007	1f	f0	80	fd	Nv-BdIzc	TAY	T0+T2	e0	e0	e0
8	0007	ea	1		0007	1f	f0	80	fd	Nv-BdIzc	TAY	T0+T2	e0	e0	c0
9	0007	ea	1	NOP	0007	1f	f0	1f	fd	nv-BdIzc	TAY	T1	1f	1f	c0
9	0007	ea	1	NOP	0007	1f	f0	1f	fd	nv-BdIzc	TAY	T1	1f	1f	3e
10	0008	ea	1		0008	1f	f0	1f	fd	nv-BdIzc	NOP	T0+T2	3e	3e	3e
10	0008	ea	1		0008	1f	f0	1f	fd	nv-BdIzc	NOP	T0+T2	3e	3e	7c

LDA (A9)		Cycles: 2	Size: 2
Immediate			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Operand.	
2.1		Inc. PC . A = Operand. N flag = (A & \$80), Z flag = ! A .	
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	a9	1	LDA #	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	a9	1	LDA #	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	0f	1		0007	1f	f0	80	fd	Nv-BdIzc	LDA #	T0+T2	e0	e0	e0
8	0007	0f	1		0007	1f	f0	80	fd	Nv-BdIzc	LDA #	T0+T2	e0	e0	c0
9	0008	ea	1	NOP	0008	0f	f0	80	fd	nv-BdIzc	LDA #	T1	0f	0f	c0
9	0008	ea	1	NOP	0008	0f	f0	80	fd	nv-BdIzc	LDA #	T1	0f	0f	1e
10	0009	ea	1		0009	0f	f0	80	fd	nv-BdIzc	NOP	T0+T2	1e	1e	1e
10	0009	ea	1		0009	0f	f0	80	fd	nv-BdIzc	NOP	T0+T2	1e	1e	3c

TAX (AA)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Discard.	
2.1		X = A . N flag = (X & \$80), Z flag = ! X .	
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	aa	1	TAX	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	aa	1	TAX	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ea	1		0007	1f	f0	80	fd	Nv-BdIzc	TAX	T0+T2	e0	e0	e0
8	0007	ea	1		0007	1f	f0	80	fd	Nv-BdIzc	TAX	T0+T2	e0	e0	c0
9	0007	ea	1	NOP	0007	1f	1f	80	fd	nv-BdIzc	TAX	T1	1f	1f	c0
9	0007	ea	1	NOP	0007	1f	1f	80	fd	nv-BdIzc	TAX	T1	1f	1f	3e
10	0008	ea	1		0008	1f	1f	80	fd	nv-BdIzc	NOP	T0+T2	3e	3e	3e
10	0008	ea	1		0008	1f	1f	80	fd	nv-BdIzc	NOP	T0+T2	3e	3e	7c

LXA (AB)	Cycles: 2	Size: 2
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC.
1.2	Read PC	Store as Operand.
2.1		Inc. PC. For Single-Step Tests: A = X = ((A \$EE) & Operand). Otherwise: A = X = ((A \$FF) & Operand). N flag = (A & \$80), Z flag = !A.
2.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	ab	1	unknown	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	ab	1	unknown	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	fe	1		0007	1f	f0	80	fd	Nv-BdIzc	unknown	T0+T2	e0	e0	e0
8	0007	fe	1		0007	1f	f0	80	fd	Nv-BdIzc	unknown	T0+T2	e0	e0	c0
9	0008	ea	1	NOP	0008	1e	1e	80	fd	nv-BdIzc	unknown	T1	1e	1e	c0
9	0008	ea	1	NOP	0008	1e	1e	80	fd	nv-BdIzc	unknown	T1	1e	1e	3c
10	0009	ea	1		0009	1e	1e	80	fd	nv-BdIzc	NOP	T0+T2	3c	3c	3c
10	0009	ea	1		0009	1e	1e	80	fd	nv-BdIzc	NOP	T0+T2	3c	3c	78

LAX #imm (ATX, LXA, OAL, ANX)

Type: Combination of an immediate and an implied command

OpC.	Mnemonic	Function	Size	Cycles	N	V	-	B	D	I	Z	C
\$AB	LAX #imm	A,X = (A {CONST}) & #{imm}	2	2	o						o	

Operation: This opcode ORs the A register with CONST, ANDs the result with an immediate value, and then stores the result in both A and X.

Instability: CONST is chip- and/or temperature dependent (common values may be \$ee, \$00, \$ff, ...). Some dependency on the RDY line. Bit 0 and Bit 4 are “weaker” than the other bits, and may drop to 0 in the first cycle of DMA when RDY goes low.

Do not use LAX #imm with any immediate value other than 0, or when the accumulator value is Sff (both take the magic constant out of the equation)! (Or, more accurately, these are safe if all bits that could be 0 in A are 0 in the immediate value.)

LDY (AC)	Cycles: 4	Size: 3
Absolute (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address	Store as Operand.
4.1		Y = Operand. N flag = (Y & \$80), Z flag = ! Y .
4.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	ac	1	LDY Abs	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	ac	1	LDY Abs	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	70	1		0007	1f	f0	80	fd	Nv-BdIzc	LDY Abs	T2	e0	e0	e0
8	0007	70	1		0007	1f	f0	80	fd	Nv-BdIzc	LDY Abs	T2	e0	e0	c0
9	0008	01	1		0008	1f	f0	80	fd	Nv-BdIzc	LDY Abs	T3	00	70	c0
9	0008	01	1		0008	1f	f0	80	fd	Nv-BdIzc	LDY Abs	T3	00	70	70
10	0170	66	1		0009	1f	f0	80	fd	Nv-BdIzc	LDY Abs	T0	ff	ff	70
10	0170	66	1		0009	1f	f0	80	fd	Nv-BdIzc	LDY Abs	T0	ff	ff	fe
11	0009	ea	1	NOP	0009	1f	f0	66	fd	nv-BdIzc	LDY Abs	T1	66	66	fe
11	0009	ea	1	NOP	0009	1f	f0	66	fd	nv-BdIzc	LDY Abs	T1	66	66	cc
12	000a	15	1		000a	1f	f0	66	fd	nv-BdIzc	NOP	T0+T2	cc	cc	cc
12	000a	15	1		000a	1f	f0	66	fd	nv-BdIzc	NOP	T0+T2	cc	cc	98

LDA (AD)	Cycles: 4	Size: 3
Absolute (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address	Store as Operand.
4.1		A = Operand. N flag = (A & \$80), Z flag = ! A .
4.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	ad	1	LDA Abs	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	ad	1	LDA Abs	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	70	1		0007	1f	f0	80	fd	Nv-BdIzc	LDA Abs	T2	e0	e0	e0
8	0007	70	1		0007	1f	f0	80	fd	Nv-BdIzc	LDA Abs	T2	e0	e0	c0
9	0008	01	1		0008	1f	f0	80	fd	Nv-BdIzc	LDA Abs	T3	00	70	c0
9	0008	01	1		0008	1f	f0	80	fd	Nv-BdIzc	LDA Abs	T3	00	70	70
10	0170	66	1		0009	1f	f0	80	fd	Nv-BdIzc	LDA Abs	T0	ff	ff	70
10	0170	66	1		0009	1f	f0	80	fd	Nv-BdIzc	LDA Abs	T0	ff	ff	fe
11	0009	ea	1	NOP	0009	66	f0	80	fd	nv-BdIzc	LDA Abs	T1	66	66	fe
11	0009	ea	1	NOP	0009	66	f0	80	fd	nv-BdIzc	LDA Abs	T1	66	66	cc
12	000a	15	1		000a	66	f0	80	fd	nv-BdIzc	NOP	T0+T2	cc	cc	cc
12	000a	15	1		000a	66	f0	80	fd	nv-BdIzc	NOP	T0+T2	cc	cc	98

LDX (AE)		Cycles: 4	Size: 3
Absolute (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read PC	Store as Address.H.	
3.1		Inc. PC .	
3.2	Read Address	Store as Operand.	
4.1		X = Operand. N flag = (X & \$80), Z flag = ! X .	
4.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	ae	1	LDX Abs	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	ae	1	LDX Abs	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	70	1		0007	1f	f0	80	fd	Nv-BdIzc	LDX Abs	T2	e0	e0	e0
8	0007	70	1		0007	1f	f0	80	fd	Nv-BdIzc	LDX Abs	T2	e0	e0	c0
9	0008	01	1		0008	1f	f0	80	fd	Nv-BdIzc	LDX Abs	T3	00	70	c0
9	0008	01	1		0008	1f	f0	80	fd	Nv-BdIzc	LDX Abs	T3	00	70	70
10	0170	66	1		0009	1f	f0	80	fd	Nv-BdIzc	LDX Abs	T0	ff	ff	70
10	0170	66	1		0009	1f	f0	80	fd	Nv-BdIzc	LDX Abs	T0	ff	ff	fe
11	0009	ea	1	NOP	0009	1f	66	80	fd	nv-BdIzc	LDX Abs	T1	66	66	fe
11	0009	ea	1	NOP	0009	1f	66	80	fd	nv-BdIzc	LDX Abs	T1	66	66	cc
12	000a	15	1		000a	1f	66	80	fd	nv-BdIzc	NOP	T0+T2	cc	cc	cc
12	000a	15	1		000a	1f	66	80	fd	nv-BdIzc	NOP	T0+T2	cc	cc	98

LAX (AF)	Cycles: 4	Size: 3
Absolute (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address	Store as Operand.
4.1		A = X = Operand. N flag = (A & \$80), Z flag = ! A .
4.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	af	1	unknown	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	af	1	unknown	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	70	1		0007	1f	f0	80	fd	Nv-BdIzc	unknown	T2	e0	e0	e0
8	0007	70	1		0007	1f	f0	80	fd	Nv-BdIzc	unknown	T2	e0	e0	c0
9	0008	01	1		0008	1f	f0	80	fd	Nv-BdIzc	unknown	T3	00	70	c0
9	0008	01	1		0008	1f	f0	80	fd	Nv-BdIzc	unknown	T3	00	70	70
10	0170	66	1		0009	1f	f0	80	fd	Nv-BdIzc	unknown	T0	ff	ff	70
10	0170	66	1		0009	1f	f0	80	fd	Nv-BdIzc	unknown	T0	ff	ff	fe
11	0009	ea	1	NOP	0009	66	66	80	fd	nv-BdIzc	unknown	T1	66	66	fe
11	0009	ea	1	NOP	0009	66	66	80	fd	nv-BdIzc	unknown	T1	66	66	cc
12	000a	15	1		000a	66	66	80	fd	nv-BdIzc	NOP	T0+T2	cc	cc	cc
12	000a	15	1		000a	66	66	80	fd	nv-BdIzc	NOP	T0+T2	cc	cc	98

BCS (B0)		Cycles: 2-4	Size: 2
Relative (Jump)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC . Jump = ((P & \$01) == \$01).	
1.2	Read PC	Store as Operand.	
		If !Jump, poll interrupts.	
		Otherwise Address = i16(i8(Operand)) + PC + 1.	
		BoundaryCrossed = Address.H != (PC + 1).H.	
		If !BoundaryCrossed, poll interrupts.	
2.1		Inc. PC . If !Jump, end (next half-cycle is 4.2).	
2.2	Read PC	Discard.	
3.1		PC .L = Address.L. If !BoundaryCrossed, end (next half-cycle is 4.2).	
3.2	Read PC	Discard. Poll interrupts.	
4.1		PC .H = Address.H.	
4.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	b0	1	BCS	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	00
7	0006	b0	1	BCS	0006	1f	f0	80	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	70	1		0007	1f	f0	80	fd	Nv-BdIzc	BCS	T2	e0	e0	e0
8	0007	70	1		0007	1f	f0	80	fd	Nv-BdIzc	BCS	T2	e0	e0	c0
9	0008	ea	1	NOP	0008	1f	f0	80	fd	Nv-BdIzc	BCS		70	08	c0
9	0008	ea	1	NOP	0008	1f	f0	80	fd	Nv-BdIzc	BCS		70	08	78
10	0009	ea	1		0009	1f	f0	80	fd	Nv-BdIzc	NOP	T0+T2	78	78	78
10	0009	ea	1		0009	1f	f0	80	fd	Nv-BdIzc	NOP	T0+T2	78	78	f0

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
9	0006	b0	1	BCS	0006	0f	00	7f	fd	nv-BdIzc	NOP	T1	ff	ff	1e
9	0006	b0	1	BCS	0006	0f	00	7f	fd	nv-BdIzc	NOP	T1	ff	ff	fe
10	0007	70	1		0007	0f	00	7f	fd	nv-BdIzc	BCS	T2	fe	fe	fe
10	0007	70	1		0007	0f	00	7f	fd	nv-BdIzc	BCS	T2	fe	fe	fc
11	0008	ea	1		0008	0f	00	7f	fd	nv-BdIzc	BCS	T3	70	08	fc
11	0008	ea	1		0008	0f	00	7f	fd	nv-BdIzc	BCS	T3	70	08	78
12	0078	ea	1	NOP	0078	0f	00	7f	fd	nv-BdIzc	BCS		00	00	78
12	0078	ea	1	NOP	0078	0f	00	7f	fd	nv-BdIzc	BCS		00	00	01
13	0079	90	1		0079	0f	00	7f	fd	nv-BdIzc	NOP	T0+T2	01	01	01
13	0079	90	1		0079	0f	00	7f	fd	nv-BdIzc	NOP	T0+T2	01	01	02

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
9	0006	b0	1	BCS	0006	0f	00	7f	fd	nv-BdIzc	NOP	T1	ff	ff	1e
9	0006	b0	1	BCS	0006	0f	00	7f	fd	nv-BdIzc	NOP	T1	ff	ff	fe
10	0007	ed	1		0007	0f	00	7f	fd	nv-BdIzc	BCS	T2	fe	fe	fe
10	0007	ed	1		0007	0f	00	7f	fd	nv-BdIzc	BCS	T2	fe	fe	fc
11	0008	ea	1		0008	0f	00	7f	fd	nv-BdIzc	BCS	T3	ed	08	fc
11	0008	ea	1		0008	0f	00	7f	fd	nv-BdIzc	BCS	T3	ed	08	f5
12	00f5	00	1		00f5	0f	00	7f	fd	nv-BdIzc	BCS	T0	00	ff	f5
12	00f5	00	1		00f5	0f	00	7f	fd	nv-BdIzc	BCS	T0	00	ff	ff
13	fff5	00	1	BRK	fff5	0f	00	7f	fd	nv-BdIzc	BCS	T1	ff	00	ff
13	fff5	00	1	BRK	fff5	0f	00	7f	fd	nv-BdIzc	BCS	T1	ff	00	ff
14	fff6	00	1		fff6	0f	00	7f	fd	nv-BdIzc	BRK	T2	ff	ff	ff
14	fff6	00	1		fff6	0f	00	7f	fd	nv-BdIzc	BRK	T2	ff	ff	fe

LDA (B1) Cycles: 5-6		Size: 2
Indirect Y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read Pointer	Store as Address.
3.1		
3.2	Read u8(Pointer + 1)	Store as Address.H.
4.1		Target = Address + Y. Pointer.L = Target.L. Pointer.H = Address.H. BoundaryCrossed = Pointer.H != Target.H.
4.2	Read Pointer	Store as Operand. If !BoundaryCrossed, skip the next cycle.
5.1		Pointer.H = Target.H.
5.2	Read Pointer	Store as Operand.
6.1		A = Operand. N flag = (A & \$80), Z flag = ! A .
6.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	b1	1	LDA (zp),Y	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	fc
7	0006	b1	1	LDA (zp),Y	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	22	1		0007	1f	f0	7f	fd	Nv-BdIzc	LDA (zp),Y	T2	e0	e0	e0
8	0007	22	1		0007	1f	f0	7f	fd	Nv-BdIzc	LDA (zp),Y	T2	e0	e0	c0
9	0022	24	1		0008	1f	f0	7f	fd	Nv-BdIzc	LDA (zp),Y	T3	00	22	c0
9	0022	24	1		0008	1f	f0	7f	fd	Nv-BdIzc	LDA (zp),Y	T3	00	22	23
10	0023	01	1		0008	1f	f0	7f	fd	Nv-BdIzc	LDA (zp),Y	T4	7f	24	23
10	0023	01	1		0008	1f	f0	7f	fd	Nv-BdIzc	LDA (zp),Y	T4	7f	24	a3
11	01a3	66	1		0008	1f	f0	7f	fd	Nv-BdIzc	LDA (zp),Y	T0	00	01	a3
11	01a3	66	1		0008	1f	f0	7f	fd	Nv-BdIzc	LDA (zp),Y	T0	00	01	02
12	0008	ea	1	NOP	0008	66	f0	7f	fd	nv-BdIzc	LDA (zp),Y	T1	66	66	02
12	0008	ea	1	NOP	0008	66	f0	7f	fd	nv-BdIzc	LDA (zp),Y	T1	66	66	cc
13	0009	ea	1		0009	66	f0	7f	fd	nv-BdIzc	NOP	T0+T2	cc	cc	cc
13	0009	ea	1		0009	66	f0	7f	fd	nv-BdIzc	NOP	T0+T2	cc	cc	98

JAM (B2)	Cycles: ∞	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Discard.
2.1		Visual6502 suggests incrementing PC , but Single-Step Tests cannot be passed this way.
2.2	Read \$FFFF	Discard.
3.1		
3.2	Read \$FFFE	Discard.
4.1		
4.2	Read \$FFFE	Discard.
5.1		Repeat 5.1 and 5.2 forever.
5.2	Read \$FFFF	Discard.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	b2	1	unknown	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	fc
7	0006	b2	1	unknown	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	22	1		0007	1f	f0	7f	fd	Nv-BdIzc	unknown	T2	e0	e0	e0
8	0007	22	1		0007	1f	f0	7f	fd	Nv-BdIzc	unknown	T2	e0	e0	c0
9	ffff	00	1		0008	1f	f0	7f	fd	Nv-BdIzc	unknown	T3	ff	ff	c0
9	ffff	00	1		0008	1f	f0	7f	fd	Nv-BdIzc	unknown	T3	ff	ff	fe
10	fffe	00	1		0008	1f	f0	7f	fd	Nv-BdIzc	unknown	T4	ff	ff	fe
10	fffe	00	1		0008	1f	f0	7f	fd	Nv-BdIzc	unknown	T4	ff	ff	fe
11	fffe	00	1		0008	1f	f0	7f	fd	Nv-BdIzc	unknown	T5	ff	ff	fe
11	fffe	00	1		0008	1f	f0	7f	fd	Nv-BdIzc	unknown	T5	ff	ff	fe
12	ffff	00	1		0008	1f	f0	7f	fd	Nv-BdIzc	unknown		ff	ff	fe
12	ffff	00	1		0008	1f	f0	7f	fd	Nv-BdIzc	unknown		ff	ff	fe
13	ffff	00	1		0008	1f	f0	7f	fd	Nv-BdIzc	unknown		ff	ff	fe
13	ffff	00	1		0008	1f	f0	7f	fd	Nv-BdIzc	unknown		ff	ff	fe
14	ffff	00	1		0008	1f	f0	7f	fd	Nv-BdIzc	unknown		ff	ff	fe
14	ffff	00	1		0008	1f	f0	7f	fd	Nv-BdIzc	unknown		ff	ff	fe

LAX (B3)	Cycles: 5-6	Size: 2
Indirect Y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read Pointer	Store as Address.
3.1		
3.2	Read u8(Pointer + 1)	Store as Address.H.
4.1		Target = Address + Y. Pointer.L = Target.L. Pointer.H = Address.H. BoundaryCrossed = Pointer.H != Target.H.
4.2	Read Pointer	Store as Operand. If !BoundaryCrossed, skip the next cycle.
5.1		Pointer.H = Target.H.
5.2	Read Pointer	Store as Operand.
6.1		A = X = Operand. N flag = (A & \$80), Z flag = ! A .
6.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	b3	1	unknown	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	fc
7	0006	b3	1	unknown	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	22	1		0007	1f	f0	7f	fd	Nv-BdIzc	unknown	T2	e0	e0	e0
8	0007	22	1		0007	1f	f0	7f	fd	Nv-BdIzc	unknown	T2	e0	e0	c0
9	0022	24	1		0008	1f	f0	7f	fd	Nv-BdIzc	unknown	T3	00	22	c0
9	0022	24	1		0008	1f	f0	7f	fd	Nv-BdIzc	unknown	T3	00	22	23
10	0023	01	1		0008	1f	f0	7f	fd	Nv-BdIzc	unknown	T4	7f	24	23
10	0023	01	1		0008	1f	f0	7f	fd	Nv-BdIzc	unknown	T4	7f	24	a3
11	01a3	66	1		0008	1f	f0	7f	fd	Nv-BdIzc	unknown	T0	00	01	a3
11	01a3	66	1		0008	1f	f0	7f	fd	Nv-BdIzc	unknown	T0	00	01	02
12	0008	ea	1	NOP	0008	66	66	7f	fd	nv-BdIzc	unknown	T1	66	66	02
12	0008	ea	1	NOP	0008	66	66	7f	fd	nv-BdIzc	unknown	T1	66	66	cc
13	0009	ea	1		0009	66	66	7f	fd	nv-BdIzc	NOP	T0+T2	cc	cc	cc
13	0009	ea	1		0009	66	66	7f	fd	nv-BdIzc	NOP	T0+T2	cc	cc	98

LDY (B4)		Cycles: 4	Size: 2
Zero Page X-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Operand.	
3.1		Address = u8(Pointer + X).	
3.2	Read Address	Store as Operand.	
4.1		Y = Operand. N flag = (Y & \$80), Z flag = ! Y .	
4.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	b4	1	LDY zp,X	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	fc
7	0006	b4	1	LDY zp,X	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	22	1		0007	1f	f0	7f	fd	Nv-BdIzc	LDY zp,X	T2	e0	e0	e0
8	0007	22	1		0007	1f	f0	7f	fd	Nv-BdIzc	LDY zp,X	T2	e0	e0	c0
9	0022	24	1		0008	1f	f0	7f	fd	Nv-BdIzc	LDY zp,X	T3	f0	22	c0
9	0022	24	1		0008	1f	f0	7f	fd	Nv-BdIzc	LDY zp,X	T3	f0	22	12
10	0012	77	1		0008	1f	f0	7f	fd	Nv-BdIzc	LDY zp,X	T0	ff	ff	12
10	0012	77	1		0008	1f	f0	7f	fd	Nv-BdIzc	LDY zp,X	T0	ff	ff	fe
11	0008	ea	1	NOP	0008	1f	f0	77	fd	nv-BdIzc	LDY zp,X	T1	77	77	fe
11	0008	ea	1	NOP	0008	1f	f0	77	fd	nv-BdIzc	LDY zp,X	T1	77	77	ee
12	0009	ea	1		0009	1f	f0	77	fd	nv-BdIzc	NOP	T0+T2	ee	ee	ee
12	0009	ea	1		0009	1f	f0	77	fd	nv-BdIzc	NOP	T0+T2	ee	ee	dc

LDA (B5)	Cycles: 4	Size: 2
Zero Page X-Indexed (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read Pointer	Store as Operand.
3.1		Address = u8(Pointer + X).
3.2	Read Address	Store as Operand.
4.1		A = Operand. N flag = (A & \$80), Z flag = ! A .
4.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	b5	1	LDA zp,X	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	fc
7	0006	b5	1	LDA zp,X	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	22	1		0007	1f	f0	7f	fd	Nv-BdIzc	LDA zp,X	T2	e0	e0	e0
8	0007	22	1		0007	1f	f0	7f	fd	Nv-BdIzc	LDA zp,X	T2	e0	e0	c0
9	0022	24	1		0008	1f	f0	7f	fd	Nv-BdIzc	LDA zp,X	T3	f0	22	c0
9	0022	24	1		0008	1f	f0	7f	fd	Nv-BdIzc	LDA zp,X	T3	f0	22	12
10	0012	77	1		0008	1f	f0	7f	fd	Nv-BdIzc	LDA zp,X	T0	ff	ff	12
10	0012	77	1		0008	1f	f0	7f	fd	Nv-BdIzc	LDA zp,X	T0	ff	ff	fe
11	0008	ea	1	NOP	0008	77	f0	7f	fd	nv-BdIzc	LDA zp,X	T1	77	77	fe
11	0008	ea	1	NOP	0008	77	f0	7f	fd	nv-BdIzc	LDA zp,X	T1	77	77	ee
12	0009	ea	1		0009	77	f0	7f	fd	nv-BdIzc	NOP	T0+T2	ee	ee	ee
12	0009	ea	1		0009	77	f0	7f	fd	nv-BdIzc	NOP	T0+T2	ee	ee	dc

LDX (B6)		Cycles: 4	Size: 2
Zero Page Y-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Operand.	
3.1		Target = Pointer + Y .	
		Address.L = Target.L.	
		Address.H = Pointer.H.	
3.2	Read Address	Store as Operand.	
4.1		X = Operand. N flag = (X & \$80), Z flag = ! X .	
4.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	b6	1	LDX zp,Y	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	fc
7	0006	b6	1	LDX zp,Y	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	22	1		0007	1f	f0	7f	fd	Nv-BdIzc	LDX zp,Y	T2	e0	e0	e0
8	0007	22	1		0007	1f	f0	7f	fd	Nv-BdIzc	LDX zp,Y	T2	e0	e0	c0
9	0022	24	1		0008	1f	f0	7f	fd	Nv-BdIzc	LDX zp,Y	T3	7f	22	c0
9	0022	24	1		0008	1f	f0	7f	fd	Nv-BdIzc	LDX zp,Y	T3	7f	22	a1
10	00a1	44	1		0008	1f	f0	7f	fd	Nv-BdIzc	LDX zp,Y	T0	ff	ff	a1
10	00a1	44	1		0008	1f	f0	7f	fd	Nv-BdIzc	LDX zp,Y	T0	ff	ff	fe
11	0008	ea	1	NOP	0008	1f	44	7f	fd	nv-BdIzc	LDX zp,Y	T1	44	44	fe
11	0008	ea	1	NOP	0008	1f	44	7f	fd	nv-BdIzc	LDX zp,Y	T1	44	44	88
12	0009	ea	1		0009	1f	44	7f	fd	nv-BdIzc	NOP	T0+T2	88	88	88
12	0009	ea	1		0009	1f	44	7f	fd	nv-BdIzc	NOP	T0+T2	88	88	10

LAX (B7)	Cycles: 4	Size: 2
Zero Page Y-Indexed (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read Pointer	Store as Operand.
3.1		Target = Pointer + Y . Address.L = Target.L. Address.H = Pointer.H.
3.2	Read Address	Store as Operand.
4.1		A = X = Operand. N flag = (A & \$80), Z flag = ! A .
4.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	b7	1	unknown	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	fc
7	0006	b7	1	unknown	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	22	1		0007	1f	f0	7f	fd	Nv-BdIzc	unknown	T2	e0	e0	e0
8	0007	22	1		0007	1f	f0	7f	fd	Nv-BdIzc	unknown	T2	e0	e0	c0
9	0022	24	1		0008	1f	f0	7f	fd	Nv-BdIzc	unknown	T3	7f	22	c0
9	0022	24	1		0008	1f	f0	7f	fd	Nv-BdIzc	unknown	T3	7f	22	a1
10	00a1	44	1		0008	1f	f0	7f	fd	Nv-BdIzc	unknown	T0	ff	ff	a1
10	00a1	44	1		0008	1f	f0	7f	fd	Nv-BdIzc	unknown	T0	ff	ff	fe
11	0008	ea	1	NOP	0008	44	44	7f	fd	nv-BdIzc	unknown	T1	44	44	fe
11	0008	ea	1	NOP	0008	44	44	7f	fd	nv-BdIzc	unknown	T1	44	44	88
12	0009	ea	1		0009	44	44	7f	fd	nv-BdIzc	NOP	T0+T2	88	88	88
12	0009	ea	1		0009	44	44	7f	fd	nv-BdIzc	NOP	T0+T2	88	88	10

CLV (B8)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Discard.	
2.1		V flag = 0.	
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	b8	1	CLV	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	fc
7	0006	b8	1	CLV	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ea	1		0007	1f	f0	7f	fd	Nv-BdIzc	CLV	T0+T2	e0	e0	e0
8	0007	ea	1		0007	1f	f0	7f	fd	Nv-BdIzc	CLV	T0+T2	e0	e0	c0
9	0007	ea	1	NOP	0007	1f	f0	7f	fd	Nv-BdIzc	CLV	T1	ff	ff	c0
9	0007	ea	1	NOP	0007	1f	f0	7f	fd	Nv-BdIzc	CLV	T1	ff	ff	fe
10	0008	ea	1		0008	1f	f0	7f	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fe
10	0008	ea	1		0008	1f	f0	7f	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fc

LDA (B9)	Cycles: 4-5	Size: 3
Absolute Y-Indexed (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + Y. Address.L = Target.L. Address.H = Pointer.H. BoundaryCrossed = Pointer.H != Target.H.
3.2	Read Address	Store as Operand. If !BoundaryCrossed, skip the next cycle.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		A = Operand. N flag = (A & \$80), Z flag = ! A .
5.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	b9	1	LDA Abs,Y	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	fc
7	0006	b9	1	LDA Abs,Y	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	77	1		0007	1f	f0	7f	fd	Nv-BdIzc	LDA Abs,Y	T2	e0	e0	e0
8	0007	77	1		0007	1f	f0	7f	fd	Nv-BdIzc	LDA Abs,Y	T2	e0	e0	c0
9	0008	01	1		0008	1f	f0	7f	fd	Nv-BdIzc	LDA Abs,Y	T3	7f	77	c0
9	0008	01	1		0008	1f	f0	7f	fd	Nv-BdIzc	LDA Abs,Y	T3	7f	77	f6
10	01f6	66	1		0009	1f	f0	7f	fd	Nv-BdIzc	LDA Abs,Y	T0	00	01	f6
10	01f6	66	1		0009	1f	f0	7f	fd	Nv-BdIzc	LDA Abs,Y	T0	00	01	02
11	0009	ea	1	NOP	0009	66	f0	7f	fd	nv-BdIzc	LDA Abs,Y	T1	66	66	02
11	0009	ea	1	NOP	0009	66	f0	7f	fd	nv-BdIzc	LDA Abs,Y	T1	66	66	cc
12	000a	15	1		000a	66	f0	7f	fd	nv-BdIzc	NOP	T0+T2	cc	cc	cc
12	000a	15	1		000a	66	f0	7f	fd	nv-BdIzc	NOP	T0+T2	cc	cc	98

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	b9	1	LDA Abs,Y	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	fc
7	0006	b9	1	LDA Abs,Y	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ff	1		0007	1f	f0	7f	fd	Nv-BdIzc	LDA Abs,Y	T2	e0	e0	e0
8	0007	ff	1		0007	1f	f0	7f	fd	Nv-BdIzc	LDA Abs,Y	T2	e0	e0	c0
9	0008	00	1		0008	1f	f0	7f	fd	Nv-BdIzc	LDA Abs,Y	T3	7f	ff	c0
9	0008	00	1		0008	1f	f0	7f	fd	Nv-BdIzc	LDA Abs,Y	T3	7f	ff	7e
10	007e	33	1		0009	1f	f0	7f	fd	Nv-BdIzc	LDA Abs,Y	T4	00	00	7e
10	007e	33	1		0009	1f	f0	7f	fd	Nv-BdIzc	LDA Abs,Y	T4	00	00	01
11	017e	66	1		0009	1f	f0	7f	fd	Nv-BdIzc	LDA Abs,Y	T0	01	ff	01
11	017e	66	1		0009	1f	f0	7f	fd	Nv-BdIzc	LDA Abs,Y	T0	01	ff	00
12	0009	ea	1	NOP	0009	66	f0	7f	fd	nv-BdIzc	LDA Abs,Y	T1	66	66	00
12	0009	ea	1	NOP	0009	66	f0	7f	fd	nv-BdIzc	LDA Abs,Y	T1	66	66	cc
13	000a	15	1		000a	66	f0	7f	fd	nv-BdIzc	NOP	T0+T2	cc	cc	cc
13	000a	15	1		000a	66	f0	7f	fd	nv-BdIzc	NOP	T0+T2	cc	cc	98

TSX (BA)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Discard.	
2.1		X = S. N flag = (X & \$80), Z flag = ! X .	
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	ba	1	TSX	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	fc
7	0006	ba	1	TSX	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ea	1		0007	1f	f0	7f	fd	Nv-BdIzc	TSX	T0+T2	e0	e0	e0
8	0007	ea	1		0007	1f	f0	7f	fd	Nv-BdIzc	TSX	T0+T2	e0	e0	c0
9	0007	ea	1	NOP	0007	1f	fd	7f	fd	Nv-BdIzc	TSX	T1	fd	fd	c0
9	0007	ea	1	NOP	0007	1f	fd	7f	fd	Nv-BdIzc	TSX	T1	fd	fd	fa
10	0008	00	1		0008	1f	fd	7f	fd	Nv-BdIzc	NOP	T0+T2	fa	fa	fa
10	0008	00	1		0008	1f	fd	7f	fd	Nv-BdIzc	NOP	T0+T2	fa	fa	f4

LAS (BB)		Cycles: 4-5	Size: 3
Absolute Y-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read PC	Store as Pointer.H.	
3.1		Inc. PC . Target = Pointer + Y.	
		Address.L = Target.L.	
		Address.H = Pointer.H.	
		BoundaryCrossed = Pointer.H != Target.H.	
3.2	Read Address	Store as Operand. If !BoundaryCrossed, skip the next cycle.	
4.1		Address.H = Target.H.	
4.2	Read Address	Store as Operand.	
5.1		* A = X = S = (Operand & S). N flag = (A & \$80), Z flag = ! A .	
5.2	Read PC	Store as OpCode.	
+X.1			

* Visual6502 omits (S &= Operand); flags are updated based off the incorrect A value.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	bb	1	unknown	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	fc
7	0006	bb	1	unknown	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ea	1		0007	1f	f0	7f	fd	Nv-BdIzc	unknown	T2	e0	e0	e0
8	0007	ea	1		0007	1f	f0	7f	fd	Nv-BdIzc	unknown	T2	e0	e0	c0
9	0008	00	1		0008	1f	f0	7f	fd	Nv-BdIzc	unknown	T3	7f	ea	c0
9	0008	00	1		0008	1f	f0	7f	fd	Nv-BdIzc	unknown	T3	7f	ea	69
10	0069	00	1		0009	1f	f0	7f	fd	Nv-BdIzc	unknown	T4	00	00	69
10	0069	00	1		0009	1f	f0	7f	fd	Nv-BdIzc	unknown	T4	00	00	01
11	0169	66	1		0009	1f	f0	7f	fd	Nv-BdIzc	unknown	T0	01	ff	01
11	0169	66	1		0009	1f	f0	7f	fd	Nv-BdIzc	unknown	T0	01	ff	00
12	0009	ea	1	NOP	0009	64	64	7f	64	nv-BdIzc	unknown	T1	64	64	00
12	0009	ea	1	NOP	0009	75	fd	7f	fd	nv-BdIzc	unknown	T1	fd	fd	fa
13	000a	15	1		000a	75	fd	7f	fd	nv-BdIzc	NOP	T0+T2	fa	fa	fa
13	000a	15	1		000a	75	fd	7f	fd	nv-BdIzc	NOP	T0+T2	fa	fa	f4

LDY (BC)	Cycles: 4-5	Size: 3
Absolute X-Indexed (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + X . Address.L = Target.L. Address.H = Pointer.H. BoundaryCrossed = Pointer.H != Target.H.
3.2	Read Address	Store as Operand. If !BoundaryCrossed, skip the next cycle.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		Y = Operand. N flag = (Y & \$80), Z flag = ! Y .
5.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	bc	1	LDY Abs,X	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	fc
7	0006	bc	1	LDY Abs,X	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	c0
8	0007	11	1		0007	1f	e0	7f	fd	Nv-BdIzc	LDY Abs,X	T2	c0	c0	c0
8	0007	11	1		0007	1f	e0	7f	fd	Nv-BdIzc	LDY Abs,X	T2	c0	c0	80
9	0008	01	1		0008	1f	e0	7f	fd	Nv-BdIzc	LDY Abs,X	T3	e0	11	80
9	0008	01	1		0008	1f	e0	7f	fd	Nv-BdIzc	LDY Abs,X	T3	e0	11	f1
10	01f1	66	1		0009	1f	e0	7f	fd	Nv-BdIzc	LDY Abs,X	T0	00	01	f1
10	01f1	66	1		0009	1f	e0	7f	fd	Nv-BdIzc	LDY Abs,X	T0	00	01	02
11	0009	ea	1	NOP	0009	1f	e0	66	fd	nv-BdIzc	LDY Abs,X	T1	66	66	02
11	0009	ea	1	NOP	0009	1f	e0	66	fd	nv-BdIzc	LDY Abs,X	T1	66	66	cc
12	000a	15	1		000a	1f	e0	66	fd	nv-BdIzc	NOP	T0+T2	cc	cc	cc
12	000a	15	1		000a	1f	e0	66	fd	nv-BdIzc	NOP	T0+T2	cc	cc	98

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	bc	1	LDY Abs,X	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	fc
7	0006	bc	1	LDY Abs,X	0006	1f	f0	7f	fd	Nv-BdIzc	LDX #	T1	f0	f0	e0
8	0007	ea	1		0007	1f	f0	7f	fd	Nv-BdIzc	LDY Abs,X	T2	e0	e0	e0
8	0007	ea	1		0007	1f	f0	7f	fd	Nv-BdIzc	LDY Abs,X	T2	e0	e0	c0
9	0008	00	1		0008	1f	f0	7f	fd	Nv-BdIzc	LDY Abs,X	T3	f0	ea	c0
9	0008	00	1		0008	1f	f0	7f	fd	Nv-BdIzc	LDY Abs,X	T3	f0	ea	da
10	00da	00	1		0009	1f	f0	7f	fd	Nv-BdIzc	LDY Abs,X	T4	00	00	da
10	00da	00	1		0009	1f	f0	7f	fd	Nv-BdIzc	LDY Abs,X	T4	00	00	01
11	01da	66	1		0009	1f	f0	7f	fd	Nv-BdIzc	LDY Abs,X	T0	01	ff	01
11	01da	66	1		0009	1f	f0	7f	fd	Nv-BdIzc	LDY Abs,X	T0	01	ff	00
12	0009	ea	1	NOP	0009	1f	f0	66	fd	nv-BdIzc	LDY Abs,X	T1	66	66	00
12	0009	ea	1	NOP	0009	1f	f0	66	fd	nv-BdIzc	LDY Abs,X	T1	66	66	cc
13	000a	15	1		000a	1f	f0	66	fd	nv-BdIzc	NOP	T0+T2	cc	cc	cc
13	000a	15	1		000a	1f	f0	66	fd	nv-BdIzc	NOP	T0+T2	cc	cc	98

LDA (BD)	Cycles: 4-5	Size: 3
Absolute X-Indexed (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + X . Address.L = Target.L. Address.H = Pointer.H. BoundaryCrossed = Pointer.H != Target.H.
3.2	Read Address	Store as Operand. If !BoundaryCrossed, skip the next cycle.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		A = Operand. N flag = (A & \$80), Z flag = ! A .
5.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	bd	1	LDA Abs,X	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	fc
7	0006	bd	1	LDA Abs,X	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	c0
8	0007	11	1		0007	1f	e0	7f	fd	Nv-BdIzc	LDA Abs,X	T2	c0	c0	c0
8	0007	11	1		0007	1f	e0	7f	fd	Nv-BdIzc	LDA Abs,X	T2	c0	c0	80
9	0008	01	1		0008	1f	e0	7f	fd	Nv-BdIzc	LDA Abs,X	T3	e0	11	80
9	0008	01	1		0008	1f	e0	7f	fd	Nv-BdIzc	LDA Abs,X	T3	e0	11	f1
10	01f1	66	1		0009	1f	e0	7f	fd	Nv-BdIzc	LDA Abs,X	T0	00	01	f1
10	01f1	66	1		0009	1f	e0	7f	fd	Nv-BdIzc	LDA Abs,X	T0	00	01	02
11	0009	ea	1	NOP	0009	66	e0	7f	fd	nv-BdIzc	LDA Abs,X	T1	66	66	02
11	0009	ea	1	NOP	0009	66	e0	7f	fd	nv-BdIzc	LDA Abs,X	T1	66	66	cc
12	000a	15	1		000a	66	e0	7f	fd	nv-BdIzc	NOP	T0+T2	cc	cc	cc
12	000a	15	1		000a	66	e0	7f	fd	nv-BdIzc	NOP	T0+T2	cc	cc	98

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	bd	1	LDA Abs,X	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	fc
7	0006	bd	1	LDA Abs,X	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	c0
8	0007	f0	1		0007	1f	e0	7f	fd	Nv-BdIzc	LDA Abs,X	T2	c0	c0	c0
8	0007	f0	1		0007	1f	e0	7f	fd	Nv-BdIzc	LDA Abs,X	T2	c0	c0	80
9	0008	01	1		0008	1f	e0	7f	fd	Nv-BdIzc	LDA Abs,X	T3	e0	f0	80
9	0008	01	1		0008	1f	e0	7f	fd	Nv-BdIzc	LDA Abs,X	T3	e0	f0	d0
10	01d0	00	1		0009	1f	e0	7f	fd	Nv-BdIzc	LDA Abs,X	T4	00	01	d0
10	01d0	00	1		0009	1f	e0	7f	fd	Nv-BdIzc	LDA Abs,X	T4	00	01	02
11	02d0	00	1		0009	1f	e0	7f	fd	Nv-BdIzc	LDA Abs,X	T0	02	ff	02
11	02d0	00	1		0009	1f	e0	7f	fd	Nv-BdIzc	LDA Abs,X	T0	02	ff	01
12	0009	ea	1	NOP	0009	00	e0	7f	fd	nv-BdIzc	LDA Abs,X	T1	00	00	01
12	0009	ea	1	NOP	0009	00	e0	7f	fd	nv-BdIzc	LDA Abs,X	T1	00	00	00
13	000a	15	1		000a	00	e0	7f	fd	nv-BdIzc	NOP	T0+T2	00	00	00
13	000a	15	1		000a	00	e0	7f	fd	nv-BdIzc	NOP	T0+T2	00	00	00

LDX (BE)	Cycles: 4-5	Size: 3
Absolute Y-Indexed (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + Y. Address.L = Target.L. Address.H = Pointer.H. BoundaryCrossed = Pointer.H != Target.H.
3.2	Read Address	Store as Operand. If !BoundaryCrossed, skip the next cycle.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		X = Operand. N flag = (X & \$80), Z flag = ! X .
5.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	be	1	LDX Abs,Y	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	fc
7	0006	be	1	LDX Abs,Y	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	c0
8	0007	11	1		0007	1f	e0	7f	fd	Nv-BdIzc	LDX Abs,Y	T2	c0	c0	c0
8	0007	11	1		0007	1f	e0	7f	fd	Nv-BdIzc	LDX Abs,Y	T2	c0	c0	80
9	0008	01	1		0008	1f	e0	7f	fd	Nv-BdIzc	LDX Abs,Y	T3	7f	11	80
9	0008	01	1		0008	1f	e0	7f	fd	Nv-BdIzc	LDX Abs,Y	T3	7f	11	90
10	0190	66	1		0009	1f	e0	7f	fd	Nv-BdIzc	LDX Abs,Y	T0	00	01	90
10	0190	66	1		0009	1f	e0	7f	fd	Nv-BdIzc	LDX Abs,Y	T0	00	01	02
11	0009	ea	1	NOP	0009	1f	66	7f	fd	nv-BdIzc	LDX Abs,Y	T1	66	66	02
11	0009	ea	1	NOP	0009	1f	66	7f	fd	nv-BdIzc	LDX Abs,Y	T1	66	66	cc
12	000a	15	1		000a	1f	66	7f	fd	nv-BdIzc	NOP	T0+T2	cc	cc	cc
12	000a	15	1		000a	1f	66	7f	fd	nv-BdIzc	NOP	T0+T2	cc	cc	98

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	be	1	LDX Abs,Y	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	fc
7	0006	be	1	LDX Abs,Y	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	c0
8	0007	f0	1		0007	1f	e0	7f	fd	Nv-BdIzc	LDX Abs,Y	T2	c0	c0	c0
8	0007	f0	1		0007	1f	e0	7f	fd	Nv-BdIzc	LDX Abs,Y	T2	c0	c0	80
9	0008	01	1		0008	1f	e0	7f	fd	Nv-BdIzc	LDX Abs,Y	T3	7f	f0	80
9	0008	01	1		0008	1f	e0	7f	fd	Nv-BdIzc	LDX Abs,Y	T3	7f	f0	6f
10	016f	00	1		0009	1f	e0	7f	fd	Nv-BdIzc	LDX Abs,Y	T4	00	01	6f
10	016f	00	1		0009	1f	e0	7f	fd	Nv-BdIzc	LDX Abs,Y	T4	00	01	02
11	026f	00	1		0009	1f	e0	7f	fd	Nv-BdIzc	LDX Abs,Y	T0	02	ff	02
11	026f	00	1		0009	1f	e0	7f	fd	Nv-BdIzc	LDX Abs,Y	T0	02	ff	01
12	0009	ea	1	NOP	0009	1f	00	7f	fd	nv-BdIzc	LDX Abs,Y	T1	00	00	01
12	0009	ea	1	NOP	0009	1f	00	7f	fd	nv-BdIzc	LDX Abs,Y	T1	00	00	00
13	000a	15	1		000a	1f	00	7f	fd	nv-BdIzc	NOP	T0+T2	00	00	00
13	000a	15	1		000a	1f	00	7f	fd	nv-BdIzc	NOP	T0+T2	00	00	00

LAX (BF)	Cycles: 4-5	Size: 3
Absolute Y-Indexed (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + Y. Address.L = Target.L. Address.H = Pointer.H. BoundaryCrossed = Pointer.H != Target.H.
3.2	Read Address	Store as Operand. If !BoundaryCrossed, skip the next cycle.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		A = X = Operand. N flag = (A & \$80), Z flag = ! A .
5.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	bf	1	unknown	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	fc
7	0006	bf	1	unknown	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	c0
8	0007	f0	1		0007	1f	e0	7f	fd	Nv-BdIzc	unknown	T2	c0	c0	c0
8	0007	f0	1		0007	1f	e0	7f	fd	Nv-BdIzc	unknown	T2	c0	c0	80
9	0008	01	1		0008	1f	e0	7f	fd	Nv-BdIzc	unknown	T3	7f	f0	80
9	0008	01	1		0008	1f	e0	7f	fd	Nv-BdIzc	unknown	T3	7f	f0	6f
10	016f	00	1		0009	1f	e0	7f	fd	Nv-BdIzc	unknown	T4	00	01	6f
10	016f	00	1		0009	1f	e0	7f	fd	Nv-BdIzc	unknown	T4	00	01	02
11	026f	00	1		0009	1f	e0	7f	fd	Nv-BdIzc	unknown	T0	02	ff	02
11	026f	00	1		0009	1f	e0	7f	fd	Nv-BdIzc	unknown	T0	02	ff	01
12	0009	ea	1	NOP	0009	00	00	7f	fd	nv-BdIZc	unknown	T1	00	00	01
12	0009	ea	1	NOP	0009	00	00	7f	fd	nv-BdIZc	unknown	T1	00	00	00
13	000a	15	1		000a	00	00	7f	fd	nv-BdIZc	NOP	T0+T2	00	00	00
13	000a	15	1		000a	00	00	7f	fd	nv-BdIZc	NOP	T0+T2	00	00	00

CPY (C0)		Cycles: 2	Size: 2
Immediate			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Operand.	
2.1		Inc. PC . C flag = (Y >= Operand), N flag = ((Y - Operand) & \$80), Z flag = (Y == Operand).	
2.2	Read PC	Store as OpCode.	
+X.1		* C flag, N flag, and Z flag applied.	

* Setting of flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	c0	1	CPY #	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	fc
7	0006	c0	1	CPY #	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	c0
8	0007	7f	1		0007	1f	e0	7f	fd	Nv-BdIzc	CPY #	T0+T2	c0	c0	c0
8	0007	7f	1		0007	1f	e0	7f	fd	Nv-BdIzc	CPY #	T0+T2	c0	c0	80
9	0008	ea	1	NOP	0008	1f	e0	7f	fd	Nv-BdIzc	CPY #	T1	7f	80	80
9	0008	ea	1	NOP	0008	1f	e0	7f	fd	Nv-BdIzc	CPY #	T1	7f	80	00
10	0009	ea	1		0009	1f	e0	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00
10	0009	ea	1		0009	1f	e0	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00

CMP (C1)		Cycles: 6	Size: 2
Indirect X (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Operand.	
2.1		Inc. PC .	
2.2	Read Operand	Discard.	
3.1		Pointer = u8(Operand + X).	
3.2	Read Pointer	Store as Address.	
4.1			
4.2	Read u8(Pointer + 1)	Store as Address.H.	
5.1			
5.2	Read Address	Store as Operand.	
6.1		C flag = (A >= Operand), N flag = ((A - Operand) & \$80), Z flag = (A == Operand).	
6.2	Read PC	Store as OpCode.	
+X.1		* C flag, N flag, and Z flag applied.	

* Setting of flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	c1	1	CMP (zp,X)	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	fc
7	0006	c1	1	CMP (zp,X)	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	c0
8	0007	7f	1		0007	1f	e0	7f	fd	Nv-BdIzc	CMP (zp,X)	T2	c0	c0	c0
8	0007	7f	1		0007	1f	e0	7f	fd	Nv-BdIzc	CMP (zp,X)	T2	c0	c0	80
9	007f	10	1		0008	1f	e0	7f	fd	Nv-BdIzc	CMP (zp,X)	T3	e0	7f	80
9	007f	10	1		0008	1f	e0	7f	fd	Nv-BdIzc	CMP (zp,X)	T3	e0	7f	5f
10	005f	70	1		0008	1f	e0	7f	fd	Nv-BdIzc	CMP (zp,X)	T4	00	5f	5f
10	005f	70	1		0008	1f	e0	7f	fd	Nv-BdIzc	CMP (zp,X)	T4	00	5f	60
11	0060	01	1		0008	1f	e0	7f	fd	Nv-BdIzc	CMP (zp,X)	T5	00	70	60
11	0060	01	1		0008	1f	e0	7f	fd	Nv-BdIzc	CMP (zp,X)	T5	00	70	70
12	0170	1f	1		0008	1f	e0	7f	fd	Nv-BdIzc	CMP (zp,X)	T0	ff	ff	70
12	0170	1f	1		0008	1f	e0	7f	fd	Nv-BdIzc	CMP (zp,X)	T0	ff	ff	fe
13	0008	ea	1	NOP	0008	1f	e0	7f	fd	Nv-BdIzc	CMP (zp,X)	T1	1f	e0	fe
13	0008	ea	1	NOP	0008	1f	e0	7f	fd	Nv-BdIzc	CMP (zp,X)	T1	1f	e0	00
14	0009	ea	1		0009	1f	e0	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00
14	0009	ea	1		0009	1f	e0	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00

NOP (C2)		Cycles: 2	Size: 2
Immediate			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Operand.	
2.1		Inc. PC .	
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	c2	1	unknown	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	fc
7	0006	c2	1	unknown	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	c0
8	0007	7f	1		0007	1f	e0	7f	fd	Nv-BdIzc	unknown	T0+T2	c0	c0	c0
8	0007	7f	1		0007	1f	e0	7f	fd	Nv-BdIzc	unknown	T0+T2	c0	c0	80
9	0008	ea	1	NOP	0008	1f	e0	7f	fd	Nv-BdIzc	unknown	T1	ff	7f	80
9	0008	ea	1	NOP	0008	1f	e0	7f	fd	Nv-BdIzc	unknown	T1	ff	7f	7e
10	0009	ea	1		0009	1f	e0	7f	fd	Nv-BdIzc	NOP	T0+T2	7e	7e	7e
10	0009	ea	1		0009	1f	e0	7f	fd	Nv-BdIzc	NOP	T0+T2	7e	7e	fc

DCP (C3)	Cycles: 8	Size: 2
Indirect X (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC .
2.2	Read Operand	Discard.
3.1		Pointer = u8(Operand + X).
3.2	Read Pointer	Store as Address.
4.1		
4.2	Read u8(Pointer + 1)	Store as Address.H.
5.1		
5.2	Read Address	Store as Operand.
6.1		
6.2	Write to Address	Write Operand.
7.1		Operand -= 1. C flag = (A >= Operand), N flag = ((A - Operand) & \$80), Z flag = (A == Operand).
7.2	Write to Address	Write Operand.
8.1		
8.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	c3	1	unknown	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	fc
7	0006	c3	1	unknown	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	c0
8	0007	7f	1		0007	1f	e0	7f	fd	Nv-BdIzc	unknown	T2	c0	c0	c0
8	0007	7f	1		0007	1f	e0	7f	fd	Nv-BdIzc	unknown	T2	c0	c0	80
9	007f	10	1		0008	1f	e0	7f	fd	Nv-BdIzc	unknown	T3	e0	7f	80
9	007f	10	1		0008	1f	e0	7f	fd	Nv-BdIzc	unknown	T3	e0	7f	5f
10	005f	70	1		0008	1f	e0	7f	fd	Nv-BdIzc	unknown	T4	00	5f	5f
10	005f	70	1		0008	1f	e0	7f	fd	Nv-BdIzc	unknown	T4	00	5f	60
11	0060	01	1		0008	1f	e0	7f	fd	Nv-BdIzc	unknown	T5	00	70	60
11	0060	01	1		0008	1f	e0	7f	fd	Nv-BdIzc	unknown	T5	00	70	70
12	0170	66	1		0008	1f	e0	7f	fd	Nv-BdIzc	unknown		ff	ff	70
12	0170	66	1		0008	1f	e0	7f	fd	Nv-BdIzc	unknown		ff	ff	fe
13	0170	66	0		0008	1f	e0	7f	fd	Nv-BdIzc	unknown		ff	66	fe
13	0170	66	0		0008	1f	e0	7f	fd	Nv-BdIzc	unknown		ff	66	65
14	0170	66	0		0008	1f	e0	7f	fd	nv-BdIzc	unknown	T0	65	65	65
14	0170	65	0		0008	1f	e0	7f	fd	nv-BdIzc	unknown	T0	65	65	ca
15	0008	ea	1	NOP	0008	1f	e0	7f	fd	nv-BdIzc	unknown	T1	1f	9a	ca
15	0008	ea	1	NOP	0008	1f	e0	7f	fd	nv-BdIzc	unknown	T1	1f	9a	ba
16	0009	ea	1		0009	1f	e0	7f	fd	Nv-BdIzc	NOP	T0+T2	ba	ba	ba
16	0009	ea	1		0009	1f	e0	7f	fd	Nv-BdIzc	NOP	T0+T2	ba	ba	74

CPY (C4)		Cycles: 3	Size: 2
Zero Page (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read Address	Store as Operand.	
3.1		C flag = (Y >= Operand), N flag = ((Y - Operand) & \$80), Z flag = (Y == Operand).	
3.2	Read PC	Store as OpCode.	
+X.1		* C flag, N flag, and Z flag applied.	

* Setting of flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	c4	1	CPY zp	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	fc
7	0006	c4	1	CPY zp	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	c0
8	0007	7f	1		0007	1f	e0	7f	fd	Nv-BdIzc	CPY zp	T2	c0	c0	c0
8	0007	7f	1		0007	1f	e0	7f	fd	Nv-BdIzc	CPY zp	T2	c0	c0	80
9	007f	10	1		0008	1f	e0	7f	fd	Nv-BdIzc	CPY zp	T0	ff	7f	80
9	007f	10	1		0008	1f	e0	7f	fd	Nv-BdIzc	CPY zp	T0	ff	7f	7e
10	0008	ea	1	NOP	0008	1f	e0	7f	fd	Nv-BdIzc	CPY zp	T1	7f	ef	7e
10	0008	ea	1	NOP	0008	1f	e0	7f	fd	Nv-BdIzc	CPY zp	T1	7f	ef	6f
11	0009	ea	1		0009	1f	e0	7f	fd	nv-BdIzC	NOP	T0+T2	6f	6f	6f
11	0009	ea	1		0009	1f	e0	7f	fd	nv-BdIzC	NOP	T0+T2	6f	6f	de

CMP (C5)		Cycles: 3	Size: 2
Zero Page (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read Address	Store as Operand.	
3.1		C flag = (A >= Operand), N flag = ((A - Operand) & \$80), Z flag = (A == Operand).	
3.2	Read PC	Store as OpCode.	
+X.1		* C flag, N flag, and Z flag applied.	

* Setting of flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	c5	1	CMP zp	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	fc
7	0006	c5	1	CMP zp	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	c0
8	0007	7f	1		0007	1f	e0	7f	fd	Nv-BdIzc	CMP zp	T2	c0	c0	c0
8	0007	7f	1		0007	1f	e0	7f	fd	Nv-BdIzc	CMP zp	T2	c0	c0	80
9	007f	10	1		0008	1f	e0	7f	fd	Nv-BdIzc	CMP zp	T0	ff	7f	80
9	007f	10	1		0008	1f	e0	7f	fd	Nv-BdIzc	CMP zp	T0	ff	7f	7e
10	0008	ea	1	NOP	0008	1f	e0	7f	fd	Nv-BdIzc	CMP zp	T1	1f	ef	7e
10	0008	ea	1	NOP	0008	1f	e0	7f	fd	Nv-BdIzc	CMP zp	T1	1f	ef	0f
11	0009	ea	1		0009	1f	e0	7f	fd	nv-BdIzC	NOP	T0+T2	0f	0f	0f
11	0009	ea	1		0009	1f	e0	7f	fd	nv-BdIzC	NOP	T0+T2	0f	0f	1e

DEC (C6)		Cycles: 5	Size: 2
Zero Page (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read Address	Store as Operand.	
3.1			
3.2	Write to Address	Write Operand.	
4.1		Operand -= 1. N flag = (Operand & \$80), Z flag = !Operand.	
4.2	Write to Address	Write Operand.	
5.1			
5.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	c6	1	DEC zp	0006	1f	e0	7f	fd	Nv-BdIZc	LDX #	T1	e0	e0	fc
7	0006	c6	1	DEC zp	0006	1f	e0	7f	fd	Nv-BdIZc	LDX #	T1	e0	e0	c0
8	0007	7f	1		0007	1f	e0	7f	fd	Nv-BdIZc	DEC zp	T2	c0	c0	c0
8	0007	7f	1		0007	1f	e0	7f	fd	Nv-BdIZc	DEC zp	T2	c0	c0	80
9	007f	01	1		0008	1f	e0	7f	fd	Nv-BdIZc	DEC zp	T3	ff	7f	80
9	007f	01	1		0008	1f	e0	7f	fd	Nv-BdIZc	DEC zp	T3	ff	7f	7e
10	007f	01	0		0008	1f	e0	7f	fd	Nv-BdIZc	DEC zp	T4	ff	01	7e
10	007f	01	0		0008	1f	e0	7f	fd	Nv-BdIZc	DEC zp	T4	ff	01	00
11	007f	01	0		0008	1f	e0	7f	fd	nv-BdIZc	DEC zp	T0	00	00	00
11	007f	00	0		0008	1f	e0	7f	fd	nv-BdIZc	DEC zp	T0	00	00	00
12	0008	ea	1	NOP	0008	1f	e0	7f	fd	nv-BdIZc	DEC zp	T1	ff	00	00
12	0008	ea	1	NOP	0008	1f	e0	7f	fd	nv-BdIZc	DEC zp	T1	ff	00	ff
13	0009	ea	1		0009	1f	e0	7f	fd	nv-BdIZc	NOP	T0+T2	ff	ff	ff
13	0009	ea	1		0009	1f	e0	7f	fd	nv-BdIZc	NOP	T0+T2	ff	ff	fe

DCP (C7)	Cycles: 5	Size: 2
Zero Page (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read Address	Store as Operand.
3.1		
3.2	Write to Address	Write Operand.
4.1		Operand -= 1. C flag = (A >= Operand), N flag = ((A - Operand) & \$80), Z flag = (A == Operand).
4.2	Write to Address	Write Operand.
5.1		
5.2	Read PC	Store as OpCode.
+X.1		* C flag, N flag, and Z flag applied.

* Setting of flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	c7	1	unknown	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	fc
7	0006	c7	1	unknown	0006	1f	e0	7f	fd	Nv-BdIzc	LDX #	T1	e0	e0	c0
8	0007	7f	1		0007	1f	e0	7f	fd	Nv-BdIzc	unknown	T2	c0	c0	c0
8	0007	7f	1		0007	1f	e0	7f	fd	Nv-BdIzc	unknown	T2	c0	c0	80
9	007f	01	1		0008	1f	e0	7f	fd	Nv-BdIzc	unknown	T3	ff	7f	80
9	007f	01	1		0008	1f	e0	7f	fd	Nv-BdIzc	unknown	T3	ff	7f	7e
10	007f	01	0		0008	1f	e0	7f	fd	Nv-BdIzc	unknown	T4	ff	01	7e
10	007f	01	0		0008	1f	e0	7f	fd	Nv-BdIzc	unknown	T4	ff	01	00
11	007f	01	0		0008	1f	e0	7f	fd	nv-BdIZc	unknown	T0	00	00	00
11	007f	00	0		0008	1f	e0	7f	fd	nv-BdIZc	unknown	T0	00	00	00
12	0008	ea	1	NOP	0008	1f	e0	7f	fd	nv-BdIZc	unknown	T1	1f	ff	00
12	0008	ea	1	NOP	0008	1f	e0	7f	fd	nv-BdIZc	unknown	T1	1f	ff	1f
13	0009	ea	1		0009	1f	e0	7f	fd	nv-BdIZc	NOP	T0+T2	1f	1f	1f
13	0009	ea	1		0009	1f	e0	7f	fd	nv-BdIZc	NOP	T0+T2	1f	1f	3e

INY (C8)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Discard.	
2.1		Y += 1. N flag = (Y & \$80), Z flag = ! Y .	
2.2	Read PC	Store as OpCode.	
+X.1		* Y , N flag, and Z flag applied.	

* Setting of Y/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
5	0004	a2	1	LDX #	0004	1f	00	7f	fd	nv-BdIZc	LDY zp	T1	7f	7f	e9
5	0004	a2	1	LDX #	0004	1f	00	7f	fd	nv-BdIZc	LDY zp	T1	7f	7f	fe
6	0005	00	1		0005	1f	00	7f	fd	nv-BdIZc	LDX #	T0+T2	fe	fe	fe
6	0005	00	1		0005	1f	00	7f	fd	nv-BdIZc	LDX #	T0+T2	fe	fe	fc
7	0006	c8	1	INY	0006	1f	00	7f	fd	nv-BdIZc	LDX #	T1	00	00	fc
7	0006	c8	1	INY	0006	1f	00	7f	fd	nv-BdIZc	LDX #	T1	00	00	00
8	0007	ea	1		0007	1f	00	7f	fd	nv-BdIZc	INY	T0+T2	00	00	00
8	0007	ea	1		0007	1f	00	7f	fd	nv-BdIZc	INY	T0+T2	00	00	00
9	0007	ea	1	NOP	0007	1f	00	7f	fd	nv-BdIZc	INY	T1	7f	00	00
9	0007	ea	1	NOP	0007	1f	00	7f	fd	nv-BdIZc	INY	T1	7f	00	80
10	0008	ea	1		0008	1f	00	80	fd	Nv-BdIZc	NOP	T0+T2	80	80	80
10	0008	ea	1		0008	1f	00	80	fd	Nv-BdIZc	NOP	T0+T2	80	80	00

CMP (C9)		Cycles: 2	Size: 2
Immediate			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Operand.	
2.1		Inc. PC . C flag = (A >= Operand), N flag = ((A - Operand) & \$80), Z flag = (A == Operand).	
2.2	Read PC	Store as OpCode.	
+X.1		* C flag, N flag, and Z flag applied.	

* Setting of flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	c9	1	CMP #	0006	1f	88	7f	fd	Nv-BdIzc	LDX #	T1	88	88	fc
7	0006	c9	1	CMP #	0006	1f	88	7f	fd	Nv-BdIzc	LDX #	T1	88	88	10
8	0007	1f	1		0007	1f	88	7f	fd	Nv-BdIzc	CMP #	T0+T2	10	10	10
8	0007	1f	1		0007	1f	88	7f	fd	Nv-BdIzc	CMP #	T0+T2	10	10	20
9	0008	ea	1	NOP	0008	1f	88	7f	fd	Nv-BdIzc	CMP #	T1	1f	e0	20
9	0008	ea	1	NOP	0008	1f	88	7f	fd	Nv-BdIzc	CMP #	T1	1f	e0	00
10	0009	ea	1		0009	1f	88	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00
10	0009	ea	1		0009	1f	88	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00

DEX (CA)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Discard.	
2.1		X -= 1. N flag = (X & \$80), Z flag = ! X .	
2.2	Read PC	Store as OpCode.	
+X.1		* X , N flag, and Z flag applied.	

* Setting of X/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	ca	1	DEX	0006	1f	88	7f	fd	Nv-BdIzc	LDX #	T1	88	88	fc
7	0006	ca	1	DEX	0006	1f	88	7f	fd	Nv-BdIzc	LDX #	T1	88	88	10
8	0007	ea	1		0007	1f	88	7f	fd	Nv-BdIzc	DEX	T0+T2	10	10	10
8	0007	ea	1		0007	1f	88	7f	fd	Nv-BdIzc	DEX	T0+T2	10	10	20
9	0007	ea	1	NOP	0007	1f	88	7f	fd	Nv-BdIzc	DEX	T1	88	ff	20
9	0007	ea	1	NOP	0007	1f	88	7f	fd	Nv-BdIzc	DEX	T1	88	ff	87
10	0008	a2	1		0008	1f	87	7f	fd	Nv-BdIzc	NOP	T0+T2	87	87	87
10	0008	a2	1		0008	1f	87	7f	fd	Nv-BdIzc	NOP	T0+T2	87	87	0e

SBX (CB)		Cycles: 2	Size: 2
Immediate			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1	Read PC	Inc. PC .	
1.2		Store as Operand.	
2.1		Inc. PC . AnX = (A & X).	
		C flag = (AnX >= Operand).	
		X = AnX - Operand.	
		N flag = (X & \$80), Z flag = ! X .	
2.2	Read PC	Store as OpCode.	
+X.1		* X , C flag, N flag, and Z flag applied.	

* Setting of X/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	cb	1	unknown	0006	1f	ff	7f	fd	Nv-BdIzc	LDX #	T1	ff	ff	fc
7	0006	cb	1	unknown	0006	1f	ff	7f	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	1f	1		0007	1f	ff	7f	fd	Nv-BdIzc	unknown	T0+T2	fe	fe	fe
8	0007	1f	1		0007	1f	ff	7f	fd	Nv-BdIzc	unknown	T0+T2	fe	fe	fc
9	0008	ea	1	NOP	0008	1f	ff	7f	fd	Nv-BdIzc	unknown	T1	1f	e0	fc
9	0008	ea	1	NOP	0008	1f	ff	7f	fd	Nv-BdIzc	unknown	T1	1f	e0	00
10	0009	a2	1		0009	1f	00	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00
10	0009	a2	1		0009	1f	00	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00

CPY (CC)	Cycles: 4	Size: 3
Absolute (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address	Store as Operand.
4.1		C flag = (Y >= Operand), N flag = ((Y - Operand) & \$80), Z flag = (Y == Operand).
4.2	Read PC	Store as OpCode.
+X.1		* C flag, N flag, and Z flag applied.

* Setting of flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	cc	1	CPY Abs	0006	1f	ff	7f	fd	Nv-BdIzc	LDX #	T1	ff	ff	fc
7	0006	cc	1	CPY Abs	0006	1f	ff	7f	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	11	1		0007	1f	ff	7f	fd	Nv-BdIzc	CPY Abs	T2	fe	fe	fe
8	0007	11	1		0007	1f	ff	7f	fd	Nv-BdIzc	CPY Abs	T2	fe	fe	fc
9	0008	01	1		0008	1f	ff	7f	fd	Nv-BdIzc	CPY Abs	T3	00	11	fc
9	0008	01	1		0008	1f	ff	7f	fd	Nv-BdIzc	CPY Abs	T3	00	11	11
10	0111	7f	1		0009	1f	ff	7f	fd	Nv-BdIzc	CPY Abs	T0	ff	ff	11
10	0111	7f	1		0009	1f	ff	7f	fd	Nv-BdIzc	CPY Abs	T0	ff	ff	fe
11	0009	a2	1	LDX #	0009	1f	ff	7f	fd	Nv-BdIzc	CPY Abs	T1	7f	80	fe
11	0009	a2	1	LDX #	0009	1f	ff	7f	fd	Nv-BdIzc	CPY Abs	T1	7f	80	00
12	000a	15	1		000a	1f	ff	7f	fd	nv-BdIZC	LDX #	T0+T2	00	00	00
12	000a	15	1		000a	1f	ff	7f	fd	nv-BdIZC	LDX #	T0+T2	00	00	00

CMP (CD)		Cycles: 4	Size: 3
Absolute (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read PC	Store as Address.H.	
3.1		Inc. PC .	
3.2	Read Address	Store as Operand.	
4.1		C flag = (A >= Operand), N flag = ((A - Operand) & \$80), Z flag = (A == Operand).	
4.2	Read PC	Store as OpCode.	
+X.1		* C flag, N flag, and Z flag applied.	

* Setting of flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	cd	1	CMP Abs	0006	1f	ff	7f	fd	Nv-BdIzc	LDX #	T1	ff	ff	fc
7	0006	cd	1	CMP Abs	0006	1f	ff	7f	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	11	1		0007	1f	ff	7f	fd	Nv-BdIzc	CMP Abs	T2	fe	fe	fe
8	0007	11	1		0007	1f	ff	7f	fd	Nv-BdIzc	CMP Abs	T2	fe	fe	fc
9	0008	01	1		0008	1f	ff	7f	fd	Nv-BdIzc	CMP Abs	T3	00	11	fc
9	0008	01	1		0008	1f	ff	7f	fd	Nv-BdIzc	CMP Abs	T3	00	11	11
10	0111	1f	1		0009	1f	ff	7f	fd	Nv-BdIzc	CMP Abs	T0	ff	ff	11
10	0111	1f	1		0009	1f	ff	7f	fd	Nv-BdIzc	CMP Abs	T0	ff	ff	fe
11	0009	ea	1	NOP	0009	1f	ff	7f	fd	Nv-BdIzc	CMP Abs	T1	1f	e0	fe
11	0009	ea	1	NOP	0009	1f	ff	7f	fd	Nv-BdIzc	CMP Abs	T1	1f	e0	00
12	000a	15	1		000a	1f	ff	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00
12	000a	15	1		000a	1f	ff	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00

DEC (CE)	Cycles: 6	Size: 3
Absolute (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.
5.1		Operand -= 1. N flag = (Operand & \$80), Z flag = !Operand.
5.2	Write to Address	Write Operand.
6.1		
6.2	Read PC	Store as OpCode.
+X.1		* N flag and Z flag applied.

* Setting of flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	ce	1	DEC Abs	0006	1f	ff	7f	fd	Nv-BdIzc	LDX #	T1	ff	ff	fc
7	0006	ce	1	DEC Abs	0006	1f	ff	7f	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	11	1		0007	1f	ff	7f	fd	Nv-BdIzc	DEC Abs	T2	fe	fe	fe
8	0007	11	1		0007	1f	ff	7f	fd	Nv-BdIzc	DEC Abs	T2	fe	fe	fc
9	0008	01	1		0008	1f	ff	7f	fd	Nv-BdIzc	DEC Abs	T3	00	11	fc
9	0008	01	1		0008	1f	ff	7f	fd	Nv-BdIzc	DEC Abs	T3	00	11	11
10	0111	01	1		0009	1f	ff	7f	fd	Nv-BdIzc	DEC Abs	T4	ff	ff	11
10	0111	01	1		0009	1f	ff	7f	fd	Nv-BdIzc	DEC Abs	T4	ff	ff	fe
11	0111	01	0		0009	1f	ff	7f	fd	Nv-BdIzc	DEC Abs	T5	ff	01	fe
11	0111	01	0		0009	1f	ff	7f	fd	Nv-BdIzc	DEC Abs	T5	ff	01	00
12	0111	01	0		0009	1f	ff	7f	fd	nv-BdIZc	DEC Abs	T0	00	00	00
12	0111	00	0		0009	1f	ff	7f	fd	nv-BdIZc	DEC Abs	T0	00	00	00
13	0009	ea	1	NOP	0009	1f	ff	7f	fd	nv-BdIZc	DEC Abs	T1	ff	00	00
13	0009	ea	1	NOP	0009	1f	ff	7f	fd	nv-BdIZc	DEC Abs	T1	ff	00	ff
14	000a	15	1		000a	1f	ff	7f	fd	nv-BdIZc	NOP	T0+T2	ff	ff	ff
14	000a	15	1		000a	1f	ff	7f	fd	nv-BdIZc	NOP	T0+T2	ff	ff	fe

DCP (CF)		Cycles: 6	Size: 3
Absolute (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read PC	Store as Address.H.	
3.1		Inc. PC .	
3.2	Read Address	Store as Operand.	
4.1			
4.2	Write to Address	Write Operand.	
5.1		Operand -= 1.	
		C flag = (A >= Operand), N flag = ((A - Operand) & \$80), Z flag = (A == Operand).	
5.2	Write to Address	Write Operand.	
6.1			
6.2	Read PC	Store as OpCode.	
+X.1		* C flag, N flag, and Z flag applied.	

* Setting of flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	cf	1	unknown	0006	1f	ff	7f	fd	Nv-BdIzc	LDX #	T1	ff	ff	fc
7	0006	cf	1	unknown	0006	1f	ff	7f	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	11	1		0007	1f	ff	7f	fd	Nv-BdIzc	unknown	T2	fe	fe	fe
8	0007	11	1		0007	1f	ff	7f	fd	Nv-BdIzc	unknown	T2	fe	fe	fc
9	0008	01	1		0008	1f	ff	7f	fd	Nv-BdIzc	unknown	T3	00	11	fc
9	0008	01	1		0008	1f	ff	7f	fd	Nv-BdIzc	unknown	T3	00	11	11
10	0111	01	1		0009	1f	ff	7f	fd	Nv-BdIzc	unknown	T4	ff	ff	11
10	0111	01	1		0009	1f	ff	7f	fd	Nv-BdIzc	unknown	T4	ff	ff	fe
11	0111	01	0		0009	1f	ff	7f	fd	Nv-BdIzc	unknown	T5	ff	01	fe
11	0111	01	0		0009	1f	ff	7f	fd	Nv-BdIzc	unknown	T5	ff	01	00
12	0111	01	0		0009	1f	ff	7f	fd	nv-BdIzc	unknown	T0	00	00	00
12	0111	00	0		0009	1f	ff	7f	fd	nv-BdIzc	unknown	T0	00	00	00
13	0009	ea	1	NOP	0009	1f	ff	7f	fd	nv-BdIzc	unknown	T1	1f	ff	00
13	0009	ea	1	NOP	0009	1f	ff	7f	fd	nv-BdIzc	unknown	T1	1f	ff	1f
14	000a	15	1		000a	1f	ff	7f	fd	nv-BdIzc	NOP	T0+T2	1f	1f	1f
14	000a	15	1		000a	1f	ff	7f	fd	nv-BdIzc	NOP	T0+T2	1f	1f	3e

BNE (D0) Cycles: 2-4		Size: 2
Relative (Jump)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC . Jump = ((P & \$02) == \$00).
1.2	Read PC	Store as Operand.
		If !Jump, poll interrupts.
		Otherwise Address = i16(i8(Operand)) + PC + 1.
		BoundaryCrossed = Address.H != PC .H.
		If !BoundaryCrossed, poll interrupts.
2.1		Inc. PC . If !Jump, end (next half-cycle is 4.2).
2.2	Read PC	Discard.
3.1		PC .L = Address.L. If !BoundaryCrossed, end (next half-cycle is 4.2).
3.2	Read PC	Discard. Poll interrupts.
4.1		PC .H = Address.H.
4.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	d0	1	BNE	0006	1f	00	7f	fd	nv-BdIZc	LDX #	T1	00	00	fc
7	0006	d0	1	BNE	0006	1f	00	7f	fd	nv-BdIZc	LDX #	T1	00	00	00
8	0007	11	1		0007	1f	00	7f	fd	nv-BdIZc	BNE	T2	00	00	00
8	0007	11	1		0007	1f	00	7f	fd	nv-BdIZc	BNE	T2	00	00	00
9	0008	ea	1	NOP	0008	1f	00	7f	fd	nv-BdIZc	BNE		11	08	00
9	0008	ea	1	NOP	0008	1f	00	7f	fd	nv-BdIZc	BNE		11	08	19
10	0009	ea	1		0009	1f	00	7f	fd	nv-BdIZc	NOP	T0+T2	19	19	19
10	0009	ea	1		0009	1f	00	7f	fd	nv-BdIZc	NOP	T0+T2	19	19	32

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	d0	1	BNE	0006	1f	ff	7f	fd	Nv-BdIZc	LDX #	T1	ff	ff	fc
7	0006	d0	1	BNE	0006	1f	ff	7f	fd	Nv-BdIZc	LDX #	T1	ff	ff	fe
8	0007	11	1		0007	1f	ff	7f	fd	Nv-BdIZc	BNE	T2	fe	fe	fe
8	0007	11	1		0007	1f	ff	7f	fd	Nv-BdIZc	BNE	T2	fe	fe	fc
9	0008	01	1		0008	1f	ff	7f	fd	Nv-BdIZc	BNE	T3	11	08	fc
9	0008	01	1		0008	1f	ff	7f	fd	Nv-BdIZc	BNE	T3	11	08	19
10	0019	ea	1	NOP	0019	1f	ff	7f	fd	Nv-BdIZc	BNE		00	00	19
10	0019	ea	1	NOP	0019	1f	ff	7f	fd	Nv-BdIZc	BNE		00	00	01
11	001a	00	1		001a	1f	ff	7f	fd	Nv-BdIZc	NOP	T0+T2	01	01	01
11	001a	00	1		001a	1f	ff	7f	fd	Nv-BdIZc	NOP	T0+T2	01	01	02

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	d0	1	BNE	0006	1f	01	7f	fd	nv-BdIZc	LDX #	T1	01	01	fc
7	0006	d0	1	BNE	0006	1f	01	7f	fd	nv-BdIZc	LDX #	T1	01	01	02
8	0007	e0	1		0007	1f	01	7f	fd	nv-BdIZc	BNE	T2	02	02	02
8	0007	e0	1		0007	1f	01	7f	fd	nv-BdIZc	BNE	T2	02	02	04
9	0008	ea	1		0008	1f	01	7f	fd	nv-BdIZc	BNE	T3	e0	08	04
9	0008	ea	1		0008	1f	01	7f	fd	nv-BdIZc	BNE	T3	e0	08	e8
10	00e8	00	1		00e8	1f	01	7f	fd	nv-BdIZc	BNE	T0	00	ff	e8
10	00e8	00	1		00e8	1f	01	7f	fd	nv-BdIZc	BNE	T0	00	ff	ff
11	ffe8	00	1	BRK	ffe8	1f	01	7f	fd	nv-BdIZc	BNE	T1	ff	00	ff
11	ffe8	00	1	BRK	ffe8	1f	01	7f	fd	nv-BdIZc	BNE	T1	ff	00	ff
12	ffe9	00	1		ffe9	1f	01	7f	fd	nv-BdIZc	BRK	T2	ff	ff	ff
12	ffe9	00	1		ffe9	1f	01	7f	fd	nv-BdIZc	BRK	T2	ff	ff	fe

CMP (D1) Cycles: 5-6		Size: 2
Indirect Y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read Pointer	Store as Address.
3.1		
3.2	Read u8(Pointer + 1)	Store as Address.H.
4.1		Target = Address + Y. Pointer.L = Target.L. Pointer.H = Address.H. BoundaryCrossed = Pointer.H != Target.H.
4.2	Read Pointer	Store as Operand. If !BoundaryCrossed, skip the next cycle.
5.1		Pointer.H = Target.H.
5.2	Read Pointer	Store as Operand.
6.1		C flag = (A >= Operand), N flag = ((A - Operand) & \$80), Z flag = (A == Operand).
6.2	Read PC	Store as OpCode.
+X.1		* C flag, N flag, and Z flag applied.

* Setting of flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	d1	1	CMP (zp),Y	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	d1	1	CMP (zp),Y	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	e0	1		0007	1f	99	7f	fd	Nv-BdIzc	CMP (zp),Y	T2	32	32	32
8	0007	e0	1		0007	1f	99	7f	fd	Nv-BdIzc	CMP (zp),Y	T2	32	32	64
9	00e0	00	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP (zp),Y	T3	00	e0	64
9	00e0	00	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP (zp),Y	T3	00	e0	e1
10	00e1	00	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP (zp),Y	T4	7f	00	e1
10	00e1	00	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP (zp),Y	T4	7f	00	7f
11	007f	1f	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP (zp),Y	T0	00	00	7f
11	007f	1f	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP (zp),Y	T0	00	00	01
12	0008	ea	1	NOP	0008	1f	99	7f	fd	Nv-BdIzc	CMP (zp),Y	T1	1f	e0	01
12	0008	ea	1	NOP	0008	1f	99	7f	fd	Nv-BdIzc	CMP (zp),Y	T1	1f	e0	00
13	0009	ea	1		0009	1f	99	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00
13	0009	ea	1		0009	1f	99	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	d1	1	CMP (zp),Y	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	d1	1	CMP (zp),Y	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	e0	1		0007	1f	99	7f	fd	Nv-BdIzc	CMP (zp),Y	T2	32	32	32
8	0007	e0	1		0007	1f	99	7f	fd	Nv-BdIzc	CMP (zp),Y	T2	32	32	64
9	00e0	9f	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP (zp),Y	T3	00	e0	64
9	00e0	9f	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP (zp),Y	T3	00	e0	e1
10	00e1	01	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP (zp),Y	T4	7f	9f	e1
10	00e1	01	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP (zp),Y	T4	7f	9f	1e
11	011e	00	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP (zp),Y	T5	00	01	1e
11	011e	00	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP (zp),Y	T5	00	01	02
12	021e	00	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP (zp),Y	T0	02	ff	02
12	021e	00	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP (zp),Y	T0	02	ff	01
13	0008	ea	1	NOP	0008	1f	99	7f	fd	Nv-BdIzc	CMP (zp),Y	T1	1f	ff	01
13	0008	ea	1	NOP	0008	1f	99	7f	fd	Nv-BdIzc	CMP (zp),Y	T1	1f	ff	1f
14	0009	ea	1		0009	1f	99	7f	fd	nv-BdIzc	NOP	T0+T2	1f	1f	1f
14	0009	ea	1		0009	1f	99	7f	fd	nv-BdIzc	NOP	T0+T2	1f	1f	3e

JAM (D2)	Cycles: ∞	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Discard.
2.1		Visual6502 suggests incrementing PC , but Single-Step Tests cannot be passed this way.
2.2	Read \$FFFF	Discard.
3.1		
3.2	Read \$FFFE	Discard.
4.1		
4.2	Read \$FFFE	Discard.
5.1		Repeat 5.1 and 5.2 forever.
5.2	Read \$FFFF	Discard.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	d2	1	unknown	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	d2	1	unknown	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	e0	1		0007	1f	99	7f	fd	Nv-BdIzc	unknown	T2	32	32	32
8	0007	e0	1		0007	1f	99	7f	fd	Nv-BdIzc	unknown	T2	32	32	64
9	ffff	00	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T3	ff	ff	64
9	ffff	00	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T3	ff	ff	fe
10	fffe	00	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T4	ff	ff	fe
10	fffe	00	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T4	ff	ff	fe
11	fffe	00	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T5	ff	ff	fe
11	fffe	00	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T5	ff	ff	fe
12	ffff	00	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown		ff	ff	fe
12	ffff	00	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown		ff	ff	fe
13	ffff	00	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown		ff	ff	fe
13	ffff	00	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown		ff	ff	fe
14	ffff	00	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown		ff	ff	fe
14	ffff	00	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown		ff	ff	fe

DCP (D3)		Cycles: 8	Size: 2
Indirect Y (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Address.	
3.1			
3.2	Read u8(Pointer + 1)	Store as Address.H.	
4.1		Target = Address + Y. Pointer.L = Target.L. Pointer.H = Address.H.	
4.2	Read Pointer	Store as Operand.	
5.1		Pointer.H = Target.H.	
5.2	Read Pointer	Store as Operand.	
6.1			
6.2	Write to Pointer	Write Operand.	
7.1		Operand -= 1. C flag = (A >= Operand), N flag = ((A - Operand) & \$80), Z flag = (A == Operand).	
7.2	Write to Pointer	Write Operand.	
8.1			
8.2	Read PC	Store as OpCode.	
+X.1		* C flag, N flag, and Z flag applied.	

* Setting of flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	d3	1	unknown	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	d3	1	unknown	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	7f	1		0007	1f	99	7f	fd	Nv-BdIzc	unknown	T2	32	32	32
8	0007	7f	1		0007	1f	99	7f	fd	Nv-BdIzc	unknown	T2	32	32	64
9	007f	1f	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T3	00	7f	64
9	007f	1f	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T3	00	7f	80
10	0080	00	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T4	7f	1f	80
10	0080	00	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T4	7f	1f	9e
11	009e	01	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T5	00	00	9e
11	009e	01	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T5	00	00	01
12	009e	01	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown		01	ff	01
12	009e	01	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown		01	ff	00
13	009e	01	0		0008	1f	99	7f	fd	Nv-BdIzc	unknown		ff	01	00
13	009e	01	0		0008	1f	99	7f	fd	Nv-BdIzc	unknown		ff	01	00
14	009e	01	0		0008	1f	99	7f	fd	nv-BdIZc	unknown	T0	00	00	00
14	009e	00	0		0008	1f	99	7f	fd	nv-BdIZc	unknown	T0	00	00	00
15	0008	ea	1	NOP	0008	1f	99	7f	fd	nv-BdIZc	unknown	T1	1f	ff	00
15	0008	ea	1	NOP	0008	1f	99	7f	fd	nv-BdIZc	unknown	T1	1f	ff	1f
16	0009	ea	1		0009	1f	99	7f	fd	nv-BdIZc	NOP	T0+T2	1f	1f	1f
16	0009	ea	1		0009	1f	99	7f	fd	nv-BdIZc	NOP	T0+T2	1f	1f	3e

NOP (D4)		Cycles: 4	Size: 2
Zero Page X-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Operand.	
3.1		Address = u8(Pointer + X).	
3.2	Read Address	Store as Operand.	
4.1			
4.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	d4	1	unknown	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	d4	1	unknown	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	7f	1		0007	1f	99	7f	fd	Nv-BdIzc	unknown	T2	32	32	32
8	0007	7f	1		0007	1f	99	7f	fd	Nv-BdIzc	unknown	T2	32	32	64
9	007f	1f	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T3	99	7f	64
9	007f	1f	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T3	99	7f	18
10	0018	48	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T0	ff	ff	18
10	0018	48	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T0	ff	ff	fe
11	0008	ea	1	NOP	0008	1f	99	7f	fd	Nv-BdIzc	unknown	T1	ff	48	fe
11	0008	ea	1	NOP	0008	1f	99	7f	fd	Nv-BdIzc	unknown	T1	ff	48	47
12	0009	ea	1		0009	1f	99	7f	fd	Nv-BdIzc	NOP	T0+T2	47	47	47
12	0009	ea	1		0009	1f	99	7f	fd	Nv-BdIzc	NOP	T0+T2	47	47	8e

CMP (D5)		Cycles: 4	Size: 2
Zero Page X-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Operand.	
3.1		Address = u8(Pointer + X).	
3.2	Read Address	Store as Operand.	
4.1		C flag = (A >= Operand), N flag = ((A - Operand) & \$80), Z flag = (A == Operand).	
4.2	Read PC	Store as OpCode.	
+X.1		* C flag, N flag, and Z flag applied.	

* Setting of flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	d5	1	CMP zp,X	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	d5	1	CMP zp,X	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	ef	1		0007	1f	99	7f	fd	Nv-BdIzc	CMP zp,X	T2	32	32	32
8	0007	ef	1		0007	1f	99	7f	fd	Nv-BdIzc	CMP zp,X	T2	32	32	64
9	00ef	33	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP zp,X	T3	99	ef	64
9	00ef	33	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP zp,X	T3	99	ef	88
10	0088	1f	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP zp,X	T0	ff	ff	88
10	0088	1f	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP zp,X	T0	ff	ff	fe
11	0008	ea	1	NOP	0008	1f	99	7f	fd	Nv-BdIzc	CMP zp,X	T1	1f	e0	fe
11	0008	ea	1	NOP	0008	1f	99	7f	fd	Nv-BdIzc	CMP zp,X	T1	1f	e0	00
12	0009	ea	1		0009	1f	99	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00
12	0009	ea	1		0009	1f	99	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00

DEC (D6)	Cycles: 6	Size: 2
Zero Page X-Indexed (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read Pointer	Store as Operand.
3.1		Address = u8(Pointer + X).
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.
5.1		Operand -= 1. N flag = (Operand & \$80), Z flag = !Operand.
5.2	Write to Address	Write Operand.
6.1		
6.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	d6	1	DEC zp,X	0006	1f	99	7f	fd	Nv-BdIZc	LDX #	T1	99	99	fc
7	0006	d6	1	DEC zp,X	0006	1f	99	7f	fd	Nv-BdIZc	LDX #	T1	99	99	32
8	0007	ef	1		0007	1f	99	7f	fd	Nv-BdIZc	DEC zp,X	T2	32	32	32
8	0007	ef	1		0007	1f	99	7f	fd	Nv-BdIZc	DEC zp,X	T2	32	32	64
9	00ef	33	1		0008	1f	99	7f	fd	Nv-BdIZc	DEC zp,X	T3	99	ef	64
9	00ef	33	1		0008	1f	99	7f	fd	Nv-BdIZc	DEC zp,X	T3	99	ef	88
10	0088	01	1		0008	1f	99	7f	fd	Nv-BdIZc	DEC zp,X	T4	ff	ff	88
10	0088	01	1		0008	1f	99	7f	fd	Nv-BdIZc	DEC zp,X	T4	ff	ff	fe
11	0088	01	0		0008	1f	99	7f	fd	Nv-BdIZc	DEC zp,X	T5	ff	01	fe
11	0088	01	0		0008	1f	99	7f	fd	Nv-BdIZc	DEC zp,X	T5	ff	01	00
12	0088	01	0		0008	1f	99	7f	fd	nv-BdIZc	DEC zp,X	T0	00	00	00
12	0088	00	0		0008	1f	99	7f	fd	nv-BdIZc	DEC zp,X	T0	00	00	00
13	0008	ea	1	NOP	0008	1f	99	7f	fd	nv-BdIZc	DEC zp,X	T1	ff	00	00
13	0008	ea	1	NOP	0008	1f	99	7f	fd	nv-BdIZc	DEC zp,X	T1	ff	00	ff
14	0009	ea	1		0009	1f	99	7f	fd	nv-BdIZc	NOP	T0+T2	ff	ff	ff
14	0009	ea	1		0009	1f	99	7f	fd	nv-BdIZc	NOP	T0+T2	ff	ff	fe

DCP (D7)		Cycles: 6	Size: 2
Zero Page X-Indexed (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Operand.	
3.1		Address = u8(Pointer + X).	
3.2	Read Address	Store as Operand.	
4.1			
4.2	Write to Address	Write Operand.	
5.1		Operand -= 1.	
		C flag = (A >= Operand), N flag = ((A - Operand) & \$80), Z flag = (A == Operand).	
5.2	Write to Address	Write Operand.	
6.1			
6.2	Read PC	Store as OpCode.	
+X.1		* C flag, N flag, and Z flag applied.	

* Setting of flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	d7	1	unknown	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	d7	1	unknown	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	ef	1		0007	1f	99	7f	fd	Nv-BdIzc	unknown	T2	32	32	32
8	0007	ef	1		0007	1f	99	7f	fd	Nv-BdIzc	unknown	T2	32	32	64
9	00ef	33	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T3	99	ef	64
9	00ef	33	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T3	99	ef	88
10	0088	01	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T4	ff	ff	88
10	0088	01	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T4	ff	ff	fe
11	0088	01	0		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T5	ff	01	fe
11	0088	01	0		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T5	ff	01	00
12	0088	01	0		0008	1f	99	7f	fd	nv-BdIZc	unknown	T0	00	00	00
12	0088	00	0		0008	1f	99	7f	fd	nv-BdIZc	unknown	T0	00	00	00
13	0008	ea	1	NOP	0008	1f	99	7f	fd	nv-BdIZc	unknown	T1	1f	ff	00
13	0008	ea	1	NOP	0008	1f	99	7f	fd	nv-BdIZc	unknown	T1	1f	ff	1f
14	0009	ea	1		0009	1f	99	7f	fd	nv-BdIZc	NOP	T0+T2	1f	1f	1f
14	0009	ea	1		0009	1f	99	7f	fd	nv-BdIZc	NOP	T0+T2	1f	1f	3e

CLD (D8)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Discard.	
2.1		D flag = 0.	
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	d8	1	CLD	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	d8	1	CLD	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	ea	1		0007	1f	99	7f	fd	Nv-BdIzc	CLD	T0+T2	32	32	32
8	0007	ea	1		0007	1f	99	7f	fd	Nv-BdIzc	CLD	T0+T2	32	32	64
9	0007	ea	1	NOP	0007	1f	99	7f	fd	Nv-BdIzc	CLD	T1	ff	ff	64
9	0007	ea	1	NOP	0007	1f	99	7f	fd	Nv-BdIzc	CLD	T1	ff	ff	fe
10	0008	ea	1		0008	1f	99	7f	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fe
10	0008	ea	1		0008	1f	99	7f	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fc

CMP (D9) Cycles: 4-5		Size: 3
Absolute Y-Indexed (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + Y. Address.L = Target.L. Address.H = Pointer.H. BoundaryCrossed = Pointer.H != Target.H.
3.2	Read Address	Store as Operand. If !BoundaryCrossed, skip the next cycle.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		C flag = (A >= Operand), N flag = ((A - Operand) & \$80), Z flag = (A == Operand).
5.2	Read PC	Store as OpCode.
+X.1		* C flag, N flag, and Z flag applied.

* Setting of flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	d9	1	CMP Abs,Y	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	d9	1	CMP Abs,Y	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	7f	1		0007	1f	99	7f	fd	Nv-BdIzc	CMP Abs,Y	T2	32	32	32
8	0007	7f	1		0007	1f	99	7f	fd	Nv-BdIzc	CMP Abs,Y	T2	32	32	64
9	0008	01	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP Abs,Y	T3	7f	7f	64
9	0008	01	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP Abs,Y	T3	7f	7f	fe
10	01fe	1f	1		0009	1f	99	7f	fd	Nv-BdIzc	CMP Abs,Y	T0	00	01	fe
10	01fe	1f	1		0009	1f	99	7f	fd	Nv-BdIzc	CMP Abs,Y	T0	00	01	02
11	0009	ea	1	NOP	0009	1f	99	7f	fd	Nv-BdIzc	CMP Abs,Y	T1	1f	e0	02
11	0009	ea	1	NOP	0009	1f	99	7f	fd	Nv-BdIzc	CMP Abs,Y	T1	1f	e0	00
12	000a	15	1		000a	1f	99	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00
12	000a	15	1		000a	1f	99	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	d9	1	CMP Abs,Y	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	d9	1	CMP Abs,Y	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	81	1		0007	1f	99	7f	fd	Nv-BdIzc	CMP Abs,Y	T2	32	32	32
8	0007	81	1		0007	1f	99	7f	fd	Nv-BdIzc	CMP Abs,Y	T2	32	32	64
9	0008	01	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP Abs,Y	T3	7f	81	64
9	0008	01	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP Abs,Y	T3	7f	81	00
10	0100	01	1		0009	1f	99	7f	fd	Nv-BdIzc	CMP Abs,Y	T4	00	01	00
10	0100	01	1		0009	1f	99	7f	fd	Nv-BdIzc	CMP Abs,Y	T4	00	01	02
11	0200	00	1		0009	1f	99	7f	fd	Nv-BdIzc	CMP Abs,Y	T0	02	ff	02
11	0200	00	1		0009	1f	99	7f	fd	Nv-BdIzc	CMP Abs,Y	T0	02	ff	01
12	0009	ea	1	NOP	0009	1f	99	7f	fd	Nv-BdIzc	CMP Abs,Y	T1	1f	ff	01
12	0009	ea	1	NOP	0009	1f	99	7f	fd	Nv-BdIzc	CMP Abs,Y	T1	1f	ff	1f
13	000a	15	1		000a	1f	99	7f	fd	nv-BdIZC	NOP	T0+T2	1f	1f	1f
13	000a	15	1		000a	1f	99	7f	fd	nv-BdIZC	NOP	T0+T2	1f	1f	3e

NOP (DA)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Discard.	
2.1			
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	da	1	unknown	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	da	1	unknown	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	ea	1		0007	1f	99	7f	fd	Nv-BdIzc	unknown	T0+T2	32	32	32
8	0007	ea	1		0007	1f	99	7f	fd	Nv-BdIzc	unknown	T0+T2	32	32	64
9	0007	ea	1	NOP	0007	1f	99	7f	fd	Nv-BdIzc	unknown	T1	ff	ff	64
9	0007	ea	1	NOP	0007	1f	99	7f	fd	Nv-BdIzc	unknown	T1	ff	ff	fe
10	0008	01	1		0008	1f	99	7f	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fe
10	0008	01	1		0008	1f	99	7f	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fc

DCP (DB)		Cycles: 7	Size: 3
Absolute Y-Indexed (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read PC	Store as Pointer.H.	
3.1		Inc. PC . Target = Pointer + Y.	
		Address.L = Target.L.	
		Address.H = Pointer.H.	
3.2	Read Address	Store as Operand.	
4.1		Address.H = Target.H.	
4.2	Read Address	Store as Operand.	
5.1			
5.2	Write to Address	Write Operand.	
6.1		Operand -= 1.	
		C flag = (A >= Operand), N flag = ((A - Operand) & \$80), Z flag = (A == Operand).	
6.2	Write to Address	Write Operand.	
7.1			
7.2	Read PC	Store as OpCode.	
+X.1		* C flag, N flag, and Z flag applied.	

* Setting of flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	db	1	unknown	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	db	1	unknown	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	ea	1		0007	1f	99	7f	fd	Nv-BdIzc	unknown	T2	32	32	32
8	0007	ea	1		0007	1f	99	7f	fd	Nv-BdIzc	unknown	T2	32	32	64
9	0008	01	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T3	7f	ea	64
9	0008	01	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T3	7f	ea	69
10	0169	66	1		0009	1f	99	7f	fd	Nv-BdIzc	unknown	T4	00	01	69
10	0169	66	1		0009	1f	99	7f	fd	Nv-BdIzc	unknown	T4	00	01	02
11	0269	00	1		0009	1f	99	7f	fd	Nv-BdIzc	unknown	T5	02	ff	02
11	0269	00	1		0009	1f	99	7f	fd	Nv-BdIzc	unknown	T5	02	ff	01
12	0269	00	0		0009	1f	99	7f	fd	Nv-BdIzc	unknown		ff	00	01
12	0269	00	0		0009	1f	99	7f	fd	Nv-BdIzc	unknown		ff	00	ff
13	0269	00	0		0009	1f	99	7f	fd	Nv-BdIzc	unknown	T0	ff	ff	ff
13	0269	ff	0		0009	1f	99	7f	fd	Nv-BdIzc	unknown	T0	ff	ff	fe
14	0009	ea	1	NOP	0009	1f	99	7f	fd	Nv-BdIzc	unknown	T1	1f	00	fe
14	0009	ea	1	NOP	0009	1f	99	7f	fd	Nv-BdIzc	unknown	T1	1f	00	20
15	000a	15	1		000a	1f	99	7f	fd	nv-BdIzc	NOP	T0+T2	20	20	20
15	000a	15	1		000a	1f	99	7f	fd	nv-BdIzc	NOP	T0+T2	20	20	40

NOP (DC)		Cycles: 4-5	Size: 3
Absolute X-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read PC	Store as Pointer.H.	
3.1		Inc. PC . Target = Pointer + X .	
		Address.L = Target.L.	
		Address.H = Pointer.H.	
		BoundaryCrossed = Pointer.H != Target.H.	
3.2	Read Address	Store as Operand. If !BoundaryCrossed, skip the next cycle.	
4.1		Address.H = Target.H.	
4.2	Read Address	Store as Operand.	
5.1			
5.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	dc	1	unknown	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	dc	1	unknown	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	ea	1		0007	1f	99	7f	fd	Nv-BdIzc	unknown	T2	32	32	32
8	0007	ea	1		0007	1f	99	7f	fd	Nv-BdIzc	unknown	T2	32	32	64
9	0008	01	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T3	99	ea	64
9	0008	01	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T3	99	ea	83
10	0183	00	1		0009	1f	99	7f	fd	Nv-BdIzc	unknown	T4	00	01	83
10	0183	00	1		0009	1f	99	7f	fd	Nv-BdIzc	unknown	T4	00	01	02
11	0283	00	1		0009	1f	99	7f	fd	Nv-BdIzc	unknown	T0	02	ff	02
11	0283	00	1		0009	1f	99	7f	fd	Nv-BdIzc	unknown	T0	02	ff	01
12	0009	ea	1	NOP	0009	1f	99	7f	fd	Nv-BdIzc	unknown	T1	ff	00	01
12	0009	ea	1	NOP	0009	1f	99	7f	fd	Nv-BdIzc	unknown	T1	ff	00	ff
13	000a	15	1		000a	1f	99	7f	fd	Nv-BdIzc	NOP	T0+T2	ff	ff	ff
13	000a	15	1		000a	1f	99	7f	fd	Nv-BdIzc	NOP	T0+T2	ff	ff	fe

CMP (DD)		Cycles: 4-5	Size: 3
Absolute X-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read PC	Store as Pointer.H.	
3.1		Inc. PC . Target = Pointer + X . Address.L = Target.L. Address.H = Pointer.H. BoundaryCrossed = Pointer.H != Target.H.	
3.2	Read Address	Store as Operand. If !BoundaryCrossed, skip the next cycle.	
4.1		Address.H = Target.H.	
4.2	Read Address	Store as Operand.	
5.1		C flag = (A >= Operand), N flag = ((A - Operand) & \$80), Z flag = (A == Operand).	
5.2	Read PC	Store as OpCode.	
+X.1		* C flag, N flag, and Z flag applied.	

* Setting of flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	dd	1	CMP Abs,X	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	dd	1	CMP Abs,X	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	66	1		0007	1f	99	7f	fd	Nv-BdIzc	CMP Abs,X	T2	32	32	32
8	0007	66	1		0007	1f	99	7f	fd	Nv-BdIzc	CMP Abs,X	T2	32	32	64
9	0008	01	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP Abs,X	T3	99	66	64
9	0008	01	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP Abs,X	T3	99	66	ff
10	01ff	1f	1		0009	1f	99	7f	fd	Nv-BdIzc	CMP Abs,X	T0	00	01	ff
10	01ff	1f	1		0009	1f	99	7f	fd	Nv-BdIzc	CMP Abs,X	T0	00	01	02
11	0009	ea	1	NOP	0009	1f	99	7f	fd	Nv-BdIzc	CMP Abs,X	T1	1f	e0	02
11	0009	ea	1	NOP	0009	1f	99	7f	fd	Nv-BdIzc	CMP Abs,X	T1	1f	e0	00
12	000a	15	1		000a	1f	99	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00
12	000a	15	1		000a	1f	99	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	dd	1	CMP Abs,X	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	dd	1	CMP Abs,X	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	ea	1		0007	1f	99	7f	fd	Nv-BdIzc	CMP Abs,X	T2	32	32	32
8	0007	ea	1		0007	1f	99	7f	fd	Nv-BdIzc	CMP Abs,X	T2	32	32	64
9	0008	01	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP Abs,X	T3	99	ea	64
9	0008	01	1		0008	1f	99	7f	fd	Nv-BdIzc	CMP Abs,X	T3	99	ea	83
10	0183	00	1		0009	1f	99	7f	fd	Nv-BdIzc	CMP Abs,X	T4	00	01	83
10	0183	00	1		0009	1f	99	7f	fd	Nv-BdIzc	CMP Abs,X	T4	00	01	02
11	0283	00	1		0009	1f	99	7f	fd	Nv-BdIzc	CMP Abs,X	T0	02	ff	02
11	0283	00	1		0009	1f	99	7f	fd	Nv-BdIzc	CMP Abs,X	T0	02	ff	01
12	0009	ea	1	NOP	0009	1f	99	7f	fd	Nv-BdIzc	CMP Abs,X	T1	1f	ff	01
12	0009	ea	1	NOP	0009	1f	99	7f	fd	Nv-BdIzc	CMP Abs,X	T1	1f	ff	1f
13	000a	15	1		000a	1f	99	7f	fd	nv-BdIZC	NOP	T0+T2	1f	1f	1f
13	000a	15	1		000a	1f	99	7f	fd	nv-BdIZC	NOP	T0+T2	1f	1f	3e

DEC (DE)	Cycles: 7	Size: 3
Absolute X-Indexed (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + X . Address.L = Target.L. Address.H = Pointer.H.
3.2	Read Address	Store as Operand.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		
5.2	Write to Address	Write Operand.
6.1		Operand -= 1. N flag = (Operand & \$80), Z flag = !Operand.
6.2	Write to Address	Write Operand.
7.1		
7.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	de	1	DEC Abs,X	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	de	1	DEC Abs,X	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	66	1		0007	1f	99	7f	fd	Nv-BdIzc	DEC Abs,X	T2	32	32	32
8	0007	66	1		0007	1f	99	7f	fd	Nv-BdIzc	DEC Abs,X	T2	32	32	64
9	0008	01	1		0008	1f	99	7f	fd	Nv-BdIzc	DEC Abs,X	T3	99	66	64
9	0008	01	1		0008	1f	99	7f	fd	Nv-BdIzc	DEC Abs,X	T3	99	66	ff
10	01ff	01	1		0009	1f	99	7f	fd	Nv-BdIzc	DEC Abs,X	T4	00	01	ff
10	01ff	01	1		0009	1f	99	7f	fd	Nv-BdIzc	DEC Abs,X	T4	00	01	02
11	01ff	01	1		0009	1f	99	7f	fd	Nv-BdIzc	DEC Abs,X	T5	02	ff	02
11	01ff	01	1		0009	1f	99	7f	fd	Nv-BdIzc	DEC Abs,X	T5	02	ff	01
12	01ff	01	0		0009	1f	99	7f	fd	Nv-BdIzc	DEC Abs,X		ff	01	01
12	01ff	01	0		0009	1f	99	7f	fd	Nv-BdIzc	DEC Abs,X		ff	01	00
13	01ff	01	0		0009	1f	99	7f	fd	nv-BdIZc	DEC Abs,X	T0	00	00	00
13	01ff	00	0		0009	1f	99	7f	fd	nv-BdIZc	DEC Abs,X	T0	00	00	00
14	0009	ea	1	NOP	0009	1f	99	7f	fd	nv-BdIZc	DEC Abs,X	T1	ff	00	00
14	0009	ea	1	NOP	0009	1f	99	7f	fd	nv-BdIZc	DEC Abs,X	T1	ff	00	ff
15	000a	15	1		000a	1f	99	7f	fd	nv-BdIZc	NOP	T0+T2	ff	ff	ff
15	000a	15	1		000a	1f	99	7f	fd	nv-BdIZc	NOP	T0+T2	ff	ff	fe

DCP (DF)		Cycles: 7	Size: 3
Absolute X-Indexed (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read PC	Store as Pointer.H.	
3.1		Inc. PC . Target = Pointer + X .	
		Address.L = Target.L.	
		Address.H = Pointer.H.	
3.2	Read Address	Store as Operand.	
4.1		Address.H = Target.H.	
4.2	Read Address	Store as Operand.	
5.1			
5.2	Write to Address	Write Operand.	
6.1		Operand -= 1.	
		C flag = (A >= Operand), N flag = ((A - Operand) & \$80), Z flag = (A == Operand).	
6.2	Write to Address	Write Operand.	
7.1			
7.2	Read PC	Store as OpCode.	
+X.1		* C flag, N flag, and Z flag applied.	

* Setting of flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	df	1	unknown	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	df	1	unknown	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	66	1		0007	1f	99	7f	fd	Nv-BdIzc	unknown	T2	32	32	32
8	0007	66	1		0007	1f	99	7f	fd	Nv-BdIzc	unknown	T2	32	32	64
9	0008	01	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T3	99	66	64
9	0008	01	1		0008	1f	99	7f	fd	Nv-BdIzc	unknown	T3	99	66	ff
10	01ff	01	1		0009	1f	99	7f	fd	Nv-BdIzc	unknown	T4	00	01	ff
10	01ff	01	1		0009	1f	99	7f	fd	Nv-BdIzc	unknown	T4	00	01	02
11	01ff	01	1		0009	1f	99	7f	fd	Nv-BdIzc	unknown	T5	02	ff	02
11	01ff	01	1		0009	1f	99	7f	fd	Nv-BdIzc	unknown	T5	02	ff	01
12	01ff	01	0		0009	1f	99	7f	fd	Nv-BdIzc	unknown		ff	01	01
12	01ff	01	0		0009	1f	99	7f	fd	Nv-BdIzc	unknown		ff	01	00
13	01ff	01	0		0009	1f	99	7f	fd	nv-BdIzc	unknown	T0	00	00	00
13	01ff	00	0		0009	1f	99	7f	fd	nv-BdIzc	unknown	T0	00	00	00
14	0009	ea	1	NOP	0009	1f	99	7f	fd	nv-BdIzc	unknown	T1	1f	ff	00
14	0009	ea	1	NOP	0009	1f	99	7f	fd	nv-BdIzc	unknown	T1	1f	ff	1f
15	000a	15	1		000a	1f	99	7f	fd	nv-BdIzc	NOP	T0+T2	1f	1f	1f
15	000a	15	1		000a	1f	99	7f	fd	nv-BdIzc	NOP	T0+T2	1f	1f	3e

CPX (E0)		Cycles: 2	Size: 2
Immediate			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Operand.	
2.1		Inc. PC . C flag = (X >= Operand), N flag = ((X - Operand) & \$80), Z flag = (X == Operand).	
2.2	Read PC	Store as OpCode.	
+X.1		* C flag, N flag, and Z flag applied.	

* Setting of flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	e0	1	CPX #	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	e0	1	CPX #	0006	1f	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	66	1		0007	1f	99	7f	fd	Nv-BdIzc	CPX #	T0+T2	32	32	32
8	0007	66	1		0007	1f	99	7f	fd	Nv-BdIzc	CPX #	T0+T2	32	32	64
9	0008	ea	1	NOP	0008	1f	99	7f	fd	Nv-BdIzc	CPX #	T1	99	99	64
9	0008	ea	1	NOP	0008	1f	99	7f	fd	Nv-BdIzc	CPX #	T1	99	99	33
10	0009	ea	1		0009	1f	99	7f	fd	nv-BdIzC	NOP	T0+T2	33	33	33
10	0009	ea	1		0009	1f	99	7f	fd	nv-BdIzC	NOP	T0+T2	33	33	66

SBC (E1)	Cycles: 6	Size: 2
Indirect X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC .
2.2	Read Operand	Discard.
3.1		Pointer = u8(Operand + X).
3.2	Read Pointer	Store as Address.
4.1		
4.2	Read u8(Pointer + 1)	Store as Address.H.
5.1		
5.2	Read Address	Store as Operand.
6.1		Val = u16(Operand) ^ \$00FF. Result16 = u16(A) + Val + C flag. V flag = ((u16(A) ^ Result16) & (Val ^ Result16) & \$0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (A & \$80), Z flag = ! A .
6.2	Read PC	Store as OpCode.
+X.1		* A , C flag, N flag, V flag, and Z flag applied.

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	e1	1	SBC (zp,X)	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	e1	1	SBC (zp,X)	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	65	1		0007	20	99	7f	fd	Nv-BdIzc	SBC (zp,X)	T2	32	32	32
8	0007	65	1		0007	20	99	7f	fd	Nv-BdIzc	SBC (zp,X)	T2	32	32	64
9	0065	00	1		0008	20	99	7f	fd	Nv-BdIzc	SBC (zp,X)	T3	99	65	64
9	0065	00	1		0008	20	99	7f	fd	Nv-BdIzc	SBC (zp,X)	T3	99	65	fe
10	00fe	70	1		0008	20	99	7f	fd	Nv-BdIzc	SBC (zp,X)	T4	00	fe	fe
10	00fe	70	1		0008	20	99	7f	fd	Nv-BdIzc	SBC (zp,X)	T4	00	fe	ff
11	00ff	01	1		0008	20	99	7f	fd	Nv-BdIzc	SBC (zp,X)	T5	00	70	ff
11	00ff	01	1		0008	20	99	7f	fd	Nv-BdIzc	SBC (zp,X)	T5	00	70	70
12	0170	1f	1		0008	20	99	7f	fd	Nv-BdIzc	SBC (zp,X)	T0	ff	ff	70
12	0170	1f	1		0008	20	99	7f	fd	Nv-BdIzc	SBC (zp,X)	T0	ff	ff	fe
13	0008	ea	1	NOP	0008	20	99	7f	fd	Nv-BdIzc	SBC (zp,X)	T1	20	e0	fe
13	0008	ea	1	NOP	0008	20	99	7f	fd	Nv-BdIzc	SBC (zp,X)	T1	20	e0	00
14	0009	ea	1		0009	00	99	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00
14	0009	ea	1		0009	00	99	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00

NOP (E2)		Cycles: 2	Size: 2
Immediate			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Operand.	
2.1		Inc. PC .	
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	e2	1	unknown	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	e2	1	unknown	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	65	1		0007	20	99	7f	fd	Nv-BdIzc	unknown	T0+T2	32	32	32
8	0007	65	1		0007	20	99	7f	fd	Nv-BdIzc	unknown	T0+T2	32	32	64
9	0008	ea	1	NOP	0008	20	99	7f	fd	Nv-BdIzc	unknown	T1	ff	65	64
9	0008	ea	1	NOP	0008	20	99	7f	fd	Nv-BdIzc	unknown	T1	ff	65	64
10	0009	ea	1		0009	20	99	7f	fd	Nv-BdIzc	NOP	T0+T2	64	64	64
10	0009	ea	1		0009	20	99	7f	fd	Nv-BdIzc	NOP	T0+T2	64	64	c8

ISC (E3)	Cycles: 8	Size: 2
Indirect X (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC .
2.2	Read Operand	Discard.
3.1		Pointer = u8(Operand + X).
3.2	Read Pointer	Store as Address.
4.1		
4.2	Read u8(Pointer + 1)	Store as Address.H.
5.1		
5.2	Read Address	Store as Operand.
6.1		
6.2	Write to Address	Write Operand.
7.1		Operand += 1. Val = u16(Operand) ^ \$00FF. Result16 = u16(A) + Val + C flag. V flag = ((u16(A) ^ Result16) & (Val ^ Result16) & \$0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (A & \$80), Z flag = ! A .
7.2	Write to Address	Write Operand.
8.1		
8.2	Read PC	Store as OpCode.
+X.1		* A , C flag, N flag, V flag, and Z flag applied.

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	e3	1	unknown	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	e3	1	unknown	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	65	1		0007	20	99	7f	fd	Nv-BdIzc	unknown	T2	32	32	32
8	0007	65	1		0007	20	99	7f	fd	Nv-BdIzc	unknown	T2	32	32	64
9	0065	00	1		0008	20	99	7f	fd	Nv-BdIzc	unknown	T3	99	65	64
9	0065	00	1		0008	20	99	7f	fd	Nv-BdIzc	unknown	T3	99	65	fe
10	00fe	70	1		0008	20	99	7f	fd	Nv-BdIzc	unknown	T4	00	fe	fe
10	00fe	70	1		0008	20	99	7f	fd	Nv-BdIzc	unknown	T4	00	fe	ff
11	00ff	01	1		0008	20	99	7f	fd	Nv-BdIzc	unknown	T5	00	70	ff
11	00ff	01	1		0008	20	99	7f	fd	Nv-BdIzc	unknown	T5	00	70	70
12	0170	1f	1		0008	20	99	7f	fd	Nv-BdIzc	unknown		ff	ff	70
12	0170	1f	1		0008	20	99	7f	fd	Nv-BdIzc	unknown		ff	ff	fe
13	0170	1f	0		0008	20	99	7f	fd	Nv-BdIzc	unknown		00	1f	fe
13	0170	1f	0		0008	20	99	7f	fd	Nv-BdIzc	unknown		00	1f	20
14	0170	1f	0		0008	20	99	7f	fd	nv-BdIzc	unknown	T0	20	20	20
14	0170	20	0		0008	20	99	7f	fd	nv-BdIzc	unknown	T0	20	20	40
15	0008	ea	1	NOP	0008	20	99	7f	fd	nv-BdIzc	unknown	T1	20	df	40
15	0008	ea	1	NOP	0008	20	99	7f	fd	nv-BdIzc	unknown	T1	20	df	ff
16	0009	ea	1		0009	ff	99	7f	fd	Nv-BdIzc	NOP	T0+T2	ff	ff	ff
16	0009	ea	1		0009	ff	99	7f	fd	Nv-BdIzc	NOP	T0+T2	ff	ff	fe

CPX (E4)		Cycles: 3	Size: 2
Zero Page (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read Address	Store as Operand.	
3.1		C flag = (X >= Operand), N flag = ((X - Operand) & \$80), Z flag = (X == Operand).	
3.2	Read PC	Store as OpCode.	
+X.1		* C flag, N flag, and Z flag applied.	

* Setting of flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	e4	1	CPX zp	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	e4	1	CPX zp	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	66	1		0007	20	99	7f	fd	Nv-BdIzc	CPX zp	T2	32	32	32
8	0007	66	1		0007	20	99	7f	fd	Nv-BdIzc	CPX zp	T2	32	32	64
9	0066	99	1		0008	20	99	7f	fd	Nv-BdIzc	CPX zp	T0	ff	66	64
9	0066	99	1		0008	20	99	7f	fd	Nv-BdIzc	CPX zp	T0	ff	66	65
10	0008	ea	1	NOP	0008	20	99	7f	fd	Nv-BdIzc	CPX zp	T1	99	66	65
10	0008	ea	1	NOP	0008	20	99	7f	fd	Nv-BdIzc	CPX zp	T1	99	66	00
11	0009	ea	1		0009	20	99	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00
11	0009	ea	1		0009	20	99	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00

SBC (E5)	Cycles: 3	Size: 2
Zero Page (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read Address	Store as Operand.
3.1		Val = u16(Operand) ^ \$00FF. Result16 = u16(A) + Val + C flag. V flag = ((u16(A) ^ Result16) & (Val ^ Result16) & \$0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (A & \$80), Z flag = ! A .
3.2	Read PC	Store as OpCode.
+X.1		* A , C flag, N flag, V flag, and Z flag applied.

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	e5	1	SBC zp	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	e5	1	SBC zp	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	66	1		0007	20	99	7f	fd	Nv-BdIzc	SBC zp	T2	32	32	32
8	0007	66	1		0007	20	99	7f	fd	Nv-BdIzc	SBC zp	T2	32	32	64
9	0066	1f	1		0008	20	99	7f	fd	Nv-BdIzc	SBC zp	T0	ff	66	64
9	0066	1f	1		0008	20	99	7f	fd	Nv-BdIzc	SBC zp	T0	ff	66	65
10	0008	ea	1	NOP	0008	20	99	7f	fd	Nv-BdIzc	SBC zp	T1	20	e0	65
10	0008	ea	1	NOP	0008	20	99	7f	fd	Nv-BdIzc	SBC zp	T1	20	e0	00
11	0009	ea	1		0009	00	99	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00
11	0009	ea	1		0009	00	99	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00

INC (E6)		Cycles: 5	Size: 2
Zero Page (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Address.	
2.1		Inc. PC .	
2.2	Read Address	Store as Operand.	
3.1			
3.2	Write to Address	Write Operand.	
4.1		Operand += 1. N flag = (Operand & \$80), Z flag = !Operand.	
4.2	Write to Address	Write Operand.	
5.1			
5.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	e6	1	INC zp	0006	20	99	7f	fd	Nv-BdIZc	LDX #	T1	99	99	fc
7	0006	e6	1	INC zp	0006	20	99	7f	fd	Nv-BdIZc	LDX #	T1	99	99	32
8	0007	66	1		0007	20	99	7f	fd	Nv-BdIZc	INC zp	T2	32	32	32
8	0007	66	1		0007	20	99	7f	fd	Nv-BdIZc	INC zp	T2	32	32	64
9	0066	ff	1		0008	20	99	7f	fd	Nv-BdIZc	INC zp	T3	ff	66	64
9	0066	ff	1		0008	20	99	7f	fd	Nv-BdIZc	INC zp	T3	ff	66	65
10	0066	ff	0		0008	20	99	7f	fd	Nv-BdIZc	INC zp	T4	00	ff	65
10	0066	ff	0		0008	20	99	7f	fd	Nv-BdIZc	INC zp	T4	00	ff	00
11	0066	ff	0		0008	20	99	7f	fd	nv-BdIZc	INC zp	T0	00	00	00
11	0066	00	0		0008	20	99	7f	fd	nv-BdIZc	INC zp	T0	00	00	00
12	0008	ea	1	NOP	0008	20	99	7f	fd	nv-BdIZc	INC zp	T1	ff	00	00
12	0008	ea	1	NOP	0008	20	99	7f	fd	nv-BdIZc	INC zp	T1	ff	00	ff
13	0009	ea	1		0009	20	99	7f	fd	nv-BdIZc	NOP	T0+T2	ff	ff	ff
13	0009	ea	1		0009	20	99	7f	fd	nv-BdIZc	NOP	T0+T2	ff	ff	fe

ISC (E7)	Cycles: 5	Size: 2
Zero Page (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read Address	Store as Operand.
3.1		
3.2	Write to Address	Write Operand.
4.1		Operand += 1. Val = u16(Operand) ^ \$00FF. Result16 = u16(A) + Val + C flag. V flag = ((u16(A) ^ Result16) & (Val ^ Result16) & \$0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (A & \$80), Z flag = ! A .
4.2	Write to Address	Write Operand.
5.1		
5.2	Read PC	Store as OpCode.
+X.1		* A , C flag, N flag, V flag, and Z flag applied.

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	e7	1	unknown	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	e7	1	unknown	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	66	1		0007	20	99	7f	fd	Nv-BdIzc	unknown	T2	32	32	32
8	0007	66	1		0007	20	99	7f	fd	Nv-BdIzc	unknown	T2	32	32	64
9	0066	ff	1		0008	20	99	7f	fd	Nv-BdIzc	unknown	T3	ff	66	64
9	0066	ff	1		0008	20	99	7f	fd	Nv-BdIzc	unknown	T3	ff	66	65
10	0066	ff	0		0008	20	99	7f	fd	Nv-BdIzc	unknown	T4	00	ff	65
10	0066	ff	0		0008	20	99	7f	fd	Nv-BdIzc	unknown	T4	00	ff	00
11	0066	ff	0		0008	20	99	7f	fd	nv-BdIZc	unknown	T0	00	00	00
11	0066	00	0		0008	20	99	7f	fd	nv-BdIZc	unknown	T0	00	00	00
12	0008	ea	1	NOP	0008	20	99	7f	fd	nv-BdIZc	unknown	T1	20	ff	00
12	0008	ea	1	NOP	0008	20	99	7f	fd	nv-BdIZc	unknown	T1	20	ff	1f
13	0009	ea	1		0009	1f	99	7f	fd	nv-BdIzc	NOP	T0+T2	1f	1f	1f
13	0009	ea	1		0009	1f	99	7f	fd	nv-BdIzc	NOP	T0+T2	1f	1f	3e

INX (E8)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Discard.	
2.1		X += 1. N flag = (X & \$80), Z flag = ! X .	
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	e8	1	INX	0006	20	ff	7f	fd	Nv-BdIzc	LDX #	T1	ff	ff	fc
7	0006	e8	1	INX	0006	20	ff	7f	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	ea	1		0007	20	ff	7f	fd	Nv-BdIzc	INX	T0+T2	fe	fe	fe
8	0007	ea	1		0007	20	ff	7f	fd	Nv-BdIzc	INX	T0+T2	fe	fe	fc
9	0007	ea	1	NOP	0007	20	ff	7f	fd	Nv-BdIzc	INX	T1	ff	00	fc
9	0007	ea	1	NOP	0007	20	ff	7f	fd	Nv-BdIzc	INX	T1	ff	00	00
10	0008	ea	1		0008	20	00	7f	fd	nv-BdIZc	NOP	T0+T2	00	00	00
10	0008	ea	1		0008	20	00	7f	fd	nv-BdIZc	NOP	T0+T2	00	00	00

SBC (E9)	Cycles: 2	Size: 2
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Val = u16(Operand) ^ \$00FF. Result16 = u16(A) + Val + C flag. V flag = ((u16(A) ^ Result16) & (Val ^ Result16) & \$0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (A & \$80), Z flag = ! A .
2.2	Read PC	Store as OpCode.
+X.1		* A , C flag, N flag, V flag, and Z flag applied.

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	e9	1	SBC #	0006	20	ff	7f	fd	Nv-BdIzc	LDX #	T1	ff	ff	fc
7	0006	e9	1	SBC #	0006	20	ff	7f	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	1f	1		0007	20	ff	7f	fd	Nv-BdIzc	SBC #	T0+T2	fe	fe	fe
8	0007	1f	1		0007	20	ff	7f	fd	Nv-BdIzc	SBC #	T0+T2	fe	fe	fc
9	0008	ea	1	NOP	0008	20	ff	7f	fd	Nv-BdIzc	SBC #	T1	20	e0	fc
9	0008	ea	1	NOP	0008	20	ff	7f	fd	Nv-BdIzc	SBC #	T1	20	e0	00
10	0009	ea	1		0009	00	ff	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00
10	0009	ea	1		0009	00	ff	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00

NOP (EA)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Discard.	
2.1			
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	ea	1	NOP	0006	20	ff	7f	fd	Nv-BdIzc	LDX #	T1	ff	ff	fc
7	0006	ea	1	NOP	0006	20	ff	7f	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	ea	1		0007	20	ff	7f	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fe
8	0007	ea	1		0007	20	ff	7f	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fc
9	0007	ea	1	NOP	0007	20	ff	7f	fd	Nv-BdIzc	NOP	T1	ff	ff	fc
9	0007	ea	1	NOP	0007	20	ff	7f	fd	Nv-BdIzc	NOP	T1	ff	ff	fe
10	0008	ea	1		0008	20	ff	7f	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fe
10	0008	ea	1		0008	20	ff	7f	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fc

SBC (EB)	Cycles: 2	Size: 2
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Val = u16(Operand) ^ \$00FF. Result16 = u16(A) + Val + C flag. V flag = ((u16(A) ^ Result16) & (Val ^ Result16) & \$0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (A & \$80), Z flag = ! A .
2.2	Read PC	Store as OpCode.
+X.1		* A , C flag, N flag, V flag, and Z flag applied.

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	eb	1	unknown	0006	20	ff	7f	fd	Nv-BdIzc	LDX #	T1	ff	ff	fc
7	0006	eb	1	unknown	0006	20	ff	7f	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	1f	1		0007	20	ff	7f	fd	Nv-BdIzc	unknown	T0+T2	fe	fe	fe
8	0007	1f	1		0007	20	ff	7f	fd	Nv-BdIzc	unknown	T0+T2	fe	fe	fc
9	0008	ea	1	NOP	0008	20	ff	7f	fd	Nv-BdIzc	unknown	T1	20	e0	fc
9	0008	ea	1	NOP	0008	20	ff	7f	fd	Nv-BdIzc	unknown	T1	20	e0	00
10	0009	ea	1		0009	00	ff	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00
10	0009	ea	1		0009	00	ff	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00

CPX (EC)	Cycles: 4	Size: 3
Absolute (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address	Store as Operand.
4.1		C flag = (X >= Operand), N flag = ((X - Operand) & \$80), Z flag = (X == Operand).
4.2	Read PC	Store as OpCode.
+X.1		* C flag, N flag, and Z flag applied.

* Setting of flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	ec	1	CPX Abs	0006	20	ff	7f	fd	Nv-BdIzc	LDX #	T1	ff	ff	fc
7	0006	ec	1	CPX Abs	0006	20	ff	7f	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	70	1		0007	20	ff	7f	fd	Nv-BdIzc	CPX Abs	T2	fe	fe	fe
8	0007	70	1		0007	20	ff	7f	fd	Nv-BdIzc	CPX Abs	T2	fe	fe	fc
9	0008	01	1		0008	20	ff	7f	fd	Nv-BdIzc	CPX Abs	T3	00	70	fc
9	0008	01	1		0008	20	ff	7f	fd	Nv-BdIzc	CPX Abs	T3	00	70	70
10	0170	ff	1		0009	20	ff	7f	fd	Nv-BdIzc	CPX Abs	T0	ff	ff	70
10	0170	ff	1		0009	20	ff	7f	fd	Nv-BdIzc	CPX Abs	T0	ff	ff	fe
11	0009	ea	1	NOP	0009	20	ff	7f	fd	Nv-BdIzc	CPX Abs	T1	ff	00	fe
11	0009	ea	1	NOP	0009	20	ff	7f	fd	Nv-BdIzc	CPX Abs	T1	ff	00	00
12	000a	15	1		000a	20	ff	7f	fd	nv-BdIzc	NOP	T0+T2	00	00	00
12	000a	15	1		000a	20	ff	7f	fd	nv-BdIzc	NOP	T0+T2	00	00	00

SBC (ED)	Cycles: 4	Size: 3
Absolute (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address	Store as Operand.
4.1		Val = u16(Operand) ^ \$00FF. Result16 = u16(A) + Val + C flag. V flag = ((u16(A) ^ Result16) & (Val ^ Result16) & \$0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (A & \$80), Z flag = ! A .
4.2	Read PC	Store as OpCode.
+X.1		* A , C flag, N flag, V flag, and Z flag applied.

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	ed	1	SBC Abs	0006	20	ff	7f	fd	Nv-BdIzc	LDX #	T1	ff	ff	fc
7	0006	ed	1	SBC Abs	0006	20	ff	7f	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	70	1		0007	20	ff	7f	fd	Nv-BdIzc	SBC Abs	T2	fe	fe	fe
8	0007	70	1		0007	20	ff	7f	fd	Nv-BdIzc	SBC Abs	T2	fe	fe	fc
9	0008	01	1		0008	20	ff	7f	fd	Nv-BdIzc	SBC Abs	T3	00	70	fc
9	0008	01	1		0008	20	ff	7f	fd	Nv-BdIzc	SBC Abs	T3	00	70	70
10	0170	1f	1		0009	20	ff	7f	fd	Nv-BdIzc	SBC Abs	T0	ff	ff	70
10	0170	1f	1		0009	20	ff	7f	fd	Nv-BdIzc	SBC Abs	T0	ff	ff	fe
11	0009	ea	1	NOP	0009	20	ff	7f	fd	Nv-BdIzc	SBC Abs	T1	20	e0	fe
11	0009	ea	1	NOP	0009	20	ff	7f	fd	Nv-BdIzc	SBC Abs	T1	20	e0	00
12	000a	15	1		000a	00	ff	7f	fd	nv-BdIzc	NOP	T0+T2	00	00	00
12	000a	15	1		000a	00	ff	7f	fd	nv-BdIzc	NOP	T0+T2	00	00	00

INC (EE)	Cycles: 6	Size: 3
Absolute (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.
5.1		Operand += 1. N flag = (Operand & \$80), Z flag = !Operand.
5.2	Write to Address	Write Operand.
6.1		
6.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	ee	1	INC Abs	0006	20	ff	7f	fd	Nv-BdIZc	LDX #	T1	ff	ff	fc
7	0006	ee	1	INC Abs	0006	20	ff	7f	fd	Nv-BdIZc	LDX #	T1	ff	ff	fe
8	0007	70	1		0007	20	ff	7f	fd	Nv-BdIZc	INC Abs	T2	fe	fe	fe
8	0007	70	1		0007	20	ff	7f	fd	Nv-BdIZc	INC Abs	T2	fe	fe	fc
9	0008	01	1		0008	20	ff	7f	fd	Nv-BdIZc	INC Abs	T3	00	70	fc
9	0008	01	1		0008	20	ff	7f	fd	Nv-BdIZc	INC Abs	T3	00	70	70
10	0170	ff	1		0009	20	ff	7f	fd	Nv-BdIZc	INC Abs	T4	ff	ff	70
10	0170	ff	1		0009	20	ff	7f	fd	Nv-BdIZc	INC Abs	T4	ff	ff	fe
11	0170	ff	0		0009	20	ff	7f	fd	Nv-BdIZc	INC Abs	T5	00	ff	fe
11	0170	ff	0		0009	20	ff	7f	fd	Nv-BdIZc	INC Abs	T5	00	ff	00
12	0170	ff	0		0009	20	ff	7f	fd	nv-BdIZc	INC Abs	T0	00	00	00
12	0170	00	0		0009	20	ff	7f	fd	nv-BdIZc	INC Abs	T0	00	00	00
13	0009	ea	1	NOP	0009	20	ff	7f	fd	nv-BdIZc	INC Abs	T1	ff	00	00
13	0009	ea	1	NOP	0009	20	ff	7f	fd	nv-BdIZc	INC Abs	T1	ff	00	ff
14	000a	15	1		000a	20	ff	7f	fd	nv-BdIZc	NOP	T0+T2	ff	ff	ff
14	000a	15	1		000a	20	ff	7f	fd	nv-BdIZc	NOP	T0+T2	ff	ff	fe

ISC (EF)	Cycles: 6	Size: 3
Absolute (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.
5.1		Operand += 1. Val = u16(Operand) ^ \$00FF. Result16 = u16(A) + Val + C flag. V flag = ((u16(A) ^ Result16) & (Val ^ Result16) & \$0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (A & \$80), Z flag = !A.
5.2	Write to Address	Write Operand.
6.1		
6.2	Read PC	Store as OpCode.
+X.1		* A , C flag, N flag, V flag, and Z flag applied.

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	ef	1	unknown	0006	20	ff	7f	fd	Nv-BdIzc	LDX #	T1	ff	ff	fc
7	0006	ef	1	unknown	0006	20	ff	7f	fd	Nv-BdIzc	LDX #	T1	ff	ff	fe
8	0007	70	1		0007	20	ff	7f	fd	Nv-BdIzc	unknown	T2	fe	fe	fe
8	0007	70	1		0007	20	ff	7f	fd	Nv-BdIzc	unknown	T2	fe	fe	fc
9	0008	01	1		0008	20	ff	7f	fd	Nv-BdIzc	unknown	T3	00	70	fc
9	0008	01	1		0008	20	ff	7f	fd	Nv-BdIzc	unknown	T3	00	70	70
10	0170	1e	1		0009	20	ff	7f	fd	Nv-BdIzc	unknown	T4	ff	ff	70
10	0170	1e	1		0009	20	ff	7f	fd	Nv-BdIzc	unknown	T4	ff	ff	fe
11	0170	1e	0		0009	20	ff	7f	fd	Nv-BdIzc	unknown	T5	00	1e	fe
11	0170	1e	0		0009	20	ff	7f	fd	Nv-BdIzc	unknown	T5	00	1e	1f
12	0170	1e	0		0009	20	ff	7f	fd	nv-BdIzc	unknown	T0	1f	1f	1f
12	0170	1f	0		0009	20	ff	7f	fd	nv-BdIzc	unknown	T0	1f	1f	3e
13	0009	ea	1	NOP	0009	20	ff	7f	fd	nv-BdIzc	unknown	T1	20	e0	3e
13	0009	ea	1	NOP	0009	20	ff	7f	fd	nv-BdIzc	unknown	T1	20	e0	00
14	000a	15	1		000a	00	ff	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00
14	000a	15	1		000a	00	ff	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00

BEQ (F0)		Cycles: 2-4	Size: 2
Relative (Jump)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1	Read PC	Inc. PC . Jump = ((P & \$02) == \$02).	
1.2		Store as Operand.	
		If !Jump, poll interrupts.	
		Otherwise Address = i16(i8(Operand)) + PC + 1.	
	Read PC	BoundaryCrossed = Address.H != PC .H.	
		If !BoundaryCrossed, poll interrupts.	
2.1		Inc. PC . If !Jump, end (next half-cycle is 4.2).	
2.2		Discard.	
3.1	Read PC	PC .L = Address.L. If !BoundaryCrossed, end (next half-cycle is 4.2).	
3.2		Discard. Poll interrupts.	
4.1		PC .H = Address.H.	
4.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	f0	1	BEQ	0006	20	ff	7f	fd	Nv-BdIZc	LDX #	T1	ff	ff	fc
7	0006	f0	1	BEQ	0006	20	ff	7f	fd	Nv-BdIZc	LDX #	T1	ff	ff	fe
8	0007	70	1		0007	20	ff	7f	fd	Nv-BdIZc	BEQ	T2	fe	fe	fe
8	0007	70	1		0007	20	ff	7f	fd	Nv-BdIZc	BEQ	T2	fe	fe	fc
9	0008	ea	1	NOP	0008	20	ff	7f	fd	Nv-BdIZc	BEQ		70	08	fc
9	0008	ea	1	NOP	0008	20	ff	7f	fd	Nv-BdIZc	BEQ		70	08	78
10	0009	ea	1		0009	20	ff	7f	fd	Nv-BdIZc	NOP	T0+T2	78	78	78
10	0009	ea	1		0009	20	ff	7f	fd	Nv-BdIZc	NOP	T0+T2	78	78	f0

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	f0	1	BEQ	0006	20	00	7f	fd	nv-BdIZc	LDX #	T1	00	00	fc
7	0006	f0	1	BEQ	0006	20	00	7f	fd	nv-BdIZc	LDX #	T1	00	00	00
8	0007	70	1		0007	20	00	7f	fd	nv-BdIZc	BEQ	T2	00	00	00
8	0007	70	1		0007	20	00	7f	fd	nv-BdIZc	BEQ	T2	00	00	00
9	0008	ea	1		0008	20	00	7f	fd	nv-BdIZc	BEQ	T3	70	08	00
9	0008	ea	1		0008	20	00	7f	fd	nv-BdIZc	BEQ	T3	70	08	78
10	0078	ea	1	NOP	0078	20	00	7f	fd	nv-BdIZc	BEQ		00	00	78
10	0078	ea	1	NOP	0078	20	00	7f	fd	nv-BdIZc	BEQ		00	00	01
11	0079	90	1		0079	20	00	7f	fd	nv-BdIZc	NOP	T0+T2	01	01	01
11	0079	90	1		0079	20	00	7f	fd	nv-BdIZc	NOP	T0+T2	01	01	02

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	f0	1	BEQ	0006	20	00	7f	fd	nv-BdIZc	LDX #	T1	00	00	fc
7	0006	f0	1	BEQ	0006	20	00	7f	fd	nv-BdIZc	LDX #	T1	00	00	00
8	0007	e0	1		0007	20	00	7f	fd	nv-BdIZc	BEQ	T2	00	00	00
8	0007	e0	1		0007	20	00	7f	fd	nv-BdIZc	BEQ	T2	00	00	00
9	0008	ea	1		0008	20	00	7f	fd	nv-BdIZc	BEQ	T3	e0	08	00
9	0008	ea	1		0008	20	00	7f	fd	nv-BdIZc	BEQ	T3	e0	08	e8
10	00e8	00	1		00e8	20	00	7f	fd	nv-BdIZc	BEQ	T0	00	ff	e8
10	00e8	00	1		00e8	20	00	7f	fd	nv-BdIZc	BEQ	T0	00	ff	ff
11	ffe8	00	1	BRK	ffe8	20	00	7f	fd	nv-BdIZc	BEQ	T1	ff	00	ff
11	ffe8	00	1	BRK	ffe8	20	00	7f	fd	nv-BdIZc	BEQ	T1	ff	00	ff
12	ffe9	00	1		ffe9	20	00	7f	fd	nv-BdIZc	BRK	T2	ff	ff	ff
12	ffe9	00	1		ffe9	20	00	7f	fd	nv-BdIZc	BRK	T2	ff	ff	fe

SBC (F1)		Cycles: 5-6	Size: 2
Indirect Y (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Address.	
3.1			
3.2	Read u8(Pointer + 1)	Store as Address.H.	
4.1		Target = Address + Y. Pointer.L = Target.L. Pointer.H = Address.H. BoundaryCrossed = Pointer.H != Target.H.	
4.2	Read Pointer	Store as Operand. If !BoundaryCrossed, skip the next cycle.	
5.1		Pointer.H = Target.H.	
5.2	Read Pointer	Store as Operand.	
6.1		Val = u16(Operand) ^ \$00FF. Result16 = u16(A) + Val + C flag. V flag = ((u16(A) ^ Result16) & (Val ^ Result16) & \$0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (A & \$80), Z flag = ! A .	
6.2	Read PC	Store as OpCode.	
+X.1		* A , C flag, N flag, V flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	f1	1	SBC (zp),Y	0006	20	00	7f	fd	nv-BdIZc	LDX #	T1	00	00	fc
7	0006	f1	1	SBC (zp),Y	0006	20	00	7f	fd	nv-BdIZc	LDX #	T1	00	00	00
8	0007	70	1		0007	20	00	7f	fd	nv-BdIZc	SBC (zp),Y	T2	00	00	00
8	0007	70	1		0007	20	00	7f	fd	nv-BdIZc	SBC (zp),Y	T2	00	00	00
9	0070	00	1		0008	20	00	7f	fd	nv-BdIZc	SBC (zp),Y	T3	00	70	00
9	0070	00	1		0008	20	00	7f	fd	nv-BdIZc	SBC (zp),Y	T3	00	70	71
10	0071	00	1		0008	20	00	7f	fd	nv-BdIZc	SBC (zp),Y	T4	7f	00	71
10	0071	00	1		0008	20	00	7f	fd	nv-BdIZc	SBC (zp),Y	T4	7f	00	7f
11	007f	1f	1		0008	20	00	7f	fd	nv-BdIZc	SBC (zp),Y	T0	00	00	7f
11	007f	1f	1		0008	20	00	7f	fd	nv-BdIZc	SBC (zp),Y	T0	00	00	01
12	0008	ea	1	NOP	0008	20	00	7f	fd	nv-BdIZc	SBC (zp),Y	T1	20	e0	01
12	0008	ea	1	NOP	0008	20	00	7f	fd	nv-BdIZc	SBC (zp),Y	T1	20	e0	00
13	0009	ea	1		0009	00	00	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00
13	0009	ea	1		0009	00	00	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	f1	1	SBC (zp),Y	0006	20	00	7f	fd	nv-BdIZc	LDX #	T1	00	00	fc
7	0006	f1	1	SBC (zp),Y	0006	20	00	7f	fd	nv-BdIZc	LDX #	T1	00	00	00
8	0007	70	1		0007	20	00	7f	fd	nv-BdIZc	SBC (zp),Y	T2	00	00	00
8	0007	70	1		0007	20	00	7f	fd	nv-BdIZc	SBC (zp),Y	T2	00	00	00
9	0070	81	1		0008	20	00	7f	fd	nv-BdIZc	SBC (zp),Y	T3	00	70	00
9	0070	81	1		0008	20	00	7f	fd	nv-BdIZc	SBC (zp),Y	T3	00	70	71
10	0071	00	1		0008	20	00	7f	fd	nv-BdIZc	SBC (zp),Y	T4	7f	81	71
10	0071	00	1		0008	20	00	7f	fd	nv-BdIZc	SBC (zp),Y	T4	7f	81	00
11	0000	a9	1		0008	20	00	7f	fd	nv-BdIZc	SBC (zp),Y	T5	00	00	00
11	0000	a9	1		0008	20	00	7f	fd	nv-BdIZc	SBC (zp),Y	T5	00	00	01
12	0100	1f	1		0008	20	00	7f	fd	nv-BdIZc	SBC (zp),Y	T0	01	ff	01
12	0100	1f	1		0008	20	00	7f	fd	nv-BdIZc	SBC (zp),Y	T0	01	ff	00
13	0008	ea	1	NOP	0008	20	00	7f	fd	nv-BdIZc	SBC (zp),Y	T1	20	e0	00
13	0008	ea	1	NOP	0008	20	00	7f	fd	nv-BdIZc	SBC (zp),Y	T1	20	e0	00
14	0009	ea	1		0009	00	00	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00
14	0009	ea	1		0009	00	00	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00

JAM (F2)	Cycles: ∞	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Discard.
2.1		Visual6502 suggests incrementing PC , but Single-Step Tests cannot be passed this way.
2.2	Read \$FFFF	Discard.
3.1		
3.2	Read \$FFFE	Discard.
4.1		
4.2	Read \$FFFE	Discard.
5.1		Repeat 5.1 and 5.2 forever.
5.2	Read \$FFFF	Discard.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	f2	1	unknown	0006	20	00	7f	fd	nv-BdIZc	LDX #	T1	00	00	fc
7	0006	f2	1	unknown	0006	20	00	7f	fd	nv-BdIZc	LDX #	T1	00	00	00
8	0007	70	1		0007	20	00	7f	fd	nv-BdIZc	unknown	T2	00	00	00
8	0007	70	1		0007	20	00	7f	fd	nv-BdIZc	unknown	T2	00	00	00
9	ffff	00	1		0008	20	00	7f	fd	nv-BdIZc	unknown	T3	ff	ff	00
9	ffff	00	1		0008	20	00	7f	fd	nv-BdIZc	unknown	T3	ff	ff	fe
10	fffe	00	1		0008	20	00	7f	fd	nv-BdIZc	unknown	T4	ff	ff	fe
10	fffe	00	1		0008	20	00	7f	fd	nv-BdIZc	unknown	T4	ff	ff	fe
11	fffe	00	1		0008	20	00	7f	fd	nv-BdIZc	unknown	T5	ff	ff	fe
11	fffe	00	1		0008	20	00	7f	fd	nv-BdIZc	unknown	T5	ff	ff	fe
12	ffff	00	1		0008	20	00	7f	fd	nv-BdIZc	unknown		ff	ff	fe
12	ffff	00	1		0008	20	00	7f	fd	nv-BdIZc	unknown		ff	ff	fe
13	ffff	00	1		0008	20	00	7f	fd	nv-BdIZc	unknown		ff	ff	fe
13	ffff	00	1		0008	20	00	7f	fd	nv-BdIZc	unknown		ff	ff	fe
14	ffff	00	1		0008	20	00	7f	fd	nv-BdIZc	unknown		ff	ff	fe
14	ffff	00	1		0008	20	00	7f	fd	nv-BdIZc	unknown		ff	ff	fe

ISC (F3)		Cycles: 8	Size: 2
Indirect Y (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Address.	
3.1			
3.2	Read u8(Pointer + 1)	Store as Address.H.	
4.1		Target = Address + Y. Pointer.L = Target.L. Pointer.H = Address.H.	
4.2	Read Pointer	Store as Operand.	
5.1		Pointer.H = Target.H.	
5.2	Read Pointer	Store as Operand.	
6.1			
6.2	Write to Pointer	Write Operand.	
7.1		Operand += 1. Val = u16(Operand) ^ \$00FF. Result16 = u16(A) + Val + C flag. V flag = ((u16(A) ^ Result16) & (Val ^ Result16) & \$0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (A & \$80), Z flag = !A.	
7.2	Write to Pointer	Write Operand.	
8.1			
8.2	Read PC	Store as OpCode.	
+X.1		* A, C flag, N flag, V flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	f3	1	unknown	0006	20	00	7f	fd	nv-BdIZc	LDX #	T1	00	00	fc
7	0006	f3	1	unknown	0006	20	00	7f	fd	nv-BdIZc	LDX #	T1	00	00	00
8	0007	70	1		0007	20	00	7f	fd	nv-BdIZc	unknown	T2	00	00	00
8	0007	70	1		0007	20	00	7f	fd	nv-BdIZc	unknown	T2	00	00	00
9	0070	81	1		0008	20	00	7f	fd	nv-BdIZc	unknown	T3	00	70	00
9	0070	81	1		0008	20	00	7f	fd	nv-BdIZc	unknown	T3	00	70	71
10	0071	00	1		0008	20	00	7f	fd	nv-BdIZc	unknown	T4	7f	81	71
10	0071	00	1		0008	20	00	7f	fd	nv-BdIZc	unknown	T4	7f	81	00
11	0000	a9	1		0008	20	00	7f	fd	nv-BdIZc	unknown	T5	00	00	00
11	0000	a9	1		0008	20	00	7f	fd	nv-BdIZc	unknown	T5	00	00	01
12	0100	1e	1		0008	20	00	7f	fd	nv-BdIZc	unknown		01	ff	01
12	0100	1e	1		0008	20	00	7f	fd	nv-BdIZc	unknown		01	ff	00
13	0100	1e	0		0008	20	00	7f	fd	nv-BdIZc	unknown		00	1e	00
13	0100	1e	0		0008	20	00	7f	fd	nv-BdIZc	unknown		00	1e	1f
14	0100	1e	0		0008	20	00	7f	fd	nv-BdIZc	unknown	T0	1f	1f	1f
14	0100	1f	0		0008	20	00	7f	fd	nv-BdIZc	unknown	T0	1f	1f	3e
15	0008	ea	1	NOP	0008	20	00	7f	fd	nv-BdIZc	unknown	T1	20	e0	3e
15	0008	ea	1	NOP	0008	20	00	7f	fd	nv-BdIZc	unknown	T1	20	e0	00
16	0009	ea	1		0009	00	00	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00
16	0009	ea	1		0009	00	00	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00

NOP (F4)		Cycles: 4	Size: 2
Zero Page X-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Operand.	
3.1		Address = u8(Pointer + X).	
3.2	Read Address	Store as Operand.	
4.1			
4.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	f4	1	unknown	0006	20	00	7f	fd	nv-BdIZc	LDX #	T1	00	00	fc
7	0006	f4	1	unknown	0006	20	00	7f	fd	nv-BdIZc	LDX #	T1	00	00	00
8	0007	70	1		0007	20	00	7f	fd	nv-BdIZc	unknown	T2	00	00	00
8	0007	70	1		0007	20	00	7f	fd	nv-BdIZc	unknown	T2	00	00	00
9	0070	81	1		0008	20	00	7f	fd	nv-BdIZc	unknown	T3	00	70	00
9	0070	81	1		0008	20	00	7f	fd	nv-BdIZc	unknown	T3	00	70	70
10	0070	81	1		0008	20	00	7f	fd	nv-BdIZc	unknown	T0	ff	ff	70
10	0070	81	1		0008	20	00	7f	fd	nv-BdIZc	unknown	T0	ff	ff	fe
11	0008	ea	1	NOP	0008	20	00	7f	fd	nv-BdIZc	unknown	T1	ff	81	fe
11	0008	ea	1	NOP	0008	20	00	7f	fd	nv-BdIZc	unknown	T1	ff	81	80
12	0009	ea	1		0009	20	00	7f	fd	nv-BdIZc	NOP	T0+T2	80	80	80
12	0009	ea	1		0009	20	00	7f	fd	nv-BdIZc	NOP	T0+T2	80	80	00

SBC (F5)		Cycles: 4	Size: 2
Zero Page X-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read Pointer	Store as Operand.	
3.1		Address = u8(Pointer + X).	
3.2	Read Address	Store as Operand.	
4.1		Val = u16(Operand) ^ \$00FF. Result16 = u16(A) + Val + C flag. V flag = ((u16(A) ^ Result16) & (Val ^ Result16) & \$0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (A & \$80), Z flag = ! A .	
4.2	Read PC	Store as OpCode.	
+X.1		* A , C flag, N flag, V flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	f5	1	SBC zp,X	0006	20	00	7f	fd	nv-BdIZc	LDX #	T1	00	00	fc
7	0006	f5	1	SBC zp,X	0006	20	00	7f	fd	nv-BdIZc	LDX #	T1	00	00	00
8	0007	70	1		0007	20	00	7f	fd	nv-BdIZc	SBC zp,X	T2	00	00	00
8	0007	70	1		0007	20	00	7f	fd	nv-BdIZc	SBC zp,X	T2	00	00	00
9	0070	1f	1		0008	20	00	7f	fd	nv-BdIZc	SBC zp,X	T3	00	70	00
9	0070	1f	1		0008	20	00	7f	fd	nv-BdIZc	SBC zp,X	T3	00	70	70
10	0070	1f	1		0008	20	00	7f	fd	nv-BdIZc	SBC zp,X	T0	ff	ff	70
10	0070	1f	1		0008	20	00	7f	fd	nv-BdIZc	SBC zp,X	T0	ff	ff	fe
11	0008	ea	1	NOP	0008	20	00	7f	fd	nv-BdIZc	SBC zp,X	T1	20	e0	fe
11	0008	ea	1	NOP	0008	20	00	7f	fd	nv-BdIZc	SBC zp,X	T1	20	e0	00
12	0009	ea	1		0009	00	00	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00
12	0009	ea	1		0009	00	00	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00

INC (F6)	Cycles: 6	Size: 2
Zero Page X-Indexed (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read Pointer	Store as Operand.
3.1		Address = u8(Pointer + X).
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.
5.1		Operand += 1. N flag = (Operand & \$80), Z flag = !Operand.
5.2	Write to Address	Write Operand.
6.1		
6.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	f6	1	INC zp,X	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	f6	1	INC zp,X	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	70	1		0007	20	99	7f	fd	Nv-BdIzc	INC zp,X	T2	32	32	32
8	0007	70	1		0007	20	99	7f	fd	Nv-BdIzc	INC zp,X	T2	32	32	64
9	0070	ff	1		0008	20	99	7f	fd	Nv-BdIzc	INC zp,X	T3	99	70	64
9	0070	ff	1		0008	20	99	7f	fd	Nv-BdIzc	INC zp,X	T3	99	70	09
10	0009	ff	1		0008	20	99	7f	fd	Nv-BdIzc	INC zp,X	T4	ff	ff	09
10	0009	ff	1		0008	20	99	7f	fd	Nv-BdIzc	INC zp,X	T4	ff	ff	fe
11	0009	ff	0		0008	20	99	7f	fd	Nv-BdIzc	INC zp,X	T5	00	ff	fe
11	0009	ff	0		0008	20	99	7f	fd	Nv-BdIzc	INC zp,X	T5	00	ff	00
12	0009	ff	0		0008	20	99	7f	fd	nv-BdIZc	INC zp,X	T0	00	00	00
12	0009	00	0		0008	20	99	7f	fd	nv-BdIZc	INC zp,X	T0	00	00	00
13	0008	ea	1	NOP	0008	20	99	7f	fd	nv-BdIZc	INC zp,X	T1	ff	00	00
13	0008	ea	1	NOP	0008	20	99	7f	fd	nv-BdIZc	INC zp,X	T1	ff	00	ff
14	0009	00	1		0009	20	99	7f	fd	nv-BdIZc	NOP	T0+T2	ff	ff	ff
14	0009	00	1		0009	20	99	7f	fd	nv-BdIZc	NOP	T0+T2	ff	ff	fe

ISC (F7)	Cycles: 6	Size: 2
Zero Page X-Indexed (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read Pointer	Store as Operand.
3.1		Address = u8(Pointer + X).
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.
5.1		Operand += 1. Val = u16(Operand) ^ \$00FF. Result16 = u16(A) + Val + C flag. V flag = ((u16(A) ^ Result16) & (Val ^ Result16) & \$0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (A & \$80), Z flag = ! A .
5.2	Write to Address	Write Operand.
6.1		
6.2	Read PC	Store as OpCode.
+X.1		* A , C flag, N flag, V flag, and Z flag applied.

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	f7	1	unknown	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	f7	1	unknown	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	70	1		0007	20	99	7f	fd	Nv-BdIzc	unknown	T2	32	32	32
8	0007	70	1		0007	20	99	7f	fd	Nv-BdIzc	unknown	T2	32	32	64
9	0070	ff	1		0008	20	99	7f	fd	Nv-BdIzc	unknown	T3	99	70	64
9	0070	ff	1		0008	20	99	7f	fd	Nv-BdIzc	unknown	T3	99	70	09
10	0009	ff	1		0008	20	99	7f	fd	Nv-BdIzc	unknown	T4	ff	ff	09
10	0009	ff	1		0008	20	99	7f	fd	Nv-BdIzc	unknown	T4	ff	ff	fe
11	0009	ff	0		0008	20	99	7f	fd	Nv-BdIzc	unknown	T5	00	ff	fe
11	0009	ff	0		0008	20	99	7f	fd	Nv-BdIzc	unknown	T5	00	ff	00
12	0009	ff	0		0008	20	99	7f	fd	nv-BdIZc	unknown	T0	00	00	00
12	0009	00	0		0008	20	99	7f	fd	nv-BdIZc	unknown	T0	00	00	00
13	0008	ea	1	NOP	0008	20	99	7f	fd	nv-BdIZc	unknown	T1	20	ff	00
13	0008	ea	1	NOP	0008	20	99	7f	fd	nv-BdIZc	unknown	T1	20	ff	1f
14	0009	00	1		0009	1f	99	7f	fd	nv-BdIZc	NOP	T0+T2	1f	1f	1f
14	0009	00	1		0009	1f	99	7f	fd	nv-BdIZc	NOP	T0+T2	1f	1f	3e

SED (F8)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Discard.	
2.1		D flag = 1.	
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	f8	1	SED	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	f8	1	SED	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	ea	1		0007	20	99	7f	fd	Nv-BdIzc	SED	T0+T2	32	32	32
8	0007	ea	1		0007	20	99	7f	fd	Nv-BdIzc	SED	T0+T2	32	32	64
9	0007	ea	1	NOP	0007	20	99	7f	fd	Nv-BDIzc	SED	T1	ff	ff	64
9	0007	ea	1	NOP	0007	20	99	7f	fd	Nv-BDIzc	SED	T1	ff	ff	fe
10	0008	ea	1		0008	20	99	7f	fd	Nv-BDIzc	NOP	T0+T2	fe	fe	fe
10	0008	ea	1		0008	20	99	7f	fd	Nv-BDIzc	NOP	T0+T2	fe	fe	fc

SBC (F9) Cycles: 4-5		Size: 3
Absolute Y-Indexed (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + Y. Address.L = Target.L. Address.H = Pointer.H. BoundaryCrossed = Pointer.H != Target.H.
3.2	Read Address	Store as Operand. If !BoundaryCrossed, skip the next cycle.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		Val = u16(Operand) ^ \$00FF. Result16 = u16(A) + Val + C flag. V flag = ((u16(A) ^ Result16) & (Val ^ Result16) & \$0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (A & \$80), Z flag = !A.
5.2	Read PC	Store as OpCode.
+X.1		* A, C flag, N flag, V flag, and Z flag applied.

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	f9	1	SBC Abs,Y	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	f9	1	SBC Abs,Y	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	70	1		0007	20	99	7f	fd	Nv-BdIzc	SBC Abs,Y	T2	32	32	32
8	0007	70	1		0007	20	99	7f	fd	Nv-BdIzc	SBC Abs,Y	T2	32	32	64
9	0008	01	1		0008	20	99	7f	fd	Nv-BdIzc	SBC Abs,Y	T3	7f	70	64
9	0008	01	1		0008	20	99	7f	fd	Nv-BdIzc	SBC Abs,Y	T3	7f	70	ef
10	01ef	1f	1		0009	20	99	7f	fd	Nv-BdIzc	SBC Abs,Y	T0	00	01	ef
10	01ef	1f	1		0009	20	99	7f	fd	Nv-BdIzc	SBC Abs,Y	T0	00	01	02
11	0009	ea	1	NOP	0009	20	99	7f	fd	Nv-BdIzc	SBC Abs,Y	T1	20	e0	02
11	0009	ea	1	NOP	0009	20	99	7f	fd	Nv-BdIzc	SBC Abs,Y	T1	20	e0	00
12	000a	15	1		000a	00	99	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00
12	000a	15	1		000a	00	99	7f	fd	nv-BdIZC	NOP	T0+T2	00	00	00

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	f9	1	SBC Abs,Y	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	f9	1	SBC Abs,Y	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	82	1		0007	20	99	7f	fd	Nv-BdIzc	SBC Abs,Y	T2	32	32	32
8	0007	82	1		0007	20	99	7f	fd	Nv-BdIzc	SBC Abs,Y	T2	32	32	64
9	0008	01	1		0008	20	99	7f	fd	Nv-BdIzc	SBC Abs,Y	T3	7f	82	64
9	0008	01	1		0008	20	99	7f	fd	Nv-BdIzc	SBC Abs,Y	T3	7f	82	01
10	0101	80	1		0009	20	99	7f	fd	Nv-BdIzc	SBC Abs,Y	T4	00	01	01
10	0101	80	1		0009	20	99	7f	fd	Nv-BdIzc	SBC Abs,Y	T4	00	01	02
11	0201	00	1		0009	20	99	7f	fd	Nv-BdIzc	SBC Abs,Y	T0	02	ff	02
11	0201	00	1		0009	20	99	7f	fd	Nv-BdIzc	SBC Abs,Y	T0	02	ff	01
12	0009	ea	1	NOP	0009	20	99	7f	fd	Nv-BdIzc	SBC Abs,Y	T1	20	ff	01
12	0009	ea	1	NOP	0009	20	99	7f	fd	Nv-BdIzc	SBC Abs,Y	T1	20	ff	1f
13	000a	15	1		000a	1f	99	7f	fd	nv-BdIZC	NOP	T0+T2	1f	1f	1f
13	000a	15	1		000a	1f	99	7f	fd	nv-BdIZC	NOP	T0+T2	1f	1f	3e

NOP (FA)		Cycles: 2	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Discard.	
2.1			
2.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	fa	1	unknown	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	fa	1	unknown	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	ea	1		0007	20	99	7f	fd	Nv-BdIzc	unknown	T0+T2	32	32	32
8	0007	ea	1		0007	20	99	7f	fd	Nv-BdIzc	unknown	T0+T2	32	32	64
9	0007	ea	1	NOP	0007	20	99	7f	fd	Nv-BdIzc	unknown	T1	ff	ff	64
9	0007	ea	1	NOP	0007	20	99	7f	fd	Nv-BdIzc	unknown	T1	ff	ff	fe
10	0008	01	1		0008	20	99	7f	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fe
10	0008	01	1		0008	20	99	7f	fd	Nv-BdIzc	NOP	T0+T2	fe	fe	fc

ISC (FB)	Cycles: 7	Size: 3
Absolute Y-Indexed (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + Y . Address.L = Target.L. Address.H = Pointer.H.
3.2	Read Address	Store as Operand.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		
5.2	Write to Address	Write Operand.
6.1		Operand += 1. Val = u16(Operand) ^ \$00FF. Result16 = u16(A) + Val + C flag. V flag = ((u16(A) ^ Result16) & (Val ^ Result16) & \$0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (A & \$80), Z flag = ! A .
6.2	Write to Address	Write Operand.
7.1		
7.2	Read PC	Store as OpCode.
+X.1		* A , C flag, N flag, V flag, and Z flag applied.

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	fb	1	unknown	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	fb	1	unknown	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	ea	1		0007	20	99	7f	fd	Nv-BdIzc	unknown	T2	32	32	32
8	0007	ea	1		0007	20	99	7f	fd	Nv-BdIzc	unknown	T2	32	32	64
9	0008	01	1		0008	20	99	7f	fd	Nv-BdIzc	unknown	T3	7f	ea	64
9	0008	01	1		0008	20	99	7f	fd	Nv-BdIzc	unknown	T3	7f	ea	69
10	0169	66	1		0009	20	99	7f	fd	Nv-BdIzc	unknown	T4	00	01	69
10	0169	66	1		0009	20	99	7f	fd	Nv-BdIzc	unknown	T4	00	01	02
11	0269	ff	1		0009	20	99	7f	fd	Nv-BdIzc	unknown	T5	02	ff	02
11	0269	ff	1		0009	20	99	7f	fd	Nv-BdIzc	unknown	T5	02	ff	01
12	0269	ff	0		0009	20	99	7f	fd	Nv-BdIzc	unknown		00	ff	01
12	0269	ff	0		0009	20	99	7f	fd	Nv-BdIzc	unknown		00	ff	00
13	0269	ff	0		0009	20	99	7f	fd	nv-BdIZc	unknown	T0	00	00	00
13	0269	00	0		0009	20	99	7f	fd	nv-BdIZc	unknown	T0	00	00	00
14	0009	ea	1	NOP	0009	20	99	7f	fd	nv-BdIZc	unknown	T1	20	ff	00
14	0009	ea	1	NOP	0009	20	99	7f	fd	nv-BdIZc	unknown	T1	20	ff	1f
15	000a	15	1		000a	1f	99	7f	fd	nv-BdIZc	NOP	T0+T2	1f	1f	1f
15	000a	15	1		000a	1f	99	7f	fd	nv-BdIZc	NOP	T0+T2	1f	1f	3e

NOP (FC)		Cycles: 4-5	Size: 3
Absolute X-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read PC	Store as Pointer.H.	
3.1		Inc. PC . Target = Pointer + X .	
		Address.L = Target.L.	
		Address.H = Pointer.H.	
		BoundaryCrossed = Pointer.H != Target.H.	
3.2	Read Address	Store as Operand. If !BoundaryCrossed, skip the next cycle.	
4.1		Address.H = Target.H.	
4.2	Read Address	Store as Operand.	
5.1			
5.2	Read PC	Store as OpCode.	
+X.1			

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	fc	1	unknown	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	fc	1	unknown	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	ea	1		0007	20	99	7f	fd	Nv-BdIzc	unknown	T2	32	32	32
8	0007	ea	1		0007	20	99	7f	fd	Nv-BdIzc	unknown	T2	32	32	64
9	0008	01	1		0008	20	99	7f	fd	Nv-BdIzc	unknown	T3	99	ea	64
9	0008	01	1		0008	20	99	7f	fd	Nv-BdIzc	unknown	T3	99	ea	83
10	0183	00	1		0009	20	99	7f	fd	Nv-BdIzc	unknown	T4	00	01	83
10	0183	00	1		0009	20	99	7f	fd	Nv-BdIzc	unknown	T4	00	01	02
11	0283	00	1		0009	20	99	7f	fd	Nv-BdIzc	unknown	T0	02	ff	02
11	0283	00	1		0009	20	99	7f	fd	Nv-BdIzc	unknown	T0	02	ff	01
12	0009	ea	1	NOP	0009	20	99	7f	fd	Nv-BdIzc	unknown	T1	ff	00	01
12	0009	ea	1	NOP	0009	20	99	7f	fd	Nv-BdIzc	unknown	T1	ff	00	ff
13	000a	15	1		000a	20	99	7f	fd	Nv-BdIzc	NOP	T0+T2	ff	ff	ff
13	000a	15	1		000a	20	99	7f	fd	Nv-BdIzc	NOP	T0+T2	ff	ff	fe

SBC (FD)		Cycles: 4-5	Size: 3
Absolute X-Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read PC	Store as Pointer.H.	
3.1		Inc. PC . Target = Pointer + X . Address.L = Target.L. Address.H = Pointer.H. BoundaryCrossed = Pointer.H != Target.H.	
3.2	Read Address	Store as Operand. If !BoundaryCrossed, skip the next cycle.	
4.1		Address.H = Target.H.	
4.2	Read Address	Store as Operand.	
5.1		Val = u16(Operand) ^ \$00FF. Result16 = u16(A) + Val + C flag. V flag = ((u16(A) ^ Result16) & (Val ^ Result16) & \$0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (A & \$80), Z flag = ! A .	
5.2	Read PC	Store as OpCode.	
+X.1		* A , C flag, N flag, V flag, and Z flag applied.	

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	fd	1	SBC Abs,X	0006	20	33	7f	fd	nv-BdIzc	LDX #	T1	33	33	fc
7	0006	fd	1	SBC Abs,X	0006	20	33	7f	fd	nv-BdIzc	LDX #	T1	33	33	66
8	0007	11	1		0007	20	33	7f	fd	nv-BdIzc	SBC Abs,X	T2	66	66	66
8	0007	11	1		0007	20	33	7f	fd	nv-BdIzc	SBC Abs,X	T2	66	66	cc
9	0008	01	1		0008	20	33	7f	fd	nv-BdIzc	SBC Abs,X	T3	33	11	cc
9	0008	01	1		0008	20	33	7f	fd	nv-BdIzc	SBC Abs,X	T3	33	11	44
10	0144	1f	1		0009	20	33	7f	fd	nv-BdIzc	SBC Abs,X	T0	00	01	44
10	0144	1f	1		0009	20	33	7f	fd	nv-BdIzc	SBC Abs,X	T0	00	01	02
11	0009	ea	1	NOP	0009	20	33	7f	fd	nv-BdIzc	SBC Abs,X	T1	20	e0	02
11	0009	ea	1	NOP	0009	20	33	7f	fd	nv-BdIzc	SBC Abs,X	T1	20	e0	00
12	000a	15	1		000a	00	33	7f	fd	nv-BdIzc	NOP	T0+T2	00	00	00
12	000a	15	1		000a	00	33	7f	fd	nv-BdIzc	NOP	T0+T2	00	00	00

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	fd	1	SBC Abs,X	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	fc
7	0006	fd	1	SBC Abs,X	0006	20	99	7f	fd	Nv-BdIzc	LDX #	T1	99	99	32
8	0007	70	1		0007	20	99	7f	fd	Nv-BdIzc	SBC Abs,X	T2	32	32	32
8	0007	70	1		0007	20	99	7f	fd	Nv-BdIzc	SBC Abs,X	T2	32	32	64
9	0008	01	1		0008	20	99	7f	fd	Nv-BdIzc	SBC Abs,X	T3	99	70	64
9	0008	01	1		0008	20	99	7f	fd	Nv-BdIzc	SBC Abs,X	T3	99	70	09
10	0109	00	1		0009	20	99	7f	fd	Nv-BdIzc	SBC Abs,X	T4	00	01	09
10	0109	00	1		0009	20	99	7f	fd	Nv-BdIzc	SBC Abs,X	T4	00	01	02
11	0209	00	1		0009	20	99	7f	fd	Nv-BdIzc	SBC Abs,X	T0	02	ff	02
11	0209	00	1		0009	20	99	7f	fd	Nv-BdIzc	SBC Abs,X	T0	02	ff	01
12	0009	ea	1	NOP	0009	20	99	7f	fd	Nv-BdIzc	SBC Abs,X	T1	20	ff	01
12	0009	ea	1	NOP	0009	20	99	7f	fd	Nv-BdIzc	SBC Abs,X	T1	20	ff	1f
13	000a	15	1		000a	1f	99	7f	fd	nv-BdIzc	NOP	T0+T2	1f	1f	1f
13	000a	15	1		000a	1f	99	7f	fd	nv-BdIzc	NOP	T0+T2	1f	1f	3e

INC (FE)	Cycles: 7	Size: 3
Absolute X-Indexed (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + X . Address.L = Target.L. Address.H = Pointer.H.
3.2	Read Address	Store as Operand.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		
5.2	Write to Address	Write Operand.
6.1		Operand += 1. N flag = (Operand & \$80), Z flag = !Operand.
6.2	Write to Address	Write Operand.
7.1		
7.2	Read PC	Store as OpCode.
+X.1		

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	fe	1	INC Abs,X	0006	20	33	7f	fd	nv-BdIzc	LDX #	T1	33	33	fc
7	0006	fe	1	INC Abs,X	0006	20	33	7f	fd	nv-BdIzc	LDX #	T1	33	33	66
8	0007	11	1		0007	20	33	7f	fd	nv-BdIzc	INC Abs,X	T2	66	66	66
8	0007	11	1		0007	20	33	7f	fd	nv-BdIzc	INC Abs,X	T2	66	66	cc
9	0008	01	1		0008	20	33	7f	fd	nv-BdIzc	INC Abs,X	T3	33	11	cc
9	0008	01	1		0008	20	33	7f	fd	nv-BdIzc	INC Abs,X	T3	33	11	44
10	0144	7f	1		0009	20	33	7f	fd	nv-BdIzc	INC Abs,X	T4	00	01	44
10	0144	7f	1		0009	20	33	7f	fd	nv-BdIzc	INC Abs,X	T4	00	01	02
11	0144	7f	1		0009	20	33	7f	fd	nv-BdIzc	INC Abs,X	T5	02	ff	02
11	0144	7f	1		0009	20	33	7f	fd	nv-BdIzc	INC Abs,X	T5	02	ff	01
12	0144	7f	0		0009	20	33	7f	fd	nv-BdIzc	INC Abs,X		00	7f	01
12	0144	7f	0		0009	20	33	7f	fd	nv-BdIzc	INC Abs,X		00	7f	80
13	0144	7f	0		0009	20	33	7f	fd	Nv-BdIzc	INC Abs,X	T0	80	80	80
13	0144	80	0		0009	20	33	7f	fd	Nv-BdIzc	INC Abs,X	T0	80	80	00
14	0009	ea	1	NOP	0009	20	33	7f	fd	Nv-BdIzc	INC Abs,X	T1	ff	80	00
14	0009	ea	1	NOP	0009	20	33	7f	fd	Nv-BdIzc	INC Abs,X	T1	ff	80	7f
15	000a	15	1		000a	20	33	7f	fd	Nv-BdIzc	NOP	T0+T2	7f	7f	7f
15	000a	15	1		000a	20	33	7f	fd	Nv-BdIzc	NOP	T0+T2	7f	7f	fe

ISC (FF)	Cycles: 7	Size: 3
Absolute X-Indexed (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC	Store as Pointer.H.
3.1		Inc. PC . Target = Pointer + X . Address.L = Target.L. Address.H = Pointer.H.
3.2	Read Address	Store as Operand.
4.1		Address.H = Target.H.
4.2	Read Address	Store as Operand.
5.1		
5.2	Write to Address	Write Operand.
6.1		Operand += 1. Val = u16(Operand) ^ \$00FF. Result16 = u16(A) + Val + C flag. V flag = ((u16(A) ^ Result16) & (Val ^ Result16) & \$0080) != 0. A = u8(Result16). C flag = (Result16 > \$FF), N flag = (A & \$80), Z flag = ! A .
6.2	Write to Address	Write Operand.
7.1		
7.2	Read PC	Store as OpCode.
+X.1		* A , C flag, N flag, V flag, and Z flag applied.

* Setting of A/flags is delayed by 1 cycle until the start of the following instruction.

cycle	ab	db	rw	Fetch	pc	a	x	y	s	p	Execute	State	alua	alub	alu
7	0006	ff	1	unknown	0006	20	33	7f	fd	nv-BdIzc	LDX #	T1	33	33	fc
7	0006	ff	1	unknown	0006	20	33	7f	fd	nv-BdIzc	LDX #	T1	33	33	66
8	0007	11	1		0007	20	33	7f	fd	nv-BdIzc	unknown	T2	66	66	66
8	0007	11	1		0007	20	33	7f	fd	nv-BdIzc	unknown	T2	66	66	cc
9	0008	01	1		0008	20	33	7f	fd	nv-BdIzc	unknown	T3	33	11	cc
9	0008	01	1		0008	20	33	7f	fd	nv-BdIzc	unknown	T3	33	11	44
10	0144	7f	1		0009	20	33	7f	fd	nv-BdIzc	unknown	T4	00	01	44
10	0144	7f	1		0009	20	33	7f	fd	nv-BdIzc	unknown	T4	00	01	02
11	0144	7f	1		0009	20	33	7f	fd	nv-BdIzc	unknown	T5	02	ff	02
11	0144	7f	1		0009	20	33	7f	fd	nv-BdIzc	unknown	T5	02	ff	01
12	0144	7f	0		0009	20	33	7f	fd	nv-BdIzc	unknown		00	7f	01
12	0144	7f	0		0009	20	33	7f	fd	nv-BdIzc	unknown		00	7f	80
13	0144	7f	0		0009	20	33	7f	fd	Nv-BdIzc	unknown	T0	80	80	80
13	0144	80	0		0009	20	33	7f	fd	Nv-BdIzc	unknown	T0	80	80	00
14	0009	ea	1	NOP	0009	20	33	7f	fd	Nv-BdIzc	unknown	T1	20	7f	00
14	0009	ea	1	NOP	0009	20	33	7f	fd	Nv-BdIzc	unknown	T1	20	7f	9f
15	000a	15	1		000a	9f	33	7f	fd	NV-BdIzc	NOP	T0+T2	9f	9f	9f
15	000a	15	1		000a	9f	33	7f	fd	NV-BdIzc	NOP	T0+T2	9f	9f	3e